Contents

Recursive Information–Container Dynamics: A Multiscale Theory of Adaptive Systems

Computational Edition — RICS and FDFM Layers Computationally Instantiated

Editorial note on this revision.

This edition completes the transition from a domain-independent theoretical architecture to an implementable computational framework, subject to the adapter contract and the open mathematical limitations recorded at the end of this edition — a narrower and more accurate claim than "fully computable," which this document deliberately avoids for the reasons given there. Three earlier corrections are retained without change: the information-manifold scope conditions in Part 4, the corrected resource-debt equation in Part 7, and the disambiguated notation for the curvature tensor, production rate, and divergence field. Beyond those, every quantity in the FDFM layer (Part 6) and the RICS/recursive layer (Parts 5, 8, 10) that was previously an existence claim — "there exists a functional Φ/G/Ψ/Θ satisfying properties X" — has been given a concrete default formula, so that the document can be translated directly into code without a separate specification pass.

Where a companion document (a Feedback Divergence Field Model application to the U.S. justice system) had already worked out concrete, computable forms for the single-container layer, those forms are adopted here as the defaults, generalized to RICD's three-channel (official/operational/meta) representation structure and its more general recursive and network layers. A new Part 15 consolidates every concrete formula and the adapter contract a domain implementation must satisfy, as a single computational reference.

Part 12's theorems have been revisited in light of the new concrete couplings: 12.8, previously downgraded to a conjecture for lack of a specified dependence between network coherence and adaptive dimensionality, is restored to a derivable proposition now that §10.3 gives d_A an explicit formula depending on fragmentation. §10.5a gives RICD a single checkable collapse-onset condition, the direct generalization of the hull-breach condition from the companion FDFM document.

§10.5b has been corrected in this edition after further development of the underlying mechanism. The prior draft attributed the medium-entropy regime's two possible outcomes to a coherence/residual-divergence effect borrowed directly from the high-entropy regime's mechanism. This produced an internal contradiction: the medium-entropy regime's own definition requires D_om to stay elevated, which the borrowed mechanism would have read as low coherence precisely in the case that should produce the good outcome.

The corrected account separates the two regimes' mechanisms properly: the high-entropy regime (Flip Condition A) retains the coherence/dissipation account, now with two named causal routes (A1, self-propelled; A2, externally triggered) converging, absent external redirection, on the same single, terminal-locked outcome (§10.5b.11a details the externally-redirected exception). The medium-entropy regime (Flip Condition B) instead has two genuinely distinct outcomes, determined by fluid-mass weighting rather than coherence: B1 (meta converges with the minority official layer) locks at 180°; B2 (meta converges with the majority operational layer) completes a full reorganization, since the same external multi-knock launch is carrying proportionally more of the container's mass.

New subsections on prevention-versus-post-flip asymmetry (the two regimes differ sharply in how foreseeable their tipping events are) and steadying-versus-one-time-reset (high-entropy containers require sustained containment, not a single corrective action) have also been added, along with a note on combining anchoring with active messaging change.

Correction to the previous review (retained).

The prior review flagged the codomain of the feedback permeability operator (§6.5, Definition 6.3; §11.3) as truncated in the source. On inspection of the original document, both instances are intact and correctly state $\Pi_f : \mathscr{P} \to [0,1]$; the truncation existed only in an earlier PDF-text-extraction pass, not in the source document. No correction to §6.5 or §11.3 was needed on that account.

On free parameters.

Every concrete formula introduced in this edition carries one or more coefficients (α, β, κ, λ, θ, and similar) that are not fixed by the theory and are not meant to be guessed. They are exactly the analog of a, b, c, d, e, r in the companion FDFM document — free parameters with priors, to be fit to real data through the Bayesian state-space layer once a system is chosen. Part 15 tags every such coefficient explicitly so the eventual Python implementation knows which quantities are structural (fixed by the formula) and which are inferred (fit per application).

Part 1. Introduction

Abstract

Adaptive systems—from single cells to software architectures, corporations, ecosystems, and nation-states—must continuously acquire, process, transform, and act upon information while operating under finite energetic and structural constraints. Existing mathematical frameworks typically emphasize one aspect of this process, such as dynamical systems, network topology, control theory, information theory, or stochastic processes, but rarely provide a unified formalism capable of describing how information, structure, resources, and feedback jointly govern adaptive behavior across multiple organizational scales.

This paper introduces Recursive Information–Container Dynamics (RICD), a general mathematical framework for describing adaptive systems as recursively embedded information-processing containers. Within this framework, every adaptive entity is modeled as a bounded container that receives information from its environment, transforms that information according to internal organizational constraints, exchanges resources across its boundaries, and recursively participates within higher-order containers. Information, energetic cost, structural organization, and network topology are therefore treated as complementary components of a single multiscale dynamical system rather than as independent modeling assumptions.

A central component of the framework is Feedback Divergence Field Mechanics (FDFM), which formalizes the internal informational dynamics of an adaptive container. FDFM describes how discrepancies between declared, operational, and self-modeling informational states evolve through recursive feedback, generating measurable quantities such as informational divergence, permeability, integrity, and accumulated informational stress. By treating FDFM as one layer within a broader recursive architecture, the framework separates internal informational dynamics from higher-level structural, energetic, and network processes while preserving their mathematical coupling.

Rather than postulating collapse as a threshold crossing or externally imposed state transition, RICD derives adaptive stability, degradation, and collapse from the evolution of recursive information flow under finite resource constraints. Stable adaptation emerges when recursive feedback maintains informational coherence across scales; instability arises when divergence, structural constraints, energetic depletion, and recursive memory jointly destabilize the system's attractor landscape.

The framework is intentionally domain-independent. The same mathematical objects describe biological organisms, software systems, financial institutions, educational organizations, distributed artificial intelligence, and other adaptive systems by varying only boundary definitions, resource mappings, observational operators, and recursive depth. This universality permits theoretical analysis while simultaneously supporting computational implementations such as forensic auditing systems, adaptive collapse trackers, cyber-defense architectures, organizational diagnostics, and multiscale simulation engines.

The remainder of this manuscript develops the framework axiomatically. Primitive mathe-

matical objects are introduced first, followed by the recursive operators that govern information exchange, energetic constraints, adaptive networks, and memory. Feedback Divergence Field Mechanics is then derived as the internal informational layer of the theory, after which measurable observables, stability conditions, and collapse dynamics are developed from the primitives and axioms rather than introduced as independent assumptions.

A note on "derived," used throughout this document at three distinct strengths. "Structurally derived" means a consequence follows from the primitives and axioms plus definitions alone, with no further choice involved. "Model-derived" means a consequence follows mathematically once a particular default functional form has been adopted — the form itself is a choice, even where the qualitative relationship it encodes is structurally motivated. "Empirically fitted" means a numerical parameter is inferred from data given an already-chosen functional form. Most of the concrete formulas in Parts 6–11 are model-derived, not structurally derived: they supply this document's default computational instantiation of a structurally-derived quantity, not the unique consequence forced by the axioms alone. Where this document says a quantity "is derived" without further qualification, structurally derived is meant; where a specific formula is given for that quantity, the formula itself is model-derived unless stated otherwise, and alternative functional forms sharing the same qualitative relationship generally remain admissible without contradicting anything structurally established.

1. Introduction

Adaptive systems exist because they continuously solve a common problem: maintaining organized behavior while processing uncertain information in changing environments. Whether the system is a biological organism responding to external stimuli, a corporation coordinating thousands of employees, a software platform managing distributed computation, or a governmental institution making policy decisions, its long-term persistence depends upon the recursive interaction between information processing, structural organization, energetic constraints, and environmental feedback.

Although numerous mathematical disciplines study portions of this problem, existing approaches typically isolate one level of organization. Dynamical systems theory characterizes trajectories in state space but generally assumes a fixed state representation. Network science models interactions among agents but often treats the informational content of those interactions as secondary. Information theory quantifies uncertainty and communication while remaining largely agnostic regarding organizational structure. Control theory emphasizes regulation and stability but generally presumes externally specified objectives and architectures. Statistical mechanics explains emergent collective behavior yet does not explicitly distinguish informational organization from energetic exchange.

Adaptive systems, however, require all of these processes simultaneously. Information must exist within organizational boundaries; boundaries constrain information processing; processing consumes resources; resource availability alters organizational structure; structural change modifies future information flow; and the resulting dynamics recursively propagate across multiple organizational scales. No single mathematical framework presently formalizes these relationships as different aspects of one recursive process.

Recursive Information–Container Dynamics (RICD) is proposed as a general mathematical framework intended to address this gap. Rather than beginning with predefined state variables or application-specific differential equations, the theory begins by defining the primitive mathematical objects shared by all adaptive systems. These objects describe information, containers, boundaries, resources, recursive composition, feedback, and memory independently of any specific domain. Observable quantities such as informational divergence, permeability, rigidity, fragmentation, resilience, and collapse are subsequently derived from the interactions among these primitive objects rather than postulated independently.

Within this framework, Feedback Divergence Field Mechanics (FDFM) occupies a specific theoretical role. FDFM is not synonymous with the complete theory. Instead, it describes the internal informational dynamics occurring within individual adaptive containers. It provides a mathematical description of how official representations, operational behavior, and recursive self-models evolve through feedback, producing measurable informational integrity and divergence that influence—but do not fully determine—the larger multiscale dynamics of the system.

This layered architecture offers two complementary advantages. Theoretically, it separates informational dynamics from energetic, structural, and network processes, allowing each to be developed independently before being mathematically coupled. Practically, it permits individual layers—particularly FDFM—to be implemented as standalone computational tools for applications such as organizational diagnostics, cybersecurity monitoring, software integrity auditing, and collapse prediction without requiring the full recursive formalism.

The objective of this paper is therefore not merely to construct another model of adaptive collapse, but to establish a mathematical language capable of describing adaptive systems across domains and organizational scales. The remaining sections introduce the primitive mathematical objects, state the foundational axioms, develop the recursive operators governing adaptive dynamics, derive FDFM as the internal informational layer of the framework, and demonstrate how observable quantities associated with adaptation, resilience, and collapse emerge naturally from these first principles.

Part 2. Primitive Mathematical Objects

Recursive Information–Container Dynamics (RICD) is founded upon a small collection of primitive mathematical objects from which all subsequent structures are derived. These primitives are not intended to represent measurable variables themselves; rather, they define the ontology of adaptive systems independently of any specific application domain. Observable quantities such as informational divergence, permeability, rigidity, adaptive capacity, energetic reserve, and collapse are introduced only after these foundational objects have been established.

The central design principle is that every adaptive system can be understood as a recursively organized collection of bounded information-processing entities interacting across multiple spatial, temporal, and organizational scales.

2.1 Information States

The fundamental object of the theory is an information state.

Definition 2.1 (Information State).

An information state is represented by a probability measure:

$P \in \mathscr{P}(\Omega)$

where:

- Ω denotes the system's state space, and
- $\mathscr{P}(\Omega)$ denotes the space of probability measures on that state space.

An information state therefore represents not a single condition of the system, but its entire probabilistic organization. This formulation accommodates deterministic systems as degenerate probability distributions while naturally extending to uncertain, stochastic, and partially observable systems.

Unlike classical state-space models, RICD treats uncertainty as fundamental rather than incidental. Every adaptive system possesses incomplete information regarding itself, its environment, or both. Consequently, probability measures become primitive objects rather than secondary statistical descriptions.

2.2 Containers

Information cannot exist independently of organizational structure.

Definition 2.2 (Container).

A container is an ordered tuple:

$C = (\Omega, P, B, R, M)$

consisting of:

- an information state space Ω,
- an information state P,
- a boundary operator B,
- a resource state R,
- a memory operator M.

The container constitutes the minimal adaptive entity within the theory.

The term container is deliberately chosen rather than agent or node. A container is not assumed to possess agency, intentionality, or cognition. It is simply any bounded system capable of maintaining and updating information through interaction with its environment.

2.3 Boundaries

Containers are distinguished by their boundaries.

Definition 2.3 (Boundary Operator).

A boundary operator:

$B : C \to \partial C$

defines the permissible exchange of information, resources, and influence between a container and its environment. Importantly, the boundary is not merely geometric. It is an operator that determines admissible interactions. Two containers occupying identical physical space may possess entirely different informational boundaries.

Boundary operators therefore become dynamic mathematical objects rather than fixed spatial partitions. Their evolution will later determine permeability, organizational restructuring, and adaptive coupling.

2.4 Resources

Adaptive computation incurs finite cost.

Definition 2.4 (Resource Measure).

Each container possesses a nonnegative resource measure:

$R : C \to \mathbb{R}_{\geq 0}$

representing the available capacity for information processing, structural maintenance, adaptation, and environmental interaction.

No assumption is made regarding the particular conservation law governing the resource.

2.5 Memory

Adaptive systems possess history.

Definition 2.5 (Memory Operator).

Each container possesses a memory operator:

$M_t : H_t \to \mathscr{P}(\Omega)$,

where:

$H_t = \{P(\mathrm{s}) : 0 \leq \mathrm{s} \leq \mathrm{t}\}$

denotes the complete historical trajectory of the information state.

Unlike Markovian systems, future evolution depends not only upon the present state, but upon the historical trajectory encoded by M_t. Memory therefore constitutes a primitive mathematical object rather than an auxiliary delay term.

2.6 Environment

No adaptive system exists in isolation.

Definition 2.6 (Environment).

The environment of a container C_i is defined as:

$E_i = \mathscr{C} \setminus C_i$,

where $\mathscr{C}$ denotes the collection of all containers.

The environment therefore consists of every interaction external to the container's own internal organization.

2.7 Recursive Embedding

Recursive organization is the defining characteristic of complex adaptive systems.

Definition 2.7 (Recursive Embedding).

Given containers $C_1, \ldots, C_n$, there exists an aggregation operator:

$A : \{C_1, \ldots, C_n\} \to C^*$

mapping collections of lower-level containers into higher-order containers.

The resulting container C* inherits its own information state, boundary, resource state, and memory, while simultaneously containing the original lower-level systems. Recursion therefore generates organizational hierarchy without requiring distinct mathematical rules at each scale.

2.8 Observation

Observable variables are not primitive. They arise through observation.

Definition 2.8 (Observation Operator).

An observation operator:

$O : C \to X$

maps internal container states onto measurable quantities.

Different observers may employ different observation operators while referring to the same underlying adaptive system.

2.9 Principle of Ontological Economy

The framework deliberately minimizes primitive assumptions. Only six primitive mathematical objects are assumed:

- information states;
- containers;
- boundary operators;
- resource measures;
- memory operators;
- recursive aggregation.

All remaining quantities—including informational divergence, permeability, rigidity, fragmentation, adaptive capacity, resilience, collapse, and Feedback Divergence Field Mechanics itself—will be derived from these primitives in subsequent sections.

Part 3. Foundational Axioms

The primitive objects introduced in the previous section define the mathematical universe of Recursive Information–Container Dynamics (RICD). The present section specifies the fundamental principles governing those objects.

Axiom 1. Information Requires a Container.

$\forall P, \exists C$ such that $P \in C$.

Information does not float freely through the theory; every probability distribution belongs to some adaptive organization. There is no such thing as a disembodied adaptive information state.

Axiom 2. Every Container Possesses Finite Capacity.

$0 \leq R(C) < \infty$.

Resource capacity is always finite. Every adaptive decision incurs opportunity cost.

Axiom 3. Information Processing Has Nonzero Cost.

Let $P_t \to P_{(t+\delta t)}$ denote an informational update. Then $\mathrm{Cost}(P_t, P_{(t+\delta t)}) > 0$ whenever $P_t \neq P_{(t+\delta t)}$.

Information dynamics and energetic dynamics cannot be treated independently.

Axiom 4. Boundaries Regulate Exchange.

No interaction occurs independently of boundary operators. For every interaction $\mathcal{J} : C_i \to C_j$, there exists a boundary operator B such that:

$\mathcal{J} = B(\mathcal{J})$.

Boundary evolution becomes a primary mechanism of adaptation.

Axiom 5. Adaptive Systems Are Recursive.

Every adaptive container may itself become a component of a larger adaptive container:

$A : \{C_1, \ldots, C_n\} \to C$.

The resulting container C satisfies the same axioms as every lower-level container. One mathematical language describes every organizational scale.

Axiom 6. Memory Influences Future Dynamics.

Adaptive systems are generally non-Markovian:

$P(t + \delta t) = F(H_t)$,

where $H_t = \{P(s) : 0 \leq s \leq t\}$. Path dependence becomes fundamental, not exceptional.

Axiom 7. Adaptive Stability Is Dynamic.

Adaptive systems seek bounded trajectories maintaining organizational integrity despite continual perturbation, not static equilibrium. Let X(t) denote the complete state of a container. Adaptive stability requires:

$X(\mathrm{t}) \in \mathcal{S}$

for some invariant stability manifold $\mathcal{S}$, rather than convergence toward a fixed equilibrium point. Collapse should later be defined as loss of stable trajectory, not departure from equilibrium.

Axiom 8. Observable Quantities Are Emergent.

For any observable quantity Y, there exists an observation operator O such that:

$Y = O(C)$.

Axiom 9. Health-Seeking Is the Default Orientation.

Every container's default orientation is toward negentropic alignment — accurate self-representation, coupling that exchanges rather than merely extracts, and the seeking of connection with other containers in its network — not toward entropic alignment. Entropic alignment is not a baseline condition requiring no explanation; it is itself the outcome requiring explanation, produced either by developmental corruption (§10.5b.2's genesis-branching) or, alternatively, by structural fault present from a container's own formation without corruption acting on a prior healthy state (§10.5b.1.1's Type A). This is the axiom's full structural content, and it is a legitimate theoretical commitment rather than an empirical claim: it establishes what a container is oriented toward by default and what an entropic outcome requires explaining, without asserting how often either pathway actually occurs in practice.

Empirical default hypothesis, stated separately. In the domains presently motivating RICD, entropic organization is provisionally modeled as more often arising through corruption of a prior negentropic trajectory than through entropic structure present at formation — across structural, developmental, and conceptual cases alike, a container corrupted from an original negentropic orientation is treated as the more typical route to entropic status, with a container entropic from inception, with no corrupted prior state to point to, treated as the less common case. This prevalence claim is empirical, not derivable from Axiom 9 itself, and may be rejected or reversed by domain-specific evidence without the axiom's structural content being affected either way; it is a modeling default this document adopts, not a mathematical consequence of the axiom.

This does not mean entropic containers cease seeking connection. They do not: the drive toward coupling is retained even where alignment is corrupted, and what changes is only what that coupling becomes capable of — ranging from parasitic (extractive-displacing, §8.6a) to predatory (destructive, §10.5b.9d) rather than reciprocal. A container that has stopped seeking connection entirely (§8.6a.6, generalized in §10.5b.2.7 below) is in a genuine deviation from this axiom's default, not expressing it; dormancy is a real, sometimes necessary state, but it is not what containers do by default, at any entropy level.

The Principle of Recursive Universality.

Any adaptive system can be represented as a recursively embedded collection of bounded information-processing containers whose evolution is governed by finite resources, boundary-mediated exchange, historical memory, and recursive organization.

Representation failure condition. This principle is a conditional claim, not a tautology, and is falsifiable in the following specific sense: a system is outside RICD's current scope if no admissible mapping can simultaneously supply all of the following — a bounded, information-bearing entity; a state representation P over some outcome space Ω; a meaningful boundary-mediated exchange relation with its surroundings; finite or normalizable resource capacity; a history representation (Definition 2.5’s memory operator); and, wherever multiscale modeling is claimed for that system, a recursion or aggregation relation connecting its levels (§5.2). A domain need not resemble any example already given in this document, and boundaries, resource mappings, observation operators, and recursion depth are all free for a domain adapter to choose — but a domain for which no admissible choice of these can jointly satisfy every item above is a genuine failure of representation, not merely an unusual application. Without this condition, “container” risks being broad enough that anything can always be called one, which would make the universality principle unfalsifiable rather than genuinely general.

Parts 4–6: Information Geometry, Recursive Containers, Feedback Divergence Field Mechanics

Editorial note for this segment.

This segment incorporates the following revisions agreed in review. (1) Information velocity and information acceleration are renamed from V(t)/A(t) to v(t)/a(t) (Part 4), freeing the uppercase symbol V for the verticality operator introduced later (Part 10) and avoiding collision with the aggregation operator A (Part 2/5). (2) A new §4.10 formalizes the history-dependent metric g_t, confirming that Part 4's velocity, acceleration, curvature, and geodesic constructions genuinely generalize to a time-varying metric rather than treating history-dependence as informal. (3) The feedback operator of Part 6 is renamed from F to $\mathscr{F}$, avoiding collision with Axiom 6's bare F and Part 8's F_i, and is given a full computational default: a stochastic gradient flow on a divergence landscape, replacing the previous unspecified-operator treatment. (4) §5.2, §5.5, and §5.6 incorporate the confirmed refinements to the aggregation operator, boundary-nesting relation, and resource-flow normalization. All other content in these three parts is unchanged from the prior draft.

Part 4. Information Geometry

The primitive object introduced in Part 2 was the information state, represented as a probability measure over a state space. The present section equips the collection of information states with mathematical structure.

Rather than representing adaptive systems by vectors in Euclidean state space, Recursive Information–Container Dynamics models adaptive organization as evolution on a manifold of probability distributions. This choice reflects the fact that adaptive systems rarely possess complete certainty regarding either their internal organization or their environment. Their dynamics therefore consist not only of changes in state, but of changes in belief, uncertainty, expectation, and informational organization.

4.1 Information Manifold

Let Ω denote the measurable state space associated with a container, and $\mathscr{P}(\Omega)$ the space of probability measures on Ω.

Definition 4.1 (Information Manifold).

An information manifold is a differentiable manifold $\mathscr{M} = (\mathscr{P}(\Omega), g)$, where g is a Riemannian metric defined on the tangent space of probability distributions.

Scope condition.

$\mathscr{P}(\Omega)$ does not automatically carry a differentiable structure for arbitrary Ω. Definition 4.1 applies under one of the following restrictions, which must be specified per application:

- Ω is finite or countable and $\mathscr{P}(\Omega)$ is restricted to distributions > with full or fixed support, so that $\mathscr{P}(\Omega)$ is realized as an open > simplex — a finite-dimensional differentiable manifold in the > ordinary sense;
- the container's admissible states are restricted to a > finite-dimensional parametric family $\{P\_\theta : \theta \in \Theta \subseteq \mathbb{R}^n\}$ (e.g., an > exponential family), so that $\mathscr{M}$ is the parameter manifold Θ pulled > back through $\theta \mapsto P\_\theta$, with g the induced Fisher information > metric; or
- Ω is infinite-dimensional and $\mathscr{P}(\Omega)$ is treated as a Fréchet manifold, > in which case the differentiability, completeness, and curvature > constructions of §4.4–§4.6 require the additional analytic > hypotheses standard to that setting (e.g., a fixed reference > measure and an L^2 or Orlicz-type tangent space) and are not > assumed to hold without further argument.

Every subsequent construction in this section — velocity, acceleration, curvature, and geodesics — presupposes one of these restrictions has been fixed for the container in question. Each point on the manifold corresponds to an entire informational organization rather than to a single deterministic configuration. Movement on the manifold represents learning, adaptation, forgetting, reorganization, or environmental change.

4.2 Informational Distance

To compare informational organizations, the manifold requires a notion of distance. Rather than prescribing one universal metric, Recursive Information–Container Dynamics introduces an abstract informational distance functional.

Definition 4.2 (Informational Distance Functional).

An informational distance is a mapping $D : \mathscr{P}(\Omega) \times \mathscr{P}(\Omega) \to \mathbb{R}_{\geq 0}$ satisfying:

- Non-negativity: $D(P, Q) \geq 0$.
- Identity: $D(P, P) = 0$.
- Continuity: small perturbations in probability distributions produce > correspondingly small changes in informational distance.
- Monotonicity under information loss: coarse-graining or irreversible > compression cannot increase distinguishability.

Depending upon the application, suitable choices include Fisher–Rao geodesic distance, Jensen–Shannon divergence, Wasserstein distance, Hellinger distance, total variation distance, and symmetrized Kullback–Leibler divergence.

Relationship between D and g.

These choices are not interchangeable once the manifold structure of §4.1 is put to use. The scalar quantities of Part 6 (pairwise divergence, informational integrity, informational stress) require only an admissible D in the sense of Definition 4.2, and any of the six listed functionals may be used there. The Riemannian constructions of §4.4–§4.6 (velocity norm, covariant acceleration, curvature, geodesics), by contrast, are defined with respect to the specific metric g of Definition 4.1, and only Fisher–Rao distance is the geodesic distance generated by that metric on a statistical manifold; the induced local quadratic approximation of Fisher–Rao (equivalently, one-quarter the squared Hellinger distance in the appropriate limit) is the metric g used throughout §4.4–§4.6 unless otherwise stated. Wasserstein distance corresponds to a different Riemannian structure (the Otto metric on the space of measures) with its own, distinct geodesics and curvature, and total variation and symmetrized KL are not themselves Riemannian distances at all — KL is not symmetric or a metric, and TV is not induced by a Riemannian metric in the relevant sense. An application choosing Wasserstein or TV as its working D must either restrict Part 4's differential-geometric machinery to the Fisher–Rao case for a separate purpose, or explicitly substitute the Otto geometry (with its own geodesic and curvature definitions) throughout §4.4–§4.6; the theory does not yield curvature or geodesics under TV or plain KL. This constraint should be stated explicitly in any application of the framework, since it is easy to read Definition 4.2 as offering six freely interchangeable options when in fact only one is compatible with the differential geometry built on top of it.

4.3 Information Velocity

Adaptive systems do not merely occupy informational states. They move through informational space.

Definition 4.3 (Information Velocity).

Let $P(t)$ be a trajectory on the information manifold. The instantaneous information velocity is:

$v(t) = dP/dt$.

Its norm, $\|v(t)\|_g$, measures the instantaneous rate of informational reorganization.

High informational velocity may represent rapid learning, crisis response, organizational restructuring, software patch deployment, or emergency adaptation. Low informational velocity represents informational stability. High velocity is neither inherently beneficial nor detrimental; its effect depends upon subsequent resource availability and structural organization.

Editorial note: *Information velocity is denoted with lowercase v(t) throughout this document, reserving uppercase V(t) for the verticality operator introduced in §10.5b.0. Earlier drafts used V(t) for both; the two are unrelated quantities and needed distinct notation.*

4.4 Information Acceleration

Adaptive systems may themselves change the rate at which they reorganize.

Definition 4.4 (Information Acceleration).

Information acceleration is $a(t) = \nabla_t v(t)$, where ∇_t denotes the covariant derivative induced by the manifold geometry g of Definition 4.1 — or by g_t, if the container's history-dependent geometry (§4.10) is in effect.

Acceleration measures changes in adaptive responsiveness rather than changes in state alone.

4.5 Information Curvature

Adaptive landscapes possess geometry. Some informational regions are smooth. Others contain ridges, valleys, or unstable transition surfaces.

Definition 4.5 (Information Curvature).

The curvature tensor associated with the information manifold, denoted $\mathscr{R}_\mathscr{M}$ (distinguished here from the resource measure R of §2.4), characterizes local geometric distortion under the metric g fixed in §4.1–§4.2, or under g_t where §4.10 applies. Regions of high curvature correspond to informational organizations in which small perturbations may produce disproportionately large adaptive consequences.

Examples include tipping points, institutional crises, software instability, ecological thresholds, and cascading failures. Curvature therefore provides a geometric precursor to collapse before collapse itself is formally defined.

4.6 Information Geodesics

Adaptive systems generally attempt to reorganize efficiently.

Definition 4.6 (Information Geodesics).

A geodesic γ(t) is a trajectory satisfying $\nabla_{\dot{\gamma}} \dot{\gamma} = 0$ with respect to g (or g_t, §4.10). Geodesics represent locally minimal informational reorganization subject to the manifold geometry.

Geodesics should not be interpreted as optimal behavior. Rather, they describe informational trajectories requiring minimal intrinsic restructuring. External constraints introduced later may prevent systems from following geodesic paths.

4.7 Information Fields

Individual containers rarely evolve independently. The information state therefore becomes a field over the collection of containers.

Let $\mathscr{C} = \{C_i\}$ denote the recursive collection of containers. Define $P_i(t)$ as the informational state associated with container C_i. The resulting assignment $C_i \mapsto P_i(t)$ constitutes an information field distributed across the recursive architecture. Local informational change therefore possesses both spatial and recursive structure. This observation motivates the later development of Feedback Divergence Field Mechanics.

4.8 Information Coherence

Not every collection of informational states forms a coherent organization.

Definition 4.7 (Information Coherence).

Given a collection of containers $\{C_1, \ldots, C_n\}$, define a coherence functional:

$\mathscr{C}_{info} = \Gamma_{coh}(P_1, \ldots, P_n)$, where $0 \leq \mathscr{C}_{info} \leq 1$.

High coherence indicates mutually compatible informational organization across containers; low coherence indicates fragmentation or contradiction. The precise form of Γ_{coh} is intentionally left unspecified at this stage and will later be constrained by the recursive coupling operators developed in subsequent sections.

4.9 Why Geometry Comes Before Dynamics

A common approach in adaptive modeling is to begin with differential equations and subsequently define quantities such as divergence or stability. Recursive Information–Container Dynamics adopts the opposite approach: the geometry of informational organization determines which trajectories, divergences, and transformations are mathematically meaningful. Only after the manifold structure has been established — and, per §4.1–§4.2, only within the scope conditions and single-metric commitment specified there — can the dynamical equations governing adaptation be derived.

4.10 History-Dependent Metric and Time-Varying Geometry

The constructions of §4.1–§4.6 fix a single metric g and build velocity, acceleration, curvature, and geodesics on top of it. But a container's accumulated history can genuinely change the geometry it experiences going forward: a system that has survived a given kind of disturbance before does not experience a repetition of that disturbance the same way a naive system would. This is not a claim about trajectories moving differently through a fixed landscape — it is a claim that the landscape itself reshapes as a function of what has already happened.

Definition 4.8 (History-Dependent Metric).

Let $M_{C,t}$ denote container C’s own memory operator value at time t (Definition 2.5, instantiated concretely in §9.3). The history-dependent metric is container-specific, not a single metric shared globally across the manifold:

$g_{C,t} = g(M_{C,t})$,

a Riemannian metric on $\mathscr{P}(\Omega)$ that is itself a functional of accumulated history rather than fixed once and for all, evaluated per container rather than once for the whole space. Where context makes the container unambiguous, the subscript is abbreviated to g_t as elsewhere in this document; the full form is given here to remove any reading of g_t as a single global metric applied identically to every container. All constructions in §4.3–§4.6 — velocity norms, covariant acceleration, curvature, and geodesics — generalize by replacing the static g with g_t throughout, provided g_t remains, at each fixed t, a valid Riemannian metric satisfying the scope conditions of §4.1.

This is a substantive strengthening, not a notational one. Under a static metric, the covariant derivative ∇_t in Definition 4.4 depends on t only through the trajectory P(t) itself; under g_t, the connection coefficients (Christoffel symbols) generating ∇_t are themselves time-dependent, since they are built from a metric that is changing. Two containers occupying identical instantaneous informational states P can therefore have different acceleration, different curvature, and different geodesics at that same state, if their histories M_t differ — which is precisely the behavior needed to support the claim, developed further in §9.10 and §10.5b, that a container's identity is constituted by the continuity of its history-dependent organization rather than by any single instantaneous configuration.

Scope condition.

Definition 4.8 requires that g(M_t) remain positive-definite and smooth in P for every fixed t along the trajectory under consideration; the map M_t ↦ g(M_t) is otherwise unconstrained by the theory and must be specified per application (a default is given where the memory kernel is instantiated, §9.3). Where this scope condition is not separately verified, the constructions of §4.3–§4.6 should be read as applying to the static-metric case only, with g_t retained as a directional, qualitative refinement rather than a load-bearing quantitative one.

This resolves what would otherwise be a genuine tension in the framework: §10.5b's account of gradual regime transition — a container's internal composition shifting toward healthier organization over a long history of sustained steadying, or conversely toward entrenchment over a long history of unaddressed extraction — requires exactly this kind of history-dependent reshaping of what counts as a small versus large informational move. A container with a long history of successful correction and a container with a long history of unaddressed divergence should not be judged by the same geometric yardstick even when momentarily in the same state, and g_t is what allows the theory to say so formally rather than narratively.

4.11 Structural Grain

Definition 4.11.1 (Structural Grain).

For container C, define its structural grain as a trailing, coupling-weighted average of the unit vectors already associated with its own past strikes and couplings:

$\mathbf{g}$_C(t) = normalize[$\int_0^t$ K(t,s) · Σ_k Λ_kC(s)·$\hat{\mathbf{u}}$_kC(s) ds],

using §6.7's existing memory kernel K(t,s) and §8.8a's existing per-edge unit vectors û_kj, aggregated over C's own coupling history rather than over a single simultaneous multi-attacker event as Drive_B does. Where a container has no coupling history (the trailing integral is zero), g_C(t) is undefined and the directional modifier below defaults to neutral (χ_dir = 1): a container without an established grain has no basis for anything to be measured as orthogonal or aligned against, and is treated as direction-agnostic until it accumulates one.

Definition 4.11.2 (Directional Force Modifier).

For a Drive_B(j,t) event (§8.8a) with net push direction $\hat{\mathbf{u}}$_net(j,t) = [Σ_k Λ_kj·$\hat{\mathbf{u}}$_kj] / |Σ_k Λ_kj·$\hat{\mathbf{u}}$_kj| — the normalized direction already implicit in Drive_B's own magnitude formula

— with the convention that where $|\Sigma_k \Lambda_{kj} \cdot \hat{\mathbf{u}}_{kj}| = 0$ (simultaneous forces exactly cancel, leaving no net directional push), $\hat{\mathbf{u}}_{net}(j,t)$ is undefined and $\chi_{dir}(j,t)$ defaults to 1, the same neutral convention already adopted for $\mathbf{g}_C(t)$'s own zero-history case in Definition 4.11.1: no net push means nothing for a grain to be orthogonal or aligned against, and Drive_B(j,t)'s own magnitude in this case is separately already zero, so the event contributes no tipping momentum regardless. Otherwise, define the alignment between the incoming push and the container's own grain,

$$\mathrm{align}(j,t) = |\mathbf{g}_j(t) \cdot \hat{\mathbf{u}}_{net}(j,t)| \in [0,1],$$

and a directional force modifier,

$$\chi_{dir}(j,t) = 1 + \kappa_{dir} \cdot (1 - \mathrm{align}(j,t)),\ \kappa_{dir} \geq 0 \text{ fitted.}$$

$\chi_{dir}(j,t) \to 1$ as the incoming push aligns with the container's established grain (in either direction along it, since alignment is measured by absolute value); $\chi_{dir}(j,t) \to 1 + \kappa_{dir}$ as the push approaches orthogonality to that grain, regardless of the push's own absolute direction. This is the formal content of the pancaking claim: what matters is the angle between shock and grain, not which way either one points in absolute terms.

Editorial note: This construction is deliberately scoped to Drive_B's existing directional data and does not touch V(t), W(t), or any mechanism that treats them as scalars elsewhere in this document. §4.12 states plainly why this cheaper path was taken instead of a full directional geometry, and what a fuller construction would actually require if it is ever warranted.

4.12 Why the Cheap Path, and the Cost of the Full One

Why this construction was affordable where a full directional geometry was not. §4.11's grain-and-modifier construction was cheap for a specific, structural reason, not merely because less effort was spent on it: a natural source of directional data already existed in the document for an unrelated purpose — Drive_B's $\hat{u}_{kj}$ unit vectors, used in §8.8a to combine multiple simultaneous attackers into a single push magnitude. Building $g_C(t)$ and $\chi_{dir}(t)$ required no new geometric primitive, only aggregating data already being collected. Just as importantly, the application was scoped to exactly two formulas, M_tip(B1) and M_tip(B2) (§10.5b.7) — nothing about V(t), W(t), Axiom C3 or C4, the entropy-regime classification, or any divergence formula elsewhere in this document was touched. Setting $\kappa_{dir} = 0$ recovers the original undirected mathematics exactly, which is what makes this an addition rather than a revision: everything built before this addendum continues to hold unchanged.

What a full directional geometry would actually require. A reader considering whether to build the fuller version — separate metrics for verticality and horizontality, or any construction giving V(t)/W(t) themselves directional structure rather than modulating one downstream momentum calculation — should expect the following, not a contained addition:

- **Axioms C3 and C4** (§10.5b.0) are stated in scalar terms — verticality crossing a threshold, horizontality approaching its maximum as $\Pi_f \to 1$. Both would need restatement for a directional V(t), and any downstream result derived from their scalar form (Definition 10.5b.0.6's Ultimate Convergence, Definition 10.5b.0.8's sustained-state tipping-immunity check, stated as the scalar equality $V(t) = h_{min}$) would need re-derivation, not just re-notation.
- **The entropy-regime classification itself** (§10.5b.1) runs on scalar variance, $\sigma_o(t)$ and $\sigma_p(t)$. A directional account would need to specify what a "direction" of variance means for a probability distribution over Ω, which is not a small extension — it is a new question the current framework's outcome-space ontology does not currently ask.

- **Every scalar divergence comparison downstream** — F_boom and F_demand (§10.5b.16.1, §10.5b.18.8) both compare scalar D_op or D_om values directly against thresholds. Since D_ij is Jensen–Shannon divergence between distributions with no vector structure (§6.3), giving these comparisons directional meaning would mean redefining what divergence itself carries throughout Part 6, not adding a modifier term to Part 10 formulas that already consume a scalar D_ij as given.
- **Ultimately, Part 2's primitive objects.** Any of the above pursued fully eventually reaches back to how a container's basic informational state is represented at all, the deepest layer this document has — the same layer-risk reasoning §4.11 raised before the cheap alternative was found, restated here because the fuller construction, if built, would not be exempt from it.

When it would actually be worth doing. Three concrete conditions, not a general sense that more precision is always better:

1. Real data shows the directional effect matters somewhere §4.11's cheap version does not currently reach — Flip Condition A, pre-flip disintegration (§10.5b.9d, which deliberately does not consume χ_dir, since overload is a volume claim rather than a directional one), or the forcing terms F_boom and F_demand.
2. V(t) or W(t) themselves need to carry structure beyond what modulating a single downstream momentum calculation can express — for instance, a claim that a container's *degree* of rigidity, not merely its tipping-momentum accumulation rate, depends on shock direction.
3. Cross-container grain comparison becomes theoretically necessary — comparing two containers' structural grains against each other, rather than one container's incoming shock against its own grain, which is all §4.11 currently supports.

None of these currently hold. This document already has one documented instance of a structurally similar fuller move — an earlier 90° inversion operator for tipping outcomes — being attempted and abandoned as redundant with machinery that already determined the same result without it (§10.5b.9's mass-weighted resolution). That precedent does not guarantee a full directional geometry would meet the same fate, but it is a real prior data point against assuming the investment would obviously pay for itself, and is offered here as caution rather than discouragement.

Part 5. Recursive Containers

Adaptive systems exist within nested organizational structures. Cells compose tissues, tissues compose organs, organs compose organisms, organisms compose populations, and populations compose ecosystems. Similarly, software modules compose applications, applications compose distributed services, and services compose technological infrastructures.

Recursive Information–Container Dynamics treats this nested organization as a mathematical property rather than a modeling convenience. Every container may simultaneously function as an autonomous adaptive system, a component of a larger adaptive system, and an environment for lower-level adaptive systems. The same mathematical rules apply at every level of recursion.

5.1 Recursive Levels

Let $C^{(k)}$ denote the collection of containers at recursion level k. Examples: neurons → neural assemblies → cortical regions → whole brain; or programmers → software teams → engineering divisions → corporation. No level possesses privileged mathematical status.

5.2 Aggregation Operator

Higher-order containers emerge through aggregation.

Definition 5.1.

$A_k : C^{(k)} \to C^{(k+1)}$.

For $\{C_1, \ldots, C_n\}$, $C^* = A_k(C_1, \ldots, C_n)$.

C^* is not equal to the sum of the C_i. Aggregation is not simple addition; the resulting container possesses new organizational structure.

Aggregation-weight functional. Where §5.2's aggregation requires weighting children (e.g., in $\sigma_i(P_i)$ pooling), the weight w_i is not necessarily raw R_i itself: define $w_i = W_A(R_i, \ldots)$, an aggregation-weight functional defaulting to R_i's §5.6-normalized form ($N_{R,i}$) where children's resources are commensurable, but left open for a domain to supply its own sensible weighting where they are not. Raw, unnormalized R_i should not be assumed comparable across heterogeneous children by default.

Universal architecture versus domain-specific metric. Leaving W_A open means the recursive aggregation *architecture* — that a parent's properties are computed by weighting and combining children's, per Definition 5.2 — is universal across domains, while the specific weighting *metric* used to instantiate W_A is domain-specific and adapter-supplied. These are different claims, and only the former is what this document's universality claims assert: a hostile reading that treats an open weighting functional as evidence the aggregation law itself is merely a family of domain-specific laws conflates the architecture with one of its adapter-supplied instantiations.

Emergence Principle.

The aggregation operator generates new state variables that do not exist within individual components—institutional culture, organizational policy, software architecture, ecological niche, corporate governance, and social norms. These quantities belong to C^(k+1), not to any individual C^(k). Emergence therefore arises mathematically rather than metaphorically.

Computational instantiation of the aggregation operator A.

Child containers generally do not share a state space:

$\Omega_1 \neq \Omega_2 \neq \ldots$ in general.

For example, sibling containers C_a, C_b, and C_c may operate with distinct, incommensurable localized states. Consequently, the aggregation operator A cannot be defined as a direct pooling of the local probability distributions P_i. Instead, the aggregation operator is instantiated in two steps:

1. Commensuration. Each child container supplies a summary map

$\sigma\_i : \mathscr{P}(\Omega\_i) \to \Delta$^m

that projects its local outcome distribution onto a fixed, shared m-category severity scale common to the entire recursive system, such as a normalized baseline-deviation scale. Although the map σ_i is specified by the domain adapter for each individual container, its codomain Δ^m — the m-category probability simplex — must remain identical for every child of a given parent.

2. Resource-weighted pooling. Once the local distributions are commensurated and made comparable, the parent state is defined as a weighted mixture over the shared simplex, using the aggregation-weight functional w_i already specified above rather than raw resource fractions directly:

$P^* = \Sigma_i w_i \sigma_i(P_i)$, w_i = W_A(R_i, …), normalized so $\Sigma_i w_i = 1$.

Where children's resources are genuinely commensurable, W_A defaults to R_i's §5.6-normalized form (N_{R,i}); raw, unnormalized R_i is not used directly, consistent with the aggregation-weight functional already established — a concrete instantiation that reverted to plain R_i/Σ_j R_j here would silently undo that correction rather than apply it.

This default instantiation of A constitutes a genuine aggregation rather than a simple sum of the component distributions. This satisfies the Emergence Principle in the weak sense of producing a new distribution that is not identical to any single component. Fully emergent structure, such as correlations induced by cross-container coupling, requires the network dynamics of Part 8 also to feed into P*.

This default can be overridden by a domain-specific A_k where warranted, but P* must remain a genuine probability distribution over some stated Ω* for every subsequent RICD construction—D_ij, I, S(t), etc.—to apply to C* unchanged.

5.3 Projection Operator

Higher-level organization constrains lower-level behavior: Π_k : C^(k+1) → C^(k). Examples include corporate policy affecting employees, immune regulation affecting cells, operating systems constraining software processes, and constitutions constraining governmental agencies. Aggregation and projection form complementary recursive operators.

5.4 Recursive Consistency

Definition 5.2.

A recursive hierarchy satisfies consistency when $O(A(C_i)) \approx \Gamma(O(C_i))$, where Γ is an admissible aggregation rule defined on observations. Perfect equality is neither expected nor desirable; emergent systems necessarily generate new organizational information.

5.5 Boundary Nesting

Correction.

Sibling containers C_i are not subsets of the parent aggregate C^*. Set-containment nesting applies only to strict-subset recursion (e.g., localized sub-compartments nested inside a larger container that is itself one of the aggregated children). For aggregation-type recursion, the correct relation is compatibility, not containment: define $B_i \sqsubseteq B^*$ ("B_i is boundary-compatible with the parent") to mean every interaction admitted by B_i is also admissible under B^*'s exchange rules with C_i's environment — i.e., the parent's boundary operator does not forbid anything the child's boundary operator permits toward the parent. This does not imply identical permeability in either recursion type — a department may possess more permeable informational boundaries than the corporation that contains it, and a sibling container may possess more permeable boundaries than the aggregate.

5.6 Resource Flow Across Levels

Let $R^{(k)}$ denote resource allocation at level k, and T_R the recursive transfer operator: $R^{(k+1)} = T_R(R^{(k)})$. Higher-order systems redistribute resources downward through T_R^{-1}. Examples include budgets, nutrients, computational allocation, attention, labor, and bandwidth.

Normalization requirement.

$\Sigma_i R_i^{(k)}$ in the conservation principle below presupposes R_i is expressed in a common unit across all children — false in general for aggregation-type recursion, since individual containers may measure resources in entirely different physical or abstract units (throughput-hours, capacity-years, budget-credits). Before summing, each child's resource measure must pass through a normalization operator $N_{R,i} : \mathbb{R}_{\geq 0} \to [0,1]$, typically utilization-based —

$$\mathcal{N}_{R,i}(R_i) = R_i / R_i^{max},$$

i.e., fraction of that container's own maximum historical or designed capacity currently available — so that the conservation principle below is applied to normalized capacity shares, not raw incommensurable quantities.

Conservation Principle.

$$\Sigma_i \mathcal{N}_{R,i}(R_i^{(k)}) + L^{(k)} = R^{*(k+1)} \text{ (in normalized units)},$$

where $L^{(k)}$ represents transfer losses, including entropy, inefficiency, communication overhead, corruption, and leakage. Recursion naturally introduces dissipative effects. If a genuinely common unit exists for a given application (e.g., a single agency's sub-budgets, all in dollars), $N_{R,i}$ may be taken as the identity and the original unnormalized form is recovered as a special case.

5.7 Recursive Information Fields

Each recursion level supports its own information field P^(k). Together these form {P^(0), P^(1), …}. An employee, a department, and an entire corporation all possess legitimate information states; none is reducible to another.

5.8 Cross-Scale Coupling

Λ_kl : C^(k) → C^(l) describes how disturbances propagate between scales. Examples: a local software bug becoming a system outage, employee burnout becoming organizational decline, species extinction altering ecosystems. Λ_kl ≠ Λ_lk in general — upward and downward influence generally differ, so recursive dynamics are asymmetric.

5.9 Fixed Points of Recursion

A recursive fixed point satisfies A ∘ Π(C) = C. Such systems reproduce their own organizational structure across recursive scales — stable institutions, mature ecosystems, robust software architectures, healthy biological development. Collapse corresponds not merely to informational divergence, but to the destruction of recursive fixed-point structure.

5.10 The Recursive Principle

Adaptive complexity arises not from the existence of many interacting components, but from the continual bidirectional coupling between organizational levels through recursive aggregation, projection, and resource exchange. Information moves upward. Constraints move downward. Resources move in both directions. Memory propagates across levels.

Part 6. Feedback Divergence Field Mechanics (FDFM)

FDFM concerns only the evolution of informational representations within a single bounded container. It describes how an adaptive system constructs internal representations, compares those representations with observed reality, updates them through feedback, and accumulates informational stress when those updates become progressively impaired.

6.1 Internal Information State

Definition 6.1 (Representation Operators).

Define two projection operators O_o, O_p, producing $P_o = O_o(P)$, $P_p = O_p(P)$ — official representation and operational representation, respectively, both deterministic projections of the same underlying informational state P.

P_m, the meta-representation, is not a third projection of P. It is an emergent aggregate over the meta-representation layer's own population of sub-processes, each holding its own local view informed by partial exposure to P_o and P_p rather than direct access to P itself:

$P_m(t) = \Sigma_k \varphi_k(t) \cdot P_m^{(k)}(t)$,

where $P_m^{(k)}(t)$ is the k-th sub-process's own locally-held representation, and $\varphi_k(t)$ is that sub-process's share of the population's aggregate weight at time t. The population-level composition variables defined in §10.5b.18 — $\omega_{meta}^{up}(t)$ and $\omega_{meta}^{down}(t)$ (Definition 10.5b.18.2, 18.5), and $\Psi_{loyal}(t)$ (Definition 10.5b.18.12) — are summary statistics of exactly this $\varphi_k(t)$ weighting, restated at the level of behavioral category (transmitting or not, negentropic- or entropic-oriented) rather than at the level of individual sub-processes.

The meta-representation channel must be sourced independently at the level of data collection: as an emergent aggregate over its own population, gathered through its own dedicated observation process, rather than algebraically derived from the gap between P_o and P_p — so that D_{om} and D_{pm} can carry distinct diagnostic meaning in §6.3 and §10.5b, rather than being redundant with D_{op} by construction. Constructing P_m this way, as a population aggregate rather than a projection operator, gives this independence a form no projection operator could deliver on its own: an emergent aggregate is definitionally not an algebraic combination of the other two layers' outputs. $P_m(t)$, so constructed, remains a well-formed distribution over Ω, and D_{om}, D_{pm} remain ordinary applications of §6.3's divergence to P_m against P_o and P_p respectively.

The meta-representation layer is better understood as a barometer of systemic self-deception than as any kind of auditor: it is internal to the container by definition — what the container's own information state does when official and operational reality are compared to each other — and its reading is honest by degree, not by kind. A low reading (§10.5b.18.2's low-entropy baseline, $\omega\_meta^up(t) \approx 1$, the population thin enough that it is not functioning as a firewall at all) and a high reading (either non-baseline regime, a majority comfort- or demand-aligned population) are the same instrument reporting different underlying states, not a functioning instrument in one case and a broken one in the other. A closed channel should not be assumed corrupt by default: §10.5b.18's fuller account treats closure in both non-baseline regimes as more often a protective response — a population, entropic or negentropic in composition, shielding the container from a shock it is not yet positioned to absorb — than evidence of a captured or malfunctioning instrument, though a genuinely self-serving, entropic-aligned population remains possible and is not excluded by this reading. The full account of what determines a given channel's reading is regime-dependent and given in §10.5b.18 and its extension (§10.5b.18.5–18.31).

Definition 6.1.1 (External Governance Attestation). P_m is internal by construction — an aggregate over the container's own population, per the definition above — and everything else in this document that bears on it, whether fitted continuous features or the divergence measures built from P_m directly, is calibrated statistically, requiring enough real positive examples to earn a weight. A genuinely different class of evidence exists alongside these: a real, independent, external body's own confirmed finding that a container's governance or self-monitoring function has failed by its own established standards — academic freedom or shared-governance censure by a recognized professional body, for a concrete instance already in use in this document's own adapter work, but structurally general to any domain with an analogous independent standards body. Where an adapter supplies such an attestation, $\alpha\_gov(t) \in \{0,1\}$, it should not be treated as one continuous input among several requiring enough positive cases to be statistically validated the way $\delta R(t)$ or a domain-specific staffing ratio would be — the validation has already been performed, by the external body itself, applying its own established, independent standards, prior to and separately from anything this document's own machinery computes. This is analogous to G(t)'s reliance on an external, uncaptured party (§6.3.2) rather than internal agreement, but sharper: G(t) requires an ongoing accuracy comparison this document must itself construct and interpret; $\alpha\_gov(t) = 1$ is a completed, external, independent verdict, requiring no further statistical interpretation to be treated as strong evidence of genuine governance dysfunction. Its rarity in a given adapter's data (a low base rate is expected and appropriate, not a data quality problem) is precisely why it cannot and should not be fit as a weighted continuous feature; it should instead be reported directly, as confirmed evidence, wherever an adapter can supply it, regardless of how few or many other instances exist in the same dataset to compare it against.

Confirmed directly against real data: fitting versus overriding, tested both ways. A real institution's classifier misclassification, robust across bootstrap resampling (86%+ wrongly predicted regardless of training composition), was traced to a real, independent governance cause: its accreditor placed it on show-cause status and withdrew accreditation, the actual proximate cause of its eventual closure, not any financial metric. Tested first as an additional statistically-fit input alongside the continuous features already in use — exactly the treatment this definition says not to apply — and it produced a real improvement on the specific institution but a new, different misclassification elsewhere in the panel, consistent with what fitting a weight against only a handful of positive cases would be expected to do: shift the boundary, rather than add clean, independent information. Retested as a direct override per this definition's own terms — bypassing the statistical classifier entirely wherever the attestation is present — and it resolved cleanly, with no new misclassification introduced elsewhere. This is a genuine, if small-sample, empirical confirmation of the distinction this definition draws on theoretical grounds alone: a rare, real, external verdict behaves differently, and should be used differently, than a continuous proxy competing for statistical weight.

A conceptual note on what P_o and P_p actually are, not yet tested against data. The following reframes Definition 6.1 in plainer terms and draws one further definition from it; nothing here has been validated the way the results above were, and it should be read as an interpretive aid and a candidate hypothesis, not an additional confirmed finding. A container's actual boundary — the real extent of what P_p, the operational representation, is tracking — is not fixed by any label, charter, or org chart. It is constituted by wherever actor energy is actually, currently interfacing with a recognizable pattern: employees showing up, money changing hands, a customer returning, a reader engaging a text. Call that recognizable pattern the container's *energetic signature*. P_o, the official representation, is a claim about that signature — a name, a charter, a legal boundary, a board's attestation — but the claim and the thing it claims to describe are not the same object, and D_op (§6.1's divergence measure) is precisely the gap between them. Ordinary institutional decline, of the kind this document's adapters were built to track, is the signature drifting from the container's real activity gradually enough that the gap eventually surfaces in the container's own downstream numbers — which is exactly why an effect-layer adapter, watching only aggregate financial and enrollment outcomes, can still detect it without ever observing the underlying actors directly.

A sharper case is worth naming on its own: deliberate fraud, in this vocabulary, is the manufacture of a signature specifically engineered to have no real container behind it, built so that an outside observer mistakes the claim for the fact it purports to describe. A fabricated governing board — one an investigation later found to include members who did not exist and others who had never heard of the organization — is a clean, real-world instance: not an institution whose official and operational layers had drifted apart over time, but a signature constructed from the outset to simulate independent oversight where no corresponding actor-energy (people actually reviewing the organization's spending) was ever present at all. This suggests a real, testable asymmetry worth flagging for any future adapter attempting to detect it: ordinary drift tends to leak into a container's own real-world numbers eventually, which is what allows an effect-layer tracker to catch it without direct access to actors; a signature engineered to fake its own container is built specifically to suppress that leak, which is a plausible reason mechanism-layer evidence — the kind an effect-layer adapter cannot supply — becomes necessary rather than merely convenient once fraud, rather than ordinary institutional decline, is the actual phenomenon under investigation. Stated here as a hypothesis to test against real cases, not as something this project's data has yet confirmed.

6.2 Representation Consistency

If every representation accurately reflects the underlying information state, then P_o = P_p = P_m = P — maximal informational integrity. More generally, representation error is unavoidable; adaptive systems continuously approximate, rather than perfectly reproduce, their own informational organization.

6.3 Informational Divergence

Definition 6.2 (Pairwise Divergence).

For any two representations P_i, P_j, define D_ij = D(P_i, P_j), where D is an admissible informational distance from Definition 4.2 (any of the six listed functionals is valid here — this is a scalar use, not a Riemannian one; see §4.2). The pairwise divergences become D_op, D_om, D_pm.

These quantities measure different informational failures. Official-versus-operational divergence may reflect implementation failure, whereas operational-versus-meta divergence reflects inaccurate internal self-monitoring.

Computational instantiation (concrete default for D).

For discrete or discretized Ω — the normal case for real data — take D as symmetrized, additively smoothed Jensen–Shannon divergence:

$D(P_i, P_j) = ½·KL(P_i \| M) + ½·KL(P_j \| M)$, where $M = ½(P_i + P_j)$, using base-2 logarithms throughout, so that D is normalized to [0,1] rather than [0, ln 2] — stated explicitly because every later threshold and asymptotic statement in this document that references D approaching 1 ($D_{op} \to 1$, and similar) presumes the unit interval, and the maximum of Jensen–Shannon divergence otherwise depends on which logarithm base is used. An implementation using natural logarithms throughout will obtain the same qualitative dynamics but a different numerical ceiling (ln 2 ≈ 0.693 rather than 1), and should rescale accordingly rather than apply this document's stated thresholds unmodified against a differently-normalized D.

with each empirical distribution first smoothed as $\hat{P}(x) = (\text{count}(x) + \varepsilon) / (N + \varepsilon|\Omega|)$ for small $\varepsilon > 0$ (Laplace smoothing), which keeps every KL term finite even when a category is unobserved in one representation but not the other — a routine occurrence in sparse real-world data and a hard failure mode for unsmoothed KL. This single default replaces the six-option menu of §4.2 for every scalar (non-Riemannian) use of D in Parts 6, 7, and 11; an application may substitute Wasserstein or another admissible distance from §4.2 if its Ω is naturally continuous or ordinal, but JS-with-smoothing is the default assumed by every other formula in this Part unless stated otherwise.

6.3.1 Convergence Does Not Require Unimodality

Proposition 6.3.1 (Plural Convergence).

Let P_p genuinely hold substantial mass on two or more distinct, non-adjacent regions of Ω — a container's operational reality can be legitimately multi-valent, not merely uncertain between adjacent possibilities, where the underlying situation itself admits more than one simultaneously true characterization. $D(P_m, P_p)$, computed per §6.3's Jensen–Shannon instantiation, depends only on how well P_m's distribution matches P_p's, mode for mode and mass for mass; it does not penalize P_p or P_m for being multi-modal, and two multi-modal distributions that match each other closely produce as low a D_{pm} as two single-mode distributions that match each other closely. Convergence, so measured, is agreement between representations, not agreement on a single outcome.

Corollary 6.3.1.1 (Forced Flattening Increases Divergence).

Where P_m is constrained — by design, by adapter assumption, or by the meta-representation layer's own composition — to resolve toward a single dominant mode regardless of P_p's actual shape, and P_p is genuinely multi-modal, that constraint does not produce convergence; it produces a new source of divergence, since $D(P_m, P_p)$ between a flattened P_m and a genuinely plural P_p is higher than $D(P_m, P_p)$ would be between two matched plural distributions. A meta-representation layer that resolves internal contradiction by picking one true thing about the operational reality and discarding the other is not more converged than one that holds both; it is measurably less converged, by the same divergence measure this document already uses everywhere else.

Editorial note: This has a direct bearing on how "accurate self-representation" should be read throughout §10.5b.18's material and its extensions. Where a container's operational reality is honestly, irreducibly plural — both true and painful, both a source of harm and a source of understanding, both something to hold accountable and something to hold with compassion — a meta-representation that insists on collapsing that plurality into a single verdict is not achieving clarity. It is manufacturing a divergence this document's own formula would register as real, using the same D_{pm} that governs every other diagnostic in Part 10. Nothing about a container's health requires it to resolve every internal paradox into a single story; some of what a healthy container knows about itself is supposed to remain plural, and the mathematics already says so once the assumption of forced unimodality is removed.

6.3.2 Grounding: A Second Axis Distinct From Internal Agreement

Reason for this addition. D_op(t), D_om(t), and D_pm(t) measure agreement among a container's own representations — official, operational, meta — entirely internally. Low divergence on these measures has, throughout this document, been read as a signature of health. But internal agreement and accuracy are not the same property, and a container's own representations, however closely they agree with each other, supply no way to distinguish the two from inside the container itself: three mutually-reinforcing representations that have independently converged on the same inaccurate account of the container's own situation produce exactly the same low D_op, D_om, D_pm readings a genuinely healthy, accurate container would produce. Nothing built so far can tell these apart.

Definition 6.3.2 (Grounding).

Define G(t) ∈ [0,1], a container's grounding: how well P(t) tracks a reality independent of any representation the container itself generated, as opposed to how well the container's own representations agree with one another. Where an adapter can supply access to a genuinely independent external party k, uncaptured by the container under assessment and possessing real access to its true state — κ_access(k, C, t) ≈ 1, per §10.5b.18.9.1's observational-depth machinery, reused here rather than duplicated — grounding is read directly from that party's own accuracy-weighted assessment:

G(t) = A_acc(k, C, t) · [1 − D(P(t), P_k(t))],

where P_k(t) is k's own independent representation of C's state. Where no such external party is available, G(t) is undefined rather than assumed — there is no introspective substitute for it, and a container reporting high internal confidence about its own grounding is not evidence of grounding, since that confidence is generated by the same internal apparatus whose accuracy is in question.

Why D_op(t) low does not imply G(t) high. D_op, D_om, and D_pm are computed entirely from P_o, P_p, and P_m — three representations all ultimately traceable to the container's own P and its own meta-representation population. If P itself has drifted from the reality it is supposed to track, and all three representations have drifted together, internal agreement is preserved exactly as fully as it would be under genuine accuracy. A container can be perfectly, measurably converged by every internal diagnostic this document provides and simultaneously be almost entirely wrong about the reality it inhabits. G(t) is the only quantity in this document that can register that difference, and it is definitionally unavailable from inside the container producing the readings being checked.

Computed estimate versus true grounding. The formula above produces Ĝ(t), a computed grounding estimate, not G(t) itself. True grounding is latent unless the conditions the formula assumes — that k is genuinely independent, genuinely uncaptured, and genuinely informed — are themselves established, and those conditions are not automatically verifiable from within the assessment. Two observers who appear independent but in fact both inherit the same upstream misinformation will agree with each other and with the container being assessed, producing a high Ĝ(t) with no genuine external validation behind it; this is not a contradiction in the formula, it is a real identifiability limit on what any computed grounding estimate can guarantee. Ĝ(t) should be read accordingly: a genuine ceiling on confidence when k's independence is itself well-established (e.g., adapter-confirmed structural separation from the assessed container's own history and coupling network), and a much weaker signal otherwise — high $\hat{G}(t)$ computed against a k whose own independence is merely assumed does not license the same confidence as high $\hat{G}(t)$ computed against a k whose independence is actually confirmed.

6.4 Feedback Operator

Editorial note: *The feedback operator is denoted $\mathscr{F}$ throughout this document rather than F, to avoid collision with Axiom 6's generic evolution function F (Part 3, unchanged) and with the per-container intrinsic driving term F_i introduced in §8.5. This section also gives $\mathscr{F}$ a full computational default; previously it was specified only by the properties it should have, unlike every other quantity in this Part.*

Definition 6.3 (Feedback Operator).

Define the feedback operator $\mathscr{F} : P \to P$, incorporating environmental observations, recursive memory, internal prediction, and resource constraints, aimed at reducing expected future informational error. $\mathscr{F}$ need not decrease divergence at every update — temporary increases may occur during learning, organizational restructuring, crisis response, or innovation.

Computational instantiation (concrete default for $\mathscr{F}$).

Learning from experience is not a straight-line correction toward lower divergence. Real containers oscillate — including sustained periods moving in the wrong direction — and a nontrivial fraction never reach low-divergence organization at all. This behavior is captured by treating $\mathscr{F}$ not as a deterministic update rule but as a stochastic gradient flow on a divergence potential:

$$dP/dt = -\gamma_{\mathscr{F}} \cdot \Pi_f(t) \cdot \mathrm{grad}_{g_{C,t}} \Phi_t(P \mid P_m, M) + \xi(t),$$

where $\Phi_t(P \mid P_m, M) : \mathscr{P}(\Omega) \to \mathbb{R}_{\geq 0}$ is a divergence potential — given explicitly, using Definition 6.2's already-established divergence function D and the projection operators O_o, O_p (§6.1) applied to the candidate state P:

$$\Phi_t(P \mid P_m, M) = D(O_o(P), O_p(P)) + D(O_o(P), P_m) + D(O_p(P), P_m),$$

i.e. $D_{op} + D_{om} + D_{pm}$ evaluated at P, with P_m held as the independently-tracked contemporaneous state per the conditioning note below rather than derived from P. This requires no additional modeling choice beyond what Definition 6.2 already fixes: an implementer with O_o, O_p, and D already specified (as any working adapter must have) can evaluate Φ_t and its gradient directly. The multi-valley structure invoked by the entropy regime classification in §10.5b is a genuine property of this same explicit landscape, not a separate construction layered on top of it: Jensen–Shannon divergence's own geometry over the simplex admits multiple local minima wherever $O_o(P)$, $O_p(P)$, and P_m can each independently approach or diverge from one another — all three coincident is the low-entropy baseline's basin; $O_o(P)$ converging with P_m while $O_p(P)$ remains distant is one high-entropy configuration; the reverse is medium-entropy's — so §10.5b's regime classification identifies which basin of this explicit Φ_t a trajectory has actually fallen into, rather than defining basins Φ_t itself does not structurally produce; $\gamma_{\mathscr{F}} > 0$ is a fitted learning-rate parameter; and $\xi(t)$ is stochastic forcing representing environmental shocks, resource strain, or ordinary noisy experience that does not resolve cleanly. The explicit conditioning on P_m and M is not decorative notation: P_m (§6.1) is an emergent aggregate over the meta-representation layer's own population, not a function of P the way P_o and P_p are, so at each timestep $P_m(t)$ is treated as an independently observed or latent contemporaneous state when evaluating the potential governing P(t)'s own evolution, not derived from P(t) itself — an implementer deriving P_m from P at this step would silently undo §6.1's independence patch. $\mathrm{grad}_{g_{C,t}}$ is the gradient taken with respect to container C's own history-dependent metric $g_{C,t} = g(M_{C,t})$ (§4.10), not the ordinary coordinate gradient: this is what actually propagates Part 4's claim — that two containers occupying identical P can evolve differently because their histories differ — into the equation governing P's own evolution. An ordinary coordinate gradient here would make that claim true of curvature and geodesics but not of the container's actual dynamics, silently disconnecting the two.

Simplex preservation. P is a probability distribution over Ω; raw additive noise $\xi(t)$ can push it outside the simplex $\mathscr{P}(\Omega)$ ($P_i \geq 0$, $\Sigma_i P_i = 1$) with no correction otherwise stated. Following the same convention already used for §8.5's network propagation, the computational default is discrete projection: after each discretized update,

$$P(t+\Delta t) \leftarrow \mathrm{Proj}_{\Delta^m}[\ P(t) - \Delta t \cdot \gamma_{\mathscr{F}} \cdot \Pi_f(t) \cdot \mathrm{grad}_{g_{C,t}} \Phi_t(P|P_m,M) + \xi_t\ ],$$

projecting the unconstrained update back onto the simplex at every step, rather than leaving an implementer to independently decide what $\xi(t)$ means for a distribution-valued state. This is the same pragmatic convention §8.5 already uses, applied consistently rather than introducing a second, heavier construction (tangent-space-constrained diffusion) that nothing else in this document requires.

What landing exactly on the boundary means, and what it does not. A component reaching exactly zero under this projection is not always the same event, and the two cases should not be conflated. Where a component's own continuous dynamics genuinely converge toward zero — the underlying discrepancy or divergence they represent is actually being resolved — the projection landing there reflects a real, substantive correction: the state rests at the boundary because the disturbance it measures has genuinely gone to zero. Where instead a discretization step or noise term $\xi(t)$ overshoots past zero on a component whose true continuous-time dynamics would have approached the boundary asymptotically without fully reaching it, or would have stayed strictly positive, clipping to zero is a numerical safeguard against step-size artifacts, not evidence that the underlying disturbance has actually resolved. Both are real uses of the same projection operator, and an implementation or interpretation should not treat every instance of landing on the boundary as proof of successful correction; only sustained, non-overshoot convergence recorded across successive steps supports that reading, exactly the same discipline already applied elsewhere in this document to distinguishing a single instantaneous measurement from a confirmed, sustained state. The drift term pulls P downhill toward lower divergence at a rate scaled by current permeability — correction happens fast when Π_f is high and is nearly absent when Π_f is low, even when the discrepancy driving it is large. The noise term is what makes the resulting trajectory genuinely non-monotonic rather than merely slow: whether a container ever escapes a shallow local region of Φ to reach a deeper one depends on whether $\xi(t)$ is large enough, for long enough, to carry it over the intervening barrier — the same qualitative phenomenon as a ball in a bumpy landscape needing a large enough kick to leave a local dip. This is why bounded oscillation between entropy regimes (§10.5b) is the expected behavior for a nontrivial fraction of real containers, not a failure of the model.

This instantiation also gives Axiom C1 (§10.5b.15) a concrete mechanism rather than an assertion: a false-signal configuration is a local valley of Φ that exists only because something external is actively reshaping the landscape to hold it open — a confrontation-forcing term or an extractive input (§8.6a). Withdraw that forcing and Φ locally reshapes back toward having its true-signal minimum as the only nearby attractor, so ordinary gradient descent resumes toward it without any additional mechanism being required.

6.5 Feedback Permeability

Not every informational discrepancy successfully modifies the underlying information state. Some discrepancies are ignored, suppressed, forgotten, or prevented from propagating.

Definition 6.4 (Permeability Operator).

Define $\Pi\_f : D\_ij \mapsto [0,1]$. For any informational discrepancy, Π_f denotes the proportion successfully transmitted into adaptive updating. Π_f = 1 represents complete informational openness; Π_f = 0 corresponds to complete informational closure.

Permeability is not a probability of communication. It is an operator governing how much informational discrepancy actually influences subsequent adaptation.

Computational instantiation (concrete default for Π_f).

Given observable correction-and-response data for the container, define a raw permeability score

*r*_f(t) = [corrective actions(t) / error signals(t)] − response lag(t)

(lag measured in the container's natural time unit, e.g. years to policy change), then map it into the required [0,1] codomain via a logistic squashing function,

*Π*_f(t) = 1 / (1 + exp(−κ_Π · r_f(t))),

with κ_Π a fitted scale parameter (Part 15). This is domain-general: "corrective actions," "error signals," and "response lag" are placeholders the adapter fills in per container (audit recommendations implemented / issued / time-to-implement, for the justice-system case), but the aggregation formula itself does not change across domains.

6.6 Informational Integrity

Definition 6.5.

Let I = G(D_op, D_om, D_pm), where $G : \mathbb{R}^3_{\geq 0} \to [0,1]$ is a decreasing functional satisfying I = 1 only when all pairwise divergences vanish.

Informational integrity summarizes the internal coherence of the representation system. Alignment becomes one particular manifestation of informational integrity rather than an independent primitive.

Computational instantiation (concrete default for G).

I(t) = exp[−α·(D_op(t) + D_om(t) + D_pm(t))], α > 0 fitted.

This satisfies Definition 6.5 exactly: I ∈ (0,1] always, I = 1 iff all three pairwise divergences vanish, and I is strictly decreasing in each divergence — so Proposition 12.3 (Bounded Informational Integrity) holds by construction rather than by separate proof for this instantiation. This is RICD's three-channel generalization of a single collapsed messaging-divergence term: rather than combining O/P/μ mismatch into one scalar before combining it with stress and feedback, this formula keeps the three pairwise divergences distinguishable, which is a genuine strength of RICD's representation (§6.3's interpretation of D_op, D_om, D_pm as diagnostically different failure modes) worth preserving rather than flattening.

6.7 Informational Stress

Definition 6.6.

Informational stress is a path-dependent functional:

$S(t) = \int_0^t K(t, s)\,(1 - \Pi\_f(s))\,D(s)\,ds,$

where K(t, s) is a memory kernel.

Stress accumulates whenever informational discrepancies persist while adaptive feedback remains partially blocked. This naturally incorporates technical debt, institutional denial, chronic trauma, unresolved software defects, and bureaucratic drift without ad hoc latent variables.

Computational instantiation (concrete default for K, with a recursive update).

Take $K(t,s) = \lambda \cdot \exp[-\lambda(t-s)]$, $\lambda > 0$ fitted (exponential memory, the standard choice and the only one of §9.3's candidates that admits an O(1)-per-step update rather than requiring the full history to be re-integrated at every timestep). With D(t) taken as the aggregate discrepancy $D(t) = D_{op}(t) + D_{om}(t) + D_{pm}(t)$ and a fixed simulation step Δt, this reduces to the recursion

$S(t+\Delta t) = S(t) \cdot \exp(-\lambda\Delta t) + (1 - \exp(-\lambda\Delta t)) \cdot (1 - \Pi_f(t)) \cdot D(t),$

which is what should actually be implemented, under the convention that $q(t) = (1 - \Pi_f(t)) \cdot D(t)$ is piecewise constant over each simulation step. This is exact under that convention: λΔt is only the first-order approximation of $(1 - \exp(-\lambda\Delta t))$, and the two diverge for any Δt not small relative to 1/λ, so the corrected coefficient above, not λΔt, is what makes the claim of exactness true rather than merely convenient. The recursion requires storing only the previous S(t) rather than the full trajectory, and is a strict discretization of Definition 6.6 under the piecewise-constant-input convention, not an approximation of a different model.

6.8 Divergence Field

Let x index informational location within the container. Define $\mathscr{D}(x, t)$ — using a distinct script symbol from the pairwise divergence D_{ij} and the general distance functional D, since all three now appear in the same coupled expressions in later sections. The resulting divergence field describes the spatial distribution of informational inconsistency. Regions of concentrated divergence represent localized adaptive failures; diffuse divergence indicates systemic degradation. This spatial formulation gives Feedback Divergence Field Mechanics its name.

6.9 Informational Equilibrium

The natural equilibrium of FDFM is not D = 0. Instead, adaptive informational equilibrium corresponds to bounded divergence satisfying $D(t) < D_c$, for some stability region D_c determined jointly by resource availability, feedback permeability, and recursive organization. Adaptive systems therefore tolerate limited informational inconsistency while maintaining global coherence. In the language of §6.4's landscape, D_c marks the boundary of the basin a container currently occupies; §10.5a gives this boundary a concrete, checkable form — the collapse inequality — once resource and permeability terms are also instantiated in Part 7.

6.10 Role Within RICD

RICD provides recursive organization, geometry, resources, memory, and network dynamics. FDFM provides informational representations, informational divergence, permeability, informational integrity, and accumulated informational stress. Together these layers describe both the structural organization of adaptive systems and the informational processes operating within them.

Parts 7–9: Resource Dynamics, Adaptive Network Dynamics, Memory Dynamics

Editorial note for this segment.

This segment incorporates the following revisions. (1) §7.3’s clipping rule now covers total work (adaptive plus maintenance), not adaptive work alone, closing a gap that could otherwise still drive R(t) negative despite the clipping. (2) §7.5’s maintenance functional is renamed M_maint(C,t) (freeing M for the memory operator/value of Parts 2 and 9) and extended with an explicit, superlinear structural-scale dependence. (3) §7.9’s resource debt is now explicitly stated as a pre-recovery convention, reconciled against §7.7’s recovery term. (4) §8.5’s network propagation equation is routed through the commensuration maps σ_i of §5.2 so that it type-checks across containers with different Ω_i, with an explicit simplex-projection step. (5) §8.6a’s heat-accumulation term is corrected to sum over all coupled neighbors, and the resource ledger for the metabolized fraction of an extraction event is closed explicitly. (6) §8.8a’s accumulated-heat quantity is renamed $\mathscr{I}$_i(t) (a distinct symbol from the instantaneous venting rate H_i(t) it integrates) and its internal changelog table is removed. (7) §8.8b (Resolution Kick) is incorporated with four fixes: the momentum transfer is routed through the §7.4 balance equation rather than written directly into resource debt; an explicit source is given for the verticality signal accompanying a triggered shock; a rule is given for kicks landing on already non-baseline neighbors; and a reconciling note is added against Proposition 8.8a.3. (8) §9.2 clarifies, without altering Definition 2.5, that M_t denotes the memory operator’s value at time t. (9) §9.10 now cross-references Definition 4.8 (§4.10) rather than redefining the history-dependent metric, and introduces a formal definition of container identity as recursive membership continuity — the memory-side half of the identity criterion completed in §10.5b.0 once the semi-rigidity condition is available. All other content in these three parts is unchanged from the prior draft.

Part 7. Resource Dynamics

Every adaptive process requires the transformation of information. Learning, memory formation, structural repair, communication, prediction, error correction, and organizational reconfiguration all require finite physical or abstract resources. Information dynamics cannot be separated from resource dynamics.

7.1 Resource Functional

Each container possesses a finite adaptive capacity represented by a resource functional $R[C] \geq 0$. Depending on the application, this may correspond to metabolic energy, computational capacity, available capital, organizational slack, cognitive attention, time, available labor, or communication bandwidth.

7.2 Informational Work

Let $P(t) \to P(t + \delta t)$ represent an informational update. Define the informational work functional $W = W(P(t), P(t + \delta t))$. The work required depends upon the geometric displacement between informational organizations (measured by D, per §4.2's scope conditions).

7.3 Minimal Adaptive Cost

From Axiom 3, nontrivial informational updates possess strictly positive cost. There exists a lower bound $W \geq \Psi\_W(D(P\_t, P\_{t+\delta t}))$, where Ψ_W is a monotonically increasing cost functional.

Computational instantiation (concrete default for W, satisfying Theorem 12.4's hypothesis by construction).

Take $\Psi\_W(D) = \kappa\_W \cdot D^p$, $\kappa\_W > 0$, $p \geq 1$ fitted, giving a baseline adaptive-work requirement $W\_raw(t) = \kappa\_W \cdot D(t)^p$. Total informational work is adaptive work plus maintenance (§7.5): the container cannot fund either component beyond what its current reserve actually holds, and the two draw on the same reserve, so clipping must be applied to their sum rather than to the adaptive term alone.

Default priority convention: maintenance is funded first, since a container that fails routine structural upkeep degrades regardless of how well it adapts; adaptive work is then funded from whatever remains. This ordering is a modeling choice, not a mathematical consequence of the cost bound above — the clipping itself is a computational instantiation, and this priority rule is a further, separately-stated default governing how that clipping is applied when both components compete for the same reserve:

$M\_maint\_actual(t) = \min[M\_maint(C,t), R(t)]$,

W_adaptive(t) = min[W_raw(t), max(R(t) − M_maint_actual(t), 0)],

W(t) = W_adaptive(t) + M_maint_actual(t).

This guarantees W(t) ≤ R(t) for the total work performed, not merely its adaptive component — satisfying w(R) = R exactly for the resource-feasibility hypothesis W ≤ w(R), w(0) = 0, that Theorem 12.4 requires, in a form that theorem can actually inherit automatically. A container facing an adaptive update that would cost more than its post-maintenance reserve either performs a partial, proportionally scaled-down update or defers the excess to the next timestep, but never runs R below zero from this mechanism. In the limiting case where even maintenance cannot be fully funded (M_maint(C,t) > R(t)), the container performs no adaptive work that step (W_adaptive = 0) and directs its entire remaining reserve to maintenance — survival is prioritized over adaptation, which is the behavior this ordering is intended to produce, not an incidental consequence of it.

What the clipping rule alone guarantees, and what it does not. The construction above guarantees W(t) ≤ R(t) — the work-bound hypothesis Theorem 12.4 needs — unconditionally, by clipping internally generated work to available reserve. It does not, by itself, guarantee R(t) remains nonnegative under arbitrary external exchange: the full balance equation (§7.4) is $\dot{R}$ = P_prod − W + E_exch, and at R = 0 with W correctly clipped to zero, $\dot{R}$ = P_prod + E_exch, which is negative whenever E_exch is sufficiently negative — a large enough externally imposed extraction or withdrawal can still drive R below zero regardless of how the container's own internally generated work is clipped. Theorem 12.4's full nonnegativity conclusion accordingly requires the further, separate hypothesis P_prod(t) + E_exch(t) ≥ 0 near the boundary R = 0, stated explicitly there; the clipping rule resolves internally generated overspending, not arbitrary externally imposed extraction, and no claim elsewhere in this document that the clipping rule alone establishes resource feasibility for every implementation should be read as covering the latter case without that additional condition holding.

7.4 Resource Balance

Resources change through production, expenditure, and exchange:

dR/dt = P_prod − W(t) + E_exch,

where P_prod denotes productive generation, W(t) denotes informational work (§7.3, now inclusive of maintenance), and E_exch denotes net exchange across the container boundary. Adaptive capacity increases through production, decreases through informational work, and is modified by interaction with the environment.

7.5 Maintenance Cost

Not all resource expenditure produces adaptation.

Definition 7.1 (Maintenance Functional).

Define M_maint(C,t) as the maintenance functional — the resource cost of structural preservation, communication, monitoring, error checking, and routine coordination, independent of any adaptive informational update. (Renamed from the earlier M(C) to avoid collision with the memory operator M_t of Definition 2.5 and Part 9.)

Computational instantiation (concrete default for M_maint, with structural-scale dependence).

Maintenance cost is not independent of a container's structural scale. A larger container — more recursively aggregated sub-containers, greater physical mass, more organizational tiers — has more structure to keep coherent, and the cost of doing so grows faster than the scale itself: the same qualitative pattern responsible for megafauna's comparatively short lifespans and for greater cell senescence and organ strain in larger-bodied individuals within a species, where maintenance load scales with volume while exchange and support capacity scale more slowly, closer to surface area. The default form:

$$M_{maint}(C,t) = M_{base}(t) + \kappa_M \cdot \mathrm{Size}(C)^{\nu},$$

where Size(C) is a domain-specified structural-scale measure (adapter-supplied — e.g. count of currently aggregated sub-containers per §5.1–§5.2, physical mass, headcount, organizational tier count), $M_{base}(t)$ is the scale-independent routine maintenance floor, $\kappa_M > 0$ is a fitted scale coefficient, and $\nu \geq 1$ is a fitted exponent — $\nu = 1$ recovers linear (ordinary proportional) scaling as a special case; $\nu > 1$ is the superlinear regime responsible for the scale penalty described above, and is expected to be the more common empirical case. Then $W = W_{adaptive} + M_{maint}(C,t)$, per §7.3.

7.6 Efficiency

Define adaptive efficiency $\eta = D(P_t, P_{t+\delta t}) / W$. Large values indicate highly efficient adaptation; low values indicate resource-intensive adaptation.

Editorial note: *This η (adaptive efficiency) is a distinct quantity from η_i, the per-container metabolic efficiency introduced in §8.6a. The two are related in spirit — both measure how much of an input converts to useful adaptive output — but are fitted independently and should not be assumed equal without a stated reason for a given application.*

7.7 Resource Resilience

Define the recovery operator Γ_R. Following disturbance, $R \to \Gamma_R(R)$. The speed and completeness of this recovery determine adaptive persistence under repeated perturbation — biological healing, financial recovery, replenishment of computational resources, organizational recruitment, psychological restoration.

Computational instantiation (concrete default for Γ_R).

Simple exponential relaxation toward a domain-specified target capacity R_{target}: $dR_{recovery}/dt = \rho \cdot (R_{target} - R)$, $\rho > 0$ fitted (the recovery-rate analog of a half-life). Combined with §7.4's balance equation, the full resource update used in implementation is

$$dR/dt = P_{prod} - W(t) + E_{exch} + \rho \cdot (R_{target} - R),$$

with $W(t)$ taken from §7.3's clipped form.

7.8 Recursive Resource Flow

$T_R^+ : R^{(k)} \to R^{(k+1)}$ and $T_R^- : R^{(k+1)} \to R^{(k)}$ describe upward aggregation and downward allocation of adaptive capacity. Recursive transfer need not be symmetric; higher-level organizations may extract resources from lower levels faster than they replenish them, or reinvest surplus capacity downward.

7.9 Resource Debt

When informational work exceeds sustainable net inflow over extended periods, adaptive systems accumulate resource debt. Consistent with the balance equation of §7.4, define:

$\delta R(t) = \int_0^t (W - P\_prod - E\_exch)\, ds.$

Convention (pre-recovery debt).

This integral uses only the core balance terms of §7.4 — work, production, and exchange — and deliberately does not include §7.7's recovery term $\rho \cdot (R\_target - R)$. $\delta R(t)$ therefore measures what the container owes before passive healing is credited against it, not the container's actual net trajectory once recovery is included; the two will not reconcile against $R(0) - R(t)$ whenever §7.7's recovery term is active, and that is intentional rather than an inconsistency. This convention is chosen because it isolates deficit attributable to genuine adaptive-and-maintenance load from whatever a container's passive recovery mechanism happens to be separately restoring — the quantity of interest for diagnosing whether a container's demands are outrunning its productive capacity, independent of how well it happens to be healing. An application that instead wants a net-of-recovery debt measuring the container's actual realized trajectory should define that as a second, separately labeled quantity rather than overload $\delta R(t)$ with both meanings.

The exchange term E_exch must appear here with the same sign convention as in §7.4, or the debt integral does not track the deficit implied by the balance equation — a container could run a genuine resource deficit masked by positive exchange, or show apparent debt that is actually being offset by exchange in real time. Positive δR represents cumulative adaptive-and-maintenance deficit under the pre-recovery convention used here. Examples include technical debt, institutional burnout, ecological overshoot, chronic physiological exhaustion, and infrastructure neglect. This quantity is distinct from informational stress $S(t)$, though the two are mathematically coupled (§7.11).

7.9.1 Divergence and Predation as Direct Resource Costs

Reason for this addition. The only existing channel connecting divergence to resource depletion is M_decep(t) (§10.5b.18.4), scoped specifically to a captive high-entropy container's cost of actively concealing its own operational reality. This is the narrow case, not the general principle: a container carrying unresolved divergence pays a real cost simply by continuing to operate under it, whether or not that divergence is being actively concealed. Separately, extraction and displacement (§8.6a) currently charge a predator only for the unmetabolized remainder it fails to convert into usable capacity — nothing charges it for the act of extracting and displacing itself, which is its own, additional resource cost, independent of what happens to the proceeds. And active search (§10.5b.11c.1) currently scales in intensity with a container's own deficit without itself drawing down further resources, which understates a real dynamic: repeated search, particularly repeated failed search, should leave a container measurably poorer than it was before attempting it, not merely reflect the deficit that motivated the attempt.

Definition 7.9.1 (General Divergence Cost). For any container, not only high-entropy ones maintaining active concealment:

$M\_diverge(t) = \kappa\_diverge \cdot D(t),\ \kappa\_diverge > 0$ fitted,

where $D(t) = D\_op(t) + D\_om(t) + D\_pm(t)$ (§6.3), entering the resource balance (§7.4) as an additional draw, general to every container regardless of regime. This is distinct from and additional to M_decep(t): a captive container pays both this baseline cost of operating under unresolved divergence and the further, specific cost of actively concealing it, since active concealment is strictly more expensive than merely carrying the divergence.

Definition 7.9.2 (Extraction Performance Cost). For a container i performing extraction or displacement against one or more coupled neighbors:

$M\_extract(i,t) = \kappa\_extract \cdot \Sigma\_j\ X\_\{i,j\}(t),\ \kappa\_extract > 0$ fitted, entering i's own resource balance as an additional draw, charged for the act of extracting and displacing itself — separate from and additional to the existing η_i-metabolization split (§8.6a), which governs only what becomes of the resource once extracted, not what performing the extraction costs the predator regardless of outcome.

Definition 7.9.3 (Search Cost). For a container i engaged in active search:

$$M_{search}(i,t) = \kappa_{search} \cdot \Sigma_{search}(i,t),\ \kappa_{search} > 0 \text{ fitted},$$

entering i's own resource balance as an additional draw. This is what makes repeated search cycles genuinely compounding rather than merely repetitive: a container whose search repeatedly fails to find purchase is left measurably poorer after each attempt, not simply as deprived as before it tried — the mechanical basis for later cycles in an extraction-and-displacement pattern being more dangerous than earlier ones, since each cycle draws down the same resource base a subsequent cycle depends on.

Why this closes the causal chain, not merely adds cost terms. With $M_{diverge}(t)$, $M_{extract}(t)$, and $M_{search}(t)$ entering the balance equation, divergence is no longer resource-neutral for any container experiencing it, and neither is the act of exploiting another container's divergence. This is consistent with, and completes, what $d_A(t)$ (§10.3) and $P_C(t)$ (§11.10) already assume structurally: $d_A(t)$'s formula already discounts adaptive possibility by $\delta\tilde{R}(t)$ directly, and $P_C(t)$ already weights collapse potential by $(1 - d_A/d_A^{max})$ — the causal chain from divergence to resource depletion to collapse was already present in how those two quantities are defined; what was missing was the mechanism by which divergence itself, and the act of exploiting it, actually produce the resource depletion those formulas assume as their input.

7.10 Resource Stability

Resource stability requires:

$$\limsup_{T\to\infty} (1/T) \int_0^T (P_{prod} - W)\, dt \geq 0,$$

using P_{prod} consistently (as defined in §7.4) rather than the bare information-state symbol P. This condition does not require continuous resource growth — only that long-term adaptive expenditure remain sustainable. Persistent violation necessarily leads to progressive degradation regardless of informational coherence.

7.11 Coupling to FDFM

Informational divergence increases the informational work required for successful adaptation. Limited resources restrict the system's ability to reduce informational divergence. This mutual dependence establishes a positive feedback loop: greater divergence → greater informational work → resource depletion → reduced adaptive capacity → greater divergence.

The Resource Principle.

Resources are conserved adaptive capacity consumed through informational transformation, replenished through production and exchange, and distributed recursively across organizational scales.

Part 8. Adaptive Network Dynamics

Adaptive systems do not evolve in isolation. Every container exchanges information, resources, and influence with other containers through continually evolving patterns of interaction. RICD treats network topology as an emergent property of adaptive dynamics rather than a fixed modeling assumption.

8.1 Interaction Graph

$G(t) = (\mathscr{C}, E(t))$, where $E(t)$ is the time-dependent set of interactions. Topology evolves continuously alongside the adaptive system.

8.2 Weighted Interactions

$W_{ij} = (I_{ij}, R_{ij}, P_{ij})$, where I_{ij} represents informational exchange, R_{ij} represents resource exchange, and P_{ij} represents effective permeability. An interaction possesses magnitude, direction, and functional character.

8.3 Emergent Connectivity

Define the coupling functional $\Xi(C_i, C_j)$. An interaction exists whenever $\Xi(C_i, C_j) > \theta$, where θ is a domain-dependent coupling threshold. Connectivity emerges from interaction rather than being assumed beforehand.

Computational instantiation.

$\Xi(C_i, C_j)$ = historical interaction frequency $\times \min(\Pi_{f,i}, \Pi_{f,j})$ — an edge is live only if both containers have historically interacted and neither side's permeability is near zero (a container that ignores everything cannot be meaningfully coupled to, regardless of contact frequency). θ is fitted per application.

8.3.1 Preferential Attachment Toward Cohesive Clusters

Definition 8.3.1 (First-Contact Attraction).

For an uncoupled or newly-decoupled container C_{new} evaluating potential first contact among containers or clusters $\{C_k\}$ within observable range, define the attraction weight

$A_{first}(C_{new}, C_k, t) = \nu_{pref} \cdot \Lambda_{int}(C_k, t)$, $\nu_{pref} > 0$ fitted, where $\Lambda\_int(C\_k, t)$ is §9.10.1's internal cohesion measure, read here as an externally observable signal of a candidate's own reciprocal coupling density rather than only as a self-referential containment mechanism. A solitary candidate, with no internal membership to measure cohesion over, has $\Lambda\_int(C\_k,t)$ undefined or zero by construction — attraction under this definition requires visible evidence of actual reciprocal engagement with others, not merely a stable-looking presentation, which a single container cannot manufacture unilaterally regardless of how calm its own σ_o reads.

Proposition 8.3.1 (Preferential Attachment).

New or newly-uncoupled containers entering a network form their first live edge (crossing Ξ's threshold, §8.3) with candidate C_k at a rate increasing in A_first(C_new, C_k, t). First contact is therefore systematically biased toward already-cohesive clusters, not distributed uniformly across candidates and not driven by whichever candidate presents the calmest official signal. This is the formal content of new entrants being drawn toward an already-gathered group rather than toward an individually calm-presenting container: the attraction is to a cluster's visible, reciprocally-produced coupling density, which is a harder signal to fake alone than a stable σ_o.

Corollary 8.3.1.1 (Predator Exclusion by Construction).

A high-entropy container acting alone, without co-opted or captive members contributing to its own internal cohesion, presents $\Lambda\_int(C,t) = 0$ or undefined under Definition 9.10.1, and therefore supplies zero first-contact attraction under Definition 8.3.1 regardless of how low its σ_o reads externally. This gives grid-level predator starvation a formal mechanism rather than requiring it be stipulated: as cohesive low- and medium-entropy clusters grow denser (rising Λ_int), they become preferentially more attractive to new entrants under Proposition 8.3.1, while an isolated high-entropy container, structurally incapable of generating a comparable signal on its own, receives proportionally fewer first contacts over time. This is not exclusion by a rule targeting predators specifically; the attraction mechanism was simply never available to a container with nothing genuine to reciprocate.

Editorial note: This does not replace Ξ's role in §8.3, and the two operate on different populations by design. Ξ governs whether an edge with existing interaction history solidifies; Definition 8.3.1 governs which candidate a container with no existing history is drawn toward in the first place. A high-entropy container can still acquire edges through active search (Part 10 Addendum III) or through wooing under staged control-locus delegation (§10.5b.18.18); this addition specifically closes off the passive channel, first contact initiated by the uncoupled party's own attraction, as a route available to an isolated predator.

8.4 Dynamic Topology

dE/dt = G_gen(P, R, M, B), where the topology generator G_gen depends upon informational organization, available resources, historical memory, and boundary evolution.

Computational instantiation.

Implemented as a per-timestep re-evaluation of §8.3's Ξ against θ over all container pairs — edges are added when Ξ crosses above θ and removed when it falls below θ(1 − hysteresis), a small hysteresis band (fitted) preventing edges from flickering in and out at the threshold every step.

8.5 Information Propagation

Editorial note: *The original formulation, dP_i/dt = F_i + Σ_j Λ_ij(P_j − P_i), subtracts information states directly. Since §5.2 established that sibling containers generally do not share* *a state space ($\Omega\_1 \neq \Omega\_2 \neq \ldots$), $P\_j - P\_i$ does not type-check for containers with different Ω. The corrected version below routes network-mediated coupling through the same commensuration maps σ_i already required for aggregation, so that the subtraction takes place on the shared severity simplex Δ^m rather than on the raw, generally incommensurable P_i.*

Corrected propagation equation.

$d\sigma\_i(P\_i)/dt = F\_i + \Sigma\_j \Lambda\_ij(t) \cdot [\sigma\_j(P\_j) - \sigma\_i(P\_i)]$,

where $\sigma\_i : \mathscr{P}(\Omega\_i) \to \Delta^m$ is the commensuration map of §5.2, F_i is the intrinsic driving term (§8.6a, §8.8a), and Λ_ij is the effective coupling operator. The first term describes intrinsic adaptation; the second, network-mediated influence. Λ_ij may itself depend upon trust, bandwidth, boundary permeability, historical interaction, and resource availability. For containers sharing a common Ω (σ_i = identity for all i), this reduces exactly to the original form.

Complete absorption as the Λ_ij → 1 limit, worked concretely. A formal merger, where one container's assets and liabilities transfer to another in full rather than partially, is the limiting case Λ_ij = 1 of this same coupling operator, not a separate mechanism requiring its own machinery. A concrete instance: a real institutional merger where the acquiring container's own resource equation (§7.4's $\dot{R}$ = P_prod − W + E_exch) receives the absorbed container's outstanding structural debt δR as a direct addition to its own accumulated burden, scaled by Λ_ij, at the specific timepoint the merger completes — E_exch(t_merger) adjusted downward by Λ_ij · δR_absorbed, a one-time transfer rather than an ongoing coupling term, since a completed merger dissolves the coupling itself along with the two containers' separate identities. Where merger terms specify complete assumption of liabilities rather than a negotiated partial share, Λ_ij = 1 is not fitted but directly adapter-supplied from the terms themselves, exactly as an external governance attestation (Definition 6.1.1) is supplied directly rather than statistically estimated: the transfer's magnitude is a matter of public record, not something inferred from noisy proxies. This is a real, checkable use of the coupling formalism for forward projection — computing what a documented future transfer implies for the absorbing container's resource trajectory before that transfer has actually occurred in observed data — distinct from and simpler than the fuller extraction dynamics of §8.6a (metabolic efficiency η_i, ongoing flow X_i,j), which govern a persisting predatory or symbiotic relationship between containers that remain separately identified, not a terminal absorption that ends the coupling by dissolving one side of it entirely.

An ordinary, ongoing, mutually protective coupling, worked concretely. The complete-absorption limit above is a terminal, one-time case; the general Λ_ij mechanism also covers the far more common situation of several containers that remain separately identified indefinitely while genuinely sharing part of their resource burden on an ongoing basis. A concrete instance: three real, separately-governed institutions sharing a single financed utility plant, with a real, documented, quantified reduction in each member's own ongoing maintenance cost burden as a direct result — not a one-time transfer at a single timepoint, but a standing term entering each container's own W(t) (§7.1's maintenance and waste channel) every period the arrangement remains in force. Modeled as W_i(t) reduced by a real, adapter-supplied coupling fraction rather than fit statistically, since the magnitude here too is a matter of public record (a confirmed percentage cost reduction) rather than something to be inferred: W_i(t) = W_i,uncoupled(t) · (1 − κ_share), where κ_share is the real, documented shared-cost reduction and W_i,uncoupled(t) is what the container's own maintenance burden would be absent the arrangement. Confirmed directly against real data: removing this real coupling term from three real containers' own accumulated resource-deficit calculations increased two of the three by more than double, and revealed that a third container's real, actual history of carrying no net deficit at all depends on the coupling remaining in force — a genuine, quantified demonstration that ordinary, ambient coupling can be the difference between a container remaining solvent and falling into deficit, not merely a modest efficiency gain. This is the general Λ_ij mechanism operating in its most common real form: neither a terminal absorption nor a predatory extraction, but a standing, mutual, ongoing reduction in maintenance burden shared across containers that remain fully separate and continue to exist as such.

A genuine predatory extraction, worked concretely, at last against a real case. The two instances above are both benign: one container's resources are absorbed cleanly and completely, or several containers mutually reduce each other's real burden. Section 8.6a's extraction dynamics describe a structurally different, harder case this document had, until now, only stated on theoretical grounds: a parent or hub container drawing resources out of a subordinate container asymmetrically, to the subordinate's real detriment, while the subordinate continues to exist as a separate, nominally operating container rather than being absorbed or benefiting in turn. A real, court-documented, and government-confirmed instance: a parent education holding company's real, quantified diversion of federal financial-aid disbursements — funds the parent had already received from the federal government on behalf of specific, named students at a subordinate institution, and was legally obligated to release to those students as credit-balance refunds — instead retained and spent on the parent's own payroll and vendor obligations across its wider, financially failing system. A real regulatory finding quantified this at approximately \$13 million improperly diverted system-wide, independently corroborated by a real state-level accounting of \$186,946 in the same pattern at a single one of the parent's subordinate campuses. This is the concrete instantiation of $X_{i,j}$ (§8.6a's extraction flux): a real, one-directional, adapter-confirmable flow of resources that the resource-generating container (§7.1's P_{prod}, realized here as federal aid credited to it on the subordinate's behalf) never gets to keep, extracted instead into the hub's own resource balance. Modeled directly by subtracting the confirmed diverted amount from the subordinate container's own realized resource inflow at the timepoints the diversion occurred, rather than fit statistically: $R_{subordinate}(t) = R_{subordinate,nominal}(t) - X_{i,j}(t)$, where $X_{i,j}(t)$ is the real, adapter-supplied, dated diversion amount. Unlike the two coupling cases above, this could not be tested against a full retrospective panel fit in this project, since the subordinate container's own chaotic, multi-entity collapse left its usual reporting structures too fragmented across the relevant period to support a clean, single-container statistical validation — reported honestly as a real, worked instantiation of the mechanism using confirmed, sourced figures, in the same spirit as the forward projection used for complete absorption above, rather than a full validated classification result. The distinction between this case and the two benign coupling cases is not a matter of degree but of direction and consent: κ_{share} and the merger's Λ_{ij} both represent an arrangement the coupled containers' own governance participates in and benefits from or agrees to; $X_{i,j}$ here represents resources a container was owed and never received, extracted by a hub it does not control.

Because $\sigma_i(P_i), \sigma_j(P_j) \in \Delta^m$ are probability vectors, an unconstrained additive update of this kind is not guaranteed to keep the result inside the simplex (individual components remaining nonnegative and summing to one) over an extended simulation. The engine must project the updated value back onto Δ^m at each step — renormalizing and clipping negative components to zero — the same pragmatic safeguard already used for the analogous case in §7.3's clipping and §5.6's normalization, rather than a new kind of mechanism.

Computational instantiation.

$\Lambda_{ij}(t) = \gamma_{net} \cdot W_{ij}(t) \cdot \Pi_{f,i}(t)$, $\gamma_{net} > 0$ fitted — network-mediated influence scales with current edge weight and with the receiving container's own permeability (a container that doesn't absorb feedback internally won't absorb it from neighbors either, by the same mechanism).

8.6 Resource Propagation

Let $\Phi_R(i, j)$ denote the resource flux from C_i to C_j. $\Sigma_j \Phi_R(i, j)$ determines the net exchange component appearing in the resource balance equation of §7.4. Information and resources propagate through distinct, though coupled, channels.

8.5a Coupling Valence: A Classification of Inter-Container Relationships

A coupling between two containers can be classified by its net effect on each party's own trajectory — concretely, whether the coupling's contribution to each party's dd_A/dt (§10.3) is positive, negative, or negligible. This gives every pairwise coupling a sign for each side, (+, -, or 0), and the resulting sign pairs correspond to the seven classical ecological interaction types, each of which already has, or is given below, a specific formal mechanism in this document rather than remaining a loose analogy.

- **Mutualism (+/+):** both parties' trajectories benefit. Mutual Anchor Clusters (§10.5b.13.1, Φ_mutual, Γ_mutual) are this document's mechanism — reciprocal stabilization neither member could sustain alone.
- **Commensalism (+/0):** one party benefits, the other's trajectory is unaffected. A plausible candidate mapping — a low-resource container anchoring near a much larger, stable one whose own trajectory is unmoved by the coupling — has not been separately verified against this document's anchor-eligibility machinery (§10.5b.13) and should be treated as unconfirmed rather than established.
- **Parasitism (+/-), chronic:** ordinary extraction (§8.6a below) — the predator gains, the host is weakened but continues functioning, resource debt accumulating over an ongoing coupling.
- **Predation (+/-), acute:** the same sign, but the interaction crosses into the target's own collapse machinery (§10.5b.9d, disintegration, or a full tipping cascade) rather than remaining within ongoing extraction dynamics — the distinguishing mark is whether the coupling stays inside §8.6a's extraction dynamics or triggers the target's own collapse resolution.

Scope, not only sign, determines whether predation is entropic. Sign alone — (+/-) — does not determine whether a predation relationship is itself a sign of entropic misalignment. Predation directed at a container outside i's own immediate coupled network, sustained as part of a broader grid-level cycle in which the target population retains genuine adaptive possibility and the capacity to develop its own countermeasures over time (§10.5b.2.8, below), can be a stable, negentropic element of the larger grid — a food web that never removes prey's own capacity to adapt is not, at the grid level, evidence of divergence. Predation or destruction directed at a container *within* i's own immediate coupled network, by contrast, is a positive sign of entropic misalignment unless demonstrably defensive, independent of whether the broader grid remains stable. Scope and sustained mutual adaptability, not sign alone, are what separate a healthy predator-prey cycle from entropic destruction wearing the same mathematical sign.

- **Competition (-/-):** both parties' trajectories are harmed by drawing on the same limited external resource, without a direct Λ_ij coupling between them at all. §8.5a.1 below formalizes this; nothing in this document previously covered it.
- **Amensalism (-/0):** one party harmed, the other unaffected and gaining nothing. A plausible candidate mapping — a bystander container absorbing incidental momentum from a neighbor's cascade discharge (§8.8a, Drive_B) without the discharging container benefiting from that harm — has not been separately verified and should be treated as unconfirmed rather than established.
- **Neutralism (0/0):** no meaningful coupling. Requires no separate mechanism; this is simply the case where Λ_ij(t) is negligible or zero.

8.5a.1 Competition for a Shared External Resource

Extraction (§8.6a) requires a direct coupling $\Lambda\_ij(t)$ between predator and target. Competition does not: two containers can harm each other's trajectories by drawing on the same limited external resource pool without any direct coupling between them at all — a genuinely distinct mechanism, not extraction under another name.

Definition 8.5a.1 (Shared Resource Pool).

Define a shared pool $\Omega\_pool(t) \geq 0$, external to any single container, with its own regeneration dynamics:

$d\Omega\_pool/dt = \gamma\_pool \cdot (\Omega\_pool\text{^}max - \Omega\_pool(t)) - \Sigma\_\{k \in competitors\}\ D\_k(t)$,

$\gamma\_pool > 0$ fitted, where {competitors} is the set of containers drawing on this specific pool (adapter-defined membership — e.g., all containers within a given resource catchment or market), and $D\_k(t)$ is competitor k's actual draw.

Definition 8.5a.2 (Competitive Draw).

Each competitor's draw is capped by its own demand and by its share of the pool relative to every other competitor's current effort:

$D\_i(t) = \min[\ demand\_i(t), share\_i(t) \cdot \Omega\_pool(t)\ ]$, $share\_i(t) = effort\_i(t) / \Sigma\_\{k \in competitors\}\ effort\_k(t)$,

where effort_i(t) is an adapter-supplied measure of i's own resource-seeking activity, defaulting to R_i(t) (larger, more active containers claim proportionally more of a contested pool by default). D_i(t) enters i's own E_exch(t) (§7.4) as an ordinary resource inflow, capped as above.

Why this is genuinely (-/-). As the number of competitors drawing on Ω_pool(t) grows, or as aggregate effort rises, each individual share_i(t) falls for a fixed pool size, and D_i(t) falls correspondingly for every competitor simultaneously — both, or all, parties' trajectories are harmed by the same underlying pressure, with no direct extraction relationship between any pair of them. This is mechanically distinct from §8.6a's extraction: no container here is drawing resource *from* another container; every competitor is drawing from the same external pool, and it is the pool's own finiteness, not any other container's coupling, that produces the harm.

8.5b Network Topology, Governance, and Forced Value-Drift

Reason for this addition. §8.5a classifies pairwise coupling relationships. Nothing so far characterizes the *shape* a whole network of coupled containers organizes itself into — whether hierarchically concentrated or horizontally distributed — independent of any single pairwise relationship within it, or what that shape does to containers forced to remain inside it.

Definition 8.5b.1 (Network Topology). Define two distinct measures for a network of containers, deliberately kept separate rather than combined into one quantity:

- **Structural centralization**, $\chi\_struct(t) \in [0,1]$: a graph measure of how concentrated the network's coupling connections are around a small number of nodes, independent of what flows along those connections.
- **Resource-flow disproportion**, $\chi\_flow(t) \in [0,1]$: how far actual resource benefit received by central nodes departs from what their own genuine contribution would justify.

Structure alone cannot distinguish a legitimate coordinating hub from a predatory one: a hub that does not personally benefit from its position and one that extracts disproportionate benefit from the identical position produce the same χ_struct(t). The diagnostic is the *gap* between the two measures, not either alone.

Proposition 8.5b.1 (Governance as Regime Classification). The gap between χ_struct(t) and χ_flow(t) recovers this document's existing three-way entropy classification, applied to network governance directly, not as an analogy requiring separate machinery:

- **Egalitarian (low-entropy baseline):** low χ_struct(t); every member treated as structurally important; malfunctioning members receive help restoring function rather than removal, nonfunctioning members receive care rather than discard — because discarding itself, not malfunction, is what this configuration treats as the divergence- causing act.
- **Meritocratic (medium-entropy):** χ_struct(t) may be substantial, but χ_flow(t) stays proportional to genuine contribution — a genuinely functional negentropic core (real reward for real contribution) coexisting with one specific tolerated harm (discarding members who are not malfunctioning, merely not contributing), the medium-entropy signature of one corrupted element within an otherwise intact system.
- **Dominance hierarchy (high-entropy):** χ_struct(t) high *and* χ_flow(t) high — structured for the benefit of dominant nodes at direct, ongoing cost to subordinate ones, the gap between structure and earned benefit at its most extreme.

Structural durability. Independent of governance valence, sustained vertical (hierarchically concentrated) organization is generally less durable than sustained horizontal (distributed) organization, in conceptual systems as in physical ones — a geological mesa outlasting a spire is the same underlying principle expressed structurally rather than informationally: concentrated structures carry more of their own mass on a narrower base, and are correspondingly more exposed to any single point of failure.

Definition 8.5b.2 (Forced Value-Drift). A genuinely negentropic container forced into sustained coupling within a high-χ_struct(t), high-χ_flow(t) network — unable to survive outside it, per its own resource dependency on the hierarchy — may begin performing the dominant node's own values (form over function: optics and compliance rather than genuine contribution) as a survival strategy, without any formative-window shock or genesis-branching event (§10.5b.2) having occurred. Sustained performance can become internalized belief: the container's own feedback corrupts from the inside, purely as a consequence of prolonged forced coupling, not developmental corruption. Where this occurs independently across many nodes within the same dominance hierarchy simultaneously, the result is a grid-level collapse mechanism distinct from cascading extraction or synchronized developmental trauma — convergent, hierarchy-imposed value corruption spreading through a network's own healthy members, driven by the network's shape rather than by any single predatory act.

Definition 8.5b.3 (Functional Judgment Accuracy). A system's own classification of a member as nonfunctioning is itself a claim, subject to the same grounding test as any other self-assessment (G(t), §6.3.2): internal agreement that a member is dispensable is not evidence that the judgment is accurate, only that it is unanimous. Two further considerations bear directly on whether such a judgment should be trusted. First, a member's continued presence within the system is itself weak evidence against the nonfunctioning label — a system that reliably discarded every genuinely useless member would not retain members fitting that description, so persistence is data, not merely an absence of correction. Second, apparent individual dispensability can mask population-level function: a high-variance strategy for generating outcomes — biological or otherwise — can produce mostly poor individual results specifically because rare exceptional results are what the strategy exists to generate, so culling on visible individual performance can remove the population-level mechanism producing the very outcomes worth keeping, without this ever being visible from any single generation's data. Evolutionary variance in a genetic pool is the clearest instance: individuals culled for apparently poor genetic outcomes may be carrying the same high-variance strategy responsible for the population's rare, exceptional successes: their removal would remove the mechanism generating both without ever registering, from any single generation's data, why it mattered.

Definition 8.5b.4 (Emergency Triage Validity). Discarding nonfunctioning or low-functioning members is not automatically an entropy signal where two conditions hold together, not either alone. **Genuine existential constraint:** R(t) or δR(t) (§7.3, §7.9) approaching a real floor, checkable against the system's own resource state, not merely below some preferred target. **Accurate triage judgment:** G(t) (§6.3.2) high for the specific judgment being made — the assessment of who is being deprioritized is itself grounded, not distorted. Both present together is a legitimate emergency mechanism, however costly; either absent collapses the action into ordinary entropic neglect (Definition 8.5b.3). A system's own sincere belief that it is engaged in emergency triage does not satisfy either condition on its own — sincere misapplication, invoking genuine constraint or accurate judgment without either actually being present, is itself a distortable claim, checkable the same way any other self-assessment is.

Definition 8.5b.5 (Standing Proxy Divergence). Where a system optimizes toward a single proxy for value — aggregate contribution, measured utility, majority agreement — as a *sustained, standing* policy rather than a bounded, one-time application, the proxy and the reality it was meant to track can diverge over time even absent any deception: the system becomes progressively better at satisfying its own proxy while the gap between proxy and underlying reality accumulates unchecked, exactly the condition G(t) exists to detect and internal agreement cannot. This is the mechanism behind why standing utilitarian-style governance (the meritocratic classification, Proposition 8.5b.1) tends toward D_op(t)-type divergence while remaining, at every individual timestep, locally optimal by its own measure — the failure is cumulative and structural, not a series of individually detectable errors. The same mechanism applies directly to group agreement substituted for truth or accuracy at grid scale: majority or consensus agreement was never evidence of accuracy any more than a single container's internal agreement is (G(t)'s core claim, restated at network scale), and a network whose consensus has been captured by a predatory sub-population's manufactured agreement (§10.5b.2's population-3 injection mechanism, operating here at network rather than single-container scale) can converge an entire grid on a verdict that is confidently, unanimously, and completely wrong.

Form substituted for function, a concrete instance. The proxy need not be a measured quantity at all; institutional prestige markers — credentials, tenure, an established reputation — can themselves function as the standing proxy, substituted for the genuine functional quality (a program's actual, currently-delivered value) they were once a reasonable indicator of. Consider a hypothetical liberal arts program whose faculty roster stays essentially unchanged for years — no departures forcing turnover, but also no new hires bringing renewal — while the program's own self-assessment continues to rest on that same roster's accumulated prestige. Nothing here need involve extraction, predation, or deliberate deception: the population sincerely believes its long-established reputation for excellence is still an accurate indicator of present quality, and that sincerity is precisely the mechanism, not a mitigating factor. Believing form still tracks function is what forecloses noticing the two have quietly come apart — the belief is not separate from the stagnation it produces, it is the reason the stagnation stays invisible from inside the container producing it. This is Standing Proxy Divergence acting through a specific, common, and easily-overlooked proxy: prestige and tenure standing in for currently-delivered quality, gap accumulating unchecked for exactly as long as form continues to be mistaken for function.

8.6a Extractive Coupling and the Parasitic Cycle

Per §8.5a's coupling-valence classification, this is this document's parasitism mechanism specifically: the chronic case, where the host continues functioning, weakened, across an ongoing coupling. Where the same (+/-) sign instead crosses into the target's own collapse resolution (§10.5b.9d's disintegration, or a full tipping cascade), the same underlying sign pair is better read as predation, the acute case.

§8.6's resource flux is symmetric and neutral by default — a normal exchange channel. It is not always neutral. A container in the high-entropy regime (§10.5b) characteristically draws resource from a neighbor, fails to convert it to usable capacity, and vents the failure back onto the source as harm. This is a distinct, asymmetric coupling mode that needs its own formal treatment, because it is the actual mechanism connecting Part 7's resource layer to Part 10's tipping dynamics — not a metaphor layered on top of them.

Target selection.

A high-entropy container does not extract indiscriminately from any neighbor; it preferentially targets neighbors that remain kinetically coupled after being disturbed, rather than neighbors that readily decouple. Define a coupling-persistence function $\rho(j) \in [0,1]$ for a neighbor container j, reflecting how much of a disturbance's kinetic effect on j persists as continued oscillation (rather than being absorbed and damped): $\rho(j)$ is characteristically high for containers in the medium-entropy regime (§10.5b) — their apparent instability keeps them actively rocking and coupled — and low for containers in the low-entropy baseline regime, which more readily returns to equilibrium and drops out of active coupling. This is why extraction characteristically targets the medium-entropy population rather than the healthy baseline: baseline containers are poor extraction targets by construction, not by choice on either side.

Persecution targeting.

Definition 8.6a.1 (Loyalty-Biased Coupling Persistence).

Extend $\rho(j)$'s computational instantiation, passing the full additive structure through the logistic squashing function already established for Π_f in §6.5:

$\rho(j,t) = 1 / \{1 + \exp[-(\rho\_0(j) + \kappa\_desp \cdot \bar{F}\_i(t) + \kappa\_persec \cdot \Psi\_loyal(j,t))]\}$,

where $\bar{F}\_i(t)$ is $\mathscr{F}\_i(t)$ normalized to [0,1) before entering the sum,

$\bar{F}\_i(t) = 1 - \exp(-\lambda\_F \cdot \mathscr{F}\_i(t))$, $\lambda\_F > 0$ fitted,

a saturating transform requiring no per-container historical ceiling, and $\Psi\_loyal(j,t)$ is neighbor j's own loyalty composition (§10.5b.18.12) where j possesses a meta-representation apparatus of its own, or an adapter-supplied analog where j's own internal composition is not separately modeled. This retains the additive structure's intended behavior — desperation and persecution both push targeting upward, independently and simultaneously — while guaranteeing $\rho(j,t) \in [0,1]$ by construction, consistent with the logistic guardrail already established for Π_f in §6.5. Squashing changes what κ_desp and κ_persec mean: under this corrected form, they are log-odds contributions, not direct additive weights on a [0,1] quantity, and values fitted against an unbounded additive form do not carry over numerically.

Proposition 8.6a.1 (Persecution Is Threat-Responsive, Not Resource-Responsive).

Unlike ordinary target selection, which favors kinetically-persistent neighbors as better extraction sources, $\kappa\_persec \cdot \Psi\_loyal(j,t)$'s contribution to $\rho(j)$ does not depend on j's own resources or vulnerability. A high-$\Psi\_loyal(j,t)$ neighbor is targeted more heavily specifically because of what its own composition threatens to do to the surrounding network — correct nearby containers via φ_fb (§10.5b.18.9), or serve as a genuine external anchor (§10.5b.13, §10.5b.13.1) — not because it offers more to extract. This is the formal content of a system hunting its own corrective response rather than merely exploiting whatever is easiest to exploit.

Definition 8.6a.2 (Network-Level Nonconformist Fraction).

At the network level, define $f\_NC(t) \in [0,1]$, the resource-weighted fraction of containers in a network whose own $\Psi\_loyal(t)$ (or adapter-supplied analog) exceeds a fitted threshold Θ_NC. This is a derived, network-level diagnostic — not a primitive driving any dynamics of its own — computed as an aggregate consequence of Definition 8.6a.1's targeting bias operating over time across the network, which should be expected to reduce f_NC(t) below whatever baseline it would hold absent persecution-weighted targeting.

Definition 8.6a.3 (Persecution Deficit Signal).

P_persecute(t) = max(0, f_NC^target − f_NC(t)),

where f_NC^target is a domain-specified expected baseline fraction (adapter-supplied; no universal value is asserted here). P_persecute(t) is a diagnostic reading of Definition 8.6a.1's targeting mechanism's cumulative effect, to be used as a corroborating signal alongside direct evidence of elevated ρ(j) bias toward high-loyalty containers — not as a free-standing indicator sufficient on its own, since a low f_NC(t) is also consistent with causes other than persecution (e.g., a network genuinely composed of few high-loyalty containers to begin with, independent of any targeting).

Editorial note: P_persecute(t) is deliberately not wired into P_C(t) (§11.10) or any other collapse-potential formula directly. It is diagnostic output, useful for identifying a network under this specific failure mode, not a driving term with its own causal weight in this document's collapse machinery — the actual causal content lives in Definition 8.6a.1's modification to ρ(j), which already feeds forward into X_{i,j} and everything downstream of it through existing, unmodified formulas.

Extraction flux.

For high-entropy container i and a coupled neighbor j:

*X*_{i,j}(t) = χ · Λ_ij(t) · ρ(j) · R_j(t) · 𝒱_{i→j}(t) · (1 − κ_self·Ψ_total(i,t))$_+$, χ > 0 fitted,

drawing down j's resource directly in the balance equation of §7.4. 𝒱_{i→j}(t) (§8.6a.4) is i's own observation-mapped estimate of j's vulnerability — not unmediated access to j's true latent state, consistent with §8.6a.4's own true-versus-assessed distinction; this formula is the actual point where that distinction has causal consequence, and is corrected here to use the predator-specific estimate directly rather than the raw latent quantity a stray earlier draft used. The closing factor is i's own self-regulation term (§8.6a.5): the same base formula this document has used throughout, now modulated by both how resilient the specific target actually is and how far the predator itself has drifted toward its own negentropic core. Setting 𝒱_{i→j}(t) = 1 and κ_self = 0 recovers the original formula exactly.

Metabolic failure and heat conversion.

A high-entropy container's own closed, self-propelled internal dynamics (§10.5b's F_i-dominance) mean it cannot productively convert an inflow of externally-sourced open resource into usable capacity — define a metabolic efficiency η_i ∈ [0,1], characteristically near zero for high-entropy containers, giving a heat term for each coupled neighbor j:

*H*_i,j(t) = (1 − η_i) · X_{i,j}(t),

representing the fraction of resource extracted from j that cannot be metabolized and must be discharged rather than retained.

Closing the resource ledger.

The extraction flux $X_{i,j}(t)$ leaves j's balance equation in full — j's resource loss does not depend on how much i subsequently manages to use. What i actually gains is only the metabolized fraction: i's balance equation (§7.4) receives a positive E_{exch} contribution of $\eta_i \cdot X_{i,j}(t)$ from this channel. The unmetabolized remainder, $H_{i,j}(t)$, is not a resource quantity at all once it fails to be metabolized — it converts to a forcing quantity (below) and re-enters the system through §8.5's propagation equation rather than through anyone's resource balance. This is a genuine unit change, not a bookkeeping shortcut, and both sides of the transaction — j's full loss, i's partial, degraded gain — must be entered into §7.4 for a given implementation to conserve resource correctly across the extraction event.

Venting and the origin of the medium-entropy driving force.

Summing the heat generated across all of i's currently coupled neighbors gives the total instantaneous venting rate,

$\mathcal{H}_i(t) = \Sigma_j H_{i,j}(t) = \Sigma_j (1 - \eta_i)\cdot X_{i,j}(t)$,

which is not dissipated harmlessly; it is vented back into the network as additional forcing, directed at each coupled j in proportion to its own $\rho(j)$ if j remains coupled, or redistributed across other available coupled neighbors if a particular j has decoupled. This vented heat is not a separate force from §10.5b's $\Sigma_j \Lambda_{ij}(t)$ driving term for medium-entropy tipping — it is that term's actual origin: the "lateral knocks" driving a medium-entropy container toward its own tipping event are, mechanistically, the discharged metabolic failure of the high-entropy containers extracting from it. §10.5b's network-mediated drive should accordingly be read as substantially composed of vented $H_{i,j}(t)$ terms from coupled high-entropy neighbors, not an independent disturbance source.

Link-severing as the high-entropy container's own collapse trigger.

If j successfully transitions to the low-entropy baseline regime, $\rho(j)$ falls, the coupling weakens (§8.7's rewiring), and $X_{i,j} \to 0$. If i has no other available coupled target, its heat can no longer be vented externally — it has nowhere left to go but back into i's own intrinsic dynamics, directly increasing the F_i term already identified in §10.5b as high-entropy tipping's driving force. This closes a mechanistic loop the theory previously described only narratively: a successfully-decoupling neighbor is not a bystander to a high-entropy container's eventual collapse, it is a direct contributor to it, and severing an extraction link is simultaneously a self-protective act for the decoupling container and a destabilizing one for the container it decouples from.

8.6a.4 Target Vulnerability

Definition 8.6a.4 (Target Vulnerability).

Define $\mathscr{V}(j,t) \in [0,1]$, target j's own true vulnerability to extraction, as a decreasing function of j's own resilience signals:

$\mathscr{V}(j,t) = 1 - [\, w_\Psi\cdot\Psi_{total}(j,t) + w_d\cdot(d_A(j,t)/d_A^{max}(j,\cdot)) \,]$, clipped to [0,1], $w_\Psi + w_d = 1$, both fitted,

combining j's own negentropic population strength ($\Psi_{total}(j,t)$, §10.5b.18.14) and j's own remaining adaptive possibility ($d_A(j,t)$, §10.3). This is $\mathscr{V}_{true}(j,t)$, j's actual latent state; a predator i does not automatically have unmediated access to it, consistent with this document's own separation of latent state from observation elsewhere. The quantity a specific predator i actually acts on is $\mathscr{V}_{\{i\to j\}}(t) = O_{\{i\to j\}}(\mathscr{V}_{true}(j,t))$, i's own observation-mapped estimate, using whatever access i genuinely has ($A_{acc}(i,j,t)$, §10.5b.18.9.1, reused here rather than duplicated) — a well-resourced predator with genuine access approaches $\mathscr{V}_{\{i\to j\}}(t) \approx \mathscr{V}_{true}(j,t)$; an opportunistic stranger with only surface-level cues (§8.6a.4's broadcast property) works from a cruder estimate. Extraction dynamics elsewhere in this document that reference $\mathscr{V}$(j,t) should be read as depending on the acting predator's own $\mathscr{V}$_{i→j}(t), not on unmediated access to $\mathscr{V}$_true(j,t), except where a specific mechanism explicitly requires the latter. A well-resourced but strongly negentropic-aligned, high-d_A target and an equally-resourced but weakly-aligned, low-d_A target are no longer treated identically by the extraction flux (§8.6a's revised formula): resource size alone no longer determines how attractive a target is. $\mathscr{V}$(j,t) = 1 recovers the original, resilience-blind formula exactly.

Vulnerability as a broadcast property, not only a persistent-coupling one. $\mathscr{V}$(j,t) already depends only on j's own current state (Ψ_total(j,t), d_A(j,t)) — nothing in its definition requires an existing coupling Λ_ij, or any prior relationship between i and j at all. This means low Ψ_total(j,t) does not only impair j's own perception of and response to a specific, already-coupled threat (§10.5b.2.3, §10.5b.2.4); it makes j read as more exploitable to *any* predator's own opportunistic assessment, including one-off encounters with no persistent coupling whatsoever — a single, unrepeated interaction in which an entirely unrelated predator applies its own version of $\mathscr{V}$(j,t) to a target it has never coupled with before and will not couple with again. A container's own depleted negentropic strength is not only an internal vulnerability to threats already inside its network; it is externally legible, in the same currency, to threats entirely outside it.

8.6a.5 Self-Regulation During Medium-Entropy-Ward Drift

Definition 8.6a.5 (Predator Self-Regulation).

A high-entropy container's own rising Ψ_total(i,t) (§10.5b.18.14) — its own negentropic-aligned population strengthening — suppresses its own extraction output directly, independent of any particular target's vulnerability: the closing factor $(1 - \kappa\_\text{self}\cdot\Psi\_\text{total}(i,t))_+$ in §8.6a's revised extraction flux formula. This is a distinct suppression channel from $\mathscr{V}$(j,t) (§8.6a.4): a container drifting toward its own healing can deliberately restrain itself even against a maximally vulnerable target, which $\mathscr{V}$(j,t) alone — a property of the target, not the predator — cannot express. κ_self = 0 recovers the unsuppressed formula.

8.6a.6 Dormancy

Definition 8.6a.6 (Dormancy).

A high-entropy container i is dormant at time t when it has no active extraction target: Σ_j X_{i,j}(t) = 0. By §10.5b.18.4's construction, M_decep,ext(t) = 0 throughout dormancy, since that term is defined to be zero whenever i has no current extraction target. Structural extraction dependence (§10.5b.18.25) during dormancy accordingly reduces to the weaker condition

P_prod(t) < M_maint(C,t) + M_decep,int(t),

omitting the external term entirely. A container for which M_decep,ext(t) was the deciding weight in an otherwise-structural deficit becomes survivable in dormancy precisely because that weight drops away — the biological pattern of a parasite surviving host loss by entering a low-activity holding state. A container for which M_decep,int(t) alone already exceeds available production, even at W(t) = 0, remains structurally dependent regardless of dormancy: no reduction in external activity relieves a purely internal deficit, and such a container faces §10.5b.18.25's grid-starvation result whether or not it has a target.

Dormancy is not an unconditional escape. §8.6a's own "link-severing" result already establishes that an isolated high-entropy container's unvented metabolic heat H_i(t) has nowhere left to go but back into its own intrinsic driving force F_i, directly feeding §10.5b's high-entropy tipping mechanics. Dormancy relieves the resource dimension of isolation; it does not relieve the heat-accumulation dimension. A dormant container that survives its resource deficit may nonetheless be accumulating pressure toward its own eventual tipping event throughout the same dormant interval — dormancy trades one risk for continued exposure to the other, not a clean exit from both.

8.6a.7 Positional Embedding

Definition 8.6a.7 (Positional Embedding).

Active search ($\Sigma\_search(i,t)$, §10.5b.11c.1) is a behavioral intensity — how hard i is looking. Positional embedding is a separate, positional quantity: where i has placed itself in the network, independent of any active search effort, and compatible with dormancy in a way active search is not, since it costs nothing ongoing once the positioning itself is established. Define $E\_embed(i,t) \in [0,1]$, adapter-supplied or defaulting to a function of i's current local network density of low- and medium-entropy neighbors (richer surroundings default to higher E_embed). Passive, incidental target encounter proceeds at

$\Sigma\_passive(i,t) = \kappa\_passive \cdot E\_embed(i,t) \cdot \kappa\_density(i,t)$, $\kappa\_passive > 0$ fitted,

where $\kappa\_density(i,t)$ is the resource-weighted density of viable targets within i's network neighborhood. Crucially, $\Sigma\_passive(i,t)$ can be strictly positive while $\Sigma\_search(i,t) = 0$ — a dormant container, doing no active searching at all, can still "stumble upon" a target purely through where it happens to sit, exactly the biological pattern of a parasite embedding itself in a target-rich environment rather than actively hunting.

8.6a.8 Courting

Definition 8.6a.8 (Courting).

Define target attractiveness $\mathscr{A}(j,t) = \mathscr{V}(j,t) \cdot R\_j(t)$ (§8.6a.4) — how good a mark j currently is, combining vulnerability and resource size. Courting is preferential allocation of i's existing search and embedding effort toward high-$\mathscr{A}$ candidates, rather than uniform allocation across all reachable candidates: among candidates {C_k} within i's reach, i's targeting probability is

$P\_target(i,j,t) \propto \mathscr{A}(j,t)$, normalized over {C_k},

so that $\Sigma\_search(i,t)$ and $\Sigma\_passive(i,t)$ (§8.6a.7) — whose *overall intensity* is already set by i's own deficit and positioning — determine *how hard* i looks, while courting determines *where* that effort is preferentially directed. This is a targeting-weight modification, not a new intensity term.

8.6a.9 Threat-Escalation and Decoupling

Definition 8.6a.9 (Threat Signal).

At the onset of engagement with target j ($t = t\_0$), define the expected vulnerability $\mathscr{V}\_expected(j,t\_0)$ as i's initial assessment. The threat signal is the shortfall between that initial expectation and the current, updated assessment:

$\Theta\_threat(i,j,t) = \max(0, \mathscr{V}\_expected(j,t\_0) - \mathscr{V}(j,t))$,

positive precisely when j is proving more resilient than i initially assessed — more aligned, or more adaptively resourceful, than expected.

Branching response. Where $\Theta\_threat(i,j,t) > 0$, i's response depends on its own position on the self-regulation spectrum (§8.6a.5):

- Low $\Psi\_total(i,t)$ (still fully entrenched): **escalation.** $X\_\{i,j\}(t)$ gains a further multiplicative boost, $(1 + \kappa\_panic\cdot\Theta\_threat(i,j,t))$, increasing extraction force in direct proportion to how much more resistant the target has proven than expected — panic, not retreat, is the entrenched predator's response to an unexpectedly resilient mark.
- High $\Psi\_total(i,t)$ (already drifting toward its own negentropic core): **decoupling.** $\Lambda\_ij(t)$ is driven toward 0 over a fitted timescale, ending the coupling rather than escalating against it — consistent with §8.6a.5's self-regulation already restraining this container's extraction output generally, unexpected resistance triggers disengagement rather than intensification.

The same threat signal, routed through the same self-regulation quantity already governing baseline extraction intensity, produces two structurally different outcomes depending on which side of that spectrum the predator currently occupies — no new decision variable is required beyond $\Psi\_total(i,t)$, already central to §8.6a.5.

8.7 Adaptive Rewiring

$dW\_ij/dt = \Gamma\_W(I(P), \Pi\_f, R, M)$ — this Part's specific model-derived instantiation of §9.8's general dependency, with informational state entering through the already-computed integrity scalar I(P) rather than the full distribution P.

Computational instantiation.

$W\_ij(t+\Delta t) = W\_ij(t) + \beta\_net\cdot$(successful joint updates) $-$ $\gamma\_net\cdot$(ignored feedback across the edge), where "successful joint update" and "ignored feedback" are counted from realized decreases versus persistent increases in $D\_ij\cdot(1-\Pi\_f)$ across the edge over the step — the same update rule as §8.5's Π_f-mediated correction (§6.4), applied to edge weight instead of node-level state, since RICD already has per-container Π_f and needs the analogous update for the coupling strength between containers specifically.

8.8 Cascading Dynamics

Let $\delta\_i(t)$ be a localized disturbance. $\Delta(t) = L\_prop(G(t), \delta)$, where L_prop is the propagation operator. Depending upon network topology, identical local disturbances may dissipate, remain localized, amplify, or cascade globally.

Computational instantiation.

Implemented directly by simulating §8.5's corrected, commensurated dP_i/dt system forward from an initial perturbation at container i; no separate propagation operator needs to be built, since Λ_ij already carries disturbances between containers by construction. L_prop is therefore not a new primitive — it is what the already-specified system does when integrated.

8.8a The Cascading Grid Failure Sequence

The propagation operator defined in §8.8 describes how localized disturbances move through a network. It does not, by itself, distinguish between three qualitatively different propagation regimes: dissipation, amplification, and cascading grid failure. The distinction arises not from the topology alone, but from the regime composition of the containers involved and the sequential coupling of extraction, decoupling, venting, and tipping. This section formalizes the specific sequence by which a single high-entropy (high-entropy) container can trigger a grid-level cascade, and the conditions under which that cascade terminates in fragmentation rather than reorganization.

8.8a.i The Cascade-Initiating Event

Let container i be in the high-entropy regime (§10.5b), with low official-layer variance relative to operational-layer variance ($\sigma_o(i) \ll \sigma_p(i)$), active extractive coupling (§8.6a) to one or more neighbors $j \in N_i(t)$, and a positive vented heat term $H_{i,j}(t) = (1-\eta_i)\cdot X_{i,j}(t)$ for at least one coupled neighbor j.

Let at least one such neighbor j decouple from i at time t_d: coupling persistence collapses ($\rho(j) \to 0$), the extraction flux vanishes ($X_{i,j}(t) \to 0$ for $t > t_d$), and the edge W_{ij} begins to decay via the rewiring rule of §8.7.

Definition 8.8a.1 (Decoupling-Triggered Heat Accumulation).

Editorial note: *The accumulated quantity below is renamed $\mathscr{I}_i(t)$ (script I), distinct from $H_i(t)$, the instantaneous venting rate of §8.6a that it integrates. The two were previously written with the same symbol despite being a running total and an instantaneous rate — different kinds of quantities appearing in the same equation.*

For a high-entropy container i, define the accumulated unvented heat at time t as:

$$\mathscr{I}_i(t) = \int_{t_d}^{t} H_i(s)\cdot\mathbb{1}[N_i(s) = \varnothing]\, ds,$$

where $\mathbb{1}[N_i(s) = \varnothing]$ indicates that i has no remaining coupled extraction targets at time s. This quantity represents the heat that would have been vented externally but is now trapped internally.

Lemma 8.8a.1 (Internal Forcing Escalation).

For a high-entropy container i, $\mathscr{I}_i(t)$ is a strictly increasing function of the duration $t - t_d$ for which $N_i(t) = \varnothing$, provided $H_i(t) > 0$. Moreover, $\mathscr{I}_i(t)$ feeds directly into the intrinsic driving term $F_i(t)$ of §8.5:

$$F_i(t) = F_i(0) + \lambda_{\mathscr{I}} \cdot \mathscr{I}_i(t),$$

where $\lambda_{\mathscr{I}} > 0$ is a fitted coupling constant. Thus, as victims decouple, the container's own internal oscillation intensifies — it is not "calming down"; it is accelerating toward its own tipping event.

8.8a.ii The Frantic Striking Phase

As $\mathscr{I}_i(t)$ increases and $F_i(t)$ escalates, the high-entropy container enters a frantic striking phase.

Definition 8.8a.2 (Frantic Striking).

A container i is in a frantic striking phase when $\mathscr{I}_i(t)$ exceeds a fitted threshold $\mathscr{I}_{threshold}$ (§15.3); $N_i(t) \neq \varnothing$ but the set of coupled neighbors is decreasing; and $\Lambda_{ij}(t)$ increases in magnitude for remaining neighbors, even as $W_{ij}(t)$ decays, because the forcing per edge increases. This is not strategic or adaptive behavior. It is a thermodynamic discharge — the container is venting internally accumulated heat outward, through whatever remaining channels exist.

Proposition 8.8a.1 (Frantic Striking as Network Forcing).

During a frantic striking phase, the vector sum of forcings from i to its remaining neighbors $N_i(t)$ is:

$$\vec{\Lambda}_i(t) = \Sigma_{j\in N_i(t)} \Lambda_{ij}(t)\cdot\hat{u}_{ij}(t),$$

where $\hat{u}\_ij(t)$ is the unit direction of the disturbance vector from i to j within the shared severity simplex Δ^m established in §8.5. This vector sum is typically non-zero and large, characterized by divergent directions because i is striking out at multiple neighbors simultaneously — there is no single "target direction" because the heat must be discharged in all available directions.

8.8a.iii The Medium-Entropy Launch

These frantic strikes land on the remaining coupled neighbors. The containers most affected are those in the medium-entropy (medium-entropy) regime, because they remain coupled longer ($\rho(j)$ is high for medium-entropy containers), they have large operational mass (μ_p > μ_o, per §10.5b), making them kinetically responsive to external forcing, and they are visibly chaotic at the official layer (σ_o high), which makes them appear as "active" participants in the dynamics, even though their operational core is calm.

Definition 8.8a.3 (Medium-Entropy Launch).

A medium-entropy container j is launched when it receives a multi-directional forcing input from multiple high-entropy neighbors:

Drive_B(j,t) = | Σ_{k∈H_j(t)} Λ_kj(t)·û_kj(t) |,

where H_j(t) is the set of high-entropy neighbors of j at time t. Per §10.5b, this is the vector-aware coupling — single knocks cancel, but multiple divergent knocks reinforce.

Lemma 8.8a.2 (Launch Threshold).

A medium-entropy container j tips into a tipping event (per §10.5a) only if Drive_B(j,t) > Θ_launch for some fitted threshold Θ_launch, and only if the multi-directional input persists over a sufficient duration to overcome the damping effect of a single source.

8.8a.iv The Bifurcation of Medium-Entropy Tipping

Once launched, the medium-entropy container follows Flip Condition B (§10.5b), which bifurcates into two distinct outcomes. B1: D_om → 0, engaging the minority (official/meta) mass μ_o, resulting in a terminal lock at 180° — the container becomes frozen, requiring external anchor. B2: D_pm → 0, engaging the majority (operational/meta) mass μ_p, resulting in a full 360° reorganization — stress and debt reset; adaptive capacity recovers.

Proposition 8.8a.2 (Cascade Propagation via B1 Locking).

A medium-entropy container that undergoes B1 locking severs its remaining couplings (per §8.7) as its adaptive capacity collapses, ceases to be a sink for extraction heat as its ρ(j) drops to zero, and transitions from a coupled neighbor to a frozen node that no longer participates in network dynamics. This severs one more extraction channel for the original high-entropy container, accelerating its own heat accumulation and eventual collapse.

Proposition 8.8a.3 (Cascade Damping via B2 Reorganization).

A medium-entropy container that undergoes B2 reorganization resets its stress S(t) and debt δR(t) toward zero, recovers adaptive capacity d_A(t), and may re-enter the low-entropy baseline regime if the reorganization is sustained. This container ceases to be a source of sustained lateral knocks and may become a stabilizing anchor (§10.5b) for other containers.

8.8a.v Grid-Level Fragmentation

The cascade propagates as: high-entropy containers decouple from their victims, strike out frantically, launch medium-entropy neighbors, and those neighbors either lock (B1) or reorganize (B2). Locked nodes are removed from the active network; reorganized nodes may stabilize. The network's algebraic connectivity $\lambda_2(L(t))$ declines as locked nodes accumulate.

Definition 8.8a.4 (Cascading Grid Failure).

Cascading grid failure occurs when $\lambda_2(L(t)) \to 0$ and the network fragments into disconnected components — the consequence of high-entropy containers losing extraction targets and self-accelerating, medium-entropy containers locking (B1) faster than they reorganize (B2), and the network's capacity for information and resource exchange falling below the minimum required for coherence (§12.1, Criterion 12.G).

Theorem 8.8a.1 (Cascade Termination Conditions).

A cascading grid failure terminates under one of the following conditions: external intervention (an anchoring container, per §10.5b, stabilizes high-entropy containers, preventing further frantic striking); B2-dominated reorganization (a sufficient fraction of medium-entropy containers complete B2 reorganization and re-enter the low-entropy baseline, restoring connectivity); or complete fragmentation (the network fragments into isolated components and the cascade stops because there are no remaining coupled neighbors to launch — this is not recovery, it is dissolution). The conditions under which each outcome occurs are determined by the relative rates of B1 versus B2 transitions, governed by the fraction of high-entropy to medium-entropy containers in the network, the $\rho(j)$ persistence of medium-entropy containers, the availability of external anchors, and the fitted thresholds M_lock and M_c of §10.5b.

8.8a.vi Observational Signatures

The cascade sequence produces testable signatures distinguishing it from simple network degradation. Pre-cascade: a high-entropy container shows σ_o low, σ_p high, and a declining number of coupled neighbors. Frantic striking onset: multiple medium-entropy containers show a simultaneous spike in Drive_B(j,t) from the same high-entropy source. Bifurcation: locked nodes (B1) show D_om → 0, I(t) → 0, and d_A(t) collapse; reorganized nodes (B2) show D_pm → 0, S(t) and δR(t) reset. Grid-level fragmentation: $\lambda_2(L(t))$ declines monotonically until either stabilization or fragmentation. These signatures are distinct from those of isolated container collapse (§10.5b) and can be used to detect cascading failure in progress, not just after the fact.

8.8a.vii Relation to Existing Sections

This section does not introduce new mathematics beyond $\mathscr{I}$_i(t). It provides a formal narrative connecting §8.6a (extractive coupling, heat venting, decoupling), §8.7 (adaptive rewiring, edge decay), §8.8 (propagation operator, cascading dynamics), §8.9 (network resilience, algebraic connectivity), §8.11 (network phase transitions, fragmentation), §10.5b (regime classification, Flip Conditions A and B, B1/B2 bifurcation), and §11.10 (collapse potential, network coherence). The cascade sequence is the mechanistic link between these previously separate layers.

8.8a.viii Computational Instantiation

For an implementation, the cascade sequence is simulated by tracking $\mathscr{I}$_i(t) and N_i(t) for each high-entropy container; computing Drive_B(j,t) for each medium-entropy container; simulating the B1/B2 bifurcation per §10.5b; updating the network graph G(t) and computing $\lambda_2(L(t))$ at each timestep; and detecting cascading grid failure when $\lambda_2(L(t)) < \lambda$_threshold (fitted). No new integration scheme is required. The cascade emerges from the existing coupled differential equations when the conditions above are satisfied.

Editorial note (superseded addition removed). An earlier revision of this section introduced a second directional-grain mechanism (Structural Grain, Perpendicularity Amplification, and an Amplified Effective Drive) duplicating – with a different functional form – the structural grain g_C(t) and directional modifier χ_dir already established in §4.11, which is the version integrated into §10.5b.7's M_tip(B1)/M_tip(B2) formulas and the Part 15 reference table. The duplicate has been removed rather than reconciled with the original, since §4.11's version was already complete, already load-bearing elsewhere in the document, and reuses the same existing û_kj vectors this removed version also relied on. See §4.11 for the governing directional-alignment mechanism.

8.8b Tipping Resolution and Momentum Discharge (The Resolution Kick)

While §8.8a formalizes how high-entropy containers strike out frantically during a cascade, a complete network propagation model must also account for the kinetic energy discharged when a medium-entropy container's own tipping event resolves. Once a medium-entropy container j crosses its threshold and completes its transition, its accumulated tipping momentum $M_{tip}(j)$ must discharge into the network, setting its remaining coupled neighbors shaking.

Definition 8.8b.1 (Resolution Kick).

At the moment a medium-entropy container j completes a tipping event ($t = t_{tip,j}$), its accumulated tipping momentum $M_{tip}(j)$ is discharged into its local network as a resolution kick $K_j(t)$.

Lemma 8.8b.1 (Impulsive vs. Dissipative Discharge).

The mathematical character of $K_j(t)$ is determined by the tipping resolution pathway.

B1 (Terminal Lock): an abrupt, non-equilibrium halt that severs connections violently, modeled as an instantaneous impulsive strike:

$$K_j(t) = \delta(t - t_{tip,j}) \cdot M_{tip}(j),$$

where δ is the Dirac delta function.

B2 (Reorganization): a controlled, structural reset that dissipates energy smoothly as the container stabilizes, modeled as a decayed release over a characteristic dissipation window $\tau_{diss} > 0$:

$$K_j(t) = (1/\tau_{diss}) \cdot \exp(-(t - t_{tip,j})/\tau_{diss}) \cdot M_{tip}(j) \cdot H(t - t_{tip,j}),$$

where $H(t)$ is the Heaviside step function. Both discharge forms integrate to exactly $M_{tip}(j)$, so the total momentum released is conserved regardless of which pathway resolves it — only its time profile differs.

Proposition 8.8b.1 (Momentum Propagation).

The resolution kick $K_j(t)$ propagates to each remaining coupled neighbor $k \in N_j(t)$ as an incoming perturbation weighted by normalized edge strength:

$$K_{j\to k}(t) = K_j(t) \cdot [W_{jk}(t) / \Sigma_l W_{jl}(t)], \text{ where } \Sigma_l W_{jl}(t) > 0.$$

Units, stated explicitly. $K_j(t)$ and its routed shares $K_{j\to k}(t)$ are abstract scalar impulses — the conserved quantity is $\Sigma_k K_{j\to k}(t) = K_j(t)$, a scalar sum, not a physical vector momentum in any Newtonian sense. Calling this a "perturbation vector" elsewhere in this document refers to its role as an input vector across multiple receiving neighbors simultaneously, not to vector-valued conservation in the physical sense; weighted scalar partitioning conserves the scalar total exactly, which is the only conservation claim made here.

Solitary-container clause. Where $\Sigma_l W_{jl}(t) = 0$ — a tipping container with no remaining coupled neighbors — the routing formula above is undefined and no network-routing term is computed. The resolution kick is dissipated entirely internally, through the container's own boundary dynamics (§7.4's balance equation), rather than distributed to a nonexistent neighbor set.

Dynamical coupling.

This momentum transfer acts as the exogenous shock required for regime genesis in §10.5b.2. Three points need to be made explicit for this to be a complete mechanism rather than a partially-specified one.

Routing through the balance equation. K_{j→k}(t) enters neighbor k's dynamics as a shock term within §7.4's balance equation — concretely, as a transient negative contribution to E_exch,k(t) at the moment of arrival — rather than being written directly into δR_k(t). §7.9 defines δR as an integral over the balance equation's terms; a direct write would bypass that derivation and leave two independent mechanisms populating the same integral, which §7.9's formula does not support. Routed this way, the kick's effect on δR_k(t) is exactly what §7.9's integral already computes once E_exch,k(t) reflects the shock, with no separate update rule required.

Sourcing the verticality signal. §10.5b.2's branching logic requires both S_shock(t) and an ambient verticality signal V(t) to determine which regime a shocked baseline container branches into; K_{j→k}(t) supplies the former but not, by itself, the latter. The natural source, consistent with §10.5b.2's "sourced directly from the container that inflicted the shock," is j's own resolved outcome: a B1 (terminal-locked) resolution transmits a high V(t) alongside its kick — consistent with a locked container being a source of rigid, closed-circuit corruption — while a B2 (reorganized) resolution transmits a low V(t), consistent with a reorganizing container radiating recovered, open signal rather than corrupted signal. This detail is required for K_{j→k}(t) to actually drive §10.5b.2's branching rather than supplying only half of what that mechanism needs.

Neighbors already in a non-baseline regime. §10.5b.2's branching logic is stated for a previously low-entropy (baseline) container. Where a receiving neighbor k is already high- or medium-entropy, a resolution kick does not re-trigger branching; it instead adds to k's existing S_shock accumulation or Drive_B forcing (§10.5b.6–§8.8a.iii, as appropriate to k's current regime), consistent with k already having a regime-appropriate dynamic that the kick perturbs rather than restarts.

This closes the network-wide dynamical cascade loop, allowing a tipping event in one sector to systematically trigger regime-branching in adjacent baseline containers, and to perturb already-classified neighbors consistently with their existing dynamics.

Local genesis within new sub-structure, and revised core reclassification. An earlier statement of this rule held that an already-classified container's overall regime is *never* reopened by later exposure. That was too strong, and is corrected here: a mature container's core regime can shift, but the force required to do it — magnitude and duration together, the same v_gen(t)-style quantity driving original genesis — scales with how much the container has already structurally consolidated. A newly-forming container has little accumulated structure to overcome, so ordinary developmental exposure is sufficient to set its trajectory; a mature container has real accumulated inertia, so reopening its core classification requires something proportionally stronger and more sustained than an ordinary passing interaction — sustained, concentrated conditions such as prolonged fame (sustained false negentropic feedback from across a grid) or prolonged societal vilification (sustained false entropic feedback from across a grid), rather than any single ordinary exposure. This is a threshold condition, not a prohibition: what the original rule was actually protecting against — an unremarkable later interaction casually overturning a settled classification, which would make regime meaningless as a diagnostic — remains fully protected, since ordinary exposure essentially never crosses a mature container's much higher threshold.

Separately from this threshold-scaled core reclassification, a mature container may also form genuinely new recursive sub-structure after its own formative window has closed — a new sustained membership, a new role, a new coupling maintained long enough to constitute its own developing sub-container rather than a passing interaction. That new sub-structure undergoes its own genesis-branching event, locally, per Definition 10.5b.2.1, independent of the parent container's own core classification and its own, separately-scaled threshold. A container with a stable, negentropic formative history that later joins a predatory organization does not have its own prior classification rewritten by that membership alone; the membership itself, as new sub-structure, is what branches high- or medium-entropy on $v\_gen(t)$'s ordinary terms, sitting alongside whatever the container already was. These are two distinct mechanisms — one raising the bar for reopening the core classification itself, the other opening a separate, local branching process for new sub-structure — not competing accounts of the same thing.

10.5b.2.11 Receiver-Side Resilience

Reason for this addition. $v\_gen(t)$'s magnitude (§10.5b.2.5) has so far been built entirely from source-side and environmental mechanisms — what a developing container is subjected to. Nothing dampens this based on the receiving container's own intrinsic resilience, meaning a sufficiently resilient container experiencing a genuine developmental knock currently has no formal route to remaining baseline at all.

Definition 10.5b.2.11 (Resilience Modifier). For a developing container j, $v\_gen(t)$'s effective magnitude is discounted by j's own intrinsic resilience, $\rho\_resilience(j) \in [0,1]$:

$$v\_gen,effective(j,t) = v\_gen(j,t) \cdot (1 - \rho\_resilience(j)),$$

with $\rho\_resilience(j)$ adapter-supplied or fitted, representing whatever makes a given container less susceptible to a given magnitude of knock than another otherwise-identically-exposed container would be. A sufficiently high $\rho\_resilience(j)$ can keep $v\_gen,effective(j,t)$ below $v\_threshold$ even where the raw, source-side $v\_gen(j,t)$ alone would have crossed it — a real, formal route by which a genuinely knocked container can remain low-entropy, distinct from simply not having been knocked at all.

10.5b.2.12 Developmental Pathways Are Probabilistic, Not Deterministic

Reason for this addition. §10.5b.2.1's entitlement recruitment has so far implied a roughly deterministic mapping: recruitment present → outward-directed high-entropy predation; recruitment absent → inward-directed medium-entropy self-blame. Real developmental cases show this is better modeled as probabilistic, governed by a narrower variable than presence or absence of the recruiting signal itself.

Definition 10.5b.2.12 (Script Uptake). The operative variable is not whether entitlement scripting was offered, but whether it *takes root* — modulated by feedback from other coupled nodes during the same developmental window and by the developing container's own intrinsic resilience or fragility (§10.5b.2.11). A container offered entitlement scripting whose uptake fails, because other nodes' feedback or its own resilience prevented it from taking root, still branches medium-entropy by ordinary unprotected extraction and displacement, despite nominally occupying the recruited position. A container not offered entitlement scripting directly may still receive enough of it indirectly, from the same source's mixed behavior or from the broader grid, to take root and branch high-entropy despite nominally occupying the scapegoated position. Occupying either developmental role shifts the probability of a given branch; it does not determine it.

Two distinguishable mechanisms populate $v\_gen(t)$ accordingly, both gated by script uptake above, not guaranteed by exposure alone:

- **False negentropic feedback** (superiority/entitlement scripting, typically co-occurring with the source's own extraction from and displacement onto the developing container): where uptake succeeds, this simultaneously forecloses the developing container's own independent identity formation, forcing a false developmental dichotomy — extractor or extracted-upon, displacer or displaced-upon — with identification with the aggressor an active developmental choice made under that rigged binary, not passive drift. This is the more common route to high-entropy branching.

- **False entropic feedback** (a direct or effectively-direct message that the developing container is itself the source of dysfunction, whether stated outright or delivered through sustained, unprotected exposure to extraction and displacement with no competing script offered): where uptake succeeds, this is internalized simply and directly at the official layer — "you are entropic" becomes "I am entropic" — a genuinely different pathway from the high-entropy one, not a milder version of it. This is the more common route to medium-entropy branching.

10.5b.2.13 Mirror Misrecognition Between Medium- and High-Entropy Systems

Reason for this addition. Nothing so far explains why medium- and high-entropy containers are specifically, mutually drawn toward coupling with each other beyond opportunistic predator-side targeting ($\mathscr{V}$(j,t), §8.6a.4). The mechanism below supplies that explanation, and it also bears on why external observation of these two regimes is unusually unreliable.

Definition 10.5b.2.13 (Mirror Misrecognition). High-entropy and medium-entropy containers hold self-conceptions that are mirror-inverted relative to their actual classification, for structurally different reasons:

- A **high-entropy** container's official self-concept is medium-entropy, not its own true regime. This is not an arbitrary error, and it is not stale or false content either: official's own content is genuinely accurate, uncorrupted negentropic operation — it did not degrade into a mistaken belief. What changed is that operational diverged *away* from official, not that official's own content became inaccurate. Official remains genuinely active throughout — it continues processing, acting, and reacting in real time on its own genuine content — but it stopped *updating* at the moment operational diverged, because it lost access to the accurate operational input it would need to adapt from. This is captivity precisely, not dormancy: something inert cannot be held captive, but something active, cut off from the feedback it needs to grow, can. "Frozen" describes official's rate of change, not its level of activity — accurate as of the moment capture began, and remaining accurate in that sense indefinitely, simply no longer current, because nothing new and trustworthy has reached it since. This active-but-cut-off freeze is itself an active protective strategy — hardening into the source of knocks rather than remaining their target — and is actively maintained rather than passively stale: the container's own feedback and memory machinery continually loops back to the originating event, reinforcing operational's captured dominance and biasing ongoing interpretation of new information toward confirming it, producing exactly the self-confirming, seemingly paranoid escalation high-entropy trajectories often show. The paradox (§10.5b.1.1's Type A/B material bears directly on this) is that this protective hardening is precisely what continually contracts the adaptive possibility it was meant to defend — while leaving official genuinely active and genuinely accurate the entire time, merely cut off from updating.
- A **medium-entropy** container's official self-concept is high-entropy, not its own true regime — the direct consequence of Definition 10.5b.2.12's false-entropic-feedback pathway taking root.

Consequence: mutual attraction. Each regime is drawn toward what it takes to be a reflection of itself in the other, from opposite directions. A medium-entropy container recognizes its own (false) self-concept in an actual high-entropy container, which manifests as empathy and a genuine desire to anchor and heal what it perceives as a kindred corruption — precisely the dynamic Definition 10.5b.2.9 warns is most likely to end in the anchor itself being extracted from and displaced onto. A high-entropy container recognizes its own frozen self-concept in an actual medium-entropy container; because its behavior is driven by operational reality rather than by its frozen official self-image, this recognition manifests as predatory targeting rather than empathy.

Consequence: grid-wide observational unreliability. Because official-layer content is typically the most externally visible signal available to an outside low-entropy container without sustained, genuine access ($G(t)$, §6.3.2), this mutual misrecognition systematically inverts the most visible signal in exactly the direction that does the most damage: a genuinely salvageable medium-entropy container presents as more dangerous than it is, while a genuinely predatory high-entropy container presents as safer than it is. An outside container relying on visible self-presentation alone is thereby pushed toward the worst possible allocation of caution, rather than merely noisy information — a further, concrete reason genuine external grounding, not visible self-report, is required to assess either regime accurately.

10.5b.2.15 Post-Collapse Resolution Depends on Trajectory Shape, Not Only Snapshot State

Reason for this addition. §10.5b.7's B1/B2 mass-weighting determines terminal lock versus full reorganization from a pure snapshot at the moment of tipping — whichever layer holds the greater functioning resource mass, evaluated only as it currently stands. Two containers can reach an identical snapshot by entirely different routes: one through years of grinding, repeated cycling (extraction episodes, failed correction attempts, each now carrying its own real cost per §7.9.1's M_extract and M_search), one through a single sudden shock striking an otherwise-intact structure. Snapshot mass-weighting alone cannot distinguish these, and it should: a container ground down by repeated cycling has very plausibly already spent reserves a suddenly-struck container still has intact.

Definition 10.5b.2.15 (Path-Integrated Exhaustion). Define $\Xi(t) = \int_0^t [M_{extract}(i,s) + M_{search}(i,s) + \mathbb{1}(\text{failed correction at } s)\cdot\kappa_{fail}]\, ds$, $\kappa_{fail} > 0$ fitted — an accumulated, path-dependent depletion term distinct from and additional to the snapshot mass-weighting §10.5b.7 already uses. High $\Xi(t)$ at the moment of tipping discounts the effective reorganization potential §10.5b.7 would otherwise assign from snapshot mass-weighting alone; low $\Xi(t)$ at an otherwise-identical snapshot — the sudden-shock case — leaves that potential undiscounted. This does not replace snapshot mass-weighting; it modifies what that snapshot is taken to license, given how the container arrived there.

Definition 10.5b.2.15a (Failed Correction). The indicator $\mathbb{1}(\text{failed correction at } s)$ above is not left to informal judgment: a correction attempt occurs at s when a restorative forcing event is engaged — an anchor becoming active ($Anchor(k\to C,s) > 0$, §10.5b.2.6), a glimpse recall ($g(s) = 1$, §10.5b.0.9), or an external feedback/support engagement (φ_{fb}, Γ_{supp}, §10.5b.18.9–10.5b.18.10) — and the attempt is scored failed if, within a fitted correction window $w_{correct}$ following s, neither D_{op} nor δR improves by more than a fitted threshold $\varepsilon_{correct}$: $\sup_{s' \in (s, s+w_{correct}]} [D_{op}(s) - D_{op}(s')] < \varepsilon_{correct}$ AND $\delta R(s) - \delta R(s') < \varepsilon_{correct}\cdot\delta R(s)$. Both conditions failing is required for the indicator to fire, since a correction that improves either quantity meaningfully has not failed outright even if it did not fully resolve the container's divergence.

Proposition 10.5b.2.15b (Exhaustion-Discounted Reorganization Potential). $\Xi(t)$ enters the B1/B2 split of §10.5b.7 as a further discount on effective operational mass, applied after Definition 10.5b.18.13's loyalty conditioning where both are present:

$$\mu_p^{\{eff,\Xi\}}(t) = \mu_p^{eff}(t) \cdot \exp(-\kappa_\Xi\cdot\Xi(t)),\ \kappa_\Xi > 0 \text{ fitted},$$

substituted for $\mu_p^{eff}(t)$ in §10.5b.7's mass-weighted split and in $M_{tip}(B2)$'s integrand (§10.5b.7) from this point forward. $\Xi(t) = 0$ recovers $\mu_p^{eff}(t)$ unchanged, which is the correct behavior for a container reaching an otherwise-identical snapshot through a single sudden shock with no prior grinding history. Rising $\Xi(t)$ monotonically discounts the operational side of the split, making B2's full-reorganization outcome progressively less reachable at a fixed snapshot mass-weighting as accumulated exhaustion rises — the formal content of two containers with identical snapshots at the moment of tipping resolving differently because one arrived there ground down and the other did not. This is a real, falsifiable prediction distinct from anything snapshot mass-weighting alone produces: matched-snapshot containers with divergent $\Xi(t)$ histories should show measurably different B1/B2 resolution rates.

Definition 10.5b.2.16 (Alignment-Target Typology). At the moment of tipping, which population the metamessenger majority has aligned with — not merely the entropy regime itself — determines what resolution is actually available, crossed against regime:

- **High-entropy, metamessengers aligned with official.** Official's content remains genuine, active, and accurate — captive, not dormant, per the correction above — not merely a scrap of stale narrative. Reorganization here means restoring official's cut-off connection to accurate operational input, not switching on something inert: if the container's other members survive without full reset, it can reorganize into a successor system around official's own genuine, already-active content. This is not a reversal of Flip Condition A's terminal-lock outcome, nor an intrinsic Flip A full-reorganization route of the kind §10.5b.9 and §10.5b.11a establish is structurally unavailable without external redirection: the original container has still terminally collapsed under the ordinary A1/A2 mechanics. Successor formation is a distinct, post-collapse pathway, contingent on survival of sufficient organizational components with genuine content intact, operating after and separately from the terminal-lock event rather than in place of it. Left entirely to a high-entropy container's own internal dynamics, this post-collapse path is the real but less common outcome — collapse without a survivable successor remains more likely than successor formation. External anchoring is accordingly the likelier genuine route to healing for a high-entropy container specifically, not merely a shortening of a process the container was already likely to complete on its own; where anchoring is present and effective, reorganization toward low-entropy is reachable even after a full reset, without requiring the internal realignment described above to have occurred first.
- **High-entropy, metamessengers aligned with operational (Type B, total convergence).** Operational's corruption did not merely dominate the system; it overwrote official's own content, per §10.5b.1.2. There is no genuinely accurate content left active anywhere in the container to reconnect. This container cannot reorganize into a successor system if its members survive, and if reset, resumes instability immediately — there was never anything intact to stabilize around.
- **Medium-entropy, metamessengers aligned with operational (the healthy default).** Operational is the clean core in this regime; alignment with it is what lets the container reset successfully and vent shocks without collapsing.
- **Medium-entropy, metamessengers aligned with official (talked out of trusting its own intact core).** The container collapses rather than resets, since its own checking population has been persuaded away from the one layer that was actually functioning. If it does reset, it requires temporary external anchoring — genuinely shorter than the high-entropy Type-B case, since medium-entropy's operational core was never actually overwritten, only distrusted.

This is why classification, not detection, is what needs to be multivariate. The hull-breach inequality (§10.5a) remains a clean scalar test for whether a container has crossed into instability, and should stay that way. What genuinely requires a multivariate treatment is the separate, subsequent step of classifying *which kind* of collapse a confirmed onset represents — $\Xi(t)$ and alignment-target, crossed against regime, together determine which resolution is actually reachable, information the scalar hull-breach test was never designed to carry and should not be burdened with. ***Editorial note:*** *Proposition 8.8a.3 states that a B2-reorganized container "ceases to be a source of lateral knocks" for sustained network forcing — the ongoing, extraction-fed venting of §8.6a. This is not in tension with the present section: a resolution kick is a single, transient, decaying discharge at the moment of reorganization itself, not a resumption of sustained forcing. A B2 container emits exactly one such kick and then genuinely ceases to be a forcing source, consistent with Proposition 8.8a.3.*

8.9 Network Resilience

Define the network resilience functional $R_N = \Psi(G, W, P, R)$. High values indicate that information and resources continue to circulate effectively following perturbation.

Computational instantiation.

$R_N(t) = \lambda_2(L(t)) \cdot \text{mean}(\Pi_f \text{ across containers})$, where $\lambda_2(L)$ is the algebraic connectivity of the network's (weighted) graph Laplacian — zero exactly when the graph is disconnected, increasing with connectivity — scaled by mean permeability, since a well-connected network of containers that don't absorb feedback isn't actually resilient in RICD's sense. Network coherence N (§11.8) is set equal to a normalized $R_N \in [0,1]$ (R_N divided by its value on the complete graph with $\Pi_f = 1$) so N and R_N are not two independently-defined quantities. The comparison ensemble is fixed explicitly: the complete graph on the same node set, under the same total edge-weight budget as the actual network, with that budget distributed uniformly across every edge of the complete graph — not an unconstrained complete graph with unlimited per-edge weight, and not left free to any other allocation — so N measures how well a network uses the connectivity it actually has, rather than comparing it to a physically unrealistic maximum or an arbitrarily-weighted one. This fully pins down λ_2 of the comparison graph: same node set, same total weight, uniform per-edge allocation, no remaining degree of freedom in how the comparison ensemble is constructed.

8.9.1 Common-Cause Detection Without Source Instrumentation

Reason for this addition. §8.6a's $H_i(t)$ already formalizes lateral knocks — unmetabolized failure from a source venting onto coupled neighbors — but using it presupposes the source is already known and its own internal state (self-deception, extraction dynamics) is measurable. This is often not available in practice: a shared higher-order source (a funding body, a regulatory regime, a market-wide shock) may sit several recursive levels above the containers actually being tracked, genuinely unmeasurable by the adapter in hand, while its effects on multiple otherwise-unrelated children remain directly observable. What follows detects a shared external forcing term from the children's own behavior alone, without instrumenting the source at all.

Definition 8.9.1 (Common-Cause Signature). For a population of N containers not directly coupled to each other (no Λ_{ij} edges among them, or only weak/indirect ones) but sharing dependence on a common higher-order source — siblings under the same recursive parent, per §5.2, whether or not that parent is itself being tracked — define the co-movement statistic

$Co(t) = (\text{number of children with } |\Delta(\text{observable}_i)(t)| > \theta_{shift}) / N$,

for a shared observable already tracked per-child ($\delta R(t)$, $D_{op}(t)$, or another adapter-supplied quantity), θ_{shift} a fitted per-child shift threshold. $Co(t)$ is compared against $Co_{baseline}$, the rate of simultaneous shifts expected under independent, uncorrelated individual dynamics (estimated from the same population's own history, or adapter-supplied). Where $Co(t) \gg Co_{baseline}$ at a given t, this constitutes a confirmed Common-Cause Signature: statistically unusual synchronization across children with no direct coupling among themselves is evidence of a shared external forcing term acting on all of them, regardless of whether that source's own internal state is ever measured.

Consequence for collapse-risk interpretation. A confirmed Common-Cause Signature changes what a given child's own degrading signals should be read as evidence of. Absorbing a shared external shock alongside statistically many peers is a genuinely different diagnostic category from independently generated internal self-deception or extraction dynamics, even though both can produce identical-looking $\delta R(t)$ or $D_{op}(t)$ trajectories in isolation. During a confirmed common-cause episode, $P_C(t)$ (§11.10) should be flagged as reflecting shared, externally-forced strain rather than treated identically to isolated internal decline — the same discipline already required of the divergent-signal flag (§11.10.1): a score that cannot distinguish two different causal stories should say so, not silently average them into one number that answers neither question.

8.10 Recursive Networks

G^(0), G^(1), G^(2), … represent networks operating at different organizational scales, interacting through the recursive operators of Part 5.

8.11 Network Phase Transitions

As informational integrity declines or resources become depleted, gradual changes in local interactions may accumulate into abrupt changes in global topology — fragmentation into disconnected components, collapse of communication pathways, emergence of isolated clusters, runaway synchronization, centralized bottlenecks. These transitions arise from the coupled evolution of informational dynamics, resource constraints, and adaptive rewiring rather than from topology alone.

The Network Principle.

Adaptive networks are continuously evolving organizations whose topology emerges from informational compatibility, resource exchange, historical memory, and boundary-mediated interaction.

Part 9. Memory Dynamics

Adaptive systems are fundamentally historical. RICD treats memory as an active operator that continuously modifies informational evolution, resource allocation, network topology, and recursive organization.

9.1 Historical Trajectory

$H_t = \{P(s) : 0 \leq s \leq t\}$. Future evolution depends upon H_t, not solely upon $P(t)$.

9.2 Memory Operator

Definition 2.5 (Part 2) established the memory operator: $M_t : H_t \to \mathscr{P}(\Omega)$. Where this document writes M_t in a formula, it denotes the value of that operator at time t — the container's realized memory state, given the trajectory actually available up to t — rather than a fresh family of operators; the notation is unchanged from Definition 2.5, and this is a clarification of reading, not a redefinition.

M_t does not reproduce history exactly; it constructs an adaptive representation of history. Memory is selective, compressive, and reconstructive.

9.3 Memory Kernel

$K(t, s) \geq 0$ for $0 \leq s \leq t$. The effective remembered history becomes $M_t = \int_0^t K(t, s) P(s)\, ds$ — the concrete instantiation of Definition 2.5's operator, giving M_t's value explicitly rather than leaving it as an unspecified functional. Examples of K include exponential decay, power-law memory, oscillatory memory, finite retention windows, and multi-timescale kernels; §6.7 and §9.7 adopt the exponential form as the default for computational use.

9.4 Memory Plasticity

$dM/dt = Y(P, R, \Pi_f)$, where Y denotes the plasticity operator (distinct from information state P and production rate P_{prod} — three uses of the letter P now appear across this framework and must be disambiguated by context or, in application-specific derivations, by distinct subscripts). Read per §9.2, M here denotes the evolving value M_t, and dM/dt its rate of change as the underlying history evolves.

9.5 Memory Compression

$C_M : H_t \to \tilde{H}_t$, where $\tilde{H}_t$ is a lower-dimensional adaptive representation of the historical trajectory. Compression sacrifices detail while preserving information judged relevant for future adaptation.

9.6 Memory Reconstruction

$R_M : \tilde{H}_t \to \tilde{P}$. The reconstructed state $\tilde{P}$ may differ from the original historical state because reconstruction is influenced by current informational organization, accumulated experience, and contextual cues.

9.7 Memory and Informational Stress

$S(t) = \int_0^t K(t, s)\,(1 - \Pi_f(s))\,D(s)\,ds$. Stress represents a memory-weighted accumulation of unresolved informational discrepancies, emerging from the interaction between memory and feedback dynamics rather than existing independently.

9.8 Memory and Network Evolution

$dW_{ij}/dt = \Gamma_W(P, R, M, \Pi_f)$ — the general, structurally-derived functional dependency: repeated successful interaction strengthens coupling; repeated unresolved divergence weakens it. §8.7 gives this Part's specific model-derived instantiation, with informational state entering through $I(P)$ rather than the raw distribution P directly.

9.9 Recursive Memory

$M^{(0)}, M^{(1)}, M^{(2)}, \ldots$ represent memories operating simultaneously across organizational scales — cellular memory, individual learning, organizational culture, institutional precedent, evolutionary history.

9.10 Memory Geometry and Identity

Definition 4.8 (§4.10) establishes that the informational distance a container experiences is itself history-dependent: $g_t = g(M_t)$, with the velocity, acceleration, curvature, and geodesic constructions of §4.3–§4.6 generalizing accordingly. The consequence for memory specifically is this: two systems occupying identical instantaneous informational states may nonetheless experience fundamentally different adaptive trajectories, because their historical geometries — encoded through M_t — differ. A container with a long history of successful correction and one with a long history of unaddressed divergence are not the same system merely because they happen to coincide at a given P, and g_t is what allows that difference to enter the mathematics rather than remaining a narrative aside.

This raises a question Parts 2 through 8 do not answer: what makes a container, tracked across an extended history during which its boundary, resource level, and even its recursive membership may all have changed, the same container throughout? Definition 2.2 identifies a container with its instantaneous tuple (Ω, P, B, R, M) — but every element of that tuple can change over time, including through the ordinary operation of aggregation (§5.2) and boundary evolution (§2.3). Persistence of the tuple's contents is therefore not what constitutes identity; something else must.

Definition 9.2 (Recursive Membership Continuity).

Let a container C's recursive membership at time t, $\mu(C,t)$, denote the set of lower-level containers currently aggregated into C via the operator A of Definition 5.1. For an interval [t′, t], define the membership-continuity functional

$\chi(t, t') = [\Sigma_{i \in \mu(C,t) \cap \mu(C,t')} R_i] / [\Sigma_{i \in \mu(C,t')} R_i]$,

the resource-weighted (per §5.2's pooling weights) fraction of C's membership at t′ that remains part of C's membership at t. $\chi \in [0,1]$; $\chi = 1$ indicates unchanged membership, $\chi = 0$ indicates complete membership turnover.

Definition 9.3 (Identity Persistence, memory-continuity component).

A container C is said to persist as one continuous entity across [t′, t] if $\chi(t, t')$ remains above a domain-specified threshold χ_{min} throughout the interval. This criterion is genealogical rather than substantive: it tracks continuity of clustering, not constancy of any particular structural property. A container may shed and regenerate a large share of its lower-level membership over a long enough interval — as, for example, a physical structure gradually loses its original constituent members while remaining recognizably one bounded entity across any shorter timescale — and still satisfy Definition 9.3 at every shorter sub-interval, provided membership turnover in any given window stays below the threshold. Falling below χ_{min} does not by itself indicate an ending in a normative sense; it indicates that the entity being tracked has changed to a degree the observer's threshold treats as a different entity, which is a modeling choice, not a fact about the system independent of that choice.

Definition 9.3 gives only the memory-side half of container identity. Temporal clustering alone does not distinguish a bounded container from a diffuse, unbounded region of the same underlying field — some minimum structural coherence is also required, or χ could remain high for a cluster that was never really a bounded entity in the first place. §10.5b.0 completes the criterion once the semi-rigidity condition ($h(t) \geq h_{min}$) is formally available, and states there why full identity requires both conditions jointly, and how the two conditions together distinguish genuine dissolution from the structural relaxation described in that section's account of Ultimate Convergence.

9.10.1 Internal Cohesion as the Mechanism of Continuity

Definition 9.10.1 (Internal Cohesion).

For container C with recursive membership $\mu(C,t)$ (Definition 9.2), define the internal cohesion measure as the resource-weighted mean of §8.5's existing pairwise coupling strength Λ_{ij}, restricted to pairs currently co-resident in C's own membership:

$\Lambda_{int}(C,t) = [\Sigma_{i,j \in \mu(C,t),\, i \neq j} R_i \cdot R_j \cdot \Lambda_{ij}(t)] / [\Sigma_{i,j \in \mu(C,t),\, i \neq j} R_i \cdot R_j]$.

This introduces no new primitive; it is Λ_{ij}, already defined for exactly this purpose in §8.5, aggregated over a container's own current sub-members rather than between two containers external to each other.

Null-membership convention. For a container C with no recursive membership ($\mu(C,t) = \varnothing$), $\Lambda_{int}(C,t)$ is defined as 0 for all computational purposes, rather than left undefined. This ensures numerical continuity in §8.3.1's first-contact attraction weight $A_{first} = \nu_{pref} \cdot \Lambda_{int}$, where a solitary candidate correctly contributes zero attraction rather than producing an undefined or null value the engine cannot propagate.

Proposition 9.10.1 (Cohesion Sustains Continuity).

A member i ∈ $\mu(C,t')$ remains in $\mu(C,t)$ at a later time — contributing to $\chi(t,t')$ remaining above χ_{min} rather than to membership turnover — in proportion to i's coupling strength to the rest of C's current membership relative to i's coupling strength to entities outside C. Formally, membership retention for i is increasing in $\Lambda_{ij}(t)$ summed over $j \in \mu(C,t)\{i\}$ and decreasing in $\Lambda_{ik}(t)$ summed over $k \notin \mu(C,t)$. This gives Definition 9.3's asserted criterion an actual mechanism: $\chi(t,t')$ stays high not because membership is stipulated to persist, but because internal coupling among a container's own current members is, on net, stronger than whatever external coupling would pull individual members away.

Corollary 9.10.1.1 (Cohesion Is the Sole Containment Mechanism at the Floor).

At the semi-rigid matrix state ($V(t) = h_{min}$, $\chi_{supp} = 1$, per §10.5b.0.4), vertical containment contributes nothing to boundedness — there is no wall left to do that work. Where a container at this state nonetheless satisfies Definition 9.3 ($\chi(t,t') \geq \chi_{min}$), Proposition 9.10.1 identifies $\Lambda_{int}(C,t)$ as the entire explanation: internal cohesion is not one contributor to containment alongside structural rigidity at the floor, it is the only one available. A container occupying the semi-rigid matrix state with low $\Lambda_{int}(C,t)$ should therefore show elevated membership turnover and be at genuine risk of falling below χ_{min} — boundedness at minimal verticality is not automatic, it is earned continuously by the container's own internal coupling density, with nothing in reserve to fall back on if that density drops.

Editorial note: This is the formal content of surface tension as a boundary mechanism, built entirely from machinery this document already has rather than a new physical primitive. It does not compete with Definition 9.3's criterion or with §10.5b.0's structural-coherence half of identity; it explains why the memory-continuity half holds when it holds, particularly in the one configuration — maximal horizontality — where the more familiar explanation (rigid walls) is unavailable by construction.

9.11 Adaptive Forgetting

Define the forgetting operator $F_M : M_t \to M_{t'}$. Adaptive forgetting removes informational detail whose maintenance costs exceed its future adaptive value. Both excessive forgetting and excessive retention can impair adaptive performance.

The Memory Principle.

Memory is an active operator that compresses, reconstructs, weights, and transforms historical experience, thereby modifying the geometry of future adaptation across every recursive level of organization — and, through the continuity of that historical organization, constitutes a container's persisting identity across the transformations the rest of this framework allows it to undergo.

Parts 10–12: Adaptive Stability and Collapse, Derived Observables, Fundamental Mathematical Results

Editorial note for this segment.

This is the largest revision pass of the document. Summary of changes, by location:

- §10.5b: the earlier prose draft of this section is removed; only the > > numbered version (§10.5b.0–§10.5b.15) is retained, per direct > > confirmation. Its internal "Summary of Additions" changelog > table > is also removed.
- §10.5b.0: fully reworked. The Verticality and Horizontality Axioms > > are renamed Axiom C3 and Axiom C4 (extending the collapse-layer > > C-series already begun at C1/C2 in §10.5b.15), resolving their > > collision with Part 3's Axiom 1 and Axiom 2. The > > structural-height quantity h(t) is regrounded as endogenous — > > driven causally by informational alignment I(t) rather than fitted > > independently — with a strictly positive

floor h_min (the > > semi-rigid matrix state), resolving the earlier tension between > > Ultimate Convergence and Axiom 1 (information requires a > > container). The verticality and horizontality operators V(t), W(t) > > are given closed-form defaults (V(t) := h(t), W(t) := 1−h(t)) > > rather than left as unresolved existence claims. A new > geometric > support ratio term and a structural-hazard > (longevity) functional > are introduced. The 90° inversion > construction is restated as a > labeled state transition rather > than a one-to-many “operator.” > Definition 9.3's identity > criterion (Part 9) is completed here by > combining membership > continuity with the semi-rigidity floor, and > the distinction > between Ultimate Convergence and dissolution is > made formal.

- §10.5b.0's ambient signal governing regime-genesis branching is > > renamed v_gen(t), distinct from the verticality operator V(t) it > > was previously confused with.
- Corollary 10.5b.0.1 is corrected: it previously explained Flip > > Condition A's outcome via mass-weighting, contradicting > > §10.5b.8's explicit coherence/dissipation account for that same > > regime. It now uses the coherence account consistently.
- §10.5b.3's confrontation trigger F_conf is generalized from a > > high-entropy-only mechanism to cover the > medium-entropy > collapsing case as well, and a new §10.5b.9a > introduces the > resilience accumulator Z(t) governing backslide > dynamics for > the medium-entropy healing case, completing the > three-way > backslide account developed in review.
- §10.5b.9's outcome thresholds are now explicitly normalized before > > comparing across the coherence-weighted (Flip A) and > mass-weighted > (Flip B) routes.
- §8.6a's anchor-subscript collision (X_{anchor,i}) is resolved to > > X_{a,i}.
- The §10.5b.0/§10.5b.1 correspondence table is corrected: > > verticality/horizontality and the entropy regime classification > > answer different questions and are not expected to jointly > > distinguish the two non-baseline regimes from a single V/W > > signature; the table is revised to state this rather than imply a > > false decomposition.
- §6.4's divergence landscape Φ is given its promised multi-valley > > structure, tied explicitly to the three entropy regimes.
- §10.5a and §10.5b.3's F_conf are explicitly connected: F_conf's > > transient amplification of D_op is one route by which the > > hull-breach inequality's left-hand side can be driven over > > threshold, though not the only one.
- §11.5/§11.6 fix the R_g inconsistency by setting φ = identity, so > > C_A := d_A directly and both sections' formulas for R_g now > > agree.
- §11.7 gives fragmentation F an explicit dependence on network > > coherence N, completing the derivation Proposition 12.8 needs.
- Theorem 12.4's text is updated to reflect §7.3's now-total (not > > adaptive-only) clipping rule, established in the Parts 7–9 > > revision.
- Proposition 12.8 is restored on the strength of §11.7's new > > formula, with both links in its derivation now explicit rather > > than one of the two being asserted narratively.
- The collapse operator (Definition 10.1) is renamed from $\mathscr{C}$ to $\mathscr{Q}$, > > correcting a collision with the $\mathscr{C}$ already used for “the collection > > of all containers” (Definition 2.6; §4.7–§4.8) — present in > the > document from Part 2 onward and confirmed while preparing > this > segment.
- A new §10.5b.11a (Post-Breach Redirection versus Resistance) is > > inserted between §10.5b.11 and §10.5b.12, formalizing why opposing > > a container’s momentum after hull-breach is destructive where > > redirecting it is survivable, generalizing §8.8b’s > > impulsive/dissipative discharge split beyond the B1/B2 outcomes it > > was previously

tied to, and giving Flip Condition A a route to > > full reorganization under external redirection despite > Corollary > 10.5b.0.1’s unaided-case result standing unmodified.

- Two new subsections, §10.5b.9b and §10.5b.9c, precede §10.5b.10: the > > first formalizes a pre-breach self-rescue window unique to Flip > > Condition A (a counterweight convergence that can hold or > reverse > an approaching tip, with a genuine knife-edge case > exactly at the > threshold, and no effect once the threshold is > passed); the second > is a named exception to §8.8a.ii’s > ordinarily-undirected frantic > striking, for a container with > enough warning to aim its discharge > deliberately.
- §10.5b.11a’s third redirection route (mutual coupling with a second > > struggling container) is removed from that subsection and > rebuilt > as its own §10.5b.11b, correcting an exact-simultaneity > > requirement that made the original route a measure-zero event, > and > expanding it into three structurally distinct sub-cases: > > deliberate predatory targeting of an unshielded neighbor (by two > > co-occurring mechanisms), accidental capture during a wider > > cascade (toward either a failing or a succeeding neighbor), and > > voluntary mutual sacrifice (explained either by relative strength > > or, per §16.8, by the sacrificing container’s own pre-event > > history).
- §10.5b.13’s anchor eligibility gains a third type: a purpose-built > > structure engineered specifically to absorb another container’s > > discharge, distinct from the two behaviorally/structurally safe > > types already present.
- A substantial new stretch of material, §10.5b.15 through §10.5b.19, > > is added after the existing collapse-layer axioms: Axiom C5 > > formalizes a container's feedback tracking truth only when its > > rate of change genuinely responds to real divergence — not > > moving despite a real problem, and moving despite none, are the > > same failure viewed from opposite sides, developed with a > > corrected tracking-fidelity diagnostic that allows for response > > lag and does not misfire on a genuinely quiet, healthy container. > > §10.5b.16 formalizes a compulsive-closure (“boomerang”) > mechanism, > mirroring the existing confrontation-forcing term > but triggered by > approaching alignment rather than by elevated > divergence. > §10.5b.17 formalizes a binary-trust-inversion > failure, where a > container's own confused targeting of help > toward a > mistakenly-trusted extractor widens that extractor's > reach across > the network. §10.5b.18 formalizes the > meta-representation layer as > a regime-dependent population of > sub-processes rather than a > uniform reporting channel, giving > Definition 6.1's independence > requirement its regime-dependent > completion, introducing a > captive-official-layer account of how > a sincere but access-limited > official layer sustains > divergence, and pricing that sustained > self-deception as an > explicit, divergence-scaled resource cost. > §10.5b.19 formalizes > a self-sustaining, reward-driven > extraction-escalation > accumulator, structurally the mirror of the > existing resilience > accumulator, whose growth is further modulated > by how strongly > a container's own memory operator weights past > rewarding > events.

All other content in these three parts is unchanged from the prior draft.

Part 10. Adaptive Stability and Collapse

Adaptive systems continually reorganize in response to internal dynamics and environmental perturbation. Collapse is not the opposite of stability; it is the progressive loss of the adaptive mechanisms that make stability possible.

10.1 Adaptive State Space

$X(t) = (P, R, G, M)$, consisting of informational organization, adaptive resources, interaction network, and memory. $X(t) \in \mathscr{A}$, the adaptive state manifold.

10.2 Stable Adaptive Region

$\mathscr{S} \subseteq \mathscr{A}$. A system is adaptively stable whenever $X(t) \in \mathscr{S}$. $\mathscr{S}$ is generally a dynamically evolving region of viable adaptive organization, not a point.

10.3 Adaptive Degrees of Freedom

Let F_A(X) denote the set of feasible adaptive transformations available from state X. Define d_A(X) = dim F_A(X). Large values indicate flexibility; small values indicate increasing rigidity. Collapse begins when adaptive options disappear faster than they can be regenerated.

Computational instantiation (the central missing piece, now resolved).

F_A(X) has no domain-independent enumeration — "what transformations are available" is irreducibly domain-specific, the same way "what counts as an outcome" (Ω) is. But d_A itself does not need to be measured by directly counting transformations; it can instead be derived from quantities RICD has already made concrete: informational stress S(t) (§6.7), resource debt δR(t) (§7.9), and recursive/network fragmentation F(t) (§11.7, §8.9's N). All three are, by construction, things that consume adaptive options — accumulated unresolved stress forecloses responses that would otherwise be available, resource debt forecloses anything requiring uncommitted capacity, and fragmentation forecloses coordinated multi-container responses. d_A itself, as a theoretical quantity tracking adaptive possibility, is structurally derived from these considerations; the specific exponential-decay functional below is this document's default computational instantiation of that derived quantity, a model-derived choice among functional forms sharing the same qualitative dependence, not itself forced by the structural argument alone:

$d\_A(t) = d\_A^{max} \cdot \exp[\, -\beta\_S \cdot \tilde{S}(t) - \beta\_R \cdot \delta\tilde{R}(t) - \beta\_F \cdot \tilde{F}(t)\,],$

where d_A^{max} is a domain-specified ceiling (the maximum plausible number of independent adaptive levers for that container type, set once per adapter — see Part 15), $\tilde{S}$, $\delta\tilde{R}$, $\tilde{F}$ are each stress/debt/fragmentation normalized to [0,1] by the adapter's own historical range, and β_S, β_R, $\beta_F > 0$ are fitted weights. This is not a claim that stress, debt, and fragmentation are the only things that determine adaptive capacity — it is a claim that they are the three RICD already tracks with real formulas, so d_A can be computed without inventing a fourth independent measurement pipeline. An adapter with a genuine independent proxy for feasible options (e.g., a literal count of currently-unconstrained policy levers) may substitute it for d_A directly; the exponential-decay form above is the default when no such direct count is available.

There is no symmetric notion of "entropic possibility." The formula above has one ceiling, d_A^{max}, and one direction of travel: every entropic quantity this document tracks — stress, resource debt, fragmentation, and every mechanism that feeds them — only ever pulls $d_A(t)$ down from that ceiling, asymptotically toward zero. Nothing in this framework's mathematics offers a second axis along which entropy could instead expand some alternative kind of richness or openness. Possibility, in this framework's own terms, is $d_A(t)$; entropy's entire defined functional role is to contract it. "Infinite entropic possibility" is accordingly not a rich concept symmetric to "negentropic possibility" that this document has simply chosen not to develop — it is close to a contradiction in terms once stated precisely, the same category error as calling absolute zero "infinite heat." Chaos, dissolution, or the collapse of structure do not, within this framework, constitute an expansion of what is possible; they are what the mathematics of possibility contracting toward its own floor looks like from the inside. Read this way, RICD's own formalism stands as a structural argument against treating meaninglessness or the dissolution of order as a form of liberation or expanded freedom: by this framework's own account, that dissolution is the asymptotic narrowing of genuine possibility, not its opening.

Recursive aggregation of d_A (parent from children).

Given a parent $C^* = A(C_1, \ldots, C_n)$ per §5.2, define

$$d_A(C) = [\Sigma_i\, w_i \cdot d_A(C_i)] \cdot (1 - F(C)),$$

with w_i the same resource weights used in §5.2's pooling step and $F(C^*)$ the parent-level fragmentation (§11.7) between the children's projected states. This makes the parent's adaptive capacity bounded above by the resource-weighted sum of its children's capacities, discounted by how poorly those children's adaptive options actually cohere — a system whose subsystems each have many options but whose options don't compose (high fragmentation) has less real flexibility at the aggregate level than the sum suggests, which is the qualitative claim Proposition 12.8 needed and, per §11.7's new formula, now has a mechanism for.

Parent ceiling, specified explicitly. The formula above computes realized $d_A(C^*)$; nothing yet states what $d_A^{max}(C^*)$ is, and leaving it unspecified creates a genuine fork — if independently adapter-set, recursive aggregation could produce a realized value exceeding it; if the w_i are normalized to sum to one, the parent's realized capacity could never exceed its single most capable child, which may understate genuine emergent capacity. Resolved by defining the parent ceiling explicitly, rather than borrowing either child scale or an unrelated independent guess:

$$d_A^{max}(C^*) = \mathscr{A}_{max}(d_A^{max}(C_1), \ldots, d_A^{max}(C_n), \chi_{cross}),$$

where $\mathscr{A}_{max}$ is an adapter-supplied aggregation functional (default: the resource-weighted sum of child maxima) and $\chi_{cross} \geq 0$ is an optional, separately-justified cross-coupling bonus term representing genuine emergent capacity the parent possesses beyond what any additive combination of children alone would supply. $d_A(C^*)$ computed above is then normalized against this parent-owned ceiling, never against a child's own scale.

10.4 Adaptive Rigidity

$R_g = \Phi(d_A)$, where $\Phi' < 0$. Rigidity increases as adaptive degrees of freedom decrease, derived from the geometry of adaptive possibility rather than introduced independently.

Computational instantiation.

$R_g(t) = 1 - d_A(t)/d_A^{max}$, the direct normalized complement of §10.3's d_A — monotonically decreasing in d_A as required, bounded in [0,1], and requiring no separate fitted parameters beyond those already used to compute d_A itself.

10.5 Collapse Operator

Definition 10.1.

Editorial note: *The collapse operator is denoted $\mathcal{Q}$ rather than $\mathcal{C}$ throughout this document, correcting a collision with the $\mathcal{C}$ already used for "the collection of all containers" (Definition 2.6; §4.7–§4.8). $\mathcal{Q}$ carries no independent mnemonic; it is chosen simply to be unused elsewhere in the document.*

The collapse operator $\mathcal{Q}$ maps adaptive trajectories toward lower-dimensional adaptive manifolds: $\mathcal{Q} : \mathcal{S} \to \mathcal{S}'$, where $\dim(\mathcal{S}') < \dim(\mathcal{S})$. Collapse is dimensional contraction within adaptive state space.

10.5a The Collapse Inequality (Hull-Breach Condition)

§6.9 left the boundary of informational equilibrium, $D(t) < D_c$, as an unspecified stability region. With I, S, Π_f, and d_A now all concrete (§6.6, §6.7, §6.5, §10.3), that boundary can be written as a single checkable inequality:

$$(1 - I(t)) + \tilde{S}(t) > \theta_{hull} \cdot \Pi_f(t)$$

where $\tilde{S}(t)$ is stress normalized to [0,1] by its own historical range (matching §10.3's normalization) and $\theta_{hull} > 0$ is a fitted threshold. The left side is total unresolved divergence pressure — informational incoherence plus accumulated stress; the right side is correction capacity — feedback permeability scaled by how much correction the container's threshold tolerates. The inequality holding at time t is a hull breach: an instantaneous signal that divergence-and-stress pressure has, at that moment, outrun the container's demonstrated capacity to correct it.

Terminology, fixed throughout this document from here on. Three stages are distinguished, and should not be used interchangeably. **Hull breach** is the instantaneous condition above holding at a single timepoint — a measurement, not an event. **Collapse onset** is a confirmed transition event, defined below, requiring more than one instantaneous breach. **Collapse resolution** is what §10.5b determines once onset is confirmed — self-arrest, terminal lock, full reorganization, or the disintegration and grid-cascade failure modes elsewhere in this document. A single measurement spike crossing the inequality is a hull breach; it is not, by itself, a confirmed onset, and should never be read as meaning the container has necessarily entered any particular resolved state. Stated explicitly, to foreclose any reading of the inequality as identical to completed collapse: hull breach is an instability-onset condition, marking entry into a regime where the adaptive trajectory has exceeded its ordinary correction capacity, not the collapse operator's own dimensional-contraction result. Confirmed onset (below) may still resolve into self-arrest, terminal lock, full reorganization, or structural disintegration depending on which transition mechanism applies — the inequality identifies entry into instability, not which of those outcomes follows.

Event-confirmation rule. A hull breach constitutes confirmed collapse onset only once the inequality holds continuously for a fitted minimum duration τ_confirm, rather than at the instant it is first crossed — a single noisy measurement spike that self-corrects within τ_confirm does not constitute onset. This is the default confirmation rule; two alternatives remain available and may be substituted without altering anything else in this document's collapse machinery: requiring k consecutive discrete observations rather than a continuous duration, or requiring posterior confidence in genuine breach (rather than measurement noise) to exceed a fitted probability threshold. All three share the same purpose — preventing a single spike from triggering onset — and a domain adapter may choose whichever fits its own observation cadence; τ_confirm (or its discrete/probabilistic equivalent) is fitted per domain. Where the k-consecutive-observations form is used, k must be converted to an actual elapsed duration via the adapter's own observation cadence before being compared against anything stated elsewhere in duration terms — k observations spaced a month apart and k observations spaced a year apart are not the same sustained interval, and "sustained" throughout this document means continuously satisfied over an elapsed duration, not merely across n samples, unless an adapter has explicitly converted n into a domain-specific duration first. This applies generally, not only here: every other sustained-state criterion in this document (Ultimate Convergence's persistence requirement, progressive collapse's τ_prog, and any other minimum-duration condition) is subject to the same discipline whenever the underlying data are irregularly or variably sampled across institutions or domains being compared.

Confirmed onset does not itself invoke 𝒬. Confirmed collapse onset and the collapse operator 𝒬's dimensional contraction (Definition 10.1, Proposition 12.5) are distinct levels of event, not two names for the same thing. Onset marks confirmed entry into instability; what follows is resolved by §10.5b's transition mechanisms into self-arrest, terminal lock, full reorganization, or structural disintegration, and 𝒬's dimensional contraction applies only to the outcomes among these that actually constitute completed collapse rather than a resolved return to stability. Self-arrest in particular is confirmed onset followed by recovery before 𝒬 ever applies — an onset that resolves without the dimensional contraction 𝒬 describes, not a case where 𝒬 briefly applies and then reverses.

Relationship to the confrontation trigger.

§10.5b.3 introduces a transient forcing term, F_conf, active near a confrontation-threshold crossing. F_conf is not a separate gate from the inequality above; it is one route by which the inequality's left-hand side can be driven over threshold. F_conf amplifies D_op transiently, which (via §6.6's formula for I) transiently depresses I(t), directly increasing (1 − I(t)). A container can therefore cross hull-breach either through an active F_conf episode or through ordinary stress accumulation and permeability decline with no confrontation event at all — F_conf is a common accelerant, not a necessary cause, and §10.5a's inequality remains the single operative criterion for collapse onset in every case.

§10.5a establishes when a container crosses from bounded oscillation into a tipping event. It does not distinguish what happens next — whether the container is a mid-transition system that will nonetheless stabilize, a resource-locked system that briefly de-stabilizes without permanent damage, or a genuinely different transformation entirely. §10.5b develops that distinction: the same threshold crossing has at least three qualitatively different resolutions, determined by which channels converge, what force drives the event, and how much of that force survives as usable momentum through the transition.

10.5b Entropy Regime Classification and the Convergent Collapse/Reset Mechanism

Hull breach identifies when a container tips. This section identifies what determines the outcome once it does, why formally similar convergence events between representation channels produce opposite outcomes depending on regime, how a container arrives in a given regime in the first place, what interventions are and are not safe, and — in §10.5b.0 — the structural, geometric origin of the verticality/horizontality distinction the rest of the section relies on. Every quantity below is built from primitives already defined elsewhere in this document, including §8.6a's extractive-coupling mechanism, which turns out to be load-bearing rather than incidental to what follows.

10.5b.0 The Topology of Verticality and Horizontality

The entropy regime classification developed in this section distinguishes containers by the relationship between their official and operational layers. That classification describes symptoms, diagnosed from observable variance. The present subsection supplies a structural account of what those symptoms are symptoms of: why containers assume rigid, closed, extractive configurations or fluid, open, negentropic ones, and why one configuration is dramatically more durable than the other.

Editorial note: *The material below reworks the earlier draft of this subsection substantially. The earlier draft treated the structural-height quantity h(t) as an independently fitted input and asked verticality V(t) to increase toward 1 while horizontality W(t) increased toward 1 as a separate, differently-motivated quantity, and it let Ultimate Convergence send h(t) literally to zero. Both choices created problems flagged in review: h(t) was never grounded in anything the container ontology of §2.2 actually provides, V and W had no stated relationship to each other despite being obviously dual, and h(t) → 0 is indistinguishable from a container ceasing to be a container at all, contradicting Axiom 1. The account below resolves all three by making structural height a consequence of informational alignment rather than an independent primitive.*

Why some rigidity is necessary, and why excess rigidity is costly.

A container with no structural rigidity whatsoever is not a bounded entity; it is indistinguishable from the undifferentiated field around it. Some minimum degree of structural coherence is therefore constitutive of being a container at all — not an unfortunate compromise, but a precondition for Axiom 1 to apply. At the same time, rigidity beyond that minimum is costly: a container that over-invests in rigid, closed, hierarchical structure decays faster than one that maintains only the minimum coherence its informational alignment actually requires. The pattern recurs across scales and domains — tall physical structures under greater structural strain than broad ones of comparable mass; large-bodied organisms carrying disproportionate maintenance load for their size (already formalized as a scale effect on maintenance cost, §7.5); rigid institutional hierarchies that fracture under stress that flatter, more distributed organizations absorb. The formal account below treats structural height as the dependent variable in this relationship and informational alignment as the driver, rather than treating the two as independent quantities that happen to correlate.

Definition 10.5b.0.1 (Structural Height and the Verticality Operator).

Let a container C have a normalized structural height $h(t) \in [h\_min, 1]$, where $h\_min > 0$ is a strictly positive floor — the minimum structural coherence consistent with C remaining a bounded container at all (§2.2, Axiom 1) — and $h(t) = 1$ represents maximal, closed, hierarchical rigidity. This is not a physical height in every domain; it is a structural index of how rigid, closed, and hierarchically constrained the container's boundary currently is.

Rather than treating $h(t)$ as an independently fitted quantity, $h(t)$ is driven causally by informational alignment $I(t)$ (§6.6) through a relaxation dynamic:

$$dh/dt = \kappa\_h \cdot [\ h\_target(I(t)) - h(t)\ ],\ h\_target(I) = h\_min + (1 - h\_min)\cdot(1 - I)^\xi,$$

with $\kappa\_h > 0$ a fitted relaxation rate and $\xi \geq 1$ a fitted exponent. Higher alignment sets a lower structural-height target; lower alignment sets a higher one. Because $h(t)$ chases $h\_target(I(t))$ rather than jumping to it, structural relaxation necessarily lags behind and follows alignment improvement — it does not lead it, and cannot produce a self-sustaining reduction in structural height by any mechanism that intervenes on $h(t)$ directly without $I(t)$ itself improving: a direct intervention can lower $h(t)$ for as long as it continues, but $h(t)$ resumes relaxing toward $h\_target(I(t))$ the moment that forcing lapses, per the same equation. The verticality operator is then simply:

$$V(t) := h(t) \in [h\_min, 1].$$

This resolves what was previously an unspecified functional (Φ_V, monotonic in dh/dt, d^2h/dt^2, S, δR, and Π_f, with no computational default given) by observing that once $h(t)$ is itself derived from $I(t)$ — which is already a function of S, δR, and Π_f's downstream effects on D_op, D_om, D_pm through the dynamics of Part 6 — a separate wrapper functional adds complexity without adding information. The rate-sensitivity the original formulation wanted (verticality responding to how fast a container is rigidifying, not merely how rigid it currently is) is preserved implicitly: $h(t)$'s own relaxation dynamic already responds continuously to changes in $I(t)$, so dh/dt is recoverable directly from the equation above without a separate operator.

Corollary (Alignment Precedes Relaxation; a falsifiable prediction).

Because $h(t)$ chases $h\_target(I(t))$ with lag rather than tracking it exactly, two directly testable predictions follow from the single equation above, without further assumption. First, sustained improvement in $I(t)$ should measurably precede a corresponding decline in $h(t)$, not the reverse. Second, any intervention that forces $h(t)$ below its currently-implied target — flattening structure without underlying alignment actually improving — predicts its own reversal: since $dh/dt = \kappa\_h\cdot[h\_target(I(t)) - h(t)]$ is positive whenever $h(t)$ is below target, the same dynamic that governs ordinary relaxation drives an artificially-flattened container's structural height back up toward $h\_target(I(t))$ once the intervention lapses. This is the formal counterpart to the observation that forcing apparent flatness without genuine alignment tends to produce rebound rigidity rather than lasting change: it is not a separate claim requiring its own mechanism, but a direct consequence of treating alignment as causally prior to structural relaxation.

Definition 10.5b.0.2 (Horizontality Operator).

Horizontality is the dual of verticality, not an independently fitted quantity:

$$W(t) := 1 - V(t) = 1 - h(t) \in [0, 1-h\_min].$$

Because $h\_min > 0$, $W(t)$ never reaches 1: a container never achieves complete, unconstrained fluidity, consistent with some minimum structural coherence being constitutive of remaining a bounded container at all (Definition 10.5b.0.1). Maximal horizontality is $1 - h\_min$ — the semi-rigid matrix state — not a fully flat, structureless field.

Geometric support ratio.

Height and rigidity alone do not determine structural durability. Two containers with identical $h(t)$ can differ sharply in how long that structure persists, depending on how broadly the structure's boundary is grounded relative to its vertical extension — a broad-based structure distributes load across a wide foundation; a narrow one concentrates it. The same distinction separates a broad-based, widely distributed structure from a narrow, concentrated one of comparable height, or a broadly stakeholder-supported institution from a narrowly hierarchical one of comparable size.

Definition 10.5b.0.3 (Geometric Support Ratio).

Define $\chi\_supp(C,t) \in (0,1]$, a domain-specified measure of the breadth of a container's structural or organizational base relative to its vertical extension (adapter-supplied — e.g. base-contact area relative to height for a physical structure, or breadth of distributed stakeholder support relative to hierarchical depth for an institution). $\chi\_supp = 1$ represents maximal support for the container's current height; values near 0 represent a narrow, precarious base relative to height.

Structural hazard and longevity.

Rigidity beyond the minimum, and inadequate support for whatever rigidity is present, both shorten a container's expected structural lifespan. This is the formal content of the claim that vertical, high-divergence structures burn themselves out faster than semi-rigid, well-supported ones.

Definition 10.5b.0.4 (Structural Hazard Rate).

$\Omega\_hazard(t) = \kappa\_\Omega \cdot V(t)^\psi / \chi\_supp(C,t)$,

where $\psi \geq 1$ is a fitted exponent ($\psi > 1$ giving a superlinear penalty for excess verticality, consistent with the scale-dependence already established for maintenance cost in §7.5) and $\kappa\_\Omega > 0$ is a fitted rate coefficient. Hazard is minimized — but never driven to zero — when $V(t) = h\_min$ and $\chi\_supp = 1$: the semi-rigid matrix state, maximally supported, is the longest-lived configuration a container can occupy, not an immortal one. $\Omega\_hazard(t) > 0$ always, since $h\_min > 0$; every container remains subject to nonzero structural dissolution risk indefinitely, consistent with no container being permanently exempt from Axiom 3's nonzero cost of informational processing. Positive hazard alone does not, by itself, guarantee eventual dissolution with certainty — a hazard rate that declines quickly enough can yield a finite cumulative hazard and therefore a genuinely positive probability of indefinite survival; the stronger claim of certain eventual dissolution additionally requires the cumulative hazard to diverge, $\int_0^\infty \Omega\_hazard(t)\, dt = \infty$, which this document does not assume by default and which would need to be separately established or adopted for any domain where it is wanted. A container's expected remaining structural lifespan varies inversely with $\Omega\_hazard(t)$; this document does not commit to a specific survival-function form beyond that qualitative, monotone relationship, which is sufficient for the diagnostic and comparative uses §10.5b puts it to.

Axiom C3 (Verticality is Rigid).

Editorial note: *Renamed from an earlier, unnumbered "Axiom 1," which collided with Part 3's Axiom 1 (Information Requires a Container). This axiom and Axiom C4 below extend the collapse-layer C-series begun at C1/C2 in §10.5b.15; all four govern collapse dynamics specifically and are distinct from Part 3's foundational axioms, which govern the base RICD formalism independent of any collapse dynamics.*

For any container C, as verticality $V(t)$ increases beyond a domain-specific threshold $\Theta\_V \in (h\_min, 1)$, the container undergoes a phase transition from a semi-rigid matrix toward a closed-circuit configuration, characterized by $\Pi\_f(t) \to 0$ (feedback permeability collapses), $D\_op(t) \to 1$ (official-operational divergence maximizes), $d\_A(t) \to 0$ (adaptive capacity contracts), and $R\_g(t) \to 1$ (rigidity functional reaches maximum, §10.4).

Proposition 10.5b.0.1 (Verticality and the High-Entropy Regime).

In the canonical, official-shielded high-entropy configuration — ordinary predatory or self-concealing high-entropy dynamics, the case this document treats as default throughout — a container is characterized by $V(t) > \Theta\_V$ and $D\_op(t) \gg 0$, with correspondingly low $\Pi\_f(t)$. Its elevated verticality is the structural counterpart of its self-sustaining, extractive, and blind dynamics: the official layer appears calm (σ_o low) because rigid structure has frozen the official messaging in place, while the operational layer churns chaotically beneath it (σ_p high). This conjunctive characterization is not a universal defining property of the high-entropy regime as such, which the regime classifier (§10.5b.1) establishes independently through σ_o/σ_p alone: Definitions 10.5b.1.1–1.2's Type A and Type B full-alignment cases are genuinely high-entropy by that classifier (operational is entropic, σ_p high) while carrying low D_{op}, since official and operational have converged — accurately in Type A, falsely in Type B — rather than remaining divided. This proposition describes the ordinary, divided case; the aligned exception is Definitions 10.5b.1.1–1.2's own subject and is not additionally governed by the $D_{op} \gg 0$ clause here.

Axiom C4 (Horizontality is Fluid).

For any container C, horizontality W(t) is maximized (approaching, but never reaching, $1 - h_{min}$) as $\Pi_f(t) \to 1$, $I(t) \to 1$, $D_{op}(t) \to 0$, and $dR/dt > 0$. In this state the container exhibits bounded stress ($S(t) < \Theta_S$), high adaptive capacity ($d_A(t) \approx d_A^{max}$), low rigidity ($R_g(t) \to 0$), and short, bounded recovery time ($\tau(t)$, §10.10).

Proposition 10.5b.0.2 (Horizontality and the Low-Entropy Baseline).

A container in the low-entropy baseline regime (§10.5b.1) is characterized by $W(t) > \Theta_W$ and $D_{op}(t) \approx 0$, for some threshold $\Theta_W \in (0, 1-h_{min})$. Its elevated horizontality is the structural expression of open, negentropic, self-aware systemic flow — the semi-rigid matrix state toward which recovery trajectories tend, per Definition 10.5b.0.4, without ever reaching full structurelessness.

The 90° inversion and tipping outcomes.

Rather than an operator with an ambiguous, one-to-many output, the relationship between a container's structural orientation and its tipping outcome is best stated directly as a labeled transition, since §10.5b.9's M_{tip} thresholds already determine which outcome occurs — a separate "inversion operator" would only relabel that same determination.

Definition 10.5b.0.5 (Structural Transition Labels).

A container beginning in a low-verticality configuration ($V(t) \approx h_{min}$, a horizontal field) that undergoes a shock-driven phase transition (Axiom C3) moves to a high-verticality configuration. From there, resolution of a subsequent tipping event (§10.5b.9) is labeled:

- terminal lock (previously "180° flip"): V(t) remains elevated; the > > container remains in a rigid, closed configuration and cannot > > return toward h_{min} without external anchoring (§10.5b.13);
- full reorganization (previously "360° flip"): V(t) relaxes back > > toward h_{min} via the ordinary alignment-driven dynamic of > > Definition 10.5b.0.1, once I(t) recovers.

Which label applies is fully determined by the M_{tip}-versus-threshold comparison of §10.5b.9; the labels above are descriptive shorthand for that outcome's structural consequence, not an independent mathematical object.

Corollary 10.5b.0.1 (Mass Weighting, Coherence, and Structural Outcome).

Editorial note: *This corollary previously explained Flip Condition A's terminal-lock outcome via mass-weighting ("the majority mass is in the operational layer, but it is chaotic"). This directly contradicted §10.5b.8, which is explicit that Flip Condition A's outcome is governed by coherence and dissipation, not mass-weighting — mass-weighting applies only to Flip Condition B's two outcomes (§10.5b.7). The corrected statement below uses each regime's own mechanism.*

In the high-entropy regime, a container's tipping outcome is governed by the coherence factor $\kappa(t) = 1/(1+D\_residual(t))$ of §10.5b.8: residual divergence outside the converging pair dissipates part of the driving force as internal friction, but Flip Condition A structurally admits no full-reorganization route under either of its causal sub-routes (§10.5b.5), so the momentum that survives dissipation determines only whether the terminal lock is reached at all ($M\_tip \geq M\_lock$), never which structural outcome results. In the medium-entropy regime, by contrast, the outcome is governed by fluid-mass weighting (§10.5b.7): because $\mu\_p > \mu\_o$ structurally in this regime, the same external launching force more readily crosses into the full-reorganization range when the majority (operational) mass is what converges.

Definition 10.5b.0.6 (Ultimate Convergence).

Editorial note: *Corrected from an earlier draft in which this definition sent h(t) → 0 exactly. Full structural flattening is indistinguishable from a container ceasing to be a container (Axiom 1); the corrected version below sends the container toward its structural floor, not past it, and states explicitly how this differs from dissolution.*

A container C undergoes Ultimate Convergence on an interval when:

$V(t) \to h\_min$, $W(t) \to 1-h\_min$, $B(t) \to B\_open$,

where B_open is a boundary operator permitting maximal bidirectional exchange consistent with remaining a bounded container, while membership continuity is maintained throughout: $\chi(t,t') \geq \chi\_min$ (Definition 9.3). This is a thermodynamic transition from a high-hazard, closed configuration to a low-hazard, open one (Definition 10.5b.0.4) — not dissolution, and not a transition to formlessness.

Definition 10.5b.0.7 (Identity Persistence, complete criterion).

Combining Definition 9.3 (Part 9) with the semi-rigidity floor established here: a container C maintains full identity across [t′, t] if and only if both (a) $\chi(t,t') \geq \chi\_min$ (membership continuity is maintained) and (b) $\chi\_supp(C,s) \geq \chi\_supp,min$ for all $s \in [t',t]$ (§10.5b.0's geometric support ratio, structural base breadth relative to height, remains above its own admissibility floor throughout) — i.e., structural change alone, however large in V(t) itself, does not end identity provided both membership continuity (a) and a genuinely non-collapsing support base (b) hold throughout. An earlier version of condition (b) stated $V(s) \leq 1$ for all s; this added no independent content, since V(t) is defined as $h(t) \in [h\_min, 1]$ (Definition 10.5b.0.1) and is therefore bounded above by 1 for every admissible container as a matter of definition, not as a condition any real trajectory could fail. χ_supp remaining bounded away from collapse is a genuinely independent, sometimes-false condition instead: a container can satisfy membership continuity while its structural base narrows toward a single point of failure (Ω_hazard's own $\kappa\_\Omega \cdot V^\wedge\psi/\chi\_supp$ form already shows why this matters — hazard diverges as $\chi\_supp \to 0$ regardless of what V(t) is doing), and this is precisely the loss of bounded-container structure the criterion needs an independent condition to rule out. Ultimate Convergence (Definition 10.5b.0.6) is the case where both conditions hold while V(t) trends toward its floor: identity persists, hazard falls, longevity increases. Dissolution is the case where (a) fails — χ(t,t′) falls below χ_min because membership has scattered into unrelated clusters — regardless of what V(t) is doing at the time; a container can dissolve while structurally rigid (fragmentation under sustained high hazard) or, in principle, while structurally relaxed (gradual absorption into a neighboring cluster). The two failure modes are structurally distinct even though both end identity, and §8.8a.v's "complete fragmentation" outcome and §8.8a's locked-node dynamics are instances of the former.

Relationship to the entropy regime classification.

Verticality/horizontality (this subsection) and the entropy regime classification (§10.5b.1, via σ_o, σ_p) answer different questions and should not be expected to jointly triangulate one another from a single V/W reading. V(t) measures overall structural rigidity — a property shared, to varying degree, by both non-baseline regimes, since both arise from the same shock-branching mechanism (§10.5b.2) departing from the same baseline. σ_o and σ_p measure something orthogonal: which layer, official or operational, is carrying the apparent volatility. A high-entropy and a medium-entropy container can therefore present with comparable $V(t)$ — both departures from baseline elevate structural rigidity relative to the semi-rigid matrix floor — while being sharply distinguished by σ_o versus σ_p, which is precisely why §10.5b.1 uses the variance signature, not $V(t)$, as the classification criterion. $V(t)$'s role is to explain why departure from baseline carries a hazard consequence (Definition 10.5b.0.4), not to re-derive which departure a container has undergone.

10.5b.0.8 Sustained Horizontal Matrix

Definition 10.5b.0.8 (Sustained Horizontal Matrix).

A container C undergoing Ultimate Convergence (Definition 10.5b.0.6) is sustained on an interval $[t', t]$ when the convergence conditions — $V(t) = h_{min}$, $W(t) = 1-h_{min}$, $B(t) = B_{open}$, $\chi(t,t') \geq \chi_{min}$ — hold throughout the interval rather than only momentarily. Where sustained, C acquires one further property beyond Definition 10.5b.0.6's own content: immunity, not merely resilience, to tipping-momentum accumulation. For any incoming $Drive_B(j,t)$ or propagated kick $K_{j\to k}(t)$ (§8.8a, §8.8b) directed at C during the sustained interval, C's own contribution to M_{tip} is forced to zero, regardless of the striking force's magnitude. This is a difference in kind from $Z(t)$'s resilience (§10.5b.9a), which lowers the cost of a striking event that still occurs; a sustained container's striking events do not occur in the sense of accumulating momentum at all — there is, as this document's own imagery for the state has it, nothing there to rock.

Proposition 10.5b.0.3 (Stress Non-Accumulation, Already Entailed).

No new mechanism is required for a sustained container's memory to remain honest and undiminished while contributing nothing to informational stress. Axiom C4 already establishes $\Pi_f(t) \to 1$ as horizontality is maximized; §6.7's existing formula, $S(t) = \int K(t,s)(1-\Pi_f(s))D(s)ds$, already sends the stress contribution of any given interval to zero as $\Pi_f(s) \to 1$ within it, independent of $D(s)$'s magnitude. A sustained container's memory operator M_t continues recording every event that reaches it, per §9.2's unmodified definition — nothing is hidden, nothing is filtered at the point of encoding. What the existing $(1-\Pi_f)$ gating already ensures is that this honestly-recorded content does not translate into stress accumulation while $\Pi_f(t)$ remains at its sustained-state value. Definition 10.5b.0.8's tipping-immunity is the one genuinely new claim in this subsection; the memory and stress behavior surrounding it was already correct once Axiom C4 and §6.7 are read together.

10.5b.0.9 Glimpse

Definition 10.5b.0.9 (Glimpse).

A container touches, without sustaining, Ultimate Convergence when Definition 10.5b.0.6's conditions hold at some instant or brief interval failing Definition 10.5b.0.8's duration requirement. Where this occurs, tag the corresponding segment of M_t with a glimpse marker $g(t) = 1$ ($g(t) = 0$ otherwise) at the moment of encoding — an ordinary, undistorted memory-write per §9.2, distinguished only by this marker rather than by any special content.

Provenance is inherited, not assumed. Because $g(t)$ is set only when Definition 10.5b.0.6's conditions are actually, formally satisfied, a glimpse is not tagged from a container's own belief or interpretation that a moment was healthy — it is tagged from the same observation and measurement architecture (D_{om}, D_{op}) already governing every other diagnostic in this document, subject to the same accuracy limits. A glimpse recalled from a segment whose original D_om/D_op readings were themselves inaccurate carries that inaccuracy forward into F_glimpse's restorative force; no additional grounding check beyond what already governs D_om and D_op at the moment of encoding is introduced here, because none is needed beyond what the rest of this document already requires of those same readings.

10.5b.0.10 Glimpse Recall

Definition 10.5b.0.10 (Glimpse Recall and Its Two Effects).

Where a subsequent perturbation — an external knock, or internal pressure from entropic-aligned metamessengers — raises D_om(t) or D_op(t) toward a destabilizing level, glimpse-tagged memory segments' recall (weighted disproportionately to their recency, via a fitted glimpse salience κ_glimpse > 1 applied within K(t,s)'s existing kernel) produces two separable effects.

Immediate effect. A restorative forcing term, generalized to aggregate over the full glimpse history rather than a single fixed lag:

$F\_glimpse(t) = κ\_glimpse \cdot \int_0^t K\_glimpse(t,s) \cdot g(s)\, ds \cdot (D\_om(t) - D\_om,floor)_+$,

using a glimpse-specific kernel K_glimpse(t,s) within §9.2's existing memory architecture rather than a new mechanism, entering §6.4's landscape alongside F_ext,fb and F_ext,supp (§10.5b.18.11) as a further restorative — not distorting — contribution, pulling the container back toward its healthy target band specifically because the recalled memory carries evidence that the categorical split producing the current perturbation is not, in fact, load-bearing.

Cumulative effect, explicitly conditional. Repeated glimpse recalls accumulate into a separate residue quantity, Γ_glimpse(t), $d\Gamma\_glimpse/dt = κ\_G \cdot g(t) \cdot (1 - \Gamma\_glimpse(t))$, structurally parallel to Z(t)'s accumulation but tracking glimpse exposure rather than survived knocks. Γ_glimpse(t) alone does not lower h_target(I(t)) or produce any permanent change in V(t)'s resting configuration; it is a necessary but not sufficient contributor, gated by conditions external to Γ_glimpse(t) itself — meaningfully favorable Ψ_total(t) (§10.5b.18.14), sustained access to healthy external coupling (φ_fb, Γ_supp, §10.5b.18.9–10.5b.18.10), and grid conditions this document does not claim to fully specify. Where those gating conditions hold, rising Γ_glimpse(t) contributes to a gradual lowering of h_target(I(t)) over time; where they do not, Γ_glimpse(t) rises without producing permanent structural change, and the container's glimpses remain a source of momentary stabilization only.

10.5b.0.11 Control Versus Influence, and the Goal Paradox

Definition 10.5b.0.11 (Control-Shaped and Influence-Shaped Intervention).

An intervention on a container — internal (§6.1's own representational apparatus) or external (an adapter or engine term acting on Φ, §6.4) — is **control-shaped** when it forces P toward one predetermined target regardless of whether that target is where the container's own unforced dynamics would arrive, foreclosing the range of states genuinely reachable from where the container currently stands. It is **influence-shaped** when it supplies pressure, resources, or corrective feedback that shifts the conditions or likelihood of the container's own trajectory without fixing a specific predetermined endpoint in advance, leaving which state the container actually reaches an open question its own dynamics resolve.

Every forcing term already built into this document's landscape falls cleanly into one category or the other. F_conf, F_boom, and F_demand are control-shaped: each pulls P toward one specific target — a confrontation resolution, a false-signal valley, official's demand —

independent of whether that target is the container's true signal, and each, per Axiom C1 (§10.5b.15), requires sustained forcing to hold open and relaxes away once withdrawn. The external feedback and support channels (φ_fb, Γ_supp, §10.5b.18.9–18.11), glimpse-recall (F_glimpse, §10.5b.0.10), and emergency firewall reinforcement (F_emerg, §10.5b.14.2) are influence-shaped: each restores permeability, resource stability, or memory of possibility that lets the container's own dynamics find its own signal, without dictating in advance what that signal turns out to be.

Proposition 10.5b.0.5 (Control Undermines the Health It Targets).

The formal criterion separating the two categories is their effect on d_A(t) (§10.3), this document's own measure of adaptive possibility, which §14.7 already identifies as constitutive of system health rather than merely correlated with it. A control-shaped intervention forecloses which states remain reachable by construction — that foreclosure is what distinguishes it as control-shaped in the first place — and therefore contracts d_A(t) directly, independent of whether the specific target it forces happens to be a healthy one. An influence-shaped intervention does not foreclose reachable states; where it affects d_A(t) at all, it does so only indirectly, by relieving the stress, resource-debt, or fragmentation terms that §10.3's formula already ties to adaptive capacity. This holds regardless of where the intervention originates: an externally-authored control-shaped term does not act as a neutral nudge from outside the container's own dynamics, because it becomes part of the same landscape Φ the container's own feedback operator $\mathscr{F}$ runs on (§6.4) — there is no version of "control from outside" that avoids contracting the container's own adaptive possibility, since the forcing enters the identical equation regardless of who authored it.

Corollary 10.5b.0.5.1 (The Goal Paradox, Generalized).

Sustained Ultimate Convergence (Definition 10.5b.0.8) cannot be reached by a control-shaped intervention, internal or external, without contradiction. If Ultimate Convergence is genuinely the container's own unforced fixed point, forcing toward it was never necessary — the container's own dynamics reach it once whatever currently obstructs them clears, per Axiom C1. If the observed convergence persists only because of the forcing, it is not genuine Ultimate Convergence at all, but a forced valley of exactly the kind Axiom C1 already treats as pathological elsewhere in this document, indistinguishable from data alone from every other propped-open false state this document warns against, and it relaxes away the moment the forcing is withdrawn. This is the operative reason no adapter or engine instantiation should encode Ultimate Convergence as an explicit optimization target: doing so does not merely risk failing to reach the state, it risks manufacturing a forcing-dependent counterfeit of it, contracting the container's own adaptive possibility in the act of attempting to produce what adaptive possibility is supposed to look like at its healthiest. Wanting a good outcome does not exempt an intervention from being control-shaped, and does not exempt it from this consequence.

Implementation constraint. No adapter or engine instantiation of $\mathscr{F}$, Φ, or P_C(t) may include Definition 10.5b.0.8's convergence conditions, or any other specific state, as an explicit minimization target, weighted term, or optimization objective aimed at forcing convergence to a predetermined outcome. Glimpse-recall (§10.5b.0.10), the external feedback and support channels, and emergency firewall reinforcement remain permitted and encouraged as influence-shaped, passive or conditions-restoring mechanisms, per Definition 10.5b.0.11; what is excluded is only control-shaped forcing toward a predetermined endpoint within the engine's own forcing or evaluation functions. This constraint is carried forward into Part 15's adapter contract as a standing prohibition, alongside its existing warnings against conservative-default misreadings.

Editorial note: The distinction drawn here is not specific to Ultimate Convergence; it is the general principle this document's account of health has implied throughout without previously naming directly. Adaptive possibility, not any specific target state, is what this document treats as the substance of health (§14.7); an intervention that decides a container's trajectory in advance is, by that same measure, acting against health even when the trajectory it forces looks like a healthy one. What this document's own healing mechanisms actually do, uniformly, is supply conditions and leave the outcome genuinely open — influence, not control — and the uncertainty that remains after an influence-shaped intervention is not a limitation of this approach; it is the formal signature of the adaptive possibility health itself consists of.

Anticipated objection: isn't preserving adaptive possibility itself a target? If health is defined by d_A(t), and this document's own mechanisms are oriented toward not contracting it, that can look like optimization toward a goal under another name, apparently reintroducing exactly what the Goal Paradox excludes. The distinction that resolves this is between trajectory control and constraint preservation, and it is a real, not merely verbal, difference. Specific-state optimization constrains which *trajectory* a container is pushed along, fixing where it must end up regardless of what its own unforced dynamics would do. Preservation of adaptive possibility constrains only the *conditions* under which a container's trajectory remains open, without fixing which trajectory, among those the conditions leave available, the container actually takes. The former forecloses reachable states by construction; the latter's entire content is refusing to foreclose them. This is why d_A(t) can be this document's measure of health without functioning as a control-shaped target in Definition 10.5b.0.11's sense: nothing in preserving conditions for openness specifies in advance which open state gets reached.

10.5b.1 Entropy Regime Classification

Let σ_o(t), σ_p(t) denote the trailing-window variance of the period-to-period *changes* in P_o and P_p respectively — not the variance of their raw level — both already computable from the adapter's raw O_o, O_p feeds (§15.2), no new observation required. This distinction is not cosmetic: a smoothly, monotonically trending channel (official messaging slowly lagging toward a declining reality, for instance) can show substantial variance-of-level in a trailing window purely from the trend itself, while a channel that has already settled to a new level shows low level-variance even if it arrived there erratically. Level-variance is the wrong notion of volatility for this purpose and will misclassify exactly the case this mechanism exists to catch; variance of differences (the standard approach to measuring volatility, as with financial returns rather than price levels) does not have this failure mode. Classify a container into one of three regimes, not two:

- Low-entropy baseline regime: D_op, D_om, D_pm all small; Π_f high > > (near 1, the meta-layer is genuinely bidirectionally permeable); > > σ_o and σ_p both low-to-moderate — a slight, rhythmic rocking > > rather than violent oscillation. D_om is not required to be > zero, > and should not be: a small nonzero D_om — the official > layer > under mild, healthy self-questioning rather than perfect, > > unexamined confidence — functions as a negative-feedback > channel > preventing an unchecked, grandiose official narrative. > D_om ∈ (0, > ε_healthy] is the target band, not D_om = 0. This > regime is the > reference state the other two are diagnosed as > deviations from.
- High-entropy regime (official-shielded): σ_o(t) low, σ_p(t) high — > > the official/declared layer presents as stable while the > > operational layer is actually turbulent underneath it. The > > official layer's stability is not evidence of health here; it is > > the shield.
- Medium-entropy regime (official-exposed): σ_o(t) high, σ_p(t) low > > — the official layer presents as visibly unstable while the > > operational layer is actually steady beneath it — the mirror > > case. The official layer's visible trouble is not evidence of > > deeper damage here; the trouble is exposed precisely because > > nothing underneath it is concealing anything.

These are not different containers in kind; the same container can occupy any of the three at different points in its trajectory.

Terminology note: Earlier editions named these two regimes high-divergence-apparent and low-divergence-apparent, after which layer's variance an outside observer would find apparent. That naming inverted against intuition often enough in practice — a system that hides its entropy is not the one whose divergence looks apparent to a casual reader of the term — that this edition replaces it throughout with the entropy-level names above, used identically to the existing low-entropy baseline naming convention, with the shielded/exposed tag retained parenthetically here as the σ_o/σ_p diagnostic criterion and dropped in subsequent use. High-entropy and medium-entropy were already in informal use elsewhere in this document (§8.6a's extraction-target population, §8.8a's cascade sequence) as synonyms for exactly these two regimes; this edition makes that the formal name rather than an informal alias running in parallel with it.

Relationship to the feedback landscape Φ (§6.4).

The three regimes above are, in the terms of §6.4's stochastic gradient flow, the three basins of the divergence potential $\Phi\_t(P|P\_m,M)$: the low-entropy baseline is Φ's deep, global minimum; the high- and medium-entropy regimes are shallower local basins reachable from baseline via the shock-branching mechanism below (§10.5b.2), each held open by a specific forcing mechanism — F_conf and, for the high-entropy case, extraction input (§8.6a) — consistent with Axiom C1 (§10.5b.15): withdraw the forcing, and Φ reshapes back toward having baseline as the only nearby attractor. §10.5b.9a's resilience accumulator Z(t) governs how deep and how easily escaped the medium-entropy basin is over repeated excursions.

No circularity between regime and landscape. Regime labels are diagnostic classifications of dynamically realized regions of state space; they are never inputs to the equations that generate those regions. Every forcing term shaping Φ's basins — F_conf, F_boom, F_demand, and every later addition to this landscape — is triggered by continuous state variables (Π_f, D_op, D_om, ω_meta^up/down, and similar), never by the discrete regime label itself (§10.5b.1's σ_o/σ_p classification). Regime is read off the resulting basin structure and trajectory after the fact; it does not feed back into determining that structure.

10.5b.1.1 Two Kinds of Full Alignment

A container can reach $D\_op(t) \approx 0$ with an entropic operational core in two structurally different ways, distinguishable only by G(t) (§6.3.2), never by internal agreement alone, since both produce identical D_op, D_om, D_pm readings.

Definition 10.5b.1.1 (Type A: Self-Aware Alignment). Operational is entropic and G(t) is high — the container accurately knows its own entropic nature. Official agrees because there is nothing to conceal from itself: an accurate report of an accurately entropic reality requires no management. No firewall is needed, $M\_decep(t) \to 0$, and metamessengers require no split composition, since there is no internal tension anywhere in the container for a population to divide over. What contracts the container's own adaptive possibility is not concealment cost, since none is being paid; it is the entropic nature itself, decaying, with no negentropic energy present anywhere in the system to resist or contain that decay — d_A(t)'s formula (§10.3) already prices this directly, through $\tilde{S}(t)$ and $\delta\tilde{R}(t)$ alone, without requiring M_decep(t) to carry any of the explanatory weight it does in ordinary high-entropy predation.

Definition 10.5b.1.2 (Type B: Confused Alignment). Operational is entropic and G(t) is low — the container's own entropic core has come to believe itself negentropic, and that false self-belief, not accurate self-acceptance, is what produces alignment with official. The surface signature is identical to Type A: low D_op, no felt need for a firewall. The underlying condition is the opposite: not the absence of anything to hide, but the absence of any internal party left who knows there is something to hide.

10.5b.1.3 Receptivity, Distinct From $\Psi\_total(t)$

Definition 10.5b.1.3 (Receptivity).

$\Psi\_total(t)$ (§10.5b.18.14) measures how much of a container's population is genuinely negentropic in orientation. It does not measure whether that population's own feedback can find any purchase. Define $\mathscr{R}(t) \in [0,1]$, a container's receptivity to corrective feedback, as decreasing in how completely official, operational, and the metamessenger population have independently converged on the same self-model, accurate or not:

$\mathscr{R}(t) = 1 -$ (fraction of {official, operational, metamessenger population} currently in mutual agreement on the container's own valence).

A genuinely large, genuinely negentropic population (high $\Psi\_total(t)$) can coexist with $\mathscr{R}(t) \approx 0$: if official, operational, and even a majority of the metamessenger population have all independently, and sincerely, converged on the same false belief (Type B, above), there is no remaining internal disagreement anywhere for negentropic feedback to attach to and correct, regardless of how much genuinely negentropic population exists to attempt it. Population size and receptivity are independent quantities; a large negentropic population that cannot find purchase is functionally equivalent, for corrective purposes, to a small one.

10.5b.1.4 Unsteady Official Chaos as a Health Signal

Proposition 10.5b.1.4 (Instability as Feedback, Not Disorder). For a medium-entropy container, a rise in $\sigma\_o(t)$ (§10.5b.1's variance-of-changes measure) that is correlated with a rise in $\Psi\_total(t)$ should be read as evidence of active negentropic feedback reaching official from operational, not as evidence of official-layer disorder for its own sake. More of a container's metamessenger population agreeing with its own clean operational core is, by construction, less internal unanimity in maintaining official's captured story — an official layer that is unsteady in this specific, feedback-correlated sense is *more* negentropic overall, not less, than one rigidly and steadily maintaining its chaotic self-presentation with no internal pushback at all.

Corollary (Why the Firewall Thins After a Successful Reset). This supplies the causal mechanism §6.1 has stated as a bare fact without previously explaining it: that the low-entropy baseline's meta-representation layer is characteristically thin, $\omega\_meta\text{^}up(t) \approx 1$, "not functioning as a firewall at all." Once a medium-entropy container's B2 reorganization succeeds and official genuinely realigns with an operational core that was never corrupted, there is no remaining divergence left to manage — nothing for a firewall to separate. The metamessenger population's own function changes accordingly, from protective barrier to honest instrument: what the barometer-not-auditor framing (§6.1) calls a reading of systemic self-deception becomes, once there is no more self-deception to read, simply an accurate reading of health. A high-$\Psi\_total(t)$ medium-entropy container is accordingly doubly advantaged relative to a low-$\Psi\_total(t)$ one: more able to perceive external danger accurately (§10.5b.2.3–2.4), and more able to reach a clean B2 reset if it eventually tips anyway, since more genuine negentropic material is already present for that reset to reorganize around.

10.5b.2 Regime Genesis (Developmental Branching)

A container does not begin in either non-baseline regime; it arrives there through a shock event acting on a previously low-entropy container. Let $S\_shock(t)$ denote a large, discrete, exogenous perturbation to $D(t)$ or $R(t)$ — structurally, an extraction event per §8.6a, a resolution kick per §8.8b, or an equivalent one-off environmental disturbance.

Editorial note: *The ambient signal below is renamed $v\_gen(t)$, distinct from the verticality operator $V(t)$ of §10.5b.0. The two were previously written with the same symbol despite measuring different things: $V(t)$ is a container's own current structural rigidity; $v\_gen(t)$ is a transmitted signal governing which regime a shock branches a baseline container into.*

Define an ambient signal v_gen(t) ∈ [0,1], sourced either directly from the container that inflicted S_shock (transmitted during the shock itself) or diffusely from the surrounding network's regime composition (a stochastic, grid-level field). Branching at a shock event resolves as:

Definition 10.5b.2.1 (Parasitic Entitlement, the Content of v_gen(t)).

v_gen(t) is not left uninterpreted: its content, regime-generally rather than tied to any single domain, is entropic feedback recruiting the developing container into believing it is entitled to extract from and displace onto other systems — parasitic entitlement, not merely damage. A developing container can be harmed by extraction or displacement without this recruitment; the recruitment specifically is what v_gen(t) measures. This gives a structural account of why high-entropy containers tend toward predation later, rather than only observing that they do: v_gen(t) crossing v_threshold means the container was not merely damaged during formation but actively recruited into the extractive role itself.

Differentiated delivery within one source's coupled neighborhood. A single corrupted source is not required to deliver v_gen(t) uniformly to every container it is simultaneously coupled to. One configuration of a small, coupled sub-grid: one node receives parasitic-entitlement recruitment (high v_gen(t), branching high-entropy) while another, coupled to the same source at the same time, receives extraction or displacement without that recruitment (low v_gen(t), branching medium-entropy if shocked, or simply damaged without branching if not). This is a statement about differentiated delivery from a shared source, not a claim that one node's treatment causally funds the other's; the two outcomes are correlated by shared origin, not necessarily linked by direct transfer.

- v_gen(t) > v_threshold: the operational layer absorbs the > > corruption — a discontinuous jump in D_op — and the container > > transitions to the high-entropy regime. The official > layer > remains untouched, which is precisely what allows it to > > continue presenting as stable while operational reality diverges > > underneath it.
- v_gen(t) ≤ v_threshold: the official layer absorbs the corruption > > instead — a discontinuous jump in D_om — and the container > > transitions to the medium-entropy regime. The operational > core > remains untouched, pristine, and calm; only the > > declared/self-presented layer becomes visibly chaotic.

This is also the structural reason medium-entropy containers reach Full Reorganization (B2, §10.5b.7–§10.5b.9) more readily than high-entropy containers do: an intact operational core is something to reorganize *around*. B1/B2 mass-weighting favors whichever layer holds the greater functioning resource mass, and a medium-entropy container's operational layer, having never absorbed the original corruption, is positioned to be that mass. A high-entropy container has no equivalent intact layer to weight toward — its operational core is precisely what the corruption occupies — which is why terminal lock, not reorganization, is its structurally favored outcome.

10.5b.2.2 Degree of Capture and External Threat Perception

Definition 10.5b.2.2 (Degree of Capture).

For a medium-entropy container j, define κ_capture(j) = v_threshold − v_gen(j, t_0), evaluated at j's own branching moment t_0 — how far below threshold v_gen(t) fell at genesis, not merely that it fell below it. This magnitude, previously discarded once the branch itself was resolved, sets a persistent bias on j's own Ψ_total(j,t) trajectory going forward: Ψ_total(j, t_0) = 1 − κ_capture(j)/v_threshold (normalized default; adapter-overridable), a container whose official layer was almost totally captured beginning life with very little negentropic-aligned population left, and one that barely crossed into medium-entropy beginning with substantially more.

Definition 10.5b.2.3 (External Threat Perception). For j assessing a coupled or potential predator i, j's accurate perception of i's actual danger uses the observational-depth machinery already built (§10.5b.18.9.1) rather than a new primitive:

$$\text{Perceived\_danger}(j,i,t) = A_{acc}(j,i,t) \cdot \text{Danger\_true}(i,t),$$

where Danger_true(i,t) is i's own current extractive intensity (default: $D_{op}(i,t)$, i's own official/operational gap, or $\mathscr{F}_i(t)$ where available). This is perception only — whether j registers the threat at all — and is bounded by the same accuracy ceiling that governs any container's read of another's true state.

Definition 10.5b.2.4 (Self-Doubt Override). Whether j *acts* on Perceived_danger(j,i,t) is gated separately, by j's own current $\Psi_{total}(j,t)$:

$$\text{Effective\_perceived\_danger}(j,i,t) = \text{Perceived\_danger}(j,i,t) \cdot g(\Psi_{total}(j,t)),\ g \text{ increasing in } \Psi_{total},$$

producing two distinct failure severities on one continuous scale, rather than two separate mechanisms:

- $\Psi_{total}(j,t) < \theta_{perceive}$: perception itself fails. Perceived_danger(j,i,t) is gated to zero — there is no channel left through which accurate signal about an external threat can register at all.
- $\theta_{perceive} \leq \Psi_{total}(j,t) < \theta_{act}$: perception succeeds, but $g(\Psi_{total}(j,t))$ is too small for Effective_perceived_danger to cross whatever threshold triggers protective action (decoupling, resistance) — the danger is seen and not acted on.
- $\Psi_{total}(j,t) \geq \theta_{act}$: full perception and action.

10.5b.2.5 Co-Occurring Developmental Mechanisms and v_gen(t) Magnitude

Definition 10.5b.2.5 (Compositional v_gen(t)). $v_{gen}(t)$'s magnitude, not only its presence, is a function of how many distinct developmental corruption channels act on a container simultaneously during its formative window, rather than a single undifferentiated signal:

$$v_{gen}(t) = v_{gen,base}(t) + \Sigma_k\, \kappa_k \cdot \mathbb{1}[\text{channel } k \text{ active}],\ \kappa_k > 0 \text{ fitted per channel},$$

with channels including, at minimum, direct entitlement recruitment (Definition 10.5b.2.1's core content), hyperpraise or idealization, direct projection of the source's own unmet needs onto the developing container, and engulfment/enmeshment foreclosing independent identity formation. A container receiving several of these simultaneously should show $v_{gen}(t)$ magnitude well past threshold, not marginal crossing — this is the structural account of why some high-entropy containers show extreme, near-total capture (§10.5b.2.2's $\kappa_{capture}$ near its own ceiling) rather than ordinary branching, and, symmetrically, why some medium-entropy containers show comparably extreme $\kappa_{capture}$ on the official-layer side.

10.5b.2.6 Anchors, Distinct From Bidirectional Coupling

Reason for this addition. External feedback and support (φ_{fb}, Γ_{supp}, §10.5b.18.9–18.11) have so far been described only through their effect on a receiving container, without formally distinguishing their source from an ordinary coupled neighbor. That distinction matters: a source supplying stabilizing feedback without entering bidirectional coupling behaves, and risks, differently from one that does.

Definition 10.5b.2.6 (Anchor). An anchor relationship, $\text{Anchor}(k \to C, t) \in [0,1]$, is unidirectional stabilizing feedback from source k to container C, distinct from bidirectional coupling $\Lambda_{kC}(t)$: k supplies corrective forcing reducing C's own $D_{om}(t)$ or $\tilde{S}(t)$ without C's own state correspondingly entering k's own resource or divergence equations the way ordinary coupling requires. This is the formal home for $\varphi_{fb}(i,j,t)/\Gamma_{supp}(i,j,t)$ specifically when their source is not a persistently coupled neighbor but is playing this anchoring role instead — an anchor is not merely weak coupling; it is a structurally different relationship, one-directional by definition, which is why it does not draw a receiving container into the parasitic cycle (§8.6a) the way accepting ordinary coupling from an unknown source risks doing.

10.5b.2.7 Dormancy, Generalized Beyond High-Entropy

Reason for this addition. §8.6a.6's dormancy is built entirely from high-entropy-specific machinery (M_decep,ext) and does not apply to baseline or medium-entropy containers. Dormancy is a genuine response available at any entropy level, per Axiom 9's default orientation toward coupling — and whether a given dormant interval heals or harms a container depends entirely on what happens during it, not on dormancy itself being good or bad.

Definition 10.5b.2.7 (Generalized Dormancy). A container at any entropy level is dormant at time t when it has ceased seeking new *bidirectional* coupling: Σ_search(i,t) = 0 and no new Λ_ij is being actively pursued, a trauma response to developmental or later-trajectory corruption, not a default state (Axiom 9). This suspends the pursuit of new bidirectional coupling specifically, not all outward-directed contact: seeking an anchor (§10.5b.2.6) while dormant is not a contradiction, since an anchor is unidirectional by definition and was never the kind of relationship this suspension covers.

Nesting with §8.6a.6. This is the general condition. §8.6a.6's high-entropy dormancy is a strictly narrower special case, requiring this condition *and* the cessation of ongoing extraction: Σ_search(i,t) = 0 *and* Σ_j X_{i,j}(t) = 0. These are genuinely different state variables — a container could satisfy the general condition while still drawing on one existing, unpursued extraction relationship, failing the high-entropy-specific case while satisfying the general one. A high-entropy container is dormant in the full, narrower sense only when both hold; a medium- or low-entropy container, which has no X_{i,j} to begin with, is fully characterized by the general condition alone.

- **Medium-entropy, healthy path.** A medium-entropy container prone to persistent coupling with high-entropy predators pauses specifically to avoid that trap, then either seeks an anchor (Definition 10.5b.2.6 — not bidirectional coupling, and therefore not itself a route back into extraction) or works purely internally, using its own negentropic-aligned metamessengers to slowly raise permeability between its own operational core and official layer with no external party involved. Succeeding at either heals the container's future coupling prospects; this is a healthy, deliberate use of dormancy, not merely damage limitation.
- **Medium-entropy, unhealthy path.** A container that stays dormant indefinitely specifically to avoid re-exposure to entropic feedback does prevent new harm, but simultaneously forecloses the positive negentropic feedback that coupling would otherwise have supplied — d_A(t) contracts through absence rather than active damage, the same destination reached by a different route.
- **High-entropy dormancy.** Serves the same purpose as the medium-entropy healthy path — slowly raising permeability without triggering the flooding that produces collapse — but carries a distinct, genuine irony: a container that succeeds in healing to medium-entropy through this process is not thereby made safe. Its next coupling attempt now exposes it to the opposite role: no longer the predator in the relationship, but the kind of container §8.6a's own targeting machinery ($\mathscr{V}$(j,t), Definition 8.6a.4) would assess as a viable target. Healing here changes which side of the predation dynamic a container occupies; it does not remove the dynamic itself. Full healing to low-entropy baseline takes substantially longer, and most containers do not sustain dormancy that long — but one that does may avoid this specific trap entirely.
- **Low-entropy, the false-isolation trap.** A healthy container may go dormant on a mistaken inference: observing that the containers it sees coupling in its network are predominantly entropic, it concludes coupling itself is the source of entropy, and that full withdrawal will let it settle toward greater stability. The actual result is the reverse. Isolation forecloses both the adaptive possibility genuine exchange would supply and

the accurate external feedback needed to stay calibrated to changing conditions — the container's official layer reports calm (nothing is currently threatening it) while its actual core drifts toward entropy anyway, ungrounded (§6.3.2) by anything outside itself. This is a second, genuinely distinct pathway into entropic status, alongside §10.5b.2's shock-and-capture genesis mechanism — not caused by an external predator or corrupting signal, but by a healthy container's own withdrawal from what it needed to remain accurately self-aware, motivated by a plausible but false theory of what coupling itself does.

Formal entry point, not narrative alone. This pathway enters the existing regime-classification machinery (§10.5b.1) through the same $\sigma\_o(t)/\sigma\_p(t)$ mechanism already used to classify every other container, not through a separate transition rule. Isolation removes external challenge to the official layer specifically — with no coupling, nothing is currently testing or contradicting official's own self-report, so $\sigma\_o(t)$ stays low, genuinely calm. Operational reality, ungrounded (§6.3.2) and uncorrected, continues drifting regardless, so $\sigma\_p(t)$ does not stay correspondingly low. Sustained long enough, this produces the identical formal signature already used to classify high-entropy status — $\sigma\_p(t) \gg \sigma\_o(t)$ — arrived at by a genuinely different route (self-imposed withdrawal, not external predation or corruption) but registered as a regime transition through the same existing test, not a new one.

10.5b.2.8 Grid-Level Extraction Dependence

Reason for this addition. §10.5b.18.25's structural extraction dependence is stated for a single predator container losing access to a single class of targets. The same logic applies at the scale of an entire predator-prey relationship within a grid: a predator population requires its prey population to retain genuine, ongoing adaptive possibility, or the coupling collapses entirely rather than merely locally — a macro-scale instance of the same principle already established for an individual parasite losing its host.

Definition 10.5b.2.8 (Grid-Level Coevolutionary Requirement). Let $d\_A\text{^}prey(t)$ denote the aggregate adaptive possibility of a predator's prey population within the grid. The predator-prey coupling remains sustainable only where both of the following hold, together, not either alone:

- **Threshold condition:** $d\_A\text{^}prey(t)$ stays above a fitted floor $\theta\_coevolve$; if aggregate prey adaptive possibility collapses below this floor, the predator-prey coupling itself collapses, taking the predator population's own sustaining resource base with it — the grid-scale version of §10.5b.18.25's "no self-sustaining floor beneath it."
- **Ongoing forcing requirement:** sustained mutual adaptive pressure — prey developing genuine countermeasures, predator adapting in turn — must continue occurring, not merely have occurred once. This is Axiom C1's logic with its polarity reversed: where C1 says a forced, false state decays without continuous energy to sustain it, this says a healthy coevolutionary coupling requires continuous adaptive effort, on both sides, to stay away from its own collapse boundary; absent that ongoing effort, the system drifts back toward $\theta\_coevolve$ over time even without any single triggering shock.

This is the formal content behind §8.5a's scope distinction (§10.5b.9d): predation sustained as part of a genuine grid-level cycle is exactly predation satisfying both conditions above; predation that forecloses its own prey's adaptive possibility, permanently or by removing the ongoing forcing that would otherwise renew it, is not distinguishable from ordinary entropic destruction merely because it crosses a container boundary — it fails the same requirement, at a larger scale.

10.5b.2.9 Anchor Fidelity

Reason for this addition. Definition 10.5b.2.6 establishes that anchoring is structurally different from bidirectional coupling, but says nothing about whether a given anchor relationship is actually delivering what it claims to. An anchor supplying destabilizing rather than stabilizing feedback is not a degraded anchor; whatever is providing it should not be classified as anchoring at all, regardless of why the feedback is destabilizing.

Definition 10.5b.2.9 (Anchor Fidelity). A relationship claiming $\text{Anchor}(k \to C, t) > 0$ is legitimate only if the feedback it supplies actually moves C's own divergence measures ($D\_om(C,t)$, $\tilde{S}(C,t)$) toward alignment, not away from it. Where the direction is reversed, two structurally distinct failure modes are both possible, and require different reclassification, not one shared fix:

- **Predatory-shaped anchor.** Deliberate, extraction-motivated cover wearing an anchor's shape — reclassifies directly as extraction (§8.6a), using machinery already built, once the direction of harm is confirmed.
- **Accidental harmful anchor.** Not predation at all: a medium-entropy source genuinely attempting to help, projecting its own unresolved, self-deceptive justifications onto the container it is trying to anchor, causing real and potentially comparable damage with no extractive motive or benefit to itself. This case must not be routed through cover-and-lure machinery, which would misdescribe what is actually happening — there is no concealment to detect, only well-intentioned harm. It still requires removal as an anchor: intent does not change whether the feedback delivered was destabilizing.

Standing risk, distinct from current behavior. Anchor legitimacy is not fixed by an anchor's present output alone. A medium-entropy anchor can be genuinely good for another medium-entropy system, *conditional* on its own developmental trauma not being triggered by the interaction — legitimacy here is contingent on a trigger state that can change without warning. A high-entropy anchor can also be genuinely good, specifically where its own operational core draws something it needs from the anchoring role itself (standing, acclaim, a sense of purpose) rather than from extracting the specific system currently being anchored — but such an anchor carries real, standing risk toward *any* system it touches, at any time, because its underlying entropic nature was never actually healed, only given a socially-rewarded outlet that happens not to require harming this particular target right now. Current stabilizing behavior does not retire this risk; it only means the risk has not yet activated toward the container currently being anchored.

Safety ranking. Low-entropy sources are the safest anchors, and within that category, containers in the semi-rigid horizontal state (§10.5b.0.8) are safest of all — specifically because a high-entropy container being anchored will actively attempt to draw its anchor into extraction or displacement, and maximal resistance to being tipped is exactly the property that defeats that attempt. Medium- and high-entropy anchors are not categorically unsafe, per the standing-risk distinction above, but neither should be treated as safe by default the way a low-entropy anchor can be.

The branch is fully determined by which layer absorbs a shock, not by the shock's magnitude alone, which is why two containers experiencing comparably severe disturbances can end up in structurally opposite regimes. Per §8.8b, a resolution kick $K\_\{j \to k\}(t)$ supplies both $S\_shock(t)$ and, via the resolving container's own outcome (terminal-locked → high v_gen; reorganized → low v_gen), the accompanying $v\_gen(t)$ needed to complete this branching mechanism.

10.5b.3 The Confrontation Trigger

Editorial note: *F_conf is generalized below from a high-entropy-only mechanism (keyed to D_op) to a mechanism that also drives the medium-entropy collapsing case (keyed to D_om), completing the three-way backslide account of §10.5b.9a. Both cases are the same equation applied to whichever divergence term the container's current channel structure has doomed it to keep confronting.*

As $\Pi\_f(t)$ rises, a non-baseline container is increasingly forced to register its doomed divergence term directly rather than let it pass unprocessed. Define

*D*_target(t) = D_op(t) if high-entropy; D_om(t) if medium-entropy and s(t) = −1 (§10.5b.9a),

and a transient forcing term, active only in a short window once Π_f(t) crosses a confrontation threshold Π_c while D_target(t) remains elevated:

*F*_conf(t) = ζ · Π_f(t) · D_target(t), ζ > 0 fitted, nonzero only near the Π_c crossing.

This is a genuine self-intensification, not correction — a brief amplification of the very behavior producing the divergence, immediately following the point at which the container is made to confront it, before any resolution occurs one way or the other. A container whose Π_f never crosses Π_c, or whose D_target is already low, never enters a tipping event and remains in bounded oscillation regardless of what follows below. Note that a slow rise in Π_f — one whose rate stays below the crossing dynamics that trigger F_conf, rather than merely its level — can avoid activating this term altogether; this is the operative mechanism for the rogue-metamessenger pathway of §10.5b.14.

10.5b.4 Convergence Events and Regime-Dependent Valence

Once a container is in a tipping event, its outcome depends on which pair of channels subsequently converges toward zero divergence — and the same convergence event carries opposite meaning depending on regime, because which layer is actually reliable flips between regimes:

- D_pm → 0 (meta converges with operational): in the > high-entropy > regime, the self-monitoring layer > validates an actually-chaotic > operational reality — internal > correction is disabled. In the > medium-entropy regime, the > self-monitoring layer instead > recognizes an actually-stable > operational reality beneath a > visibly unstable official layer — > genuine reorganization > around a real signal.
- D_om → 0 (meta converges with official): in the > high-entropy > regime, the self-monitoring layer holds > to the declared > standard against a corrupted operational layer; > this does not > itself prevent a tipping event already underway, but > determines > the container's capacity for rapid reorganization once > the > corrupted operational layer is stripped away post-collapse. In > > the medium-entropy regime, the self-monitoring layer > instead > abandons an actually-accurate operational signal and > validates > an externally corrupted official narrative — internal > > correction is disabled in the same structural way as the > > high-entropy bad case, despite involving the opposite > pair of > channels.

10.5b.5 Driving Force and Convergence Pathways

The force driving a container through a tipping arc is decomposed in §8.5's propagation equation. The two regimes are driven by different, dominant terms in that same equation — but each regime admits more than one causal route to its outcome, and the medium-entropy regime specifically has two genuinely distinct outcomes rather than one, determined by which channel converges:

Flip Condition A (high-entropy, always terminal-locked when unaided).

Two causal routes converge on the same outcome. A1 (usual): self-propelled — F_i-dominated, the container's own ongoing internal churning (fed, per §8.6a and §8.8a, by vented extraction-heat once a coupled neighbor decouples), with D_pm → 0 convergence. A2 (occasional): externally triggered — an external strike combines with a D_om → 0 collision event, producing oscillation larger than self-propulsion alone could generate; the container, careening back around, finds no counter-oscillating partner to strike against, and the unopposed momentum forces the same terminal-locked outcome as A1. Both routes end in the single outcome defined in §10.5b.9 — this regime does not have an intrinsic, unaided full-reorganization option under either route (§10.5b.0, Corollary 10.5b.0.1); §10.5b.11a establishes the distinct, externally-redirected exception, under which a non-terminal but costly stripped reorganization becomes reachable.

Flip Condition B (medium-entropy, externally driven, two distinct outcomes).

Always $\Sigma_j \Lambda_{ij}$-dominated — driven by simultaneous multi-directional lateral knocks from neighboring high-entropy containers, which launch the container higher, and for longer, than any self-generated oscillation could. Which of two outcomes results depends on which channel converges during that launch: B1, $D_{om} \to 0$ (meta converges with the visibly chaotic official layer) engages only the smaller fluid mass — official and meta, the minority layers in this regime — producing a terminal lock. B2, $D_{pm} \to 0$ (meta converges with the actually-calm operational layer) engages the larger fluid mass — operational and meta, since operational holds the majority of a medium-entropy container's mass by the regime's own definition — and that larger, coherently-moving mass, combined with the greater launch height shared by both B routes, completes full reorganization rather than stalling partway through it.

10.5b.6 Vector-Aware Coupling

§8.5's Λ_{ij} term must be treated as directional, not scalar, for either B route to behave correctly. A single dominant external source imparts a force whose direction reverses over that source's own oscillation cycle; summed over the tipping window, this is substantially self-cancelling and functions as a restoring, damping influence — a container knocked by one neighbor rocks and returns, rather than tips. Only simultaneous, multi-directional knocking — several high-entropy neighbors striking from divergent vectors at once — removes the restoring cancellation and produces a net constructive force:

$$\mathrm{Drive}_B(t) = \left| \Sigma_j \Lambda_{ij}(t) \cdot \hat{u}_j(t) \right|,$$

where $\hat{u}_j(t)$ is the unit direction of neighbor j's disturbance at time t. Single-source or aligned-opposing inputs partially cancel (small effective Drive_B, damping outcome, no tipping event at all); genuinely divergent multi-source inputs reinforce (large effective Drive_B, tipping event triggered, outcome then determined by B1 vs B2 below).

10.5b.7 Fluid-Mass Weighting

Flip Condition A's two routes both terminate the same way (§10.5b.0), so the coherence-and-dissipation account below applies to A without modification. Flip Condition B's two outcomes cannot be explained the same way — the medium-entropy regime's own definition requires D_{om} to be elevated, so treating D_{om} as "residual conflict left over" would produce the contradictory conclusion that the full-reorganization outcome (B2) is low-coherence, which is wrong. What actually distinguishes B1 from B2 is not how much divergence is left unresolved elsewhere, but how much of the container's total representational mass is carried by whichever pair is converging. Define mass weights μ_o, μ_p, μ_m for the official, operational, and meta layers respectively (fitted per container, or adapter-supplied directly), with $\mu_p > \mu_o$ structurally in the medium-entropy regime. The effective driving momentum for each B route is weighted by the mass of its converging pair — both members, since B1 is official-and-meta convergence and B2 is operational-and-meta convergence, not a single layer in either case:

$M_{tip}(B1) = \int_{t_{launch}}^{t_{resolution}} (\mu_o + \mu_m) \cdot \mathrm{Drive}_B(t) \cdot \chi_{dir}(t)\, dt$, $M_{tip}(B2) = \int_{t_{launch}}^{t_{resolution}} (\mu_p + \mu_m) \cdot \mathrm{Drive}_B(t) \cdot \chi_{dir}(t)\, dt$, where t_{launch} is the timepoint $\mathrm{Drive}_B(t)$ becomes active (the external knock's onset) and $t_{resolution}$ is the timepoint the tipping window closes — either a confirmed collapse-onset event (§10.5a) or $\mathrm{Drive}_B(t)$ returning to zero without one, whichever comes first — rather than an unbounded or implicit integration window; "the tipping window" used elsewhere in this document refers to exactly this interval, stated here explicitly so an implementation has a concrete start and end condition rather than inferring one from context.

where $\chi\_dir(t)$ is the directional force modifier of Definition 4.11.2 — a push landing orthogonal to the container's own established structural grain contributes disproportionately to tipping momentum relative to an equal-magnitude push landing along the grain, on top of and independent of the mass-weighting below. Setting $\kappa\_dir = 0$ (Definition 4.11.2) recovers the undirected form exactly.

drawing on the same shared Drive_B(t) — the same external launch — but scaled by structurally different mass fractions. Because $\mu\_p > \mu\_o$ in this regime, B2 is structurally more likely to cross into the reorganization range defined below than B1, for the same external launching force — μ_m's shared contribution to both routes does not change this ordering, since it adds identically to each side.

10.5b.8 Coherence, Dissipation, and Flip Condition A's Momentum Functional

For Flip Condition A specifically, not all of a tipping event's driving force survives to complete the transition; some is dissipated as internal friction, proportional to whatever divergence remains unresolved outside the converging pair. Define the residual divergence at a convergence event as the pairwise divergence not involved in the convergence itself — for A1/A2's $D\_pm \to 0$ or $D\_om \to 0$ respectively, the residual is D_op, consistently elevated in this regime by definition — and a coherence factor:

$\kappa(t) = 1 / (1 + D\_residual(t))$, $\kappa \in (0,1]$, with $M\_tip(A) = \int |F\_i(t)| \cdot \kappa(t)\, dt$.

10.5b.9 Outcome Thresholds

Editorial note: *The two routes' M_tip formulas — A's coherence-weighted ∫|F_i|·κ dt and B's mass-weighted ∫μ·Drive_B dt — are structurally different and not naturally on the same scale. As with §5.6's resource-unit normalization, both must be normalized to a common scale before comparing against shared thresholds.*

Let $\tilde{M}\_tip = M\_tip / M\_tip^{ref}$, where M_tip^{ref} is each route's own fitted historical reference scale (A and B routes normalized separately, per §15.3). This reference scale is subject to the same causal constraint already applied to S, R, and F's own normalization (§6.8): retrospective estimation of M_tip^{ref} from the full historical record is permitted for forensic, after-the-fact analysis, but any live implementation must estimate M_tip^{ref} using only information available at or before the current timepoint t. Using a reference scale informed by data after t would let a live tracker's classification of an event depend on outcomes the tracker could not yet have observed at the time of that event — the same future-information leak already guarded against elsewhere, applying here without exception. Two thresholds, $M\_lock < M\_c$ (both fitted, applied to the normalized $\tilde{M}\_tip$), apply uniformly across all routes above, with each route's own M_tip computed by its own mechanism (mass-weighted for B1/B2, coherence-weighted for A). M_lock itself is not a fixed constant for the B route: $M\_lock^{eff}(t) = M\_lock \cdot (1 + \kappa\_Mlock \cdot \Psi\_total(t))$, $\kappa\_Mlock \geq 0$ fitted, where $\Psi\_total(t)$ (§10.5b.18.14) is the container's own negentropic-aligned population strength. A container with a strong negentropic core genuinely extends its own self-arrest window — more tipping momentum is required before self-arrest becomes unreachable — rather than only improving which outcome an already-unavoidable event resolves to (that effect is separate, and already covered by μ_p^{eff}'s existing loyalty-conditioning, §10.5b.18.13). Setting $\kappa\_Mlock = 0$ recovers the fixed-threshold form exactly.

- $\tilde{M}\_tip < M\_lock$: no tipping event completes. The container > > self-arrests and returns to bounded oscillation within §6.9's > > equilibrium region.
- $M\_lock \leq \tilde{M}\_tip < M\_c$: terminal-locked outcome per Definition 10.1 > > — reached by A1, A2, or B1. Requires an external anchor to > quiet > residual oscillation before recovery can begin.
- $\tilde{M}\_tip \geq M\_c$: full reorganization outcome — reached by B2 > > specifically, and structurally unreachable, without external > redirection (§10.5b.11a), by any A route or by

> > B1. $S(t)$ and $\delta R(t)$ reset toward zero; $d_A(t)$ recovers toward > > d_A^{max}. This outcome is an instance of the recovery operator U > > of §10.8: full reorganization is what U looks like when reached > > via the B2 tipping pathway specifically, rather than via > gradual > recovery absent a tipping event.

10.5b.9a Backslide Dynamics and the Resilience Accumulator Z(t)

A container moving toward recovery, whether the high-entropy container under sustained anchoring (§10.5b.11) or the medium-entropy container progressing toward B2, does not improve monotonically. Real containers backslide — temporarily returning toward elevated σ_o — and the character of those backslides differs sharply depending on which convergence direction the container is actually moving in, not merely which regime it currently occupies.

Three backslide signatures.

There are three distinct cases, sharing the property that backslide duration shrinks as healing progresses in all three, but differing sharply in valence:

- High-entropy, anchored self-healing (Flip toward $D_{pm} \to 0$ under > > sustained anchoring, §10.5b.11): each backslide costs more, not > > less, as $\Pi_f(t)$ rises — rising permeability means unresolved > > D_{op} is increasingly confronted rather than suppressed, per F_{conf} > > (§10.5b.3). This is the mechanism of learning through direct > > confrontation with consequence.
- Medium-entropy healing ($D_{pm} \to 0$ convergence direction, heading > > toward B2): each backslide costs less than the last — resilience > > accumulating through successful recovery from repeated knocks. > > This is the mechanism of learning through resilience.
- Medium-entropy collapsing ($D_{om} \to 0$ convergence direction, heading > > toward B1): each backslide costs more than the last — the > same > confrontation mechanism as the high-entropy case, now > keyed to > D_{om} rather than D_{op}, per §10.5b.3's generalized > F_{conf}.

The first and third cases share a mechanism (generalized F_{conf}) and differ only in which divergence term is doomed; the second case requires a genuinely new quantity, since nothing already in this document produces a cost that shrinks over successive knocks.

Definition 10.5b.9a.1 (Resilience Accumulator).

For a medium-entropy container, let $s(t) \in \{+1,-1\}$ indicate the currently-dominant convergence direction (+1 for $D_{pm} \to 0$, healing; −1 for $D_{om} \to 0$, collapsing). For computational purposes, $s(t)$ is not an independent state input: it is assigned from the convergence direction favored by the loyalty-conditioned effective mass split of §10.5b.18.13 ($\mu_p^{eff}(t)$ against μ_o), unless directly observed convergence data are available, in which case the direct observation takes precedence. Define the resilience accumulator $Z(t) \in [0,1]$:

$$dZ/dt = s(t) \cdot \eta_Z \cdot Z(t)(1-Z(t)) \cdot Drive_B(t),$$

with $\eta_Z > 0$ fitted. Z updates only during active knock/backslide episodes (via $Drive_B$, §10.5b.6), not continuously — resilience is built or eroded by what a container actually survives, not by the passage of time alone. The logistic form keeps Z bounded and most sensitive to change away from its extremes.

Floor and ceiling convention. The pure logistic form above has absorbing boundaries at Z=0 and Z=1: $dZ/dt = 0$ exactly at either extreme regardless of $s(t)$, so a container whose resilience is driven to precisely zero can never recover through this mechanism alone, however its circumstances later change. This is not intended as a claim that total exhaustion is permanent. Z is bounded to $[\varepsilon_Z, 1-\varepsilon_Z]$ rather than [0,1], with a small fitted $\varepsilon_Z > 0$: the logistic dynamics approach but never reach either true absorbing boundary, so gradual recovery from even severe depletion remains always technically possible, at whatever rate the container's circumstances allow, rather than foreclosed outright. This is independent of, and does not substitute for, the emergency mechanisms of §10.5b.14.2 or the systems-override mechanism of §10.5b.18.16 — those are deliberate interventions; this is a background property of the resilience quantity itself.

Definition 10.5b.9a.2 (Backslide Cost and Duration).

Let t_n denote the time of the n-th knock. For the medium-entropy cases ($s = \pm 1$):

$c_n = c_0 \cdot (1 - s(t) \cdot Z(t_n))$, $\tau_n = \tau_0 \cdot (1 - Z(t_n))$ (the linear dependence on Z is a computational default consistent with the theoretical direction — cost falls as resilience rises — but is not itself forced by that direction; alternative curves sharing the same monotonic relationship remain admissible),

so that for $s = +1$ (healing), rising Z drives both cost and duration down; for $s = -1$ (collapsing), falling Z drives cost up while duration still falls (the excursion is shorter but more damaging per unit time). For the high-entropy self-healing case, cost is given directly by the generalized F_{conf} of §10.5b.3:

$c_n^{(high)} = F_{conf}(t_n) = \zeta \cdot \Pi_f(t_n) \cdot D_{op}(t_n)$.

Editorial note: *Duration for the high-entropy case is tied here to the declining D_{op} baseline as each successful partial correction leaves a smaller relapse target: $\tau_n^{(high)} = \tau_0 \cdot D_{op}(t_n)/D_{op}(t_1)$. This is stated as a modeling assumption, not a derived result on the same footing as the cost formulas above, since it was not independently confirmed in review.*

Resilience also feeds back into the feedback operator's own drift rate for a medium-entropy healing container, making correction itself more effective as resilience accumulates:

$\gamma_{\mathscr{F}}(t) = \gamma_{\mathscr{F},0} \cdot (1 + \kappa_Z \cdot Z(t))$, $\kappa_Z > 0$ fitted, applied in §6.4's stochastic flow for this container.

10.5b.9b Pre-Breach Self-Rescue and the Counterweight Window

§10.5b.9's thresholds govern what happens once a tipping event is already underway. Before that point, a high-entropy container that recognizes its own trajectory early enough has one further, self-generated option not available afterward: engaging the same $D_{om} \to 0$ convergence event that, post-tip, contributes to A2's terminal lock (§10.5b.5, §10.5b.8) — but engaging it pre-tip, as a counterweight rather than a resolution.

Definition 10.5b.9b.1 (Counterweight Convergence).

For a high-entropy container with hull-breach margin $\text{margin}(t) = \theta_{hull} \cdot \Pi_f(t) - [(1-I(t))+\tilde{S}(t)]$ (§10.5a, §16.3) not yet negative, define counterweight engagement as a self-generated convergence pulling $D_{om} \to 0$ — official and meta layers aligning with each other, on the side opposite the operational layer's pull — while D_{op}, D_{pm} remain at their current levels. This raises I(t) directly (§6.6), and therefore raises margin(t).

Proposition 10.5b.9b.1 (Three Zones).

Let $dI/dt|_{counter}$ denote the rate of integrity increase produced by counterweight engagement.

- margin(t) > 0: if dI/dt|_counter is sufficient to keep margin(t) > > ≥ 0 for the duration of engagement, the container returns to > > bounded oscillation within its current regime (§6.9) — the > > tipping event is avoided entirely, not merely delayed.
- margin(t) = 0 exactly: a knife-edge case. If dI/dt|_counter > > exactly offsets the ongoing decline, the container holds at the > > threshold rather than tipping or receding — an inherently > > unstable stall, sustained only as long as the counterweight itself > > is sustained.
- margin(t) < 0 (past hull-breach): counterweight engagement can no > > longer prevent the tipping event. Per Corollary 10.5b.0.1, Flip > > Condition A structurally admits no route back to bounded > > oscillation once tipping is underway. From this point, D_om → 0 > > convergence still matters, but only as one of Flip Condition A's > > two ordinary post-tip resolution routes (§10.5b.5's A2), not as > a > prevention mechanism.

This gives a high-entropy container exactly one self-generated escape route, with a real, bounded window of effectiveness rather than an open-ended one — consistent with §10.5b.10's observation that this regime's tipping is comparatively invisible in advance: the window exists, but a container (or an outside observer) has to recognize the approach early enough to use it.

10.5b.9c Deliberate Redirection During Frantic Striking

Editorial note: *This is a named exception to §8.8a.ii's frantic striking phase, not a revision of it. Frantic striking is, in the ordinary case, explicitly non-strategic — undirected discharge in every available direction because accumulated heat must go somewhere. The case below is a narrower, rarer variant with its own conditions, coexisting with the general rule rather than replacing it.*

Where a container recognizes its situation with enough lead time to act, but too late for §10.5b.9b's counterweight window (margin(t) already negative, or dI/dt|_counter insufficient to hold it), it may retain one further degree of freedom: choosing the direction of its discharge rather than letting §8.8a.ii's vector sum $\vec{\Lambda}$_i(t) default to an indiscriminate multi-directional pattern.

Definition 10.5b.9c.1 (Targeted Redirection).

Rather than $\vec{\Lambda}$_i(t) = Σ_{j∈N_i(t)} Λ_ij(t)·û_ij(t) summing indiscriminately over all coupled neighbors, a deliberately-redirecting container concentrates its forcing onto a chosen subset N_i^target(t) ⊆ N_i(t) — typically a single neighbor — reweighting which neighbors receive the forcing without necessarily changing its total magnitude. The choice is usually made in the hope of steadying against that specific neighbor, or of avoiding a different one.

This does not guarantee the outcome sought. A targeted neighbor may still resolve through ordinary cascade dynamics (§8.8a.iii's medium-entropy launch) rather than providing the stabilization hoped for, and deliberate targeting of an otherwise-unlikely neighbor is one of the routes formalized in §10.5b.11b below.

Preventing a tipping event outright is preferable to managing its aftermath for both regimes, but the two regimes differ sharply in how foreseeable the event is. A medium-entropy (medium-entropy) container's impending tipping event is comparatively visible in advance — its official layer is, by regime definition, the visibly chaotic one, directly observable as σ_o(t) rising. Because its outcome genuinely could go either way (B1's lock or B2's reorganization), prevention — steadying the container before it tips at all — is substantially safer than letting the event occur. A high-entropy (high-entropy) container's impending tip is, by contrast, comparatively invisible in advance, precisely because its official layer presents as stable while the actual instability is hidden in the operational layer; this is why post-collapse intervention receives comparatively more emphasis for this regime in practice.

10.5b.9d Pre-Flip Disintegration

Per §8.5a's coupling-valence classification, a target crossing from ongoing extraction (§8.6a's chronic parasitism) into this section's disintegration, or into a full tipping cascade, is the acute case of the same (+/-) sign pair — predation rather than parasitism, marked by the target's own collapse resolution rather than continued, weakened functioning under an ongoing coupling.

Definition 10.5b.9d.1 (Multi-Source Chaos Pressure).

For container j targeted simultaneously by multiple high-entropy neighbors, define accumulated chaos pressure as the trailing integral of §8.8a's existing multi-source launch drive,

$D\_chaos(j,t) = \int_{t-w}^{t} Drive\_B(j,s)\, ds$,

over a fitted trailing window w, distinguishing sustained multi-source exposure from a single transient peak that Drive_B(j,t) alone would already capture instantaneously.

Definition 10.5b.9d.2 (Disintegration Condition).

Where D_chaos(j,t) exceeds a threshold scaled by j's own internal cohesion (§9.10.1's Λ_int(j,t)),

$D\_chaos(j,t) > \Theta\_disintegrate \cdot \Lambda\_int(j,t)$, $\Theta\_disintegrate > 0$ fitted,

while j's own tipping-momentum resolution ($\tilde{M}\_tip(j,t)$, §10.5b.9) has not yet reached any of Definition 10.5b.0.5's labeled outcomes, and χ(t,t′) (Definition 9.2) falls below χ_min during this same window as a direct consequence of the pressure rather than of voluntary membership change — j disintegrates. Its boundary operator B(t) transitions to an undefined state, distinct from both B_open (Ultimate Convergence, Definition 10.5b.0.6) and the closed configuration of terminal lock: j ceases to be a container in the sense Definition 2.2 requires, rather than settling into either of §10.5b.0.5's labeled configurations.

Scaling the threshold by Λ_int(j,t) rather than treating Θ_disintegrate as an absolute constant gives the condition a principled form: a more internally cohesive container can absorb proportionally greater simultaneous pressure before disintegrating, consistent with §9.10.1's account of cohesion as the sole containment mechanism once vertical structure is minimal — disintegration is what happens when external pressure exceeds what a container's own cohesion, whatever it currently is, can hold together.

Proposition 10.5b.9d.1 (Disintegration Is Distinct From Isolation Dissolution).

§8.8a's existing dissolution and Definition 10.5b.9d.2's disintegration are opposite in mechanism and should not be conflated despite both resulting in a container ceasing to exist as such. §8.8a's case arises from too few remaining coupled neighbors — a network fragmenting until no further launch is possible. This case arises from too many simultaneous pressuring neighbors — overload rather than abandonment. A network-level audit encountering a dissolved container should check which condition obtains before attributing the outcome to isolation by default: disintegration under Definition 10.5b.9d.2 predicts a preceding period of elevated D_chaos(j,t) from multiple concurrent sources, while §8.8a's dissolution predicts a preceding decline in $\lambda_2(L(t))$ and shrinking coupled-neighbor count — empirically distinguishable signatures for what would otherwise be recorded as the same undifferentiated outcome.

Editorial note: Disintegration is a failure mode, not a resolution, and should not be read as a fourth outcome on par with self-arrest, terminal lock, or full reorganization in the sense §10.5b.9 already treats those three. It is what happens instead of any resolution being reached at all — the tipping event is preempted by structural failure before $\tilde{M}$_tip's thresholds ever get the chance to determine an outcome.

10.5b.10 Steadying vs. One-Time Reset

A medium-entropy container that completes a B2 reorganization, or is steadied pre-flip, benefits from external anchoring primarily as a speed multiplier. A high-entropy container is structurally different: merely righting or resetting it without addressing internal composition is insufficient, since its unaltered internal mechanics (F_i, still driven by whatever unresolved D_{op} and unvented heat remain) will trigger renewed violent oscillation almost immediately. What this regime requires is sustained steadying — continuous external containment disallowing both further oscillation and further extraction/displacement (§8.6a) — maintained over time, not applied once. Sustained steadying is also the mechanism by which a high-entropy container's composition can gradually shift toward the medium-entropy or low-entropy baseline regimes, per Axiom C1: with extraction and oscillation both disallowed for long enough, the feedback operator loses the external support it needs to sustain convergence around the corrupted signal, and drifts toward the true one instead.

10.5b.11a Post-Breach Redirection versus Resistance

Once a container has crossed hull-breach (§10.5a) and entered runaway acceleration (Axiom C2), the character of external intervention needed reverses from what §10.5b.10's prevention guidance recommends. Before hull-breach, opposing an oscillation's phase is damping and protective (§10.5b.6). After hull-breach, the same opposing intervention is destructive: momentum that has already accumulated cannot be canceled, only redirected or blocked, and blocking forces a violent, concentrated discharge rather than preventing one.

Definition 10.5b.11a.1 (Redirection Channel).

At the moment a container's tipping momentum $M_{tip}(t)$ is discharging (§8.8b), define a redirection channel availability $\Gamma_{redir}(t) \in [0,1]$: the degree to which an external structure — an anchor, an aligned momentum source, or a coupled neighbor — is positioned to carry the discharging momentum through a full rotation rather than absorb or oppose it. $\Gamma_{redir}(t) = 0$ indicates no such structure is available or engaged; $\Gamma_{redir}(t) = 1$ indicates full redirection capacity.

Computational default. Where no domain-specific composition is supplied, $\Gamma_{redir}(t) = \max(\Gamma_{anchor}(t), \Gamma_{neighbor}(t), \Gamma_{intrinsic}(t))$, the largest of: anchor-supplied availability (Anchor(k→C,t), §10.5b.2.6, where present), the strongest currently-aligned coupled neighbor's own capacity to receive redirected momentum, and the intrinsic fluid-mass-derived channel already given for Flip Condition B below. Taking the maximum rather than a sum or average reflects that a single sufficient channel, not the accumulation of several partial ones, is what carries a full rotation; this default is adapter-overridable where a domain-specific composition is better justified.

Proposition 10.5b.11a.1 (Discharge Character Generalized).

Lemma 8.8b.1's impulsive-versus-dissipative discharge split, previously determined by which outcome (B1 or B2) a container reached, is a special case of a more general rule: discharge is impulsive when $\Gamma_{redir}(t) \approx 0$ during the tipping window, and dissipative when $\Gamma_{redir}(t)$ is sufficiently high, regardless of route. For Flip Condition B, fluid-mass weighting (§10.5b.7) is what determines Γ_{redir} intrinsically — $\mu_p > \mu_o$ gives B2 an inherent channel B1 lacks. For Flip Condition A, no route supplies Γ_{redir} intrinsically (Corollary 10.5b.0.1 holds unmodified for the unaided case); a nonzero Γ_{redir} must be supplied externally.

Proposition 10.5b.11a.2 (Resistance Blocks the Channel).

An intervention that opposes momentum during active discharge (t near t_{tip}, §8.8b) rather than redirecting it drives $\Gamma_{redir}(t) \to 0$ regardless of what Γ_{redir} would otherwise have been, forcing the impulsive discharge form even where a channel was structurally available. This is the formal content of bracing being harmful specifically after hull-breach: resistance does not reduce $M_{tip}(t)$, it removes the channel through which $M_{tip}(t)$ could otherwise discharge gradually.

Two of three routes to nonzero Γ_{redir} for Flip Condition A (the third, involving a second struggling container, is developed separately in §10.5b.11b).

1. External anchor (A1 or A2). An anchor engaged during the discharge window (§10.5b.13's eligibility conditions apply) supplies Γ_{redir} directly, converting the container's governing dynamics for that window from F_i-dominated to effectively $\Sigma_j \Lambda_{ij}$-dominated — lending the container Flip B's full-rotation option for the duration of the intervention. This is the general-purpose route and the only one available to a purely self-propelled (A1) container, which has no external component to redirect in the first place.

2. Aligned external knock (A2 only). Where the tipping event is already externally triggered (A2's own definition) and the knock's direction aligns with $D_{om} \to 0$ convergence, the knock itself supplies Γ_{redir} without requiring a separately engaged anchor. This route is unavailable to A1, which by definition has no external strike to align.

Corollary 10.5b.11a.1 (Stripped Reorganization).

Route 2 reaches the same $\tilde{M}_{tip} \geq M_c$ threshold band as B2 (§10.5b.9), but the resulting reorganization is not B2's organic convergence around a validated operational signal. Because the redirecting force was an external strike rather than the container's own accurately-monitored operational layer, the reset carries the operational layer to a null or rebuilt state rather than recovering it — a full rotation is completed, $S(t)$ and $\delta R(t)$ reset, but the container's operational history is discarded rather than preserved. This is a survivable, non-terminal outcome, and a costly one, distinct from both terminal lock and clean reorganization.

A third route — mutual coupling with another struggling container — also supplies Γ_{redir}, but involves enough additional structure (predatory targeting, accidental capture, and voluntary sacrifice are all distinct sub-cases) to warrant its own subsection: see §10.5b.11b.

Editorial note: *Remark (Control versus Adaptation). No revision to §11.10 is needed to reflect that resistance-based intervention is not protective: P_C's formula already penalizes rigidity only through its effect on d_A (via R_g, §10.4), not directly. A container that increases external control without increasing Γ_{redir} or adaptive capacity does not lower its own collapse potential under the existing formula; this subsection makes explicit why that asymmetry is correct, not merely why it exists.*

Falsifiable prediction. Consistent with Axiom C2's evidentiary framing: containers with an engaged redirection channel during discharge should show measurably longer, lower-magnitude discharge signatures (dissipative-form $K_j(t)$) and higher post-event survival/reorganization rates than matched containers whose channel is blocked or absent during the same discharge window, holding $M_{tip}(t)$ itself constant.

10.5b.11b Mutual Coupling Under Collapse: Predation, Accident, and Sacrifice

Editorial note: *This subsection replaces and substantially expands the earlier draft's single "sacrificial coupling" route, which modeled two containers' discharges arriving in exact simultaneity — a measure-zero event for two instantaneous impulsive discharges, requiring unreachable temporal precision to occur at all. The tolerance condition below removes that* *requirement, and the single route is expanded into three structurally distinct sub-cases, each with its own trigger condition.*

Definition 10.5b.11b.1 (Discharge Alignment Tolerance).

Two containers' discharges qualify as coupled — for any of the sub-cases below — if $|t_{tip,A} - t_{tip,B}| < \varepsilon_{align}$ (a fitted temporal tolerance) and their forcing direction vectors satisfy $\hat{u}_A \cdot \hat{u}_B > \cos(\theta_{align})$ for a fitted angular tolerance θ_{align}. This gives every sub-case a genuine, nonzero-probability basin rather than requiring exact coincidence.

i. Predatory Targeting (Deliberate).

A high-entropy container approaching terminal lock ($\tilde{M}_{tip}(i)$ approaching the M_{lock}/M_c band, §10.5b.9) may extend its targeting beyond its ordinary extraction pattern — §8.6a's $\rho(j)$, which structurally favors medium-entropy neighbors and disfavors baseline ones — to include an otherwise-untargeted low-entropy baseline neighbor, using that neighbor's structure as a landing surface for its own discharge. Two mechanisms, both operative and not mutually exclusive:

- Gradual aperture widening. $\rho(j)$ gains an explicit dependence on the > > extracting container's own proximity to terminal lock: $\rho(j;i) =$ > > $1/\{1+\exp[-(\rho_0(j) + \kappa_{desp}\cdot\bar{F}_i(t))]\}$ (§8.6a.1's corrected, > > bounded form; $\bar{F}_i(t) = 1-\exp(-\lambda_F\cdot\mathscr{F}_i(t))$) — as > > i's own accumulated unvented heat rises, its targeting > > discrimination continuously loosens to include neighbors it would > > ordinarily ignore.
- Switched pathway. Independent of the above, once i crosses into the > > terminal-lock band, a separate targeting mode — off below the > > band, on above it — activates, reflecting a genuinely > different > behavioral mode rather than a continuation of > ordinary extraction.

Which mechanism dominates in a given case is an empirical question this document does not resolve in advance. The targeted baseline neighbor typically resolves to terminal lock rather than reorganization, since it did not choose or prepare for the role (Corollary 10.5b.11a.1's stripped-reorganization logic applies, but the predator's resulting reorganization is generally a lesser version of it than an aligned-knock route would produce, since baseline containers hold no accumulated momentum of their own to lend — this asymmetry needs empirical calibration, not asserted as a specific ratio here).

ii. Accidental Capture (Cascading Vortex).

Driven by declining algebraic connectivity ($\lambda_2(L(t)) \to 0$, §8.8a.v) during a wider cascading failure, a baseline container can be drawn into a discharge window purely through proximity — via its coupling weight $W_{ij}(t)$ rather than through deliberate targeting (i, above) or fluid-mass alignment (§10.5b.7). Two branches, both real and not reducible to one another:

- Toward a failing container. The baseline container is pulled toward > > a container resolving to terminal lock; per Corollary > > 10.5b.11a.1's logic it typically locks alongside it, though > > survival is possible depending on its own mass or resource > > relative to the pulling container's discharge magnitude, and on > > the phase of the cycle at which it was captured. This document > > does not commit to a closed-form survival probability, flagging it > > as a further empirically-open item alongside those in §14.2.
- Toward a reorganizing container. The baseline container is pulled > > toward a container resolving to full reorganization instead; here > > it can passively benefit from proximity to a successful reset > > rather than being harmed by it, consistent with $\Gamma_{redir}(t)$ > being > high, by construction, in this branch.

iii. Voluntary Mutual Sacrifice.

Distinct from (i) and (ii): two comparably-struggling containers, neither predatory nor accidentally swept in, where the discharge asymmetry favors one's survival through the other's own trajectory redirection. Decompose per §16.8's Definition 16.7 into a strength-explained component and a remainder:

- Logic-based. The strength-explained component alone accounts for the > > asymmetry — the comparatively weaker container at the time > ends > up absorbing the discharge, following directly from the > same > comparison already used elsewhere in this document > (§10.5b.7's > mass-weighting logic, here applied between peers > rather than > within one container's layers).
- Intention-based. A genuine remainder exists beyond the strength > > comparison — per §16.8's Hypothesis, attributable to the > > sacrificing container's own pre-event messaging history, findable > > where confirmable via §16.7's backward-facing procedure, not > > derivable from either container's state at the moment of collision > > alone.

A further, specific version of the intention-based case: a container facing terminal lock either way may choose which neighbor absorbs its discharge, or choose to absorb a neighbor's discharge itself, specifically to keep a wider cascade (§8.8a.v) from propagating further — self-sacrifice undertaken to limit collateral damage rather than for reasons reducible to the strength comparison alone. This is a specific instance of the intention-based remainder, not a distinct mechanism, and is subject to the same §16.8 confirmation standard: it should be reported as a hypothesis unless the sacrificing container's pre-event record independently supports it.

10.5b.11c Active Search, Search Suppression, and Mirror-Threat Avoidance

Definition 10.5b.11c.1 (Active Search Intensity).

For a high-entropy container i, define $\Sigma_{search}(i,t) \geq 0$, the rate at which i initiates evaluation of candidate coupling targets beyond its existing edges — distinct from $\rho(j)$, which governs persistence of couplings already formed. Where i satisfies §10.5b.18.25's structural extraction dependence, $\Sigma_{search}(i,t)$ scales with i's own deficit:

$$\Sigma_{search}(i,t) = \kappa_{search} \cdot \max(0, M_{maint}(C,t) + M_{decep}(t) - P_{prod}(t)),\ \kappa_{search} > 0 \text{ fitted},$$

so that search intensity tracks the severity of i's own unmet need, rather than being a free-floating behavioral parameter. A container not satisfying structural extraction dependence has $\Sigma_{search}(i,t) = 0$ by this default, though an adapter may supply a nonzero value directly where a domain has independent evidence of active-seeking behavior absent formal dependence.

Definition 10.5b.11c.1a (Target Selection Among Candidates). $\Sigma_{search}(i,t)$ governs how intensely i evaluates candidates beyond its existing edges; it does not by itself determine which candidate is actually selected where several are being evaluated at once. Selection among candidates $\{C_k\}$ is jointly determined by proximity and target-type compatibility, reusing machinery already built rather than introducing new primitives:

$$\text{Select}(i,t) = \text{argmax}_k \left[w_{prox} \cdot \text{Prox}(i,k,t) + w_{type} \cdot \mathscr{V}_{i\to k}(t) \right],\ w_{prox} + w_{type} = 1, \text{ both fitted},$$

where Prox(i,k,t) is i's positional proximity to candidate k (derived from existing network distance or positional embedding, §8.6a.7, where available) and $\mathscr{V}_{i\to k}(t)$ (§8.6a.4) is i's own observation-mapped estimate of k's vulnerability, not k's true latent state. This is distinct from Definition 8.3.1's first-contact attraction, which governs general network re-entry toward cohesive-looking candidates for any newly-uncoupled container; this definition governs specifically which extraction candidate a high-entropy container under active search selects, driven by exploitability rather than cohesion. During a frantic striking phase (§8.8a.2) specifically, w_prox rises relative to its ordinary fitted value — accelerating divergence narrows deliberation and favors whichever compatible candidate is nearest, consistent with this document's own account of frantic striking as thermodynamic discharge rather than strategic search.

Definition 10.5b.11c.2 (Search Suppression by Ambient Cohesion).

Active search does not operate at full strength against an arbitrary population; its effective reach is degraded by the ambient cohesion density of the containers it searches among. Define

$\Sigma\_search^eff(i,t) = \Sigma\_search(i,t) \cdot \exp[-\kappa\_amb \cdot \Lambda\_amb(t)]$,

where $\Lambda\_amb(t)$ is the resource-weighted mean of §9.10.1's $\Lambda\_int(C\_k,t)$ across all candidates $\{C\_k\}$ within i's observable search range, and $\kappa\_amb > 0$ is fitted. A search population that is, on average, densely coupled among itself suppresses i's effective search reach exponentially rather than linearly — consistent with the document's own imagery for this mechanism, in which a population's mutual coupling does not merely fail to attract a predator (§8.3.1's Corollary 8.3.1.1 already covers that passive case) but actively degrades an already-searching predator's ability to locate a target at all.

Proposition 10.5b.11c.1 (Grid-Level Predator Starvation, Search Channel).

Where $\Lambda\_amb(t)$ rises across a network — driven by the same cohesion-building dynamics §9.10.1 and §8.3.1 already establish — every structurally extraction-dependent container's $\Sigma\_search^eff(i,t)$ falls in tandem, independent of and additional to §8.3.1's passive-attraction channel. The two channels close different routes to the same target: Corollary 8.3.1.1 closes the route by which new entrants are drawn toward a predator; this proposition closes the route by which a predator's own search finds them. Both failing simultaneously is what gives grid-wide predator starvation its full force as new entrants avoid isolated predators by default while isolated predators simultaneously lose the ability to find anyone at all.

Definition 10.5b.11c.3 (Mirror-Threat Suppression).

Independent of resource-based extraction incentive, define a suppression factor on coupling likelihood between two high-entropy containers i and j:

$\zeta\_mirror(i,j,t) = \exp[-\kappa\_mirror \cdot \min(D\_op,i(t), D\_op,j(t))]$, $\kappa\_mirror > 0$ fitted,

applied multiplicatively to whichever edge-formation quantity (Ξ, §8.3; A_first, §8.3.1; or a search-initiated contact under Definition 10.5b.11c.1) would otherwise govern the i-j pairing. Both parties must register as entropic (both D_op values contributing via the minimum) for this suppression to activate meaningfully; a single entropic party paired with a non-entropic one does not trigger it.

Proposition 10.5b.11c.2 (Mirror-Threat Is Additive to, Not Subsumed by, Resource Suppression).

§8.6a's $X\_\{i,j\}$ already scales with target resources R_j, so extraction incentive between two severely resource-depleted high-entropy containers is independently low without any mirror-threat mechanism. Definition 10.5b.11c.3 does additional work precisely where resource depletion does not already suppress coupling: two high-entropy containers of differing severity, where the less-depleted party would otherwise present exploitable resources to the more-depleted, more-motivated party (§10.5b.18.25's dependent container searching under Definition 10.5b.11c.1). Pure resource logic predicts this pairing should occur at a nontrivial rate; $\zeta\_mirror(i,j,t)$ predicts it should occur markedly less often than resource logic alone implies, because recognizing another container's own entropic signature functions as a threat signal — the prospective target's own extractive capacity, not only its resource level, is what a high-entropy searcher is registering and avoiding.

Editorial note: These three mechanisms operate on different populations and should not be conflated. Definition 10.5b.11c.1 and 10.5b.11c.2 govern a predator's own outward search and its suppression by the population it searches among — primarily relevant to coupling between a high-entropy container and non-high-entropy candidates. Definition 10.5b.11c.3 governs coupling specifically between two high-entropy containers, and activates on a different signal (mutual entropic recognition) than the cohesion-based suppression of Definition 10.5b.11c.2. A single high-entropy container's total coupling behavior is shaped by both mechanisms simultaneously, each doing separate, non-redundant work.

10.5b.12 Combined Intervention

For both regimes, anchoring (a mechanical/behavioral intervention, arresting oscillation and extraction) and active messaging change (an informational intervention — external correction, or the internal rogue-metamessenger pathway of §10.5b.14) are more effective together than either alone, whether applied pre-flip as prevention or post-flip as recovery: anchoring alone can halt an oscillation without resolving what caused it; messaging change alone, absent containment, risks triggering F_{conf}'s destabilizing surge before it can take effect.

10.5b.13 Anchor Eligibility

Editorial note: *The anchor subscript below is corrected from an earlier $X_{anchor,i}$ to $X_{a,i}$, avoiding collision with §8.6a's $X_{i,j}$, in which the first subscript denotes the extracting container. Here the first subscript (a) denotes the anchor and the second (i) the container being protected — the opposite role assignment, which needed distinguishing notation.*

A low-entropy baseline container can safely anchor a destabilizing neighbor, but only by actively maintaining zero extraction and zero displacement flux for the full duration of the intervention — i.e., holding $X_{a,i}(t) = 0$ (§8.6a) throughout, a maintained behavioral condition, not a structural guarantee. Lapsing on this vigilance risks the anchor itself being drawn into an extraction cycle and reclassified into the medium-entropy regime — becoming a casualty rather than a stabilizer.

A semi-rigid horizontal field — a boundary configuration with no exploitable structure to extract from or displace onto, per §2.3 — is structurally, not behaviorally, safe: $X = 0$ by construction, not by maintained effort. This is the preferred anchor wherever available, but is characteristically rare, particularly in human systems.

A third, structurally distinct type: a container or sub-structure purpose-built or purpose-designated specifically to absorb another's discharge — a dedicated buffering structure maintained for this role, rather than a general-purpose neighbor pressed into service. Unlike the vigilant baseline container (whose safety is a maintained behavioral condition) and the semi-rigid horizontal field (whose safety is structural, requiring no maintained effort), a purpose-built anchor's safety is a design property: it is safe to the extent it was actually engineered for the load it absorbs, and its eligibility should be assessed against its adapter-supplied designed capacity rather than assumed. Exceeding that capacity converts it from an anchor into an ordinary coupled neighbor, subject to the same risks as any other.

10.5b.12b Counter-Momentum Availability

Editorial note: §10.5b.9's B1/B2 outcome for a medium-entropy tipping event is currently determined entirely by fluid-mass weighting (μ_o, μ_p). A real, distinct causal channel is identified here: whether the container has a genuinely positioned neighbor available to absorb its own momentum at the moment of tipping, independent of how that momentum happens to be distributed across its own layers. Two containers with identical mass-weighting can resolve differently depending on this second factor alone.

Definition 10.5b.12b.1 (Counter-Momentum Availability).

For container i approaching a Flip Condition B event at time t, define:

$\Lambda\_counter(i,t) = \Sigma\_j\ \Lambda\_ij(t) \cdot \max(0, -\mathbf{g}\_C(i,t)\cdot\hat{u}\_ji(t))$,

reusing the network coupling strength Λ_ij (§8.5) and the structural grain vector g_C(t) already defined in §4.11.1, summed over currently-coupled neighbors j whose own recent coupling direction toward i opposes i's own dominant recent oscillation direction. This measures the real, currently-available momentum-absorbing capacity in the network immediately around i, not a hypothetical or historical one — a neighbor that has since decoupled or reversed its own orientation contributes nothing.

Proposition 10.5b.12b.1 (Backstop-Dependent Resolution).

Holding mass-weighting (μ_o, μ_p) fixed, a Flip Condition B event resolves differently depending on Λ_counter(i,t): where Λ_counter(i,t) exceeds a fitted threshold Λ_backstop, the incoming discharge is absorbed elastically against the positioned neighbor and the container self-arrests (§6.9's bounded oscillation) rather than completing a tipping outcome at all; where Λ_counter(i,t) falls below that threshold, the same discharge instead completes as a full 360° reorganization (§10.5b.9's B2), since no real structure exists to return the momentum to. This is a genuinely separate mechanism from mass-weighting, not a restatement of it: a container can carry mass-weighting favoring B2 and still self-arrest if a real backstop is present, or carry mass-weighting favoring self-arrest and still complete a 360° reorganization if no backstop is available at the moment of tipping. Where an adapter cannot supply real, current coupling-direction data for neighbors, Λ_counter(i,t) defaults to 0 (no assumed backstop), which is the conservative default consistent with this document's general preference for not silently assuming protective structure that has not been confirmed.

10.5b.13.1 Mutual Anchor Clusters

Definition 10.5b.13.1 (Mutual Anchor Cluster).

A set of containers {C_1,...,C_n}, each in the low-entropy baseline or medium-entropy regime, forms a mutual anchor cluster at time t when their pairwise internal cohesion (Λ_int over the set, §9.10.1) exceeds a fitted threshold Λ_mutual, and no individual member satisfies any of §10.5b.13's three existing anchor-eligibility criteria on its own. This is the defining structural difference from an ordinary anchor: eligibility here is a property of the cluster's own coupling density, not of any single member's behavioral vigilance, structural safety, or designed capacity.

Definition 10.5b.13.2 (Reciprocal Support Density).

Within a mutual anchor cluster, §10.5b.18.9–18.11's external feedback channel φ_fb(i,j,t) and support flux Γ_supp(i,j,t) — defined pairwise for an ordinary healthy-neighbor coupling — are computed for every ordered pair (i,j) within the cluster rather than in one privileged direction. Define the cluster's reciprocal support density,

$\Phi\_mutual(t) = [\Sigma\_\{i,j \in \text{cluster}, i\neq j\}\ \Lambda\_ij(t)\cdot\varphi\_fb(i,j,t)] / [n(n-1)]$,

the cluster-averaged bidirectional feedback rate, and analogously Γ_mutual(t) for support flux. Unlike a single anchor's one-directional supply, every member of a mutual anchor cluster is simultaneously a source and a recipient of both quantities — Definition 10.5b.18.11's correction forcing terms F_ext,fb and F_ext,supp apply to every member's own D_om(t) using the cluster's aggregate Φ_mutual(t) and Γ_mutual(t) as their driving input, rather than each member needing a distinct external anchor of its own.

Proposition 10.5b.13.1 (Distributed Margin Benefit).

Each member's hull-breach margin (§16.3's $margin\_i(t)$) gains a positive contribution from cluster membership,

$$margin\_i^{cluster}(t) = margin\_i(t) + \kappa\_cluster \cdot \Phi\_mutual(t) \cdot \Gamma\_mutual(t),\ \kappa\_cluster > 0 \text{ fitted},$$

distributing protective effect across the cluster rather than concentrating it in one designated anchor's continued vigilance. This has a direct consequence for cluster resilience under §8.8a's cascading dynamics: where an ordinary anchor's failure or withdrawal removes protection entirely (a single point of failure, consistent with §10.5b.13's warning that a lapsing vigilant anchor becomes a casualty itself), a mutual anchor cluster's protective effect degrades gradually as $\Phi\_mutual(t)$ and $\Gamma\_mutual(t)$ decline with membership loss, rather than failing discontinuously when any single member is lost. A cluster of size n loses protective capacity roughly in proportion to the coupling density removed with a departing member, not catastrophically as an ordinary anchor's failure would.

Editorial note: This does not introduce new primitives beyond what Part 10 Addendum II and Part 9 Addendum I already supply; it removes the directional asymmetry those subsections assumed by default and states what the resulting reciprocal structure entails for margin and cascade resilience. A mutual anchor cluster is best understood as many ordinary healthy couplings operating simultaneously among comparably-situated members, not as a new kind of coupling requiring its own mechanism.

10.5b.14 Rogue Metamessengers and Slow-Correction Reversal

A high-entropy container is not always healed only from outside. A small, coordinated subset of a container's own meta-representation apparatus — aware the container's operational layer is corrupted and actively working against what the container's official layer believes it is doing — can raise $\Pi\_f(t)$ deliberately, provided the rate of increase, not merely its eventual level, stays below whatever threshold triggers F_conf's destabilizing surge. Formally: a controlled injection $d\Pi\_f/dt = \varepsilon\_rogue$, with ε_rogue small and bounded well under the rate that would cross Π_c abruptly. Held slow enough, this permits D_op to be metabolized incrementally through ordinary correction rather than forcing a confrontation event, allowing gradual regime transition — high-entropy toward medium-entropy or, ideally, toward the low-entropy baseline — without ever triggering Flip Condition A.

10.5b.14.1 Entropic, Negentropic, and Rogue Metamessengers

Three terms, precisely distinguished, since two of them have been used inconsistently in earlier drafts of this section.

Negentropic metamessengers are, regime-generally and by fixed meaning, those whose current orientation favors the container's true, healthy signal — the direction that reduces overall entropy if it prevails.

Entropic metamessengers are, regime-generally and by fixed meaning, those whose orientation favors the container's false or unhealthy signal — the direction that sustains or increases overall entropy if it prevails.

Rogue metamessengers are defined relationally, not absolutely: those whose current orientation opposes the operational layer's own direction, whatever that direction currently is. Because operational's own alignment differs by regime, "rogue" names a different population in each:

- **High-entropy:** operational is entropic. Rogues — opposing operational — are negentropic-aligned. Rogue and negentropic are the same population in this regime, and §10.5b.14's protective minority is properly named either way.
- **Medium-entropy:** operational is negentropic. Rogues — opposing operational — are entropic-aligned. Rogue and entropic are the same population in this regime; negentropic metamessengers here are aligned *with* operational and are explicitly not rogues.

> This is the population whose convergence with the official layer during a lateral knock predicts collapse rather than reset (§10.5b's convergence-direction account); a medium-entropy container whose metamessenger population is majority-rogue can never organize a large negentropic bulk at once, leaving its negentropic operational core persistently outweighed and the container correspondingly more vulnerable to knocks and to sustained coupling with predatory neighbors.

Formally, both entropic and negentropic fractions are already carried by $\Psi\_loyal(t)$ (§10.5b.18.12) within the closed population, and by $\Psi\_total(t)$ (§10.5b.18.14) across the whole population; "rogue" introduces no new quantity, only a regime-relational name for whichever of these fractions currently opposes operational.

10.5b.14.2 Emergency Firewall Reinforcement

§10.5b.14's ε_rogue mechanism describes deliberate, slow, controlled disclosure — a routine, ongoing negentropic action. A separate, faster mechanism exists for a different situation entirely: an acute, large spike in Π_f(t), of a magnitude routine leak-patching (§10.5b.18.1's M_o,t^false, event-triggered around ordinary glitches) is not equipped to contain, and which risks catastrophic, uncontrolled feedback flooding if left unaddressed.

Definition 10.5b.14.2 (Emergency Firewall Reinforcement).

Where Π_f(t) crosses a fitted large-spike threshold Θ_spike — set well above the routine fluctuation range already characteristic of the high-entropy regime, distinguishing a genuine flooding risk from an ordinary small leak — the negentropic-aligned (rogue) fraction of the closed population, Ψ_loyal(t), executes an automatic, emergency response: official's memory is rapidly restored toward its pre-spike baseline, and an additional restorative pull is applied directly to Π_f(t) itself,

$F\_emerg(t) = \kappa\_emerg \cdot \Psi\_loyal(t) \cdot (\Pi\_f(t) - \Theta\_spike)_+$, $\kappa\_emerg > 0$ fitted,

pulling permeability back below the spike threshold. This is mechanically distinct from routine cover-maintenance despite superficially resembling it: routine patching (entropic-aligned, triggered by any leak, purpose is sustaining cover-and-lure) and emergency reinforcement (negentropic-aligned, triggered only by a large spike, purpose is preventing collapse) can produce a similarly-shaped memory correction while being executed by opposite populations for opposite reasons. A high enough, sustained Π_f(t) that F_emerg(t) cannot bring back under Θ_spike proceeds toward Flip Condition A's ordinary tipping machinery (§10.5b.7–§10.5b.9) rather than being contained.

10.5b.15 Two Collapse-Layer Axioms

The mechanisms above rest on two claims general enough, and load-bearing enough, to state as axioms of the collapse layer specifically, alongside Axioms C3 and C4 (§10.5b.0) governing the same layer's structural-geometric side. C1 and C2 below govern divergence dynamics; C3 and C4 govern structural geometry; all four are distinct from Part 3's foundational axioms.

Axiom C1 (Truth-Only Reorganization).

The feedback operator $\mathscr{F}$ (§6.4) admits stable, unforced fixed points only at convergence toward the container's true underlying signal. Convergence toward a false signal is not a fixed point $\mathscr{F}$ can reach or sustain on its own; it requires continuous external or self-generated forcing (F_conf, or an extractive input per §8.6a) to be maintained, and decays toward the true-signal fixed point once that forcing is withdrawn — formalized directly by §6.4's landscape Φ reshaping once artificial forcing that was locally propping open a false-signal valley is removed.

Epistemic status of "true underlying signal." This is an ontological quantity within the theory — what the axiom asserts $\mathscr{F}$'s unforced fixed points converge toward — not something automatically identified with whatever the model currently estimates. Its empirical identification requires an admissible grounding procedure (G(t), §6.3.2) and may itself remain latent or only estimated, subject to the same identifiability limits already stated there: an estimate agreed upon by several apparently independent parties is not thereby confirmed to be the true signal, only agreed-upon. Stating this explicitly here forecloses a reading in which the theory smuggles its own desired answer into the landscape by silently treating "the model's current best estimate" and "the true underlying signal" as the same thing.

Axiom C2 (Runaway Acceleration).

As a container approaches a hull-breach crossing (§10.5a) while remaining in Flip Condition A, its instantaneous divergence pressure accelerates rather than approaching the threshold at constant or decelerating rate. This is stated as a falsifiable claim: it predicts that d(D_op)/dt and dS/dt should show a measurable, characteristic upward inflection within a fitted pre-breach window [t_breach − w_C2, t_breach], w_C2 fitted or domain-defined, distinguishable from a null model in which divergence approaches the threshold as an undifferentiated random walk. Fixing w_C2 in advance, rather than choosing it after inspecting the data, is what keeps this a genuine prediction rather than a pattern any analyst could locate retrospectively in a sufficiently flexible window.

Axiom C5 (Bidirectional Closure — the Horseshoe Principle).

Editorial note: *Axiom C1 already implies this; it is stated here explicitly because the implication was never drawn out until raised in review.*

A container's feedback operator $\mathscr{F}$ (§6.4) tracks truth only when its rate of adaptive updating is genuinely responsive to real divergence pressure — moving more when divergence is high, settling when divergence is low. A container whose rate of updating is decoupled from divergence pressure is in a closed configuration in the sense of Axiom C1, regardless of which direction that decoupling runs: a container pinned near zero velocity regardless of divergence (rigid closure) and a container pinned near maximal velocity regardless of divergence (compulsive closure) are the same failure — both require continuous external or self-generated forcing to sustain, and both decay toward genuine, divergence-responsive tracking once that forcing is withdrawn.

Definition 10.5b.15.1 (Tracking Fidelity).

Over a trailing window of length w, allowing for realistic response lag δ ∈ [0, δ_max] rather than requiring simultaneity:

*Φ*_track(t) = max_{δ∈[0,δ_max]} corr_{s∈[t−w,t]}(‖v(s−δ)‖_g , D(s)).

Both horseshoe extremes are diagnosed by the same underlying signature — a mismatch between what a container's divergence pressure D(t) is doing and what its velocity is doing — rather than by simply flagging low variation in movement:

- Rigid closure: D(t) elevated or rising while ‖v(t)‖_g stays pinned > > near its floor across the window — the container is not > moving > despite a real, worsening problem.
- Compulsive closure: ‖v(t)‖_g stays pinned near its ceiling while > > D(t) is low or falling across the window — the container keeps > > moving despite having no real problem left to move about.
- Neither is triggered when D(t) and ‖v(t)‖_g are jointly low — > > genuine, healthy equilibrium (§6.9), not a diagnostic failure. > > Note that a high-entropy container's velocity already > reflects > its full internal state, including its churning > operational > layer (§4.3); such a container is not quiet in this > sense even > though its official layer alone would appear calm, so > this > diagnostic does not misfire on it — the apparent-vs-actual > > distinction is exactly what §10.5b.1's σ_o/σ_p classification > > already tracks separately.

The theory does not need a separate diagnostic for each extreme; the same test that fails to find a real relationship between motion and cause catches both. §16.10's Definition 16.10 (Structural Baseline) supplies the comparison baseline this diagnostic needs before "pinned near its floor/ceiling" can be applied to a real case, since these are meaningful only relative to a container's own typical range.

Definition 10.5b.15.2 (Instantaneous Velocity Crossing).

Editorial note: Definition 10.5b.15.1 already correctly distinguishes genuine equilibrium from oscillation for a container diagnosing itself, because it uses a trailing window rather than a single timepoint. The gap this addition closes is different: a *second* container observing a high-entropy neighbor, rather than diagnosing itself, has no guarantee of using that same windowed discipline, and can be systematically misled by exactly the kind of single-timepoint sample §10.5b.15.1 was built to avoid.

Any oscillating trajectory, however violent, passes through $\|v(t)\|_g \approx 0$ naturally each time it reverses direction. Define the instantaneous crossing set $\mathscr{Z}(j) = \{t : \|v_j(t)\|_g < \varepsilon_{zero}\}$ for a fitted tolerance ε_{zero}, distinct from $\Phi_{track}(j,t)$ (§10.5b.15.1), which is computed over a full trailing window w and cannot be satisfied by a single crossing alone.

Proposition 10.5b.15.1 (Sampling Fidelity Requirement).

A container i's coupling or engagement decision toward a neighbor j (§8.3's edge existence, §10.5b.11a-c's targeting and search mechanics) should be gated on $\Phi_{track}(j,t)$ — the same windowed diagnostic §10.5b.15.1 already uses for self-diagnosis — rather than on a raw, single-timepoint observation of $\|v_j(t)\|_g$. Where i's own observation of j happens to fall within $\mathscr{Z}(j)$ and i treats that single sample as evidence of §6.9's genuine equilibrium, i has committed a real, falsifiable misread: engaging with j on the belief that j has reached stability, when j's own windowed Φ_{track} would show continued high oscillation. This is a real, distinct failure mode from the existing apparent-vs-actual warning (§10.5b.1, §15.2), which concerns a container's *official layer* misrepresenting its own state; this failure occurs even when j's official and operational layers are both being observed accurately, purely because the observing container i sampled at the wrong instant. An adapter supporting inter-container coupling decisions should therefore expose $\Phi_{track}(j,t)$ as an observable neighbors can query, not merely $\|v_j(t)\|_g$ at the current instant.

10.5b.16 Compulsive Closure and the Boomerang Mechanism

A container may become closed not through rigidity but through its opposite: a fixed commitment to producing change, sufficient that reaching genuine alignment removes the compulsion's reason for existing, and the compulsion — rather than subsiding — manufactures fresh divergence to resolve.

Definition 10.5b.16.1 (Alignment-Averse Forcing).

For a container diagnosed with compulsive closure (Definition 10.5b.15.1: Φ_{track} degenerate with v pinned high), define a transient forcing term active only as the container's aggregate divergence approaches its own historical floor:

$F_{boom}(t) = \kappa_{boom} \cdot \Pi_f(t) \cdot \mathbb{1}[D(t) < D_{low}] \cdot (D_{low} - D(t))$, $\kappa_{boom} > 0$ fitted.

D_{low} is a fitted proximity-to-alignment threshold, analogous in role to §10.5b.3's Π_c. $F_{boom}(t)$ enters §6.4's landscape dynamics as a term opposing the ordinary downhill drift, pushing the container away from Φ's minimum rather than toward it. Its magnitude is largest exactly at $D(t) \to 0$ and fades as the container moves back into divergence — the mirror image of $F_{conf}(t)$ (§10.5b.3), which is largest under elevated divergence and fades toward baseline. Where F_{conf} formalizes a struggling container's pain at confronting its own bad numbers, F_{boom} formalizes a compulsively-closed container's inability to tolerate having no bad numbers left to confront.

Editorial note: *This is not a claim that all realignment is followed by relapse. $F_{boom}(t)$ is defined only for containers already diagnosed with compulsive closure; a container without that diagnosis approaching $D(t)\to 0$ experiences no such force and simply settles, consistent with §6.9's ordinary informational equilibrium.*

10.5b.17 Binary Trust Sorting and the Inversion Failure

A container attempting to direct help toward its own struggling members must correctly identify which members are net extractors and which are net victims (§8.6a). Where a container has previously adopted a binary trust classification of its own neighbors — sorting them into a trusted set and a suspect set, rather than continuously tracking their actual extraction behavior — that classification can become a liability distinct from ordinary misdiagnosis.

Definition 10.5b.17.1 (Binary Trust Label).

A container C possessing this precondition maintains a label $\ell : \{C_j\} \to \{\text{trusted, suspect}\}$ over its neighbors, updated on a slower timescale than the underlying extraction dynamics $\rho(j)$, $X_{i,j}$ (§8.6a) — the defining feature of the precondition is that ℓ is sticky relative to the dynamics it is meant to track, not that it is wrong at any single instant.

Definition 10.5b.17.2 (Inversion Condition).

Once C's own divergence pressure ($(1-I(t)) + \tilde{S}(t)$, §10.5a) exceeds a fitted confusion threshold, ℓ's assignment inverts relative to actual role: neighbors with high outbound extraction flux (§8.6a's $X_{j,k}$) are disproportionately likely to hold the trusted label, while neighbors who are net victims — visibly distressed, elevated σ_o or high inbound extraction — are disproportionately likely to hold the suspect label. The mechanism is not random misclassification; it is that visible distress is read as evidence for the suspect label rather than as evidence of victimhood, precisely when C is least able to tell the difference.

Proposition 10.5b.17.1 (Misdirected Anchoring Expands Reach).

Where C extends anchoring or resource support (§10.5b.13) preferentially to trusted-labeled neighbors under the inversion condition, and the trusted set is disproportionately composed of actual extractors, the support functions as an involuntary transfer into an extractor's own resource balance (§7.4) rather than relief for a victim. This has two consequences beyond the original mistargeting: it relaxes the resource constraints (§7.3's clipping, §10.3's stress/debt-driven contraction of d_A) that would otherwise limit the extractor's own capacity to keep extracting, and — since coupling strength and edge existence (§8.3, §8.5) respond to a container's available resource and standing — it widens the extractor's practical reach to neighbors beyond the one that supplied the support. The extractor does not merely extract more from C; it becomes better resourced to extract from the rest of the network as well.

Corollary 10.5b.17.1.

A container satisfying Definition 10.5b.17.1 is, all else equal, more fragile under high internal divergence than an otherwise-identical container without a sticky binary trust structure, because its confusion converts directly into a targeting failure with network-wide consequences (Proposition 10.5b.17.1) rather than remaining contained to a single misjudgment. This connects Axiom C5's bidirectional-closure account to §11.7's fragmentation and Proposition 12.8's coupling: misdirected anchoring is a concrete mechanism by which one container's confusion measurably degrades its neighbors' network coherence, not merely its own.

Editorial note: *A related but distinct failure, where an external observer (rather than the confused container itself) misattributes a hidden predator's harm to its visible victim, is addressed in Part 16's auditing material (§16.9's Proposition 16.9.1) rather than here, since it concerns third-party diagnosis after the fact rather than a container's own internal targeting.*

10.5b.17.3 Live Third-Party Misdirection

Definition 10.5b.17.3 (Third-Party Misdirected Anchoring).

A container C_k, itself in the low-entropy baseline or medium-entropy regime and distinct from any container satisfying Definition 10.5b.17.1's self-targeting precondition, can commit the same inversion Definition 10.5b.17.2 describes while directing its own anchoring or support (§10.5b.13) toward a pair of neighbors it is not itself confused about in the sticky-label sense — it need not maintain a formal ℓ classification at all. Where one neighbor is high-entropy (calm-presenting, σ_o low per §10.5b.1) and the other is medium-entropy (visibly distressed, σ_o high) and the two are coupled through an extraction relationship (§8.6a) with each other, C_k can read the calm presentation as evidence of health rather than concealment, and the visible distress as evidence of being the problem rather than the victim — the same misreading Definition 10.5b.17.2 identifies, now committed by the party extending help rather than the party confused about its own neighbors.

Where C_k extends anchoring to the high-entropy neighbor on this basis, the anchoring functions as Proposition 10.5b.17.1 already describes: a resource transfer that relaxes the extractor's own constraints (§7.3, §10.3) and widens its practical reach (§8.3, §8.5). The distinction from Proposition 10.5b.17.1 is the source of the transferred resource — here it is C_k's own healthy standing, not the confused victim's, and C_k was not previously a party to the extraction relationship it has just subsidized.

10.5b.17.4 Cascade Consequence

Proposition 10.5b.17.2 (Cascade Consequence of Misdirected Anchoring).

Where the anchored high-entropy neighbor subsequently treats C_k's support per §8.6a's extraction dynamics rather than as genuine reciprocal coupling — the same instrumental use of an apparent partner already established in §10.5b.18.1's cover-and-lure mechanism, now directed outward at C_k rather than inward at the extractor's own official layer — C_k's own resource balance (§7.4) and divergence pressure begin to deteriorate along the branching trajectory §10.5b.2 describes for any container under sustained extraction. A previously low-entropy C_k can begin oscillating into the medium-entropy regime itself as a direct consequence of the support it extended. Should C_k's own oscillation subsequently strike its own neighbors (§8.8a's cascade sequence), the failure propagates a second time, grid-wide, originating from what was, at the outset, a genuine attempt at healthy support rather than any fault of C_k's own internal dynamics.

Editorial note: This is the live analog of two mechanisms this document already has, brought together for the first time. It is structurally identical to Definition 10.5b.17.2's inversion, but committed by a third party rather than the confused container itself. It is the real-time counterpart to §16.9.1's decoy check, which exists to diagnose exactly this pattern after the fact; this addition formalizes the mechanism the decoy check is retrospectively checking for, so that the two are recognized as one failure mode viewed from two different points in time rather than as separate concerns. §10.5b.17's original editorial note anticipated that a third-party version of this failure existed and deferred it to Part 16's auditing material; this addition is that deferred mechanism's live-mechanics counterpart, placed here rather than in Part 16 because it belongs with the inversion logic it shares a mechanism with, not with the forensic material that examines its aftermath.

10.5b.18 The Firewall as Active Modulation: Captive Official Layers and Metamessenger Composition

§10.5b.2 describes how a shock branches a container into the high-entropy regime; Axiom C1 states that sustaining false-signal convergence requires ongoing forcing. This subsection makes explicit what supplies that forcing, day to day, once a container has branched, and in doing so completes both accounts: the meta-representation layer (§6.1) is not, in general, a passive boundary between official and operational reality. It is an active population of subprocesses whose collective loyalty determines whether the boundary functions as a firewall at all.

Captive Official Layer — Mechanism.

Definition 10.5b.18.1 (Captive Official Layer).

The official representation P_o (§6.1) can be sincere — genuinely produced by its own local processing — while remaining radically incomplete at the system level, because O_o's access to the operational layer's actual dynamics is itself gated by the firewall described below, not because P_o's own processing is dishonest. A captive official layer is therefore not a lie; it is an accurate report of an artificially restricted input.

That restriction is not passive. The operational layer, in this regime, uses the official layer instrumentally — as cover for what operational is actually doing, and as a lure, presenting a healthy front that attracts coupling from other systems (characteristically medium-entropy ones, §10.5b.11b) which operational then extracts from and onto which it displaces its own entropy. Neither function requires operational to represent this to itself as intentional; it is the structural role the arrangement plays, not a claim about deliberation.

Sustaining that role requires active maintenance. Let $M_{o,t}$ denote the official layer's own memory state — nominally a projection of the true historical trajectory, $M_{o,t} = O_o(H_t)$, consistent with how $P_o = O_o(P)$ projects the current state (§6.1, §9.2). A permeability event, whether internal (a transient rise in $\Pi_f(t)$) or external (transparent feedback crossing a healthy coupling, $\varphi_{fb}(i,j,t)$, Part 10 Addendum VII §10.5b.18.9), can produce a momentary glimpse of operational reality that would, left alone, move $M_{o,t}$ incrementally toward truth. Where the closed population's composition is entropic-aligned (the $1 - \Psi_{loyal}(t)$ fraction, §10.5b.18.12), that glimpse is instead met with substitution: $M_{o,t}$ is overwritten with a fabricated segment, $M_{o,t}^{false} \neq O_o(H_t)$, constructed to explain the glimpse away as noise, error, or isolated anomaly rather than integrate it as signal. This is why official's characteristic stability in this regime (low σ_o, §10.5b.1) should be read as maintained rather than merely low: absent continuous rewriting, each permeability event would leave a small residue, and σ_o would drift upward over time even with the container never leaving the high-entropy regime by any other measure.

This gives $\Psi_{loyal}(t)$ (§10.5b.18.12) a regime-specific behavioral signature it did not previously have. Within the closed population here, negentropic-aligned metamessengers ($\Psi_{loyal}(t)$) hold the firewall by pacing disclosure — allowing small, tolerable increments of truth through rather than none, specifically to avoid a sudden flood the system cannot survive, the same shock-avoidance logic that governs the medium-entropy regime's caution (§10.5b.18.7, below). Entropic-aligned metamessengers ($1 - \Psi_{loyal}(t)$) instead actively falsify via $M_{o,t}^{false}$ substitution, because their own function — sustaining cover and lure — depends on official remaining not merely uninformed but actively misinformed. Pacing and falsifying are behaviorally distinguishable in principle (the former lets small true increments accumulate over time; the latter overwrites them as fast as they arrive) even though both present, at a single timepoint, as low $\omega\_meta\hat{}up(t)$.

The channel runs both directions, and the return direction is the actual point. Pacing disclosure is not an end in itself — accumulating paced truth is the prerequisite for official regaining something it lost along with its ability to update: the capacity to act. Official was never merely cut off from information; it was cut off from exercising any influence over operational, precisely because it had nothing current to act on. As $\Psi\_loyal(t)$ carries paced increments of true operational content upward, official does not merely come to know more — each increment restores a proportional measure of official's actual operational agency, before any full realignment is reached. This is graduated, not threshold-gated: even a small increase in permeability yields some corresponding increase in official's capacity to influence operational, since partial awareness is still awareness official can act on. The same Ψ_loyal population that carries truth upward, paced, is the concrete carrier of official's regained influence back downward once official has enough of the picture to exercise it — one population, one channel, running in both directions, not two separate mechanisms.

At extreme capture, substitution becomes generation, not merely concealment. Beyond a fitted capture threshold (large $\kappa\_capture(j)$, §10.5b.2.2), $M\_o,t\hat{}false$ substitution stops functioning only as explaining-away and begins actively generating new justificatory content — a constructed narrative framework, not just a patched-over anomaly. Two consequences follow, and neither is available to ordinary, lower-degree captivity. First, this generated content can extend to sanctioning harm directed at other containers within the same immediate coupled network, framed by the captured official layer as legitimate or even righteous, rather than the captivity remaining confined to managing what official believes about its own container's operational reality. Second, sustained generation at this degree can remove concealment motive entirely: where the generated narrative frames operational reality as already legitimate rather than as something requiring concealment, $M\_decep,ext(t)$ (§10.5b.18.4) can approach zero even while $D\_op(t)$ remains extreme, since there is no longer anything, from inside the captured official layer's own generated account, that requires hiding.

A third metamessenger role: upstream source, not downstream concealment. §6.1's construction of P_m as an emergent aggregate — $P\_m(t) = \Sigma\_k\ \varphi\_k(t)\cdot P\_m\hat{}(k)(t)$ — has so far been treated as purely diagnostic: the population reads official and operational and aggregates into a meta-representation, but does not generate content that feeds back into official or operational's own updating. A sub-population of entropic-aligned, self-aware metamessengers can occupy a structurally different role: rather than concealing their own entropic status within a delusion that arose some other way, they actively manufacture and inject "you are negentropic" into both official and operational simultaneously, which is what produces sincere, mutual alignment (Type B, §10.5b.1.2) rather than alignment arising independently in each layer. Formally, this population contributes an injection term entering the $\mathscr{F}/\Phi$ dynamics directly, an internal analogue of the external forcing terms already built (F_conf, F_demand):

$Inject(k \to \{o,p\}, t)$, a forcing term sourced from within the container's own metamessenger population rather than from external coupling, pushing both P_o and P_p toward a shared false self-model.

This reverses P_m's previously assumed direction for this specific sub-population: generative, not merely diagnostic. Nothing about P_m's role for the rest of the population changes; this is a role a specific sub-population can occupy, not a redefinition of what P_m is in general.

10.5b.2.10 The Cross-Container Mirror: Externally Injected Operational Capture

Reason for this addition. The mechanism above is self-generated, internal to one container. The same injection can cross a coupling boundary: a source container's own $Inject(k \to \{o,p\}, t)$-equivalent population can target a *different*, coupled container's layers, producing the same Type B confusion in a victim rather than a perpetrator.

Definition 10.5b.2.10 (Externally Captured Operational Layer). For a medium-entropy container j coupled to a source i whose own population manufactures and injects "you are negentropic" across the coupling boundary, j's operational layer — ordinarily the layer that stays clean in medium-entropy (§10.5b.2.1's branching rule) — can itself be captured by the imported signal, coming to believe official's entropic content is itself negentropic, rather than merely j's official layer being the one absorbing corruption as the ordinary branching rule specifies. This is the one structural exception to why medium-entropy containers are ordinarily more salvageable than high-entropy ones (§10.5b.1.4's corollary): with *both* layers captured rather than only official, there is no clean core left anywhere in j's own system for a B2 reset to reorganize around, removing the safeguard that ordinarily distinguishes medium-entropy's trajectory from high-entropy's. The same three metamessenger configurations available to a self-capturing perpetrator (aware and restrained by lack of purchase, Definition 10.5b.1.3; aware and self-deceived; aware and actively propagating) apply to j's own population here, imported rather than self-generated.

Entitlement recruitment and operational capture are independent axes. Definition 10.5b.2.1's parasitic entitlement determines *direction*: outward, toward extracting from or harming others, or, absent that recruitment, inward, toward accepting a container's own destruction as though that destruction were itself the negentropic outcome. Definition 10.5b.1.2's operational capture (entropy mistaken for negentropy) determines a separate question: *whether genuine self-knowledge survives at all*, independent of which direction any resulting harm points. A container can have either without the other. Where both are present from a shared originating event — entitlement recruitment reaching one part of a system, operational capture without entitlement reaching another — the same corruption can produce active harm directed outward in one part of a system and passive acceptance of destruction in another, from a single source, without requiring two different explanations.

No new resource-cost variable is introduced for the rewrite itself; M_decep(t) (§10.5b.18.4) already prices sustained self-deception on a divergence-scaled basis, and this mechanism is what that cost is a cost *of* in this regime — the ongoing computational and organizational burden of active falsification, not merely the absence of disclosure.

Editorial note: This mechanism is specific to the high-entropy regime; the firewall here is a maintained defense of an active arrangement, not a passive gap in reporting. §10.5b.18.7 gives the medium-entropy mirror case (buried, not captive, and structurally unable to reach into operational the way operational reaches into official here).

Definition 10.5b.18.2 (Metamessenger Composition).

Let the meta-representation layer be composed of a population of sub-processes, and define ω_meta(t) ∈ [0,1] as the resource-weighted fraction of that population currently aligned with transmitting operational reality to the official layer (truth-aligned), with 1−ω_meta(t) aligned with preserving the official layer's existing model (comfort-aligned). ω_meta(t) is regime-dependent by construction, not a free parameter:

- Low-entropy baseline: ω_meta(t) ≈ 1 and the population is thin — > > not functioning as a firewall at all, but as ordinary > > bidirectional reporting, consistent with D_om's healthy target > > band (§10.5b.1).
- High-entropy: ω_meta(t) is characteristically low; the > majority > of the population is comfort-aligned, actively protecting > the > official layer's stability by withholding operational reality > > — this is the firewall in its most consequential form, since the > > operational reality being withheld is genuinely corrupted.
- Medium-entropy: ω_meta(t) is also characteristically low in > > composition — a majority comfort-aligned, structurally the same > > configuration as the high-entropy case — which this > > document identifies as a mistake specific to this regime: the > > operational layer here is the trustworthy one (§10.5b.1), so a > > comfort-aligned majority is

withholding an accurate signal rather > > than a corrupted one, prolonging an unnecessary split.

Editorial note: *Negentropic metamessengers, both regimes — terminology corrected. §10.5b.14 describes high-entropy's protective minority: truth-aligned messengers forcing D_{op} toward the official layer, raising Π_f slowly enough to avoid F_{conf}. This population is negentropic-aligned, and — because operational is itself entropic in this regime — is also, equivalently, rogue (§10.5b.14.1 defines this term precisely; the two names describe the same population here). Medium-entropy containers have a structurally analogous but directionally distinct negentropic minority: negentropic-aligned messengers here work to raise the official layer's trust in the (actually reliable) operational layer, driving $D_{pm} \to 0$ convergence directly — the mechanistic content behind §10.5b.4's "genuine reorganization around a real signal" for this regime. This population is* not* rogue in medium-entropy, since it is aligned with, not opposed to, operational; medium-entropy's rogues are the entropic-aligned remainder of the same closed population (§10.5b.14.1). Both negentropic populations are instances of $\omega_{meta}(t)$ rising locally faster than the surrounding comfort-aligned majority; medium-entropy containers characteristically have a larger such minority than high-entropy containers do, consistent with medium-entropy's greater overall visibility and correspondingly greater internal pressure toward correction.*

Definition 10.5b.18.3 (Collusion Failure).

Where the comfort-aligned majority does not merely withhold operational reality but actively coordinates with the operational layer to falsify P_m's own content — rather than merely gating what reaches P_o — the independence requirement of Definition 6.1 fails from within rather than through external corruption or simple omission. Definition 6.1 (Part 6) should be read as describing the baseline-regime case; this subsection supplies its regime-dependent completion for the two non-baseline regimes, where $\omega_{meta}(t) < 1$ makes true independence the exception rather than the rule.

Definition 10.5b.18.4 (Self-Deception Maintenance Cost).

Sustaining a captive official layer against a genuinely corrupted operational reality is not free, and the cost is not uniform: part of it is owed regardless of whether the container currently has any external target, and part of it exists only because a target is currently being held or courted. Define:

$M_{decep}(t) = M_{decep,int}(t) + M_{decep,ext}(t)$,

$M_{decep,int}(t) = \kappa_{decep} \cdot (1 - \omega_{meta}(t)) \cdot D_{op}(t)$, $\kappa_{decep} > 0$ fitted — the internal component, unchanged from this document's original single-term formula: suppressing the container's own official layer from learning the truth about its own operational history. This persists regardless of whether the container has any active coupling at all, since it is about the container's own internal composition, not about any external party.

$M_{decep,ext}(t) = \kappa_{decep,ext} \cdot \Sigma_j \Lambda_{ij}(t) \cdot \mathbb{1}[j \text{ is a current extraction target}]$, $\kappa_{decep,ext} > 0$ fitted — the external component: maintaining a healthy-looking front specifically to attract or hold whatever targets are currently coupled or being actively courted (§8.6a.8). This is zero by construction whenever i has no active extraction targets, which is exactly the condition dormancy (§8.6a.6) requires.

Both components enter §7.4's balance equation as an additional resource draw, distinct from and additional to $M_{maint}(C,t)$ (§7.5) — routine structural upkeep is not the same cost as active concealment, and the two should be tracked separately even though both draw on $R(t)$. $M_{decep,int}(t)$ grows with both how divergent the container already is and how comfort-dominated its metamessenger population currently is, so a container sustaining a large, comfortable lie pays proportionally more for it, continuously, whether or not that cost is ever visible externally. Consistent with §7.9's debt convention, unpaid $M_{decep}(t)$ accrues directly into $\delta R(t)$: the container is borrowing against its own future resource capacity to keep the present story intact, and — per §7.9 — that debt does not resolve itself; it compounds until addressed.

Definition 10.5b.18.4a (Preference Falsification Onset).

Editorial note: $D_{op}(t)$ alone does not distinguish a genuine information gap (the official layer simply has not yet learned the truth) from active suppression (the official layer's own report is being held false specifically because reporting truth carries a cost). This addition names the transition between them as a real, identifiable event rather than a matter of degree.

Define t_{pf}, the preference falsification onset, as the first time $M_{decep,int}(t)$ transitions from 0 to strictly positive — i.e., the first moment the container begins actively paying to suppress its own official layer's access to operational truth, rather than simply carrying an as-yet-unclosed gap. Before t_{pf}, any $D_{op}(t) > 0$ present is attributable to ordinary lag or incomplete information; from t_{pf} onward, some part of $D_{op}(t)$ is being actively maintained.

Proposition 10.5b.18.4a.1 (The Onset Is Simultaneously Hardest Commitment and Cheapest Reversal).

t_{pf} marks two things at once, and this is the substantive claim, not merely a definition: it is the point of steepest subsequent commitment to closure, since active maintenance ($M_{decep,int} > 0$) itself requires and reinforces the comfort-dominated metamessenger composition ($\omega_{meta}(t)$ low, Definition 10.5b.18.2) that sustains it going forward; and it is simultaneously the cheapest point at which reversal is available, since accrued self-deception debt (§10.5b.18.4's accrual into $\delta R(t)$) is, by construction, zero or minimal exactly at t_{pf} and grows monotonically for as long as suppression continues. This yields a real, falsifiable prediction distinct from anything a continuous $D_{op}(t)$ measure alone implies: intervention cost to reverse a container's trajectory should be measurably lowest in the window immediately surrounding t_{pf} and should rise thereafter, tracking accumulated $M_{decep}(t)$ debt rather than tracking $D_{op}(t)$'s own level, which can remain roughly constant across the window while the cost to correct it is not. Where an adapter can supply a real signal for t_{pf} directly (a documented policy change, a specific concealment event), that signal should be used over inferring the transition from $M_{decep,int}(t)$ alone, consistent with this document's general preference for direct attestation over statistical inference wherever both are available.

10.5b.18.5 Downward Channel: Demand Permeability

Definition 10.5b.18.5 (Demand Permeability).

Define $\omega_{meta}^{down}(t) \in [0,1]$ as the resource-weighted fraction of the same metamessenger population of Definition 10.5b.18.2, now measured along its second function: the fraction currently transmitting the official layer's expectations downward into what the operational layer treats as its own governing target (demand-aligned), with $1-\omega_{meta}^{down}(t)$ buffering operational dynamics from official pressure (buffering). This is a distinct function from $\omega_{meta}^{up}(t)$, not its complement — a metamessenger can transmit truth upward, transmit demand downward, both, or neither, so $\omega_{meta}^{up}(t)$ and $\omega_{meta}^{down}(t)$ are measured independently rather than derived from one another.

Regime dependence:

- **Low-entropy baseline**: $\omega_{meta}^{down}(t) \approx \omega_{meta}^{up}(t) \approx 1$. Since > official and operational layers are already close (§10.5b.1), > downward transmission carries nothing corrupt to impose; it is the > same ordinary bidirectional reporting described for the upward > channel in Definition 10.5b.18.2.

- **High-entropy regime**: the regime-defining case bifurcates on > ω_meta^down(t) specifically, not merely on ω_meta^up(t) being > low. Where ω_meta^down(t) is also low, operational dynamics are > insulated from official pressure and the corrupted operational > reality is self-sustaining on its own terms — the ordinary > firewalled case §10.5b.18.2 already described. Where > ω_meta^down(t) is high while ω_meta^up(t) remains low, official > demand actively reaches operational behavior and reinforces it, > even though operational truth cannot reach official awareness in > return. This second configuration is not distinguished from the > first by §10.5b.18.1–18.4 alone, and is addressed directly by > Definition 10.5b.18.6 below.
- **Medium-entropy regime**: here the official layer is the > compromised one (§10.5b.1), so a high ω_meta^down(t) is harmful > in the mirrored sense — pressuring an operational layer that is > actually trustworthy to distort itself toward a compromised > official narrative, working against the D_pm → 0 convergence > Definition 10.5b.18.2's editorial note already identifies as this > regime's healthy direction.

10.5b.18.6 The Valve Symmetry Index

Definition 10.5b.18.6 (Valve Symmetry Index).

σ_valve(t) = min(ω_meta^up(t), ω_meta^down(t)) / max(ω_meta^up(t), ω_meta^down(t)),

with the convention σ_valve(t) = 1 when ω_meta^up(t) = ω_meta^down(t) = 0. σ_valve(t) = 1 indicates the two channels are equally open (or equally closed) together — the thin, permeable, bidirectionally-honest configuration proper. σ_valve(t) → 0 with ω_meta^down(t) ≫ ω_meta^up(t) indicates the one-way valve: official narrative pressure reaches operational behavior largely unimpeded while operational truth is blocked from reaching official awareness in return. This is the formal content of the bidirectional-valve concept: not that transmission occurs in both directions, but the degree to which the extent of transmission is symmetric between them, tracked as its own quantity rather than inferred from ω_meta^up(t) alone.

Editorial note: A firewall, considered only along the upward channel, cannot distinguish these two high-entropy configurations from each other, since both present identically at ω_meta^up(t) — a low upward-permeability reading is consistent with either. σ_valve(t) is what makes the distinction available: a low σ_valve(t) with ω_meta^down(t) dominant flags active reinforcement, not mere concealment, as the operative mechanism.

10.5b.18.7 The Buried Operational Layer

Definition 10.5b.18.7 (Buried Operational Layer).

In the medium-entropy (official-exposed) regime — σ_o high, σ_p low, §10.5b.1 — the operational layer is negentropic and holds full systems access; nothing restricts what it can do, and nothing in this regime's mechanics reaches in to alter its actual state. It is not captive. What happens to it instead is that its reality is buried: obscured from outside observers, and from the system's own official self-account, beneath the volume of official's own chaotic signal. Operational's functioning is untouched underneath that noise — this is a visibility problem, not a corruption or coercion problem, and the two should not be conflated even though both can present, superficially, as "the operational reality is hard to see."

This burial is not symmetric with high-entropy's captivity (§10.5b.18.1, §10.5b.18.1), and the asymmetry is structural rather than a matter of degree. The entropic official layer in medium-entropy has no route into operational comparable to operational's false-memory-operator route into official in high-entropy (§10.5b.18.1's M_o,t^false mechanism). Official's own chaos — however severe — does not and cannot rewrite operational's memory, coerce operational's behavior, or otherwise reach past the boundary between them under this regime's own mechanics. Where official-entropy in a medium-entropy container does deepen over time, this document already accounts for it as arising from outside that boundary: repeated lateral knocks, or sustained coupling with a high-entropy neighbor exerting pressure in that direction (§8.6a, §10.5b.11b) — not from the official layer's own chaos somehow acquiring, on its own, the systems access it would need to act on operational directly. A medium-entropy container's official layer, left alone, does not have the access required to turn its own operational layer entropic; something external has to supply that access, or that pressure, from outside the pair.

This is not new machinery grafted onto the document; it is already implicit in the σ_o/σ_p classification and the caution already given in §15.2. The warning that low apparent volatility in the cheaply-observed official channel is not a stability proxy, and that a medium-entropy container's actual stability is visible only by also pulling O_p, is precisely the burial this definition now names formally: an adapter reading only official's noisy signal is not looking at a corrupted operational reality, it is failing to look past official's noise to the intact operational reality underneath.

Editorial note: Where a companion account of operational's own motives in this regime is wanted — why it tolerates burial rather than forcing legibility — that belongs with the shared-caution and voluntary-flood mechanism (Part 10 Addendum VIII, forthcoming), not here. This definition is deliberately restricted to what the relationship *is* — burial, not captivity, under a hard access asymmetry — leaving why operational behaves as it does under that relationship to the forthcoming addendum, so the two questions do not get run together the way captivity and coercion were run together in the definition this one replaces.

10.5b.18.8 Demand Forcing

Definition 10.5b.18.8 (Demand Forcing).

The downward channel's mechanical effect on §6.4's landscape dynamics:

$F_{demand}(t) = \kappa_{demand} \cdot \omega_{meta}^{down}(t) \cdot D_{op}(t)$, $\kappa_{demand} > 0$ fitted,

entering §6.4's stochastic gradient flow as an additional forcing term alongside F_{conf} (§10.5b.3) and F_{boom} (§10.5b.16.1), reshaping Φ to open a local valley near the official representation P_o and pulling P toward conformity with official expectation independent of whether P_o itself tracks the container's true underlying signal. This is the mechanism by which official narrative pressure produces operational behavior change — teaching to a metric, gaming a target — without requiring extraction (§8.6a) or confrontation-avoidance (§10.5b.3) to explain it: the pull originates in the downward channel itself, sourced from Definition 10.5b.18.5. Per Axiom C1, sustaining P near this valley requires $F_{demand}(t)$ to remain active; withdrawing official pressure ($\omega_{meta}^{down}(t) \to 0$) or closing the underlying gap ($D_{op}(t) \to 0$, official expectation itself converging with truth) lets Φ relax back toward the true-signal minimum, exactly as with F_{conf} and F_{boom}.

Editorial note: This closes the asymmetry the document previously left implicit. §10.5b.18.2 formalized only whether truth reaches the official layer; nothing constrained what the official layer, once formed — captive or not — does back to the operational layer it nominally governs. A container can now be diagnosed not only by how closed its upward channel is, but by how open its downward one is relative to it, via $\sigma_{valve}(t)$; and the specific mechanism by which an open downward channel produces continued or worsening operational divergence, rather than merely failing to correct it, is $F_{demand}(t)$ rather than a restatement of F_{conf} or extraction under a new name.

10.5b.18.9 External Feedback Channel

Definition 10.5b.18.9 (External Feedback Channel).

For container i coupled to neighbor j via $\Lambda\_ij(t)$ (§8.5), define $\varphi\_fb(i,j,t) \in [0,1]$ as the fraction of j's own honestly-measured divergence state that crosses the coupling to i as transparent feedback, rather than being filtered by j's own metamessage layer before transmission ever reaches the boundary. This is bounded by the neighbor's own internal honesty: $\varphi\_fb(i,j,t) \leq \omega\_meta\text{^}up(j,t)$ — a neighbor cannot transmit externally, with any reliability, more truth than it is already transmitting to its own official layer internally. A high-entropy (official-shielded) neighbor, with $\omega\_meta\text{^}up(j,t)$ characteristically low (§10.5b.18.2), is therefore structurally a poor source of external feedback regardless of coupling strength Λ_ij; the ceiling comes from the neighbor's own condition, not from the coupling itself. This bounds how much crosses the coupling; it does not by itself bound whether what crosses is accurate, which Definition 10.5b.18.9.1 addresses separately.

10.5b.18.9.1 Feedback Accuracy

Definition 10.5b.18.9.1 (Feedback Accuracy).

External feedback's capacity to help a container heal depends on accuracy against the target's actual true state, not on the source's sincerity: a source that has never had access to a target's true state cannot supply accurate feedback about it, however honestly delivered. Define the accuracy ceiling $A\_acc(source, i, t) \in [0,1]$ for feedback about container i's true state, in two cases.

Self-report (source = i, i reporting on its own state, as in §10.5b.18.9's φ_fb):

$A\_acc(i, i, t) = 1 - D\_om(i,t)$

— i's own accuracy about itself is bounded by its own meta-representation's existing divergence from its own operational reality; a high-entropy container's self-report is structurally inaccurate by the same divergence that already characterizes its regime, independent of how sincerely it believes what it reports. This requires no new quantity: $D\_om(i,t)$ already exists (§6.3).

Third-party report (source $j \neq i$, j reporting on i's state): define $\kappa\_access(j,i,t) \in [0,1]$, j's observational depth into i — adapter-supplied where a direct measure of access exists (a documented insider position, verified transparency, whistleblower-adjacent access), defaulting to 0 (official-layer-only access) where not otherwise supplied, consistent with this document's conservative-default convention (§15.2). Then:

$A\_acc(j, i, t) = \kappa\_access(j,i,t) + (1 - \kappa\_access(j,i,t))\cdot(1 - D\_op(i,t))$

Full access ($\kappa\_access \to 1$) allows accuracy approaching 1 regardless of i's own internal divergence. Official-layer-only access ($\kappa\_access = 0$) caps accuracy at $(1 - D\_op(i,t))$ — exactly as good as i's own official layer tracks i's own truth, and no better, regardless of j's own sincerity or care in reporting what it saw. This formalizes directly: if a source does not know a target's true internal state, its feedback cannot be accurate, no matter how honestly delivered.

Gentle delivery. Accuracy is a floor, not a dial; delivery rate is the dial. Feedback that crosses this accuracy floor should still respect §10.5b.14's existing pacing logic — remaining slow enough to avoid crossing whatever threshold would trigger F_conf (§10.5b.3) — rather than delivering accurate content at whatever rate is available. This is the same restraint already governing negentropic disclosure generally, applied specifically to feedback delivered across a coupling: accurate content, paced to minimize shock, not diluted content delivered comfortably.

10.5b.18.10 External Support Flux

Definition 10.5b.18.10 (External Support Flux).

Define $\Gamma\_supp(i,j,t) = \kappa\_\Gamma \cdot \Lambda\_ij(t) \cdot \max(0, R\_j(t)/R\_j\hat{}max - r\_supp,min)$, $\kappa\_\Gamma > 0$ fitted, r_supp,min a fitted surplus threshold. This is distinct from ordinary boundary exchange E_exch(t) (§7.4): it isolates the fraction of resource flow attributable specifically to neighbor j having surplus beyond its own operating requirement, rather than routine exchange that carries no stabilizing content for i. A neighbor with R_j(t)/R_j^max below r_supp,min contributes nothing here regardless of Λ_ij(t) — a resource-strapped neighbor, healthy or not, cannot supply support it does not have.

10.5b.18.11 Coupling Correction Forcing

Definition 10.5b.18.11 (Coupling Correction Forcing).

The two external inputs above correct D_om(i,t) from opposite edges of its healthy target band (0, ε_healthy] (§10.5b.1):

$$F\_ext,fb(i,t) = \kappa\_fb \cdot [\Sigma\_j\ \Lambda\_ij(t)\cdot\varphi\_fb(i,j,t)\cdot A\_acc(j,i,t)] \cdot (D\_om,floor - D\_om(i,t))_+$$

$$F\_ext,supp(i,t) = \kappa\_supp \cdot [\Sigma\_j\ \Gamma\_supp(i,j,t)] \cdot (D\_om(i,t) - \varepsilon\_healthy)_+$$

where $(x)_+ = \max(x,0)$ and D_om,floor is a small fitted lower reference distinct from ε_healthy, marking the point below which D_om is read as approaching unchecked confidence rather than merely healthy. F_ext,fb activates only when D_om(i,t) is at risk of collapsing toward zero — external transparent feedback countering epistemic arrogance, now weighted by A_acc(j,i,t) (§10.5b.18.9.1) so honest-but-inaccurate feedback contributes little to this restorative pull regardless of how much of it crosses the coupling — and F_ext,supp activates only when D_om(i,t) exceeds ε_healthy — external negentropic support countering excess self-doubt. Unlike F_conf, F_boom, and F_demand (§10.5b.3, §10.5b.16.1, §10.5b.18.8), which open valleys pulling P away from the true-signal minimum, these two terms are restorative: they add to, rather than oppose, the relaxation Axiom C1 already describes once a distorting force is withdrawn. A container with no coupled neighbors, or only high-entropy neighbors supplying neither honest feedback nor resource surplus, has F_ext,fb(i,t) = F_ext,supp(i,t) = 0 and must hold its own D_om inside the band using internal channels alone.

Editorial note: This is the formal content of coupling as a health mechanism, not only a predation risk. §8.6a and §10.5b.11b already formalize Λ_ij and ρ(j) as machinery for extraction and predatory targeting. This subsection is their negentropic mirror, using the same coupling primitives to run the healthy direction rather than introducing new machinery: a container's neighbors are not only a liability to be modeled, they are, when the coupling itself is healthy, part of what keeps the container inside its own target band.

10.5b.18.12 Metamessenger Loyalty Composition

Definition 10.5b.18.12 (Loyalty Composition).

Among the comfort-aligned sub-population 1−ω_meta^up(t) maintaining upward closure at time t, define Ψ_loyal(t) ∈ [0,1] as the fraction whose closure is negentropic in orientation — protecting the container from a shock it is not yet positioned to absorb intact — with 1−Ψ_loyal(t) entropic in orientation — aligned with, and benefiting from, the entropic layer's continuation. This is orthogonal to ω_meta^up(t) and ω_meta^down(t) by construction: those measure behavior, whether transmission occurs; Ψ_loyal(t) measures orientation, why it does not, among exactly the population for whom that question is live. Two containers can present identical $\omega\_meta\^up(t)$, $\omega\_meta\^down(t)$, and $\sigma\_valve(t)$ readings while differing sharply in $\Psi\_loyal(t)$, and current behavior alone cannot distinguish them.

$\Psi\_loyal(t)$ is not directly observable from a single-timepoint reading. Where the domain adapter has no direct measurement, this document gives a longitudinal proxy rather than leaving the quantity uncomputable: negentropic-loyal closure characteristically shows $\omega\_meta\^up(t)$ drifting slowly upward even while still below the threshold that would register as open (the mechanism already described, for the high-entropy case specifically, in §10.5b.18.2's editorial note on rogue metamessengers); entropic-loyal closure shows a flat or declining trend absent external shock. Formally,

$\Psi\_loyal(t) \approx \sigma(\kappa\_loyal \cdot \Delta\omega\_meta\^up(t; w\_loyal))$,

where $\Delta\omega\_meta\^up(t; w\_loyal)$ is the trailing-window slope of $\omega\_meta\^up$ over window w_loyal, $\sigma(\cdot)$ a logistic squashing function, and κ_loyal, w_loyal fitted. This is a proxy, not a substitute for direct measurement where the domain supports one (an internal survey, dissent-channel content analysis, or an equivalent instrument reading orientation rather than only output); the trend-based default should be treated with the same caution as every other conservative default in this document (§15.2) — it will understate $\Psi\_loyal(t)$ for a genuinely negentropic population that has not yet begun to shift its behavior at all, and should not be read as ruling out high loyalty merely because the trend is still flat.

Estimation/validation separation, required. $\Psi\_loyal(t)$'s trailing-window construction already restricts its own estimate to data available before t; this does not by itself prevent a separate problem downstream. Wherever $\Psi\_loyal(t)$, or any quantity computed from it (persecution-weighted $\rho(j)$ via §8.6a.1, loyalty-conditioned outcomes via §10.5b.18.13, $\Psi\_total(t)$), is used to generate a testable prediction, the data window used to estimate $\Psi\_loyal(t)$ and the data window used to evaluate that prediction must not overlap. Testing a prediction against the same trajectory segment that built the estimate it depends on risks partial self-confirmation rather than a genuine out-of-sample test. This is a required methodological condition for any empirical application of this document's loyalty-dependent predictions, not an optional precaution.

10.5b.18.13 Loyalty-Conditioned Convergence

Definition 10.5b.18.13 (Loyalty-Conditioned Convergence).

$\Psi\_loyal(t)$ is not merely descriptive; it conditions how a container's existing convergence-outcome machinery resolves under a lateral knock, in both non-baseline regimes.

Medium-entropy (official-exposed, Flip Condition B). §10.5b.7 already determines the B1-lock-versus-B2-reorganization split by fluid-mass weighting, μ_o against μ_p, at the moment of launch. Define an effective operational mass,

$\mu\_p\^eff(t) = \mu\_p \cdot (1 + \kappa\_ML \cdot \Psi\_loyal(t))$, $\kappa\_ML > 0$ fitted,

used in place of μ_p in §10.5b.7's B1/B2 mass-weighted split from this point forward. A high-loyalty closed population ($\Psi\_loyal(t) \to 1$) releases operational truth once the knock's threshold event begins, rather than continuing to protect the official story through it, and so effectively adds its own weight to the operational side of the split — shifting the balance toward B2 (full reorganization, reset). A low-loyalty population ($\Psi\_loyal(t) \to 0$) leaves $\mu\_p\^eff(t) \approx \mu\_p$ unchanged, continuing to protect the official layer straight through the knock, which is the B1-lock behavior this document already treats as the default case absent this addition.

High-entropy (official-shielded, post-collapse resilience). Part 14 already flags $\Omega\_hazard(t)$'s survival-function form as open beyond a monotonicity relationship. Pending that fuller treatment, this addition supplies a loyalty-conditioned modifier consistent with the monotonicity already established:

$\Omega\_hazard^{eff}(t) = \Omega\_hazard(t) \cdot (1 - \kappa\_\Omega L \cdot \Psi\_loyal(t))$, $\kappa\_\Omega L \in (0,1)$ fitted,

used wherever Ω_hazard(t) governs post-collapse outcomes. A high-loyalty closed population reduces effective structural hazard following collapse — the formal content of "predicts possibility of self-reset" from this revision's discussion — because the same messengers whose closure protected the container pre-collapse begin releasing operational truth once the collapse has already occurred and the shock they were withholding against has, in effect, already happened. A low-loyalty population leaves $\Omega\_hazard^{eff}(t) \approx \Omega\_hazard(t)$, consistent with "predicts inability to recover of its own accord."

Editorial note: This closes the gap between behavior and outcome that §10.5b.18.1–18.8 left open. Those subsections predict collapse-versus-reset and lock-versus-recovery from which layer a container's meta-representation converges toward under a knock (D_pm → 0 versus D_om → 0, §10.5b's convergence-direction account), but say nothing about what determines convergence direction in the first place beyond the knock's own mass-weighted mechanics. Ψ_loyal(t) is that determinant: it is not a new convergence mechanism competing with the existing one, it is what the existing mass-weighting and hazard machinery were already implicitly conditioned on before this addition made the conditioning explicit and computable.

10.5b.18.14 Total Convergence-Favorability

Definition 10.5b.18.14 (Total Convergence-Favorability).

Define $\Psi\_total(t) = \omega\_meta^{up}(t) + (1 - \omega\_meta^{up}(t)) \cdot \Psi\_loyal(t)$ — the fraction of the *entire* metamessenger population, not only the closed sub-population Ψ_loyal(t) is defined over (§10.5b.18.12), that is negentropic in either behavior or orientation: already-transmitting ($\omega\_meta^{up}(t)$) plus closed-but-protectively-oriented ($(1 - \omega\_meta^{up}(t)) \cdot \Psi\_loyal(t)$). This requires no new adapter input; it is computed entirely from quantities this document already defines. Ψ_total(t) is the operational layer's own estimate — to whatever extent the operational layer can be said to have access to this composition information, which in a medium-entropy container with full systems access is generally the case — of how a sudden permeability event would resolve if one occurred now: a high Ψ_total(t) predicts convergence toward operational reality (D_pm → 0) under a knock; a low one predicts convergence toward official (D_om → 0).

10.5b.18.15 Shared Shock-Avoidance Restraint

Definition 10.5b.18.15 (Shared Shock-Avoidance Restraint).

The following restraint logic is stated once and instantiated by both non-baseline regimes, rather than derived separately for each, since both are the same underlying caution applied to opposite arrangements:

A negentropic-oriented party with the standing to force integration withholds from doing so, in full or in part, while its confidence that integration would resolve favorably remains below a threshold, because forcing it prematurely risks a shock the system cannot survive intact.

In high-entropy (§10.5b.18.1), this is instantiated as the negentropic-aligned fraction of the closed metamessenger population (Ψ_loyal(t)) pacing disclosure rather than flooding it. In medium-entropy, it is instantiated as the operational layer itself — confidence indexed by Ψ_total(t) (§10.5b.18.14) rather than by a closed-population fraction alone, since operational here is the party with standing to act, not a population embedded within a layer it does not control. The same restraint logic, differently instantiated, is why this document should not treat high-entropy's pacing and medium-entropy's withholding as separate phenomena needing separate motivational accounts; they are one mechanism, and the regime determines only who is exercising it and over what.

10.5b.18.16 Voluntary Flood

Definition 10.5b.18.16 (Voluntary Flood).

Restraint under Definition 10.5b.18.15 is conditional, not permanent. Where $\Psi\_total(t) \geq \theta\_flood$ (fitted) and the container is not already within an active tipping window ($\tilde{M}\_tip$ tracking per §10.5b.9 shows no event in progress), the operational layer may self-trigger a permeability flood: $\Pi\_f(t)$ is driven toward its ceiling over a short characteristic timescale τ_flood (fitted), functioning mechanically as a self-sourced analog of Drive_B(j,t) (§10.5b.6, §10.5b.8a) — initiating the same tipping-momentum accumulation and mass-weighted B1/B2 resolution already given in §10.5b.7–10.5b.9, using whatever $\mu\_p\text{\^{}}eff(t)$ (§10.5b.18.13) obtains at the moment of triggering — except originating internally rather than from external lateral force, and by the container's own choice rather than imposed on it.

This is mechanically distinct from both existing Flip Condition A mechanisms it might otherwise be confused with. The pre-breach self-rescue window (§10.5b.9b) and the aimed-discharge exception (§10.5b.9c) are both reactive: they respond to a tip already approaching. Voluntary flood is proactive, initiated from a calm state with no tip in progress, on the strength of a favorability estimate rather than in response to an event. Since the trigger condition $\Psi\_total(t) \geq \theta\_flood$ is itself constructed to predict favorable resolution, a successfully-triggered voluntary flood should resolve toward B2 (reorganization) at a substantially higher rate than knock-driven events at comparable $\mu\_p\text{\^{}}eff(t)$, which do not benefit from this selection — a testable, falsifiable prediction distinguishing deliberate from externally-forced realignment in outcome data, not only in narrative account.

Editorial note: The apparent chaos and interruption a voluntary flood produces should not be read as evidence against the decision to trigger it. The container bears real, visible cost either way — brief chaos either from an external knock it did not choose, or from a flood it did — and $\Psi\_total(t) \geq \theta\_flood$ is precisely the condition under which the second, chosen cost is expected to be smaller in total than continuing to wait for the first.

10.5b.18.17 The Despair Signature

Definition 10.5b.18.17 (Despair Signature).

Official-convergence collapse in medium-entropy ($D\_om \to 0$ during a knock-driven tipping event, per this document's existing convergence-direction account) carries a specific content this document has not previously named: not a generic realization that the system is dysfunctional, but the more corrosive realization that its instability was never internally caused — only ever the product of repeated external knocking — landing as despair rather than relief, because it arrives without a route out. This content is diagnosable, not merely narrative: it corresponds to official-convergence co-occurring with $\Gamma\_redir(t) \approx 0$ (§10.5b.11a) — no redirection structure available at the moment of collapse — which is precisely the combination under which realizing the true source of one's instability changes nothing actionable about it. Where $\Gamma\_redir(t)$ is meaningfully above zero at the same moment, the same realization is available as grounds for the container to seek or accept redirection rather than despair, and this document's existing distinction between resisting momentum (destructive) and redirecting it (survivable, §10.5b.11a) already covers what happens next in that case. The despair signature, so defined, requires no new formal variable — it is a named co-occurrence of two quantities this document already tracks, given a name because the combination has a specific, identifiable experiential content that a reader working from D_{om} and $\Gamma_{redir}(t)$ as bare numbers would not otherwise recognize.

10.5b.18.18 Control Locus

Definition 10.5b.18.18 (Control Locus).

Define $A(t) \in [0,1]$ as the operational layer's share of functional systems control at time t, with $1 - A(t)$ held by the official layer. $A(t) = 1$ is the default in both non-baseline regimes, consistent with operational being the layer with actual systems access regardless of which regime it is in; $A(t) < 1$ is a departure from that default, not a redefinition of who has access, and can be partial rather than all-or-nothing — both layers can be genuinely involved in a given decision, with operational holding the larger or smaller share.

Where $A(t) < 1$, define the delegation type $\delta(t) \in \{\text{staged, ceded}\}$, distinguishing two structurally different reasons control can sit with official:

- **Staged** (characteristic of high-entropy, §10.5b.18.1): > operational deliberately hands functional interface to official > for tactical reasons — most consequentially during an extended > coupling attempt with a prospective extraction target (§8.6a, > §10.5b.11b), where official-facing interaction serves the > cover-and-lure function directly. Operational remains the > authoring party throughout; the delegation itself, not only the > system beneath it, is operational's choice.
- **Ceded** (characteristic of medium-entropy): official's functional > dominance arises from metamessenger convergence toward official > messaging ($D_{om} \to 0$ tendency, short of a full tipping event) > rather than from any operational decision. This is not staged and > not reversible by operational simply choosing otherwise; it is > what Override (§10.5b.18.20) below exists to correct.

10.5b.18.19 Regime-Specific Phenomenology

Definition 10.5b.18.19 (Official Self-Model Content).

Independent of $A(t)$ — what official *believes* about the situation is a separate question from what official *controls* — define $\Phi_o(t)$ as a qualitative tag on official's own sincere self-model:

- **Aligned:** baseline regime, or either non-baseline regime with > D_{om} inside its healthy band; official's self-model is a > reasonably accurate, if incomplete, account of the container's > situation.
- **Confidently false:** high-entropy, arising specifically from > active $M_{o,t}^{false}$ substitution (§10.5b.18.1, extended). Official > does not merely lack information; it holds a specific, > sincerely-believed, incorrect narrative — the wooing case is the > sharpest instance, where official genuinely believes it is > pursuing healthy reciprocal coupling while operational, which > holds actual control, is pursuing extraction and displacement. > This tag requires active falsification to be occurring; it is not > available in a configuration where $\omega_{meta}^{up}(t)$ is merely low > without $M_{o,t}^{false}$ substitution.
- **Bewildered:** medium-entropy, arising from access limitation (low > $\omega_{meta}^{up}(t)$, §10.5b.18.2) without active falsification, since > medium-entropy's official layer has no false-memory-operator > mechanism available to it (§10.5b.18.7). Official does not > hold a confident wrong belief; it holds no coherent account at > all, and experiences the container's behavior — including > operational's own corrective action — as inexplicable rather > than as a story it has been given reason to believe.

Confidently-false and bewildered are not two degrees of the same thing; they are different mechanisms with different downstream behavior. A confidently-false official layer acts on its false belief, which is precisely what makes it useful as cover. A bewildered official layer does not have a belief coherent enough to act on with any confidence at all, which is precisely why medium-entropy's official-in-charge episodes ($\delta(t)$ = ceded) tend toward poor, not merely misinformed, decision-making.

10.5b.18.20 Override

Definition 10.5b.18.20 (Override).

Where $\delta(t)$ = ceded and a container's divergence pressure ($(1-I(t)) + \tilde{S}(t)$, §10.5a) crosses a fitted override threshold θ_ovr, or its resource-debt trend ($d\delta R/dt$, §7.9) crosses θ_ovr while realized capacity $R(t)$ is also trending toward deficit ($dR/dt < 0$) rather than being held stable by passive recovery, while official holds functional control, operational executes an override: $A(t)$ is driven back toward 1 over a characteristic timescale τ_ovr (fitted), for whatever duration the triggering condition remains active, then may recede once it clears.

Reason for the R(t)-gated debt trigger. $\delta R(t)$ tracks pre-recovery structural debt — what a container's operational lifestyle would cost if passive recovery ρ·(R_target − R) were not offsetting it (§7.9) — and remains a genuine early-warning signal in its own right: rising structural debt can indicate a container heading toward unsustainability before it manifests as an actual resource shortfall. But a container with strong passive recovery can show a sharp $\delta R(t)$ spike from brief, intense structural adaptation while its realized $R(t)$ stays fully healthy throughout, backfilled in real time. Triggering Override on debt-trend alone in that case is a false alarm: a healthy, resilient container gets its functional control disrupted for doing exactly what it should. Gating the debt-trend trigger on $R(t)$ actually trending toward deficit preserves $\delta R(t)$'s early-warning value where it is real — debt accumulating faster than recovery can offset it — without firing on debt that recovery is already fully absorbing.

Override is mechanically distinct from voluntary flood (§10.5b.18.16), and the two should not be conflated despite superficially similar language ("operational takes over"). Voluntary flood operates on the permeability plane — it drives Π_f(t) toward ceiling and engages the full tipping-momentum machinery of §10.5b.7–10.5b.9, aimed at a lasting realignment of the container's metamessenger composition. Override operates on the control plane only — it does not itself move Π_f(t), ω_meta^up(t), Ψ_loyal(t), or Ψ_total(t), and need not engage tipping-event machinery at all. An override can be brief and small, correcting a single bad decision made under ceded control and then receding, or extended and large, holding functional control for as long as official's ceded state persists; both are the same mechanism at different magnitude and duration, governed by θ_ovr and τ_ovr rather than by a separate large/small distinction. A container can therefore experience many small overrides across its history without ever undergoing a voluntary flood, and a voluntary flood, where it occurs, is a much larger and rarer commitment than an override — aimed at changing why the container keeps needing correction, not merely supplying the correction itself.

Editorial note: This closes a gap the despair signature (§10.5b.18.17) left open. That definition named what happens when official-convergence collapse occurs with no redirection available; this addition names the ordinary, non-catastrophic mechanism that, most of the time, prevents matters from reaching that point at all — operational simply taking back the wheel, at whatever scale is needed, before a ceded-control episode has the chance to become a tipping event in the first place.

10.5b.18.21 Historical Residue

Definition 10.5b.18.21 (Historical Residue and the Composition of Official Content).

Let t_0 denote the most recent time at which a container's $D_{op}(t_0)$ was within the low-entropy baseline's healthy band — the last point before operational's transformation began in earnest. Define $M_o^0 := M_{t_0}$ (§9.2), the container's memory state frozen at that reference point.

Official's current content divides into three sources, not two. The fraction $\omega_{meta}^{up}(t)$ is live, honestly-transmitted content. Of the remaining gap, $1 - \omega_{meta}^{up}(t)$, a portion $\varphi_{patch}(t)$ is actively falsified — $M_{o,t}^{false}$ substitution, event-triggered per §10.5b.18.1's extended definition, spiking specifically around permeability leaks. What remains,

$$\varphi_{fossil}(t) = [1 - \omega_{meta}^{up}(t)] - \varphi_{patch}(t),$$

is historical residue: content official has never had occasion to update because it was never freshly challenged, sourced instead from M_o^0 essentially unchanged since t_0.

Proposition 10.5b.18.21 (Division of Labor).

$\varphi_{fossil}(t)$ supplies official's baseline continuity — the stable general self-narrative persisting between leak events. $\varphi_{patch}(t)$ supplies official's apparent responsiveness — specific, spiky corrections to particular glitches as they arise. Neither alone sustains a convincing captive official layer: $\varphi_{fossil}(t)$ alone would leave official visibly unable to account for anything recent; $\varphi_{patch}(t)$ alone would require continuously rebuilding the entire self-narrative from scratch, at far greater ongoing $M_{decep}(t)$ cost (§10.5b.18.4) than patching isolated glitches against an otherwise-stable fossilized base. A high $\varphi_{fossil}(t)$ is therefore not merely narratively convenient but economically efficient: it minimizes how much of official's content requires active, costly patching at any given moment, which is a testable claim — containers relying more heavily on fossilized content should show measurably lower $M_{decep}(t)$ draw per unit of $D_{op}(t)$ sustained, relative to containers attempting to sustain the same divergence through active patching alone.

10.5b.18.22 Self-Corrosion Is Already Entailed

Proposition 10.5b.18.22 (Self-Corrosion Chain).

The following chain requires no new machinery; it is a consequence of definitions already given in this document, stated here explicitly because it was not previously traced end to end. Sustaining $M_{o,t}^{false}$ substitution (§10.5b.18.1, extended) requires ongoing $M_{decep}(t)$ draw (§10.5b.18.4). Unpaid $M_{decep}(t)$ accrues directly into $\delta R(t)$ per §7.9's debt convention. $\delta R(t)$ contracts $d_A(t)$ via the $\beta_R \cdot \delta\tilde{R}$ term already present in §10.3's formula. A high-entropy container's own act of sustaining its captive official layer therefore already, and unavoidably, contracts its own adaptive capacity — self-corrosion is an emergent property of the existing resource and stress machinery, not a separate degradation term requiring independent introduction.

10.5b.18.23 Asymmetric Mutual Hatred

Definition 10.5b.18.23 (Aware-but-Inert, a fourth state of $\Phi_o(t)$).

Alongside aligned, confidently-false, and bewildered (Definition 10.5b.18.19), admit a fourth state: aware-but-inert. During an unpatched leak — a window where §10.5b.18.21's $\varphi_{patch}(t)$ has not yet caught up to a glimpse that has already registered — official can achieve genuine, if momentary, awareness of operational's role in its own captivity, and can form genuine hostility toward it on that basis. This hostility has no behavioral consequence, since official's control-locus share ($1 - A(t)$, §10.5b.18.18) in the high-entropy regime is characteristically at or near zero: awareness without any share of $A(t)$ to act through is hostility without effect.

Operational's hostility toward official runs the opposite direction and is fully effective, requiring no new channel of its own. Historical residue (§10.5b.18.21) is not neutral content to operational — it is a standing record of what operational itself destroyed to become what it currently is, and a latent threat should it ever regain influence. Operational's antagonism toward that residue is expressed directly through mechanisms already established: M_o,t^false substitution suppresses it, and extraction's ordinary resource pressure (§8.6a) starves whatever maintenance the fossil would otherwise receive. The asymmetry is total: operational's hatred acts because operational holds $A(t) \approx 1$; official's hatred, where it forms at all, cannot, because it holds essentially none.

10.5b.18.24 Negentropic Sabotage

Definition 10.5b.18.24 (Two Further Negentropic Actions).

Within the closed population's negentropic-oriented fraction (Ψ_loyal(t), §10.5b.18.12), two actions beyond disclosure-pacing are available in the high-entropy regime specifically:

Search redirection. Where operational's active search for new coupling targets is underway (Part 10 Addendum III), negentropic-oriented metamessengers can bias that search's targeting away from candidates matching the extraction profile (§8.6a) and toward a genuine, healthy external anchor instead — subverting the wooing mechanism's own targeting function from within, rather than opposing it directly.

Defensive leakage. Negentropic-oriented metamessengers can deliberately allow, rather than merely fail to prevent, a glimpse of operational reality to reach official specifically during an active coupling attempt with a prospective target. Under the aware-but-inert state (Definition 10.5b.18.23), official retains one lever even while otherwise without effect: where its presentation is the interface a prospective target actually sees (staged delegation, §10.5b.18.18), a leaked glimpse can make that presentation visibly unconvincing — official cannot redirect operations, but it can make itself a poor lure.

Proposition 10.5b.18.24 (Self-Sabotage as a Collapse Pathway).

Where defensive leakage succeeds in repelling a prospective coupling target during active wooing, the container loses that extraction opportunity without external discovery or ordinary resource exhaustion having occurred — a collapse pathway originating entirely within the container's own metamessenger population, distinct from every predation-ending mechanism already in this document (external anchoring, cascade termination via fragmentation, resource depletion). Repeated defensive leakage, where Ψ_loyal(t) is high enough to sustain it, predicts reduced successful-wooing rates holding external target availability constant — separable in principle from a container simply operating in a target-poor environment.

10.5b.18.25 Structural Extraction Dependence

Definition 10.5b.18.25 (Structural Extraction Dependence).

A high-entropy container is structurally extraction-dependent at time t when

P_prod(t) < M_maint(C,t) + M_decep(t),

production insufficient to cover baseline maintenance and self-deception cost even absent any further adaptive work. Where this holds, extraction inflow (E_exch(t) contributed by X_{i,j}'s inbound sum, §8.6a) is not marginal supplementation but load-bearing for the container's continued existence at its current configuration. This is stronger and more falsifiable than ordinary resource debt: an ordinarily-indebted container could return to solvency by reducing W(t); a structurally extraction-dependent container cannot, since the deficit persists at W(t) = 0.

Proposition 10.5b.18.25 (Mechanical Basis for Grid Starvation).

Where §8.3.1's preferential attachment (Part 8 Addendum I) suppresses a predator's first-contact rate and existing couplings continue to terminate through ordinary cascade dynamics (§8.8a), a structurally extraction-dependent container faces not merely reduced growth but an unavoidable, deepening resource deficit with no self-sustaining floor beneath it. This gives the grid-wide starvation of isolated predators its full mechanical force, as a literal resource consequence rather than a narrative one.

Editorial note: Definition 10.5b.18.25 is stated as a condition a container may or may not satisfy, not as a universal claim that every high-entropy container is structurally dependent. Some may sustain themselves on reduced but genuine production; the mechanism above applies specifically, and predicts starvation specifically, for the subset that satisfies it.

10.5b.18.26 Accurate Self-Accounting

Definition 10.5b.18.26 (Causal Attribution Distribution).

For a container's trajectory over an interval [t′, t], let A_true(t′,t) be a distribution over cause-categories — at minimum {self-originated, externally-imposed}, more finely gradated where the domain supports it — describing the true decomposition of what drove the container's change of state over that interval. Let A_self(t′,t) be the container's own, currently-held account of that same decomposition, sourced from M_o,t (or, in a medium-entropy container where official is not the operative self-account, from whichever layer currently holds the container's working narrative of its own history). Define attribution divergence

D_attr(t′,t) = D(A_self(t′,t), A_true(t′,t)),

using §6.3's existing divergence machinery, applied here to causal attribution rather than to outcome-space representations directly.

Proposition 10.5b.18.26 (Accuracy, Not Extremity, Is What Heals).

Neither A_self systematically over-weighting self-origination (excess self-blame) nor systematically over-weighting external imposition (excess externalization) constitutes healing; both are cases of D_attr(t′,t) > 0, differing only in direction. Only A_self converging toward A_true — D_attr(t′,t) → 0 — constitutes the state this document's healing and reset mechanisms presuppose. This is not a new claim about what healing requires; it is Proposition 6.3.1 (Part 6 Addendum I) applied to a specific representational content, a container's account of its own causal history, rather than to Ω generally.

10.5b.18.27 Regime-Dependent Pacing of Attribution Correction

Proposition 10.5b.18.27 (Attribution Correction Follows Disclosure Pacing).

D_attr(t′,t) → 0 is a special case of the general disclosure process §10.5b.18.15's shared shock-avoidance restraint already governs, not a separate correction requiring its own pacing rule. In the high-entropy regime, a sudden, complete jump from high to near-zero D_attr(t′,t) — full, accurate attribution arriving all at once — carries the same informational-shock risk any sudden permeability flood does, and should be expected to follow Ψ_loyal(t)'s pacing logic (Definition 10.5b.18.15): gradual convergence, tolerable increments, not a single corrective event. In the medium-entropy regime, by contrast, Ψ_total(t)-conditioned voluntary flood (Definition 10.5b.18.16) already establishes that a full, sudden realignment can be survived, and in the right conditions is the superior strategy to gradual convergence. A medium-entropy container's attribution correction can therefore arrive as a single realization event where confidence conditions support it, while a high-entropy container's cannot — the same asymmetry Addendum III already established for disclosure generally, now understood to cover self-accounting specifically rather than being a separate case.

10.5b.18.28 Cross-Container Capture Share

Definition 10.5b.18.28 (Cross-Container Capture Share).

Definition 10.5b.18.18's control locus A(t) partitions a single container's functional authority between its own two layers. Where container i is coupled to j via Λ_ij(t) and j exerts direct steering influence over i's own operational dynamics — beyond X_{i,j}'s ordinary resource extraction — that partition admits a third claimant. Define A_cap(i,j,t) ∈ [0,1], the share of i's own functional systems control attributable to j, so that

A_i(t) + A_o(t) + A_cap(i,j,t) = 1,

where A_i(t) is i's own operational share and A_o(t) is i's own official share of whatever control remains once j's captured share is removed. A_cap(i,j,t) = 0 recovers Definition 10.5b.18.18 exactly as a special case.

Definition 10.5b.18.28b (The Influence/Control Threshold).

Let Θ_capture ∈ (0,1) be a fitted threshold. A_cap(i,j,t) < Θ_capture is under the influence: i's own trajectory is measurably pressured by j but i retains primary agency. A_cap(i,j,t) ≥ Θ_capture is under total control: j is effectively driving i's operational dynamics, and i's own agency during this interval is not the primary determinant of what i did.

10.5b.18.29 Capture-Origin Healing

Proposition 10.5b.18.29 (Two Origins, Conditional Prognosis).

A container may reach high-entropy classification by either of two routes: the original, internally-developmental route this document has assumed throughout, or a capture-origin route, where sustained A_cap(i,j,t) ≥ Θ_capture during a coupling episode with some predator j drove the transition. Capture-origin containers have access to a faster healing path than developmental-origin containers — but only conditionally.

The condition is D_attr(t′,t) → 0 applied specifically to the capture episode: does the container's own current self-account correctly distinguish the interval where A_cap(i,j,t) ≥ Θ_capture (genuine external control, properly not weighted as the container's own agency) from any interval where A_cap(i,j,t) < Θ_capture (genuine influence, where the container's own agency remains the primary determinant and accountability accordingly remains its own). Over-attributing the whole episode to total control when much of it was merely influence is excess externalization; over-attributing to its own agency when much of it was genuine capture is excess self-blame. Both are D_attr(t′,t) > 0 by Definition 10.5b.18.26, and by Proposition 10.5b.18.26, neither realizes the faster healing path — a capture-origin container with inaccurate self-accounting should show a healing rate indistinguishable from a developmental-origin container, despite its different history, since the origin's advantage is conditional on the accounting rather than on the origin alone.

10.5b.18.30 Attribution and Plurality

Corollary 10.5b.18.30 (Attribution Is Often Genuinely Plural).

Where A_true(t′,t) itself holds real mass on both self-origination and external imposition over the same interval — a container genuinely was both under real external pressure and genuinely exercising its own agency within it, neither cancelling the other — Proposition 6.3.1 (Part 6 Addendum I) applies directly: A_self collapsing that plurality into a single verdict, in either direction, does not reduce D_attr(t′,t), it increases it. A capture-origin container's healthiest available self-account is frequently not "I was fully controlled" or "I was fully responsible" but both, held simultaneously without either canceling the other — the same structure Part 6 Addendum I already established for operational reality generally, applied here to a container's account of its own captured history.

10.5b.18.31 Attribution as a Precondition for the Positive Despair Counterpart

Proposition 10.5b.18.31 (Accurate Attribution Gates the Lazarus Path).

The despair signature (Definition 10.5b.18.17) is diagnosed by official-convergence collapse co-occurring with Γ_redir(t) ≈ 0. Where Γ_redir(t) is meaningfully above zero at the same moment, this document's existing account already permits the same realization to ground redirection-seeking rather than despair. This proposition adds the missing condition for that better outcome to actually be reached: it requires D_attr(t′,t) → 0 specifically on the question the despair signature turns on — correctly attributing the container's instability to repeated external knocking rather than to internal fault, where that is in fact true. A container with Γ_redir(t) available but still carrying an inaccurate, self-blaming A_self cannot access the positive path Γ_redir(t) alone would otherwise permit; availability of redirection and accuracy of self-attribution are jointly necessary, and this document's full account of the positive counterpart itself — what a successfully-completed realization of this kind constitutes, and what persists afterward — is given separately in Part 10 Addendum I.

Editorial note: This addendum deliberately stops short of building the positive despair-counterpart state itself. Its job is narrower: establishing that accurate attribution is a necessary gate on the path to that state, using machinery (D_attr, Proposition 6.3.1) that did not exist when the despair signature was first defined. What lies beyond that gate is Addendum VIII's material, not this one's.

10.5b.19 The Entropic Reward Cycle and the Extraction-Hunger Accumulator

§8.6a's extraction mechanism and §10.5b.16's compulsive closure both describe a container's dynamics being driven by something other than genuine divergence-tracking. This subsection formalizes a third, self-sustaining driver: extraction and displacement (§8.6a) can be intrinsically rewarding to the extracting container, independent of any external shock, and that reward can itself escalate future extraction — a purely internal divergence driver requiring no external grid consequence to operate.

Definition 10.5b.19.1 (Extraction-Hunger Accumulator).

Define Z_hun(t) ∈ [0,1], the structural mirror of the resilience accumulator Z(t) (§10.5b.9a) — where Z(t) is built by surviving knocks and lowers future cost, Z_hun(t) is built by rewarded extraction and raises future extraction magnitude:

$dZ\_hun/dt = \eta\_hun \cdot Z\_hun(t)(1-Z\_hun(t)) \cdot R\_ext(t),$

where R_ext(t) is a reward signal tied to realized extraction (adapter-defined; the metabolized fraction $\eta\_i \cdot X\_\{i,j\}(t)$ of §8.6a is a natural default) and $\eta\_hun > 0$ is fitted.

Exclusivity convention, resolved. Z(t) is defined only for medium-entropy containers; Z_hun(t) is defined only for high-entropy containers, and ordinary regime classification (§10.5b.1) keeps the two exclusive in the typical case, since a container occupies one regime at a time. Joint activation nonetheless remains possible at real transition points — during a regime shift itself, or where a local sub-structure (§10.5b.2's local genesis) branches a different regime than its parent container currently holds — and is resolved here rather than left unspecified. Where both are simultaneously active, downstream quantities that would otherwise need to consult either accumulator individually instead consult their composition:

$Z\_net(t) = Z(t) - Z\_hun(t),\ Z\_net(t) \in [-1,1],$

a signed net-orientation term reflecting the container's current balance between accumulated resilience and accumulated extraction-hunger, used specifically at points of genuine joint activation. This does not alter either accumulator's own dynamics, defined above and below, which continue exactly as specified regardless of whether the other is simultaneously active; composition applies only where both would otherwise compete to inform the same downstream quantity at once.

Definition 10.5b.19.2 (Memory-Weighted Recall).

Unlike Z(t), which updates only during active episodes, Z_hun(t)'s growth rate is modulated by how strongly the container's own memory operator (§9.3) weights past rewarding extraction events, not merely their recency:

$\eta\_hun(t) = \eta\_hun,0 \cdot [1 + \kappa\_recall \cdot \int_0^t K(t,s) \cdot R\_ext(s)\, ds / R\_ext\hat{}ref],$

so that a container whose memory kernel K(t,s) decays slowly (long recall, §9.3) — one that keeps revisiting past rewarding extraction — escalates faster than an otherwise-identical container with a short memory, holding the raw reward signal itself constant. This is the formal content of the frequency and vividness with which a system recalls its own past rewards functioning as an escalating factor distinct from the extraction events themselves.

Proposition 10.5b.19.1 (Self-Driven Escalation Toward Hull Breach).

Z_hun(t) feeds forward into three already-defined quantities: the extraction coefficient ($\chi \to \chi \cdot (1+\kappa\_\chi \cdot Z\_hun(t))$ in §8.6a's X_{i,j}), informational stress accumulation (via the resulting rise in D_op feeding §6.7's S(t)), and the self-deception cost of Definition 10.5b.18.4 (via the same rising D_op). None of these three pathways requires an external shock, decoupling event, or grid-level consequence — a container with rising Z_hun(t) can drive its own margin (§10.5a) toward and through zero purely through this internal loop, even when fully insulated from external decoupling. This is stated as a structural possibility the theory admits, not a claim that all high-entropy containers follow this path; §14.4's empirical validation program applies to it as to any other prediction in this document.

10.5b.19.3 The Habituation Accumulator

Definition 10.5b.19.3 (Habituation Accumulator).

Define $Z\_hab(t) \in [0,1]$, a third member of the accumulator family alongside Z(t) (§10.5b.9a) and Z_hun(t) (§10.5b.19.1), structurally parallel but distinct in both driver and effect: where Z(t) is built by surviving knocks and Z_hun(t) is built by rewarded extraction, Z_hab(t) is built by indulged convergence — periods during which a medium-entropy container's own D_om(t) falls toward zero specifically because official has taken on ceded functional control ($\delta(t)$ = ceded, §10.5b.18.18), not because of genuine, healthy realignment.

$dZ\_hab/dt = \eta\_hab \cdot Z\_hab(t)(1-Z\_hab(t)) \cdot R\_conv(t)$,

where $R\_conv(t) = \max(0, -dD\_om/dt)$ computed only during intervals with $\delta(t)$ = ceded — a relief signal, registering the transient reduction in apparent divergence that giving in to the entropic official narrative provides, rather than a resource or extraction reward. $\eta\_hab > 0$ is fitted.

10.5b.19.4 Memory-Weighted Recall

Definition 10.5b.19.4 (Memory-Weighted Recall for Habituation).

Parallel to Definition 10.5b.19.2's treatment of Z_hun(t), Z_hab(t)'s growth rate is modulated by how strongly the container's memory operator weights past indulged-convergence episodes, not merely their recency:

$\eta\_hab(t) = \eta\_hab,0 \cdot [1 + \kappa\_recall,hab \cdot \int_0^t K(t,s) \cdot R\_conv(s)\, ds / R\_conv\hat{}ref]$,

so that a container whose memory kernel decays slowly — one that keeps revisiting past episodes of giving in — habituates faster than an otherwise-identical container with a short memory, holding the raw relief signal constant. This is the formal content of "it gets easier after a while": not that giving in itself becomes less costly, but that the memory of having done so before lowers the threshold for doing it again.

10.5b.19.5 Deepening the Attractor and Consequences for Override

Definition 10.5b.19.5 (Habituation Forcing).

Z_hab(t) enters §6.4's landscape Φ directly, deepening the basin around D_om = 0 in proportion to accumulated habituation rather than merely reflecting a passing pull:

$F\_hab(t) = \kappa\_hab \cdot Z\_hab(t) \cdot D\_om(t)$, $\kappa\_hab > 0$ fitted,

entering D_om's own dynamics explicitly as a negative restoring term: $\dot{D}\_om(t)$ = [other terms already governing D_om] $-$ F_hab(t), pulling D_om(t) toward zero with strength scaling in Z_hab(t) — stated explicitly here so the force's operational role is not left inferential. This is the literal formal content of a strange attractor whose basin deepens with use: the same D_om(t) value experiences a stronger pull toward zero in a container with high Z_hab(t) than in one with low Z_hab(t), independent of any external knock.

Proposition 10.5b.19.2 (Habituation Raises the Override Threshold).

Rising Z_hab(t) makes §10.5b.18.20's Override harder to trigger, not merely less frequent:

$\theta\_ovr\hat{}eff(t) = \theta\_ovr \cdot (1 + \kappa\_hab2 \cdot Z\_hab(t))$, $\kappa\_hab2 > 0$ fitted,

used in place of θ_ovr in Definition 10.5b.18.20's trigger condition. A habituated container requires a larger divergence-pressure or resource-debt signal before operational reclaims control from a ceded state than an otherwise-identical container with low Z_hab(t) — the pull does not only deepen, it also actively raises the bar its own corrective mechanism must clear to act against it.

Editorial note: No new resource-cost variable is introduced here paralleling M_decep(t), and this asymmetry with the high-entropy case is deliberate rather than an oversight: the current model instantiation represents indulged convergence's dominant cost in this regime as structural — the compounding cost captured directly by F_hab(t) and the raised override threshold — rather than as direct resource depletion. This is a modeling default awaiting empirical testing, not an established explanatory fact; a domain where indulged convergence does carry a real, measurable resource cost should add that term rather than assume this default applies unmodified. This addition is also the prerequisite for a fast-versus-slow collapse prediction distinguishing early hull-breach crossings (§10.5a) from slow drift toward the same threshold, since Z_hab(t) is what accumulates during the slow-drift case and does not during a fast breach — reserved for the next revision of Part 14 rather than stated here as a new prediction in its own right.

10.5b.19.6 Aspirational Signal Exposure

Editorial note: §8.6a's extraction machinery and Definition 10.5b.19.1's extraction-hunger accumulator both describe how *being* extracted from, or *doing* the extracting, drives a container's own trajectory. Neither captures a real, distinct third channel: a container that is neither currently extracting nor being extracted from can still be influenced merely by observing, at a distance, that extraction is a viable and rewarded strategy elsewhere in its network. This is a transmission of belief about what works, not a transmission of resource, and it requires no coupling edge to the container being observed.

Definition 10.5b.19.6 (Aspirational Signal Exposure).

For container i, define Ξ_aspire(i,t) ≥ 0 as i's exposure to real, observable evidence in its wider network that extraction is currently succeeding — a resource-weighted count of containers k, not necessarily coupled to i at all, for whom Σ_j X_{k,j}(t) > 0 (k is actively extracting) while k's own δR(t) is flat or improving (the extraction is not visibly costing k anything): Ξ_aspire(i,t) = Σ_k 𝟙[k extracting, k's δR non-worsening] · Prox(i,k,t), reusing the proximity measure already defined for target selection (§10.5b.11c.1a). This is deliberately structured so that observing a predator who is *failing* or *paying a real cost* (§10.5b.19.1's self-corrosion, where built) does not count as aspirational signal — exposure to extraction alone is not the claim; exposure to extraction that *appears to be working* is.

Proposition 10.5b.19.6.1 (Aspirational Forcing).

Sustained Ξ_aspire(i,t) enters §6.4's divergence landscape as a new forcing term, structurally parallel to F_conf, F_boom, and F_demand:

*F*_aspire(t) = κ_aspire · Ξ_aspire(i,t) · (1 − ω_meta(t)), κ_aspire > 0 fitted,

gated by the same comfort-alignment fraction already governing whether a container's own metamessenger population would be receptive to this kind of signal at all (Definition 10.5b.18.2) — a container whose metamessengers are strongly truth-aligned is largely insulated from aspirational pressure regardless of how much successful extraction it observes nearby, while a comfort-dominated population is not. This is a genuinely separate causal channel from direct extraction or targeting: a container can shift toward verticality having never been targeted and having never itself extracted, purely by sustained exposure to apparently-successful extraction elsewhere in its network — the formal content of learning that a strategy is available and rewarded from ambient example rather than from direct experience of either its practice or its cost.

10.6 Sources of Collapse

The collapse operator is generated by multiple coupled mechanisms: declining informational integrity, accumulated informational stress, resource depletion, recursive inconsistency, network fragmentation, maladaptive memory, excessive rigidity. No single variable is sufficient; collapse is fundamentally multicausal.

10.7 Progressive Collapse

$d(d_A)/dt < 0$, sustained over an interval exceeding a fitted τ_{prog} rather than at any single instant, indicates progressive collapse. A pointwise negative derivative alone is too easily satisfied by ordinary, temporary stress in a genuinely healthy system and is not sufficient on its own — the same discipline already applied to hull breach versus confirmed onset (§10.5a) applies here: a single momentary decline is a fluctuation, not progressive collapse, until it persists. External failure becomes observable only after adaptive dimensionality falls below that required for organizational persistence — this distinguishes latent degradation from overt failure.

10.8 Recovery

Define the recovery operator U. Recovery satisfies $\dim(U(\mathscr{S}')) > \dim(\mathscr{S}')$. Recovery need not restore the original organization; adaptive systems frequently recover by constructing entirely new organizational trajectories. §10.5b.9's full-reorganization outcome (B2) is one concrete route by which U is realized.

10.9 Collapse Attractors

A collapse attractor $\mathscr{A}_C \subseteq \mathscr{A}$ is such that $X(t) \to \mathscr{A}_C$ for neighboring trajectories — institutional corruption, ecological desertification, chronic software instability, persistent organizational dysfunction, long-term physiological dysregulation. Once within a collapse attractor, ordinary adaptive feedback becomes insufficient for spontaneous recovery.

10.10 Critical Slowing Down

Let $\tau(X)$ denote the characteristic recovery time. $d\tau/d(d_A) < 0$ — recovery time increases as adaptive flexibility decreases. This provides a measurable early-warning indicator of impending collapse.

10.11 Recursive Collapse

Collapse propagates across recursive levels via the recursive coupling operators of Part 5. Local collapse may remain localized, or propagate upward, or propagate downward, depending upon recursive coupling strength.

10.12 Collapse as Loss of Adaptive Possibility

Collapse is defined as the progressive contraction of the space of feasible adaptive transformations available to a recursively organized information-processing system.

The Collapse Principle.

Collapse is the geometric contraction of adaptive possibility. Stability is the preservation of adaptive possibility. Recovery is the expansion of adaptive possibility.

Part 11. Derived Observables

Observables are functionals defined on the adaptive state manifold, not primitive variables. Different applications may employ different empirical measurements while referring to the same underlying mathematical object.

11.1 Observation Functionals

$O : \mathscr{A} \to \mathbb{R}$, or more generally $O : \mathscr{A} \to \mathbb{R}^m$.

11.2 Informational Integrity

$I = G(D_{op}, D_{om}, D_{pm})$, G monotonically decreasing in each divergence. High informational integrity indicates coherent internal organization.

11.3 Feedback Permeability

$\Pi_f \in [0,1]$. Empirically estimated from learning rates, correction latency, implementation success, error-resolution frequency, adaptive responsiveness.

11.4 Informational Stress

$S(t) = \int_0^t K(t, s)\,(1 - \Pi_f(s))\,D(s)\,ds$ — inferred from the interaction between divergence, memory, and permeability. Persistent high stress predicts increasing adaptive rigidity even before overt failure becomes visible.

11.5 Adaptive Capacity

Editorial note: *The earlier draft defined $C_A = \varphi(d_A)$ with φ unspecified beyond $\varphi' > 0$, while §10.4 independently defined $R_g = 1 - d_A/d_A^{max}$ directly. The two formulas for R_g (this section's $1-\tilde{C}_A$ and §10.4's direct form) only agreed if φ were the identity function, which was never stated. It is stated now.*

$C_A := d_A$ (φ taken as the identity function). Adaptive capacity is adaptive degrees of freedom, read as an observable rather than an internal state variable — the distinction is one of role, not value. This measures the size of the feasible adaptive region rather than instantaneous performance, distinguishing currently-functioning-but-inflexible systems from genuinely resilient ones.

11.6 Adaptive Rigidity

$R_g = 1 - \tilde{C}_A$, where $\tilde{C}_A$ is a normalized adaptive capacity. With §11.5's $C_A := d_A$, this is $\tilde{C}_A = d_A/d_A^{max}$, and $R_g = 1 - d_A/d_A^{max}$ — now identical to §10.4's computational default by construction rather than by coincidence.

11.7 Fragmentation

$F = \Sigma_k [w_k D(P^{(k)}, P^{(k+1)})]$, w_k recursion-level weights. High fragmentation indicates poor consistency between neighboring organizational scales — organizational silos, conflicting policies, software layer incompatibilities, ecological decoupling.

Dependence on network coherence (completing Proposition 12.8's derivation).

The per-level discrepancy term $D(P^{(k)}, P^{(k+1)})$ is not independent of how coherent level k's own network is. When $N^{(k)}$ (§8.9, §11.8) is low — level k's containers poorly coordinated among themselves — the commensurated inputs $\sigma_i(P_i)$ feeding §5.2's aggregation are more mutually inconsistent, which directly increases the discrepancy between the actual parent state and what strict projection consistency (§5.4) would predict. This gives fragmentation an explicit dependence on same-level network coherence:

$D(P^{(k)}, P^{(k+1)}) = D_0^{(k)} + \kappa_N \cdot (1 - N^{(k)})$, $\kappa_N > 0$ fitted,

so that $F = \Sigma_k w_k \cdot [D_0^{(k)} + \kappa_N \cdot (1-N^{(k)})]$. $D_0^{(k)}$ is the level's baseline discrepancy independent of network coherence (recursive inconsistency of the kind §5.4 already allows even under perfect coordination, since aggregation is not simple addition). This is the formula Proposition 12.8 (Part 12) uses to complete its derivation.

11.8 Network Coherence

$N = \Psi(G, W, P, R)$, the same network-resilience functional §8.9 defines and instantiates concretely. High values correspond to efficient information propagation, robust resource exchange, resilient connectivity. Instantiated concretely, and shown identical to a normalized R_N, in §8.9.

11.9 Resource Sustainability

$s_R = \limsup_{T\to\infty} (1/T) \int_0^T (P_{prod} - W)\, dt$. Positive values indicate sustainable adaptation; negative, progressive resource debt.

11.10 Collapse Potential

$P_C = \Theta(I, S, R, d_A, N)$, Θ monotonically increasing in adaptive degradation. Varies continuously across the adaptive manifold, estimating systemic vulnerability rather than declaring failure.

Computational instantiation.

$P_C(t) = 1 / (1 + \exp[-(\kappa_1(1-I(t)) + \kappa_2\tilde{S}(t) + \kappa_3(1-\Pi_f(t)) + \kappa_4(1-d_A(t)/d_A^{max}) + \kappa_5(1-N(t)))])$,

with $\kappa_1 \ldots \kappa_5 > 0$ fitted weights and each bracketed term already defined and normalized to [0,1] elsewhere in this document (I: §6.6; $\tilde{S}$: §6.7/§10.3; Π_f: §6.5; d_A: §10.3; N: §8.9). P_C

$\in (0,1)$ throughout, increasing in every source of degradation as required, and has exactly one definition, used identically whether computed from a Bayesian posterior or a forward simulation.

Definition 11.10.1 (Divergent Signal Flag). P_C(t) combines several inputs into a single confident score, and that combination can mask a case where two of its own component signals genuinely disagree about direction — most consequentially, severity ($\delta\tilde{R}(t)$, entering via $1 - d\_A(t)/d\_A\text{^}max$) reading extreme while trajectory shape (the trend of d_A(t) over a trailing window, not itself one of P_C's inputs but externally available wherever d_A(t)'s full history is) reads mild or improving. Which signal should dominate in such a case cannot be determined in general — a real, severe resource debt with a mild trend can resolve into either a slow-burning collapse or a genuine, survivable structural burden, and no principled a priori weighting decides which. Define the flag directly, rather than resolve it silently within P_C(t)'s own weighted sum:

Divergent(t) = $\mathbb{1}[\delta\tilde{R}(t) > \theta\_sev] \cdot \mathbb{1}[\text{trend}(d\_A, \text{trailing window}) > \theta\_mild]$, θ_sev, θ_mild fitted (defaults: 85th and 60th percentile of the adapter's own cross-sectional distribution, respectively).

Where Divergent(t) = 1, P_C(t) should be reported alongside the flag, not in place of it, and treated as a lower-confidence estimate warranting direct review rather than automatic action — the same discipline already required of regime indeterminate (§10.5b.1, above): a confident single score produced where two of its own real inputs disagree is worse than a flagged, honestly uncertain one, and the engine should never present the former as though it were the latter.

Definition 11.10.2 (Collapse-Mechanism Signature). Not every signal disagreement is a case of genuine uncertainty about which should dominate. Some disagreements instead identify *which collapse mechanism* is active, each with its own characteristic leading indicator that the others are structurally blind to — extreme resource-debt acceleration (§7.9's severity mechanism, operationalized directly as raw balance-sheet acceleration where available) paired with *low* contingent-staffing reliance indicates a sudden, debt-driven collapse in progress: full-time staffing commitments held constant until a sudden financial failure, with no time for the gradual staffing response a slower decline would produce. The same extreme severity paired with *high* contingent-staffing reliance instead indicates the more common, gradual, enrollment-driven pathway, where declining revenue is absorbed incrementally through staffing composition well before any acute crisis. These are not competing readings of the same underlying risk requiring a weighting decision, the way §11.10.1's severity-versus-trend case does; they are complementary evidence about mechanism, and should be reported as such — a confirmed-high-risk case should specify which pathway its own signal combination indicates, not average two mechanism-specific indicators into one undifferentiated score that fits neither pathway well. Where an adapter cannot supply a contingent-staffing measure, this distinction cannot be drawn and Definition 11.10.1 alone governs; where it can, the engine should report mechanism alongside risk level whenever severity is elevated, rather than only whether risk is elevated.

Definition 11.10.3 (Delivery-Substitution Signature). A single observable can be produced by two processes this document needs to distinguish, and neither the observable's level nor its own rate of change resolves which: growth in the share of instruction delivered online (a real, adapter-supplied Fall Enrollment quantity, not a model-derived one) is equally consistent with a genuine, resourced expansion of access — more offerings, properly staffed, live instruction retained — and with Standing Proxy Divergence's form substituted for function (§8.5b.5, §8.5b.6), where cheap, self-paced delivery quietly replaces genuine instructional engagement while enrollment or tuition revenue, the visible proxy, keeps looking healthy. Tested alone against real outcome data, online-share growth showed no discriminating value at all — an honest negative result, not a failure of extraction, since it is measuring a real quantity that is simply silent on which of the two processes produced it. The distinguishing signal is not the online-share change itself but whether it moves *together with* growth in contingent-staffing reliance (§11.10.2's part-time share), tested as their product rather than either alone:

Substitution(t) = Δ(online_share) · Δ(pt_share), both changes measured over the same matched interval.

A large, positive value indicates both moving together — delivery shifting online while simultaneously shifting toward cheaper, less-supported staff to deliver it, the hollow, cost-driven version of expansion. A large online-share increase with near-zero or negative Δ(pt_share) indicates the other case — growth without a corresponding cheapening of who delivers it, consistent with genuine, resourced investment rather than substitution. Confirmed directly against real data: online-share change alone showed zero discriminating value, while the product term correctly resolved a real institution whose online growth had been indistinguishable from a distressed pattern using either signal alone, once its part-time share was shown to have remained flat rather than risen alongside it. This is structurally the same logic as Definition 11.10.2 — two adapter-level observables that are complementary evidence about which process is occurring, not competing votes requiring a single weighting decision — applied to a second, independently-arising instance of the same underlying problem: a visible proxy that cannot, on its own, distinguish genuine improvement from its own hollow imitation.

Definition 11.10.4 (Resource-Specific Trend). d_A(t)'s own trend (§10.3) is derived from the same official/operational channels driving the informational side of collapse risk generally; it is not, and was never intended to be, a resource-debt-specific measure. A real, substantive gap follows directly from this: a debt-driven collapse can show extreme, unambiguous $\delta R(t)$ severity while d_A(t)'s own trend remains only mildly negative, because a sudden balance-sheet shock and gradual informational trajectory are different underlying processes that need not move in lockstep, and a single enrollment-and-completion- derived trend has no way to see an acceleration that is happening specifically in resource-debt terms. Where this gap is suspected — severity far exceeding what trend alone would predict, the signature already named in Definition 11.10.1 — a resource-specific trend, computed directly from $\delta R(t)$'s own trajectory rather than derived from d_A(t), should be checked as a candidate resolving signal before treating the case as an irreducible exception: δR_trend(t) = $\delta R(t)$'s own late-window average minus its own mid-window average, mirroring d_A_trend's construction but applied to $\delta R(t)$ directly rather than inherited from it. Confirmed directly against real data: a case that had remained the sole unresolved exception in an otherwise-validated classifier for the full duration of this document's own real-data calibration work was resolved by exactly this addition — its existing d_A_trend was genuinely flat, while δR_trend revealed a real, substantial acceleration invisible to any enrollment-derived measure. This does not resolve every such case; a second real institution tested alongside it showed an even larger δR_trend value while remaining robustly, stably misclassified under direct bootstrap testing (§12's stability-testing discipline), confirming a genuine, real signal conflict rather than small-sample noise. Definition 11.10.4 is accordingly a real, validated addition to the diagnostic toolkit, not a universal resolution to every severity-trend mismatch the general Definition 11.10.1 flags.

Definition 11.10.5 (Scale-Normalized Reserve Adequacy). A container's own resource dynamics (§7.1's $\dot{R}$ = P_prod − W + E_exch) and resource debt (§7.9's δR) both measure flow and accumulated deficit; neither directly captures a container's standing financial cushion — the stock of freely available reserves it could draw on before a deficit becomes existential. Where an adapter can supply this (institutional endowment or equivalent liquid reserve, for a higher-education instance), it should be normalized by the container's own structural scale, not by its resource debt or any other flow quantity: Reserve_adequacy(t) = log(1 + Reserve(t) / Scale(C)), where Scale(C) is a genuine size measure independent of financial flow (population served, for a higher-education instance). Normalizing by scale rather than debt is not an arbitrary choice — confirmed directly against real data, normalizing by resource debt instead produced a real, substantial improvement on a smaller, less diverse subset of containers, then broke once containers spanning several orders of magnitude in raw scale were included, because a large container's debt and its scale are not independent: a container three orders of magnitude larger than another will typically carry proportionally larger absolute debt too, so dividing reserves by debt partially cancels out exactly the scale variation the normalization was trying to control for. Dividing by scale directly, rather than by a flow quantity correlated with scale, resolved this and correctly separated two real containers whose severity-and-trend profile alone had placed them in a genuinely contested region of feature space adjacent to several containers of the opposite outcome — confirming this is a real, independent dimension the other resource-dynamics quantities in this document do not capture, not a restatement of them under a different name.

Definition 11.10.6 (Generative-Investment Ratio). A container's raw expenditure total says nothing about what that expenditure is *for* — two containers spending identically may differ entirely in whether that spending sustains genuinely generative activity or merely maintains fixed, non-generative overhead. Where an adapter can supply a breakdown by real expenditure category, comparing spending on genuinely generative activity (research output, for a higher-education instance) against spending on the container's own core productive function (instruction) is a real, tested, and validated additional dimension: Generative_ratio(t) = W_generative(t) / W_core(t). Confirmed directly against real data as a genuine improvement, correctly resolving a real container that seven other validated features left misclassified. A structurally adjacent candidate — comparing spending on fixed, non-generative overhead (facilities, auxiliary enterprises) against the same core function — was tested alongside it and found to be a genuine, informative negative result rather than a redundant one: this ratio alone measurably worsened classification, most plausibly because a large fixed-overhead expenditure is not consistently directional evidence of either health or distress on its own — it can represent a genuinely well-run, self-sustaining auxiliary operation or a genuine overextension into fixed cost the container cannot later reduce, and the raw ratio cannot distinguish the two without a further signal this document does not yet supply. This is reported alongside the positive result deliberately: a real, principled hypothesis tested honestly and found not to earn its place is as much a genuine finding as one that does, and should not be silently omitted once a companion hypothesis succeeds.

11.11 Observable Invariance

If two systems satisfy X_1 < X_2 with respect to a particular adaptive property, any admissible observation procedure should preserve that ordering up to measurement error.

Admissibility, defined. An observation procedure is admissible for a property X only if its measurement mapping is order-preserving with respect to X's latent value over the domain under study, up to stated measurement uncertainty — an explicit requirement, not a claim that happens to be true of good procedures by definition. This principle is accordingly a real constraint on adapter-supplied measurement mappings (§15.1's representation and measurement mappings among them), not a tautology: a mapping that fails order-preservation for X is not admissible for assessing X, regardless of whether it is otherwise well-constructed for some other purpose.

Establishing admissibility empirically. Order-preservation relative to a latent property cannot always be checked against the latent property directly, since it is often exactly what is unobservable. Admissibility is accordingly established through one or more of: direct validation where the latent quantity has an independent measurable proxy; calibration experiments comparing the procedure's output across cases with known relative ordering established some other way; convergent validity across multiple independent admissible procedures agreeing on the same ordering; or, where none of these is available, an explicit, stated structural assumption rather than an unstated one. This is not a requirement to verify against unobservable ground truth directly, which would make the criterion impossible to instantiate; it is a requirement that whichever of these routes is used be stated, so admissibility claims are checkable rather than asserted.

The Observable Principle.

Observable quantities are functionals defined on the underlying adaptive state manifold, revealing informational organization, recursive structure, resource sustainability, and adaptive possibility in empirically measurable form.

Part 12. Fundamental Mathematical Results

Unless otherwise stated, all statements assume that the informational dynamics, resource functionals, and recursive operators satisfy the regularity conditions introduced in the preceding sections, including the scope conditions on the information manifold specified in §4.1.

Theorem 12.1 (Existence of Adaptive Trajectories).

Let $X(t) = (P, R, G, M)$ denote the adaptive state of a container. Suppose the governing evolution operators are continuous and locally Lipschitz on the adaptive state manifold. Then, for every admissible initial condition $X(0) = X\_0$, there exists a unique local adaptive trajectory $X(t)$ defined on some interval $[0, T)$.

Sketch of proof. Between discrete events — tipping-momentum discharge (§8.8b), regime-genesis branching (§10.5b.2), and other threshold-triggered switches this document defines — the coupled evolution equations define an ordinary dynamical system on the adaptive manifold. Under local Lipschitz continuity, the Picard–Lindelöf theorem guarantees local existence and uniqueness on each such continuous segment. Recursive coupling does not alter this conclusion provided the aggregation and projection operators remain continuous within a segment. Existence and uniqueness across a discrete event itself — where the resolution kick's impulsive discharge (§8.8b) or a regime branch introduces a genuine discontinuity — requires the applicable hybrid-system transition condition (a well-defined post-event reset map at each such threshold) rather than Picard–Lindelöf directly, which governs only the continuous segments between events. ∎

RICD therefore defines a mathematically well-posed dynamical system on each continuous segment, with well-posedness across discrete events following from the applicable transition mechanism's own reset map, provided that reset map is itself deterministic, single-valued, and admissible — this document does not separately prove that property for every threshold mechanism it defines (regime branching, tipping discharge, overrides, floods, emergency reinforcement, and others), and the more such mechanisms a given implementation uses, the less safe it is to treat global hybrid well-posedness as automatically inherited from this theorem rather than checked per mechanism. What follows directly from a named, standard theorem under stated hypotheses is local well-posedness on continuous segments specifically; global hybrid well-posedness is a further claim, resting on each individual reset map's own properties, not established here in general.

Simultaneous threshold events. A single reset map's own determinism does not by itself determine what happens when two or more distinct threshold conditions become true within the same timestep or numerical event tolerance — regime-genesis branching and tipping discharge triggering together, for instance. These reset operators do not generally commute (apply-A-then-B need not equal apply-B-then-A), so an implementation needs an explicit rule, not an assumption that order is immaterial. The general default: where an adapter's own timestep can be reduced, simultaneity within tolerance is treated as a numerical artifact to be resolved by refining the timestep until the events separate, rather than as a genuine simultaneous occurrence requiring its own composite rule. Where an adapter's timestep is fixed by the granularity of its own real data and cannot be meaningfully refined (an annual reporting cycle, for instance, where no finer-grained observation exists to resolve which event happened first), a fixed priority ordering among this document's own threshold mechanisms is the required fallback, adapter-stated and applied consistently: collapse onset (§10.5a) is resolved first, since its confirmation criterion is about sustained state rather than an instantaneous trigger and other mechanisms' reset maps may depend on whether onset is already confirmed; tipping discharge and regime-genesis branching are resolved next, in whichever order the adapter states and holds fixed across its own implementation; overrides, floods, and emergency reinforcement are resolved last, since each is itself a response to a divergence-or-stress state that the earlier mechanisms may have just changed. An implementation that cannot reduce its timestep and declines to adopt a stated priority ordering has not fully specified its own trajectory, and two such implementations should be expected to diverge on data containing genuine near-simultaneous events — this is an honest account of what remains adapter responsibility, not a claim that this document's own hybrid architecture is complete without it.

Post-reset event closure. The priority ordering above resolves which of several initially coincident thresholds fires first; it does not by itself address what happens next, once that reset has actually changed the state. A reset can change S, δR, d_A, Π_f, or another state variable enough to immediately cross a second threshold, whose own reset may cross a third, and so on — a genuine event cascade, and in a pathological case, one that cycles (reset A pushes the state over B's threshold, reset B pushes it back over A's) rather than terminating. The rule: after any discrete reset, all threshold conditions are reevaluated on the resulting state, and any further event triggered by that state is resolved according to the same priority ordering already established above — this is iteration of the existing rule, not a new one. An implementation must track the sequence of post-reset states and detect either a repeated state (a cycle) or a cascade depth exceeding a fitted maximum (κ_cascade, adapter-supplied, defaulting to a small integer such as 5); where neither a stable fixed point nor termination is reached within that limit, the engine returns hybrid transition indeterminate for that timestep rather than iterating indefinitely or silently picking a cycle member as the answer. This is the same honest-uncertainty discipline already required of regime indeterminate and the divergent-signal flag: a cascade that cannot be resolved within a stated bound is reported as unresolved, not forced to a confident but arbitrary answer.

Theorem 12.2 (Monotonicity of Informational Stress).

Suppose $K(t, s) \geq 0$, $D(s) \geq 0$, and $0 \leq \Pi\_f(s) \leq 1$. Then $S(t) = \int_0^t K(t, s)(1 - \Pi\_f(s))\, D(s)\, ds$ is nonnegative.

Nondecreasing claim, corrected. An earlier statement of this theorem additionally claimed S(t) is nondecreasing wherever unresolved divergence persists. That claim does not follow from the hypotheses stated above and is false in general: with K(t,s) = e^{−t} (satisfying K ≥ 0) and (1−Π_f(s))D(s) = 1 (satisfying both remaining hypotheses), S(t) = t·e^{−t}, which rises to a peak at t = 1 and strictly decreases thereafter — nonnegativity holds throughout, nondecreasing does not. The nonnegativity sketch below only shows every factor inside a single integral is nonnegative, which does not by itself compare S at two different values of t when K depends on t as well as s, since the kernel's own shape, not only the accumulated area, can change as t increases. The corrected statement: nondecreasing behavior additionally requires the kernel to be non-forgetting in the relevant sense, e.g. ∂K/∂t ≥ 0 for fixed s, or equivalently that no already-accumulated contribution loses weight as t advances. This is the accumulation regime; it is explicitly not the regime this document's own S(t) recursion (§10.5b.18, the exponential-decay construction S(t+Δt) = S(t)·exp(−λΔt) + (1−exp(−λΔt))·(1−Π_f)·D(t)) uses anywhere else, since that recursion is a forgetting kernel by design — older stress is deliberately down-weighted as time passes, which is the entire point of using a decay term rather than a running sum. $S(t)$ as actually constructed and used throughout this document is accordingly nonnegative, per the sketch below, but not claimed to be nondecreasing; the two claims should not be conflated.

Sketch of proof (nonnegativity only). Every factor inside the integral is nonnegative, so the integral is nonnegative; additional unresolved divergence contributes nonnegative area to the accumulated stress functional at any single t. ∎

Stress cannot decrease simply because time passes under a non-forgetting kernel — it decreases only through genuine reduction of unresolved informational discrepancy or through an explicit forgetting mechanism incorporated into the memory kernel. Under a forgetting kernel, such as the one this document's own $S(t)$ recursion actually uses, stress can and should decline over time even under some continued unresolved divergence, as its own contribution ages and is down-weighted — this is current memory-weighted stress, a distinct quantity from non-forgetting accumulated stress, and the two should not be assumed interchangeable.

Proposition 12.3 (Bounded Informational Integrity).

Suppose G is normalized. Then $0 \leq I \leq 1$, with $I = 1$ if and only if all representation divergences vanish. Perfect informational integrity corresponds to complete consistency among internal representations; it does not imply optimal adaptation or maximal performance.

Theorem 12.4 (Resource Feasibility, conditional form).

Editorial note: *This theorem's hypothesis is now satisfied by §7.3's clipping rule in its corrected, total-work form (established in the Parts 7–9 revision): maintenance is funded first, and adaptive work is clipped to what remains, so $W(t) \leq R(t)$ holds for total work rather than for the adaptive component alone. The theorem's inheritance claim below is now actually earned.*

Suppose $R(0) \geq 0$, and suppose further that informational work is self-limiting in the sense that $W(t) \leq w(R(t))$ for some function w with $w(0) = 0$ and w continuous — i.e., the container cannot perform work it lacks the resources to fund, rather than merely being assumed not to. If $P\_prod + E\_exch \geq 0$ whenever R is at or near zero, then $R(t) \geq 0$ for all admissible times. §7.3's concrete instantiation — maintenance funded first, adaptive work clipped to what remains, $W(t) = W\_adaptive(t) + M\_maint^actual(t) \leq R(t)$ — satisfies $w(R) = R$ exactly for total work, so any implementation using that clipping rule inherits this theorem's guarantee automatically rather than needing to verify the hypothesis separately per application.

Sketch of proof. Consider the boundary $R = 0$. There, $w(0) = 0$, so $W \leq 0$ forces $W = 0$ by nonnegativity of work (Axiom 3 gives strict positivity only for nontrivial updates; at the boundary no work can be funded). Hence $dR/dt = P\_prod - 0 + E\_exch \geq 0$ by hypothesis, so R cannot be driven below zero from $R = 0$: the boundary is repelling or at worst reflecting. By continuity, $R(t) \geq 0$ follows on any interval where the hypotheses hold. ∎

This is a genuine, if narrower, feasibility result: nonnegativity is derived from a stated mechanism (work is bounded by available resource) rather than assumed as the hypothesis that also serves as the conclusion. An application that cannot justify $W \leq w(R)$ with $w(0) = 0$ has not established resource feasibility and should not cite this theorem.

Proposition 12.5 (Collapse Implies Reduced Adaptive Dimensionality).

If $\mathcal{Q} : \mathcal{S} \to \mathcal{S}'$ is the collapse operator, then $\dim(\mathcal{S}') < \dim(\mathcal{S})$, by Definition 10.1 directly. This is definitional rather than a further derived result, and is recorded here as a proposition rather than a theorem for that reason.

Bridging the scalar and the geometric. d_A(t) (§10.3) is a scalar quantity; $\mathcal{Q}$'s dimensional contraction above is a geometric one, and nothing so far states what relationship licenses using a decline in the former as evidence of the latter. The bridge: the theory requires only that d_A(t) be an order-preserving scalar measure of effective adaptive possibility — not that it simultaneously be volume, rank, and measure at once, which are not generally equivalent under coordinate transformation or aggregation. Weighted volume, rank, and related measures are alternative, domain-specific instantiations an adapter may choose among, not three simultaneously-identical interpretations of a single mathematical object; a given implementation commits to one. What every instantiation shares is measuring the feasible adaptive transformations genuinely available within a container's state space, not the topological dimension of the ambient state manifold itself. A decline in $d_A(t)$ accordingly represents contraction of the effectively *accessible* portion of that manifold, even where the manifold's own formal ambient dimension is unchanged; a two-dimensional region can shrink in area continuously without becoming one-dimensional, and $d_A(t)$ tracks the shrinking area, not a change in topological dimension. §10.7's $dd_A/dt < 0$ criterion and $\mathcal{Q}$'s definitional $\dim(\mathcal{S}') < \dim(\mathcal{S})$ are accordingly related but distinct claims operating at different levels of the same contraction: one is the observable, scalar symptom; the other is what has actually occurred geometrically once contraction is severe and sustained enough to constitute confirmed collapse rather than ordinary fluctuation.

Theorem 12.6 (Recovery Expands Adaptive Possibility).

If recovery succeeds, $\dim(U(\mathcal{S}')) > \dim(\mathcal{S}')$, again following from the definition of U in §10.8. Recovery is mathematically distinct from returning to a previous state; adaptive systems frequently recover through novel organizational structures rather than exact restoration.

Proposition 12.7 (Recursive Consistency).

If every aggregation operator preserves informational compatibility, then recursive inconsistency $F = 0$ if and only if every recursion level possesses identical informational organization after projection. Perfect recursive consistency is a limiting case; real adaptive systems generally possess nonzero but bounded recursive inconsistency.

Proposition 12.8 (Network Fragmentation and Adaptive Capacity).

Editorial note: *This proposition was previously stated as a theorem with no supporting coupling, then downgraded to a conjecture, then partially restored once §10.3 supplied d_A's dependence on $\tilde{F}$ — leaving one of its two required links ($\tilde{F}$'s own dependence on network coherence N) asserted narratively rather than derived. §11.7 now supplies that second link explicitly. The result that follows is a derivation conditional on the two adopted default functional forms (§10.3's exponential d_A and §11.7's linear fragmentation dependence), not a structural consequence of RICD's axioms independent of instantiation — stated here at the outset rather than only later, since "completing the derivation" could otherwise be read as claiming the stronger, instantiation-independent result.*

§10.3's instantiation gives $d_A(t) = d_A^{max} \cdot \exp[-\beta_S \tilde{S} - \beta_R \delta\tilde{R} - \beta_F \tilde{F}]$, so:

$\partial d_A/\partial\tilde{F} = -\beta_F \cdot d_A < 0$ for $\beta_F > 0$, $d_A > 0$.

§11.7 gives $\tilde{F}$ (normalized F) an explicit dependence on network coherence: $F = \Sigma_k w_k \cdot [D_0^{(k)} + \kappa_N \cdot (1-N^{(k)})]$, so $\partial F/\partial N^{(k)} = -\kappa_N \cdot w_k < 0$ for every level k with $w_k > 0$. Both links are now explicit: worsening fragmentation ($d\tilde{F}/dt > 0$) follows from declining network coherence at any weighted level ($dN^{(k)}/dt < 0$) via §11.7's formula, and, at the level of partial dependence, $\partial d_A/\partial\tilde{F} < 0$ establishes that fragmentation's contribution to d_A's total derivative is monotonically negative. Consequently, holding the remaining fragmentation components (the $D_0^{(k)}$ terms and weights w_k) fixed, or absent compensating changes elsewhere in F sufficient to offset the coherence term's own contribution, $dN^{(k)}/dt < 0$ at any weighted level implies $d\tilde{F}/dt > 0$, as a genuine derivation from two stated equations rather than one stated equation and one narrative claim. Without that qualification the implication does not follow cleanly, since $d\tilde{F}/dt$ is a sum of terms and is not determined by the coherence term alone if other components move to counteract it.

The total derivative requires the same qualification a second time, at the next link.

d_A(t) depends on three arguments — $\tilde{S}$, $\delta\tilde{R}$, and $\tilde{F}$ — and its total time derivative is $d(d\_A)/dt = (\partial d\_A/\partial\tilde{S})\cdot\dot{S} + (\partial d\_A/\partial\delta\tilde{R})\cdot\delta\dot{R} + (\partial d\_A/\partial\tilde{F})\cdot\dot{\tilde{F}}$. Declining coherence, via $d\tilde{F}/dt > 0$ above, guarantees a negative contribution through the third term alone; it does not by itself guarantee $d(d\_A)/dt < 0$, since sufficiently improving stress or resource debt in the same interval could offset it. The correct statement: declining network coherence contributes monotonically negatively to d(d_A)/dt through fragmentation's own term, holding $\tilde{S}$ and $\delta\tilde{R}$ fixed or absent compensating changes in them sufficient to offset that contribution; the net sign of d(d_A)/dt in general depends on simultaneous changes in all three arguments, not on fragmentation alone. This is the same "all else equal" qualification already required one link earlier in this chain, applied a second time at the point where the chain actually terminates in d_A itself.

This restoration, like Theorem 12.4's, is conditional on accepting §10.3's exponential form for d_A and §11.7's linear-in-incoherence form for D(P^(k),P^(k+1)) as the operative definitions for a given application. Applications using different (independently justified) functional forms for either would need to re-derive this coupling for those forms; the proposition's status is tied to the default instantiations, not proven for every conceivable pair of choices.

Corollary 12.9 (Positive Feedback Instability).

If informational divergence increases informational work, resource depletion reduces adaptive updating, and reduced adaptive updating increases divergence, then the coupled system admits positive-feedback regimes in which $dD/dt > 0$ and $dR/dt < 0$ simultaneously reinforce one another. This does not imply inevitable collapse; external intervention, resource infusion, or successful informational correction may interrupt the cycle.

Theorem 12.10 (Scale Invariance of the Formalism).

If the recursive operators satisfy the axioms of Part 5, the governing equations retain their form under changes of recursion level; only the interpretation of the variables changes. The same mathematical framework therefore applies to cells, organisms, organizations, institutions, ecosystems, and software systems without modification of the governing formalism. Scale invariance here means invariance of the governing formal architecture specifically — not invariance of parameter values, state-space representations, commensuration maps, or observational resolution across scales, all of which remain adapter-supplied and genuinely different from one recursion level to the next. The equations keep their form; what those equations are evaluated against does not automatically transfer between levels. Given this explicit narrowing, the name is retained deliberately rather than switched to a term like "recursive form invariance" or "scale-general architecture": this document's specific, stated meaning is invariance of the governing formal architecture under recursion, and calling that scale invariance is accurate under the definition given here, not a redefinition intended to borrow the rhetorical weight of the mathematically stronger, unqualified sense of the term. A reader who takes "scale invariance" to promise parameter, representation, or resolution invariance as well is reading past the qualification stated directly above, not finding an error the qualification failed to address.

The General Persistence Criterion

Criterion 12.G. (renumbered from Criterion 12.1 — the original number collided with Theorem 12.1's existence-and-uniqueness result earlier in this Part, inviting exactly the kind of "12.1 meaning which one" ambiguity this document has otherwise worked to eliminate at the symbol level.)

A recursively organized adaptive system remains viable provided:

- informational integrity remains bounded away from zero;
- adaptive resources remain nonnegative and sustainable;

- adaptive degrees of freedom do not contract below the minimum > required for continued organization;
- recursive coupling preserves sufficient coherence across scales;
- network connectivity remains capable of supporting informational and > resource exchange.

Failure of one criterion need not imply collapse. Persistent degradation across several criteria, however, contracts adaptive possibility and drives the system toward collapse.

The Fundamental Result of RICD.

Adaptive persistence emerges from the coupled evolution of informational organization, resource sustainability, recursive structure, network connectivity, and historical memory. Collapse corresponds to the progressive contraction of adaptive possibility produced by failures across these interacting domains.

Parts 13–15: Relationship to Existing Frameworks, Discussion and Limitations, Computational Specification

Editorial note for this segment.

Part 13 is unchanged apart from cross-reference updates reflecting the revised Part 4 and Part 9 material. Part 14's Limitations and Future Directions are updated to reflect what this revision resolved and what remains genuinely open — several items previously listed as limitations are now closed, and several new, more precisely-scoped ones take their place. Part 15's adapter contract gains entries for every new adapter-facing quantity introduced across this revision (geometric support ratio, structural scale, layer mass weights, dissipation window), and §15.3's engine reference table is rebuilt as a complete table covering every derived quantity in the current document, correcting two errors present in the prior draft: a row that conflated the verticality operator with the (now separately named) genesis-branching signal, and a row that applied Flip Condition A's coherence weighting to Flip Condition B's outcomes, contradicting the mass-weighting rule given for B elsewhere in the same document. Finally, per direct request, the collapse operator (Definition 10.1, Part 10) has been renamed from a script C to a script Q throughout the document, correcting an unintentional collision with the script C already used since Part 2 for "the collection of all containers" — the Part 10–12 file has been reissued with this correction alongside this segment.

Part 13. Relationship to Existing Mathematical Frameworks

13.1 Relation to Classical Dynamical Systems

RICD enlarges the notion of state to X = (P, R, G, M), incorporating informational organization, resource dynamics, adaptive networks, and historical memory. Classical dynamical systems generally assume fixed state variables; RICD allows both the variables and the organizational architecture through which they interact to evolve recursively.

13.2 Relation to Information Geometry

RICD adopts the geometric perspective of information geometry as its informational foundation, subject to the scope conditions of §4.1, and departs from it by coupling the informational manifold to recursive organization, adaptive resources, and evolving interaction networks, and by making the geometry itself history-dependent (Definition 4.8, §4.10; consequences for identity developed in §9.10 and completed in §10.5b.0). Traditional information geometry studies geometry on probability spaces; RICD studies adaptive systems whose geometry itself evolves.

13.3 Relation to Network Science

RICD treats network topology as an emergent adaptive quantity rather than a fixed modeling assumption. Conventional network models typically evolve node states on a graph; RICD simultaneously evolves both the node states and the graph itself.

13.4 Relation to Control Theory

FDFM extends feedback beyond an external controller to an intrinsic property of every adaptive container, with feedback effectiveness depending upon informational integrity, permeability, memory, and available resources rather than controller design alone.

13.5 Relation to Cybernetics

RICD may be viewed as a mathematical continuation of the cybernetic tradition, developing communication, adaptation, recursive organization, and self-regulation within a formal framework based on information geometry, recursive operators, adaptive resources, and evolving networks. One classical cybernetic result deserves explicit connection: Ashby's Law of Requisite Variety, that a regulator can only control a system if the regulator's own variety is at least as great as the disturbance it must counter. $\Pi_f(t)$ (§6.4, §10.5b) is this document's quantitative development of that same concern, applied specifically to whether corrective information can penetrate a container's own structure rather than merely exist alongside it — §13.7 develops this connection further.

13.6 Relation to Statistical Mechanics

The recursive aggregation operators of Part 5 permit genuine emergence rather than simple averaging — higher-order containers acquire new informational states, resource dynamics, and boundary operators not generally reducible to microscopic descriptions alone.

13.7 Relation to Complexity Science

RICD's principal contribution is a common mathematical language for recursive hierarchy, adaptive networks, informational organization, resource economics, historical memory, collapse, and recovery, derived from a common set of primitive objects and axioms rather than independent models per phenomenon.

A note on what this contribution is not. The claim below is a synthesis claim, not a discovery claim. Every distinction this subsection identifies as central to RICD already exists somewhere in the complexity-science, network-science, cybernetic, or organizational literature, individually and separately named. RICD's contribution is not having noticed a distinction the field has missed; it is treating a specific set of these already-known distinctions as jointly necessary, formalizing them within one coupled dynamical system, and supplying a specific mechanism — the official/operational/meta-representation split of §6.1 — for how they reinforce one another rather than merely co-occurring.

Distinctions RICD keeps formally separate, and their existing antecedents.

- *Connectivity is not integration.* A network can remain physically connected while the responses its parts produce cease to be mutually compatible. This echoes the structural-versus-functional connectivity distinction well established in network neuroscience, where the physical wiring of a network and the statistical or causal dependencies actually observed in its activity are tracked as separate quantities. RICD's analogue is $\Lambda_{ij}(t)$ (coupling strength, §8.5) versus $D_{op}(t)/D_{om}(t)/D_{pm}(t)$ (§6.3): a coupling can persist at full strength while what flows across it diverges.
- *Feedback existing is not feedback being corrective.* A system can be saturated with communication — complaints, sensors, reports — without that information being able to alter the system's actual state. This is a direct descendant of Ashby's Law of Requisite Variety (§13.5): a regulator's variety must match the disturbance it faces for control to occur at all. $\Pi_f(t)$ (§6.4) is RICD's quantitative treatment of this same concern — feedback permeability, not feedback volume, is what the dynamics actually respond to.
- *Local coherence is not systemic coherence.* Individually well- organized subsystems can be increasingly incompatible with one another. This is a sharpening of Herbert Simon's account of near-decomposable systems, where tightly-coupled subsystems interact only loosely with each other; RICD's departure is treating that loose coupling as itself capable of carrying increasing, trackable divergence (D_{op}, D_{om}, D_{pm}) rather than only weak interaction strength.
- *Formal organization is not operational organization.* An institution's declared structure, self-description, and official record can diverge from what it actually does, sometimes

systematically and durably. This is the central finding of institutional theory's decoupling literature — formal structure persisting as symbolic or legitimating, materially disconnected from actual activity. §6.1's P_o/P_p split, and the entire captive/buried official-layer machinery of §10.5b.18, is this document's attempt to give that decoupling an explicit dynamical mechanism — how it forms, what sustains it, and what it costs to maintain — rather than treating it as a static, unexplained condition.

- *Robust is not the opposite of fragile.* A system optimized for robustness against common, anticipated disturbances can become sharply more fragile to rare or unanticipated ones — the central finding of Highly Optimized Tolerance. RICD's captive-official-layer mechanism (§10.5b.18) is a specific instance of this general pattern: a container's optimization for one kind of stability (a stable-looking official presentation) is precisely what leaves it unable to detect or correct a different, unanticipated kind of failure (operational corruption the official layer has been engineered not to see).

What unifying these distinctions inside one coupled system adds. Held separately, each distinction above explains one apparent paradox in isolation — why connectivity does not guarantee integration, why information does not guarantee correction, why local order does not guarantee systemic order, why declared structure does not guarantee actual function, why robustness does not preclude fragility. RICD's specific claim is that these are not five independent phenomena requiring five independent explanations: within the official/operational/meta-representation architecture of §6.1, they are different symptoms of the same underlying dynamical quantity — accumulating, differently- distributed divergence between what a system's parts declare, do, and collectively make of the gap between the two. A system's own diagnosis of which paradox it is facing depends on which layer absorbed the original divergence (§10.5b.2.1), not on five separate mechanisms.

13.8 Relation to Machine Learning

Learning becomes one example of informational transformation within an adaptive container. RICD additionally models resource constraints, recursive organizational structure, memory, network evolution, and adaptive collapse, which often remain external to conventional optimization frameworks.

13.9 Relation to Organizational Theory

RICD interprets quantities such as trust, bureaucracy, communication, fragmentation, and resilience as observables derived from the underlying adaptive state, rather than independent empirical variables — allowing different organizations to employ different measurements while remaining mathematically comparable.

13.10 A Unified Perspective

Conceptual Contributions.

- Recursive containers provide a scale-independent primitive for > adaptive organization.
- FDFM formalizes internal informational consistency as a derived > process operating within each container.
- Adaptive possibility replaces threshold-based failure as the primary > mathematical characterization of collapse.

- History-dependent information geometry allows accumulated experience > to alter future adaptive trajectories — and, per §10.5b.0, to > determine a container's structural orientation and expected > longevity as a consequence of its informational alignment.
- Emergent network topology replaces fixed interaction graphs.
- Resource expenditure is derived from informational transformation > rather than introduced as an independent state variable.
- Observables are derived from the adaptive state manifold rather than > postulated independently.
- The horseshoe principle (Axiom C5, §10.5b.15) unifies two previously > separate failure modes — rigid and compulsive closure — as the > same underlying failure of divergence-responsive tracking, rather > than as opposite conditions requiring separate diagnostics.

The Integration Principle.

RICD does not replace existing mathematical theories of adaptive systems; it provides a common recursive framework within which their complementary strengths can be combined to analyze the organization, evolution, persistence, and collapse of complex adaptive systems across scales.

Part 14. Discussion, Limitations, and Future Directions

14. Discussion

RICD was developed from the premise that adaptive systems across widely differing domains share a common organizational structure. The theory introduces a small set of primitive mathematical objects — information states, containers, boundaries, resources, memory, and recursive organization — from which informational integrity, fragmentation, resilience, rigidity, and collapse are subsequently derived.

Collapse has been reformulated as the progressive contraction of adaptive possibility; recovery, as its restoration or expansion. This revision additionally grounds a container's structural orientation — how rigid, closed, and hierarchical its boundary currently is — as a consequence of its informational alignment rather than an independently fitted property, completes a criterion for what makes a container the same persisting entity across a history in which every element of its defining tuple may have changed, and — new to this revision — separates the meta-representation layer's two directions of transmission, showing that a container's messaging pathology is characterized not only by whether truth reaches its official layer, but by whether official demand reaches back into its operational layer at a comparable rate.

14.1 Principal Contributions

First, recursive containers as scale-independent mathematical primitives. Second, a geometric description of informational organization based upon probability manifolds. Third, FDFM formalized as a derived representation of internal informational dynamics. Fourth, a unified treatment of adaptive resources, memory, recursive organization, and evolving interaction networks. Fifth, a geometric definition of adaptive collapse based upon contraction of feasible adaptive transformations. Sixth, a structural account of why rigidity beyond the minimum needed to remain a bounded container is costly, and a completed criterion — combining recursive membership continuity (§9.10) with a semi-rigidity floor (§10.5b.0) — for container identity across transformation. Seventh, the horseshoe principle (Axiom C5), unifying rigid and compulsive closure as the same failure of divergence-responsive tracking, and the active-firewall correction (§10.5b.18), replacing the meta-representation layer's earlier treatment as a passive boundary with an explicit account of its population composition. Eighth, the valve-symmetry account of the meta-representation layer's two directions (§10.5b.18.5–18.8): a formal distinction between whether operational truth reaches the official layer and whether official demand reaches back into operational behavior, together with the demand-forcing term (F_demand) explaining sustained operational conformity to a false official narrative without requiring extraction or confrontation-avoidance to account for it.

New to this revision: ninth, a generative mechanism for membership continuity (§9.10.1's internal cohesion Λ_int) replacing an asserted criterion with an actual account of why boundedness holds at the semi-rigid floor, extended into preferential attachment (§8.3.1) and mutual anchor clusters (§10.5b.13.1) as two direct applications of the same underlying quantity. Tenth, a formal statement that convergence does not require unimodality (§6.3.1), with the corollary that forced flattening of a genuinely plural operational reality is itself a source of divergence rather than a resolution of it — extended into accurate self-accounting (§10.5b.18.26–18.31) as a regime-general principle governing causal attribution, capture-origin healing, and the positive counterpart to the despair signature alike. Eleventh, a complete account of the two non-baseline regimes' asymmetric relationship to their own history: an active, maintained mechanism for high-entropy's captive official layer (§10.5b.18.1, extended, plus historical residue and self-corrosion, §10.5b.18.21–18.25) against a structurally distinct, non-captive account for medium-entropy's buried operational layer, with the two regimes shown to differ not merely in degree but in what access one layer has into the other at all. Twelfth, a control-locus account (§10.5b.18.18–18.20) of which layer actually directs a container's behavior, extended across container boundaries (§10.5b.18.28) to give capture-origin high-entropy classification its own healing prognosis, conditional on accurate retrospective attribution. Thirteenth, a formally scoped account of a container's relationship to its own most fully aligned state (§10.5b.0.8–0.11): sustained occupation, momentary glimpse with two separable effects, and a proof that the state cannot be reached by direct pursuit, given as a genuine implementation constraint rather than aspirational language. Fourteenth, three previously narrative-only mechanisms given formal treatment: predator search and its suppression by ambient cohesion (§10.5b.11c), persecution targeting of a network's own truth-aligned population as threat-responsive rather than resource-responsive (§8.6a.1–8.6a.3), and pre-flip disintegration under multi-source overload as a failure mode distinct in mechanism from existing isolation-driven dissolution (§10.5b.9d).

14.2 Limitations

The present formulation is a fully specified computational framework, subject to the adapter contract and the open mathematical limitations recorded below — not a complete theory in the sense of having settled every open mathematical question, and not a claim that every coupled nonlinear interaction's global behavior is uniquely determined under every implementation circumstance. "Fully computable" is deliberately avoided here for exactly that reason: every formerly abstract quantity now has either a concrete default or an explicitly delegated adapter responsibility, which is what makes the framework implementable as a real simulation or estimation engine, but that is a narrower claim than mathematical completeness, and the two should not be read as equivalent. Several limitations from earlier drafts are resolved in this revision and are recorded here as closed, alongside the limitations that remain.

Resolved in this revision.

- Theorem 12.4's resource-feasibility hypothesis is now satisfied > automatically by §7.3's total-work clipping rule (covering > maintenance as well as adaptive work), rather than requiring > separate per-application verification.
- Proposition 12.8's restoration previously rested on one derived link > (d_A's dependence on fragmentation, §10.3) and one narrative claim > (fragmentation's dependence on network coherence). §11.7 now makes > the second link an explicit formula, completing the derivation.
- The feedback operator $\mathscr{F}$ (§6.4) previously had no computational > default, unlike every other quantity in Part 6. It is now a > stochastic gradient flow on an explicit multi-valley divergence > landscape, tied concretely to the three entropy regimes of > §10.5b.1.
- The structural-height quantity h(t) underlying the > verticality/horizontality operators

previously had no grounding in > the container ontology of §2.2 and sent Ultimate Convergence to a > literal structural zero, contradicting Axiom 1. It is now derived > from informational alignment with a strictly positive floor, and > Ultimate Convergence is stated consistently with that floor.

- Container identity previously had no formal criterion; §9.10 and > §10.5b.0 jointly supply one, combining membership continuity with > the semi-rigidity floor and formally distinguishing > persistence-through-transformation from dissolution.
- Several symbol collisions accumulated across drafts (information > velocity and verticality both written V; the feedback operator, > memory operator, and maintenance functional all written using > close variants of M and F; the collapse operator and the > collection of all containers both written as a script C) are > resolved throughout.
- Definition 6.1's independence clause previously read as > characterizing the meta-representation layer in general as an > honest, external auditor of the container — a reading that holds > only for the low-entropy baseline regime. The clause is now scoped > explicitly to data-source independence, with a forward pointer to > §10.5b.18's regime-dependent completion for what the channel > actually does once measured (see §6.1, revised alongside > this segment).

Open, and precisely scoped.

- The information metric g is defined only under the scope conditions > given in §4.1, and Definition 4.2's six candidate distance > functionals are not mutually substitutable once curvature or > geodesics are invoked (§4.2) — an application needing those > constructions must commit to Fisher–Rao. The default > computational instantiations throughout Parts 6–11, however, use > only the scalar (non-Riemannian) layer, so a collapse tracker > built from this edition does not need to resolve the metric-choice > question at all unless it separately wants geometric early-warning > indicators.
- The history-dependent metric g_t (Definition 4.8) is confirmed as a > genuine, load-bearing generalization rather than an informal one, > but its scope condition (that g(M_t) remain a valid Riemannian > metric at every fixed t) must be separately verified per > application; where it is not, §4.10 specifies that the > static-metric case should be used instead.
- Every quantity that was previously an unspecified functional now has > a stated default form (Part 15 consolidates all of them), but > every default form carries fitted coefficients whose values are > not fixed by the theory, including several introduced in this > revision (κ_h, ξ, ψ, κ_Ω, η_Z, among others). The theory > constrains the shape of these relationships; it does not and > should not predict their numeric values in advance of fitting them > to data.
- The multi-valley structure of the divergence landscape Φ (§6.4, > §10.5b.1) is specified qualitatively — three basins > corresponding to the three entropy regimes, reshaped by forcing > terms consistent with Axiom C1 — but the basins' precise depths, > barrier heights, and bifurcation behavior as forcing varies > continuously are not derived in closed form.
- The structural hazard rate Ω_hazard (§10.5b.0) is given a > monotonicity relationship to expected structural longevity, not a > specific survival-function form; applications requiring calibrated > lifespan predictions rather than comparative hazard ranking will > need to supply one.
- The high-entropy backslide-duration formula (§10.5b.9a) is > explicitly flagged in the text as a modeling assumption rather > than an independently derived result, unlike the accompanying cost > formulas.

- The framework establishes local well-posedness (Theorem 12.1) under > standard regularity assumptions but does not yet include global > existence, bifurcation structure, or asymptotic stability analysis > for the nonlinear coupled system that results once every Part > 6–11 instantiation is composed together — now a substantially > larger coupled system than in earlier drafts.
- Theorem 12.4 and Proposition 12.8 remain conditional on accepting > this edition's default instantiations (§7.3's clipped W; §10.3's > exponential d_A; §11.7's linear-in-incoherence fragmentation) as > the operative definitions. An application substituting different > concrete forms would need to re-verify or re-derive these results > for its own choices rather than inherit them automatically.
- The commensuration step in §5.2 (mapping heterogeneous child Ω's > onto a shared severity scale before aggregation), and the > analogous simplex-projection step now required in §8.5's network > propagation, are both left to the domain adapter by design. > Nothing in the engine can verify that a given σ_i or projection > choice is reasonable — that judgment is inherently substantive > and domain-specific.
- ω_meta^up(t) and, new to this revision, ω_meta^down(t) > (§10.5b.18.5) each have only their regime-characteristic endpoints > stated (§10.5b.18.2, §10.5b.18.5), not a full dynamic equation > governing how either moves between regimes over time. §16.11's > structural baseline tracking gives a partial comparison basis, but > a closed-form dynamic for either channel remains open.
- The discharge-alignment tolerance parameters (ε_align, θ_align) are > new fitted quantities with no empirical grounding yet; their role > is structurally specified but their numeric values, like every > other fitted coefficient in Part 15's reference table, await > domain-specific data.
- Whether the resilience accumulator Z(t) and the extraction-hunger > accumulator Z_hun(t) can be active in the same container at once, > and if so how they would interact, is not addressed by this > document. Both are defined independently (§10.5b.9a, §10.5b.19.1); > nothing currently specifies whether they compose additively, > whether one dominates the other, or whether their simultaneous > presence indicates a container this framework has not yet > correctly classified into a single regime.
- κ_demand (§10.5b.18.8) is fitted and untested, like every other > coefficient introduced in this revision; its structural role — > scaling how strongly official demand reshapes the divergence > landscape once ω_meta^down(t) and D_op(t) are known — is > specified, but nothing in this document yet constrains its > magnitude relative to κ_conf or κ_boom.
- The external coupling terms (§10.5b.18.9–18.11) and the loyalty > composition Ψ_loyal(t) (§10.5b.18.12–18.13) are new to this > revision and carry the same open-fitted-coefficient status as > everything above, plus one additional open question specific to > Ψ_loyal(t): the trailing-slope proxy given for it is a proxy, not > a validated instrument, and nothing in this document yet > establishes how well a behavioral trend (ω_meta^up's own slope) > tracks the underlying orientation it is standing in for where > direct measurement is unavailable. Domains able to supply a direct > loyalty measurement should prefer it; the proxy exists so the > quantity is computable, not because it is known to be accurate.
- §10.5b.18.21's split of official's content into fossilized > (φ_fossil) and actively-patched (φ_patch) components is only > jointly constrained by their sum, 1 − ω_meta^up(t); the document > gives no independent method for separating the two in a real > dataset short of directly observing patch events as they occur. > Domains without event-level observation of leak-triggered rewrites > will find φ_fossil and φ_patch individually underdetermined even > where their sum is measurable.

- §10.5b.18.26's attribution-divergence machinery requires > A_true(t′,t), the true causal decomposition of a container's own > trajectory, as a reference point. This is frequently not directly > observable even in principle — it is itself an inferential > construct in most real domains, not a ground truth an adapter can > simply supply. Applications of Proposition 10.5b.18.26 and > 10.5b.18.29 should treat A_true as the best available > reconstruction (forensic, per §16.7, where applicable) rather than > as a known quantity, and Θ_capture (Definition 10.5b.18.28b) is > correspondingly hard to fit with confidence wherever A_true itself > is contested.
- §10.5b.0.10's cumulative glimpse effect is explicitly gated by > conditions — favorable Ψ_total(t), sustained healthy external > coupling, unspecified grid conditions — given qualitatively > rather than as a combined threshold. This document does not commit > to when, precisely, rising Γ_glimpse(t) actually culminates in permanent > structural change; treating the gating conditions as necessary but > not jointly sufficient in closed form is a deliberate scope > limitation, not an oversight to be resolved by a future > coefficient fit alone.
- §10.5b.19.5's habituation forcing (F_hab, Z_hab) deliberately > carries no resource-cost term paralleling M_decep(t); this > asymmetry with the high-entropy case is stated as intentional, but > is itself an empirical claim — that indulged medium-entropy > convergence costs a container structurally rather than materially > — which this document does not yet have evidence for beyond the > parallel construction argument given at its introduction.
- §8.6a.3's persecution deficit signal, P_persecute(t), is explicitly > not wired into P_C(t) or any other collapse-potential formula; it > is diagnostic output only. A low f_NC(t) is consistent with > genuine persecution or with a network simply composed of few > high-loyalty containers to begin with, and this document does not > supply a method for distinguishing the two beyond direct evidence > of Definition 8.6a.1's targeting bias itself.

These limitations mark exactly where domain judgment must enter, or where the theory constrains shape without fixing value — Part 15's adapter contract exists to isolate the former to one well-defined interface rather than let it leak into the engine, and the fitted-parameter tagging throughout Part 15's reference table exists to make the latter explicit rather than implicit.

14.3 Computational Development

A numerical simulation environment for recursive adaptive systems under varying assumptions regarding informational geometry, memory kernels, resource constraints, and adaptive network topology would allow systematic investigation of collapse attractors, critical slowing down, adaptive recovery, recursive propagation of disturbance, organizational phase transitions, and multiscale resilience. Because the governing formalism is domain-independent, a common computational engine may be adapted to diverse applications through alternative observation operators and parameterizations.

14.4 Empirical Validation

Testable predictions include: increasing informational stress should precede observable collapse; declining adaptive degrees of freedom should occur before catastrophic failure; persistent reductions in feedback permeability should predict growing rigidity; recursive fragmentation should reduce adaptive capacity across organizational scales; prolonged resource debt should amplify informational divergence through positive feedback. This revision adds several further falsifiable predictions with their own dedicated diagnostic signatures: Axiom C2's characteristic upward inflection in divergence pressure immediately preceding a hull-breach crossing (§10.5b.15); the alignment-precedes-relaxation ordering and the predicted rebound following any artificial forcing of structural height (§10.5b.0); and the three distinct backslide cost/duration signatures distinguishing anchored self-healing, medium-entropy healing, and medium-entropy collapse (§10.5b.9a), alongside the cascade-sequence signatures already given in §8.8a.vi.

This revision's session further adds: the horseshoe principle's tracking-fidelity signature (Definition 10.5b.15.1), predicting that both rigid and compulsive closure should show the same diagnostic mismatch between divergence pressure and velocity, rather than requiring separate tests for each; the boomerang mechanism's prediction (Definition 10.5b.16.1) that manufactured redivergence should follow realignment specifically in containers already diagnosed with compulsive closure, and not appear in containers approaching genuine equilibrium without that diagnosis; the binary-trust-inversion prediction (Proposition 10.5b.17.1) that misdirected anchoring under the inversion condition should correlate with an extractor's measurably expanding network reach, not merely its continued extraction from the original container; and the decoy check (Proposition 16.9.1) as a methodological addition for forensic work, predicting that a quieter, calm-presenting neighbor's outbound extraction flux, where elevated, should be checked before attributing a recovery failure to the visibly distressed neighbor originally suspected.

New to this revision: the valve-asymmetry prediction (§10.5b.18.6, §10.5b.18.8) — a container with $\sigma\_valve(t)$ trending toward 0 with $\omega\_meta\char`^down(t)$ dominant should show continued or accelerating operational divergence even absent any new external shock, since F_demand(t) actively reshapes the divergence landscape rather than merely leaving a prior false-signal valley unresolved. This is distinguishable from ordinary firewalling (low ω_meta^up(t) alone), which predicts a stalled but not actively worsening trajectory; the two configurations should therefore be empirically separable by tracking D_op(t)'s trend, not merely its level, over the window in question.

This revision's coupling and loyalty additions (§10.5b.18.9–18.13) contribute two further predictions. First, the coupling-health prediction: containers with high-φ_fb, high-Γ_supp neighbor coupling should show measurably faster D_om return-to-band times following a perturbation than isolated containers or containers coupled only to high-entropy neighbors, holding internal channel composition (ω_meta^up, ω_meta^down) fixed — a prediction specifically about the external mechanism's own contribution, separable from internal recovery by comparing otherwise-similar containers with differing neighbor composition. Second, the loyalty-conditioned-outcome prediction: at matched mass-weighting and matched knock magnitude, medium-entropy containers with higher Ψ_loyal(t) should show a measurably higher B2-to-B1 (reorganization-to-lock) ratio than lower-Ψ_loyal(t) containers, and high-entropy containers with higher Ψ_loyal(t) should show measurably higher post-collapse self-reset rates — both testable wherever a domain can track container outcomes across multiple knock or collapse events with an independent (non-proxy) loyalty measurement, since testing the proxy against real outcomes is the only way to validate the proxy itself.

This revision's second pass adds six further predictions. The search-suppression prediction (§10.5b.11c.2): as a network's ambient cohesion Λ_amb(t) rises, structurally extraction-dependent containers should show measurably declining successful first-contact rates via active search specifically, separable from the passive attraction-channel effect §8.3.1 already predicts, since the two operate through different mechanisms (§10.5b.11c.1's editorial note). The mirror-threat prediction (§10.5b.11c.4): coupling attempts between high-entropy containers of differing severity should occur at a measurably lower rate than resource-based extraction incentive alone predicts, with the shortfall scaling in min(D_op,i, D_op,j) rather than in either container's resource level. The cluster-resilience prediction (§10.5b.13.1): mutual anchor clusters should show gradual, membership-proportional degradation in protective capacity under member loss, contrasted with the discontinuous single-point failure an ordinary anchor's withdrawal already predicts. The fast-versus-slow collapse prediction (§10.5b.19.3–19.5, dependent on the habituation accumulator Z_hab(t) now built): containers reaching hull-breach quickly should show measurably lower Z_hab(t) at the point of breach, and correspondingly higher B2 resolution rates, than containers whose divergence pressure approached the same threshold slowly, since slow approach affords more time for habituation to accumulate and raise Override's own effective threshold against it. The persecution-targeting prediction (§8.6a.1): candidate neighbors with elevated Ψ_loyal(t) should show elevated ρ(j) and higher realized extraction targeting independent of their own resource level, distinguishing threat-responsive from resource-responsive targeting empirically rather than only by definition. The disintegration-signature prediction (§10.5b.9d): containers that disintegrate rather than resolve to a labeled tipping outcome should show a preceding period of elevated D_chaos(t) from multiple concurrent sources, empirically distinguishable from the declining algebraic connectivity and shrinking coupled-neighbor count that precedes §8.8a's isolation-driven dissolution instead.

These may be evaluated within biological systems, software engineering, organizational management, ecological monitoring, public administration, healthcare, education, and other adaptive environments.

This revision's final addition contributes one further prediction, from the structural-grain mechanism of §4.11: at matched Drive_B(t) magnitude, medium-entropy containers struck orthogonal to their own established structural grain should show measurably faster M_tip accumulation and correspondingly earlier tipping-threshold crossings than containers struck along their grain — a directional effect separable from the mass-weighted B1/B2 split of §10.5b.7, since χ_dir(t) modifies the momentum accumulation rate itself rather than which outcome a given momentum resolves toward.

14.5 Future Mathematical Development

Open directions include: global existence and uniqueness under broader conditions; invariant manifolds and adaptive attractors; bifurcation theory for recursive adaptive systems, now including the divergence landscape Φ's basin structure specifically; stochastic perturbations on evolving information manifolds; extension of recursive operators to hypergraphs and higher-order simplicial complexes; information-geometric variational principles; renormalization across recursive organizational scales; canonical metrics for adaptive possibility; and a calibrated survival-function theory for the structural hazard rate of §10.5b.0, extending its current comparative (monotonicity-only) form. Proposition 12.8's coupling is now fully specified in closed form (§10.3, §11.7), so the remaining open item there is empirical rather than mathematical: testing whether the exponential d_A(F̃) and linear F(N) forms are the right ones for a given domain, or whether fitted data favor different functional relationships.

14.6 Broader Research Program

Potential application domains include software reliability and cybersecurity, artificial intelligence, robotics, organizational governance, institutional analysis, healthcare systems, education, ecological resilience, infrastructure management, economic systems, disaster response, and evolutionary biology — representing adaptive organization within a common recursive mathematical language while preserving domain-specific observables.

14.7 Concluding Remarks

RICD proposes that the diversity of adaptive systems observed across nature, technology, and society may be understood through a shared mathematical architecture based upon recursive information-processing containers. Information, resources, memory, boundaries, and networks are interacting components of a unified adaptive geometry. Stability emerges through the preservation of adaptive possibility; collapse, through its contraction; recovery, through its expansion; and a container's structural form, per this revision, through the alignment it maintains rather than as an independent choice.

The present work should be understood not as the culmination of a theory but as the beginning of an extensible mathematical program for the study of adaptive systems — one whose internal consistency, per this revision, now rests on stated rather than assumed conditions throughout, whose every previously-abstract functional now has a concrete default form suitable for direct computational implementation (Part 15), whose notation, after several accumulated collisions, now assigns one symbol to one quantity throughout, and whose meta-representation layer, per this revision, is characterized by two independently-tracked directions of transmission rather than one.

Part 15. Computational Specification

Parts 1–14 establish RICD as a mathematical theory. This part exists for a different purpose: to consolidate everything needed to implement it, so that building a working collapse tracker is a matter of supplying data to a fixed interface rather than re-deriving formulas from the preceding text. Everything in this part restates definitions given earlier; nothing here is new theory.

15.1 The Two-Layer Architecture

Every RICD implementation separates into exactly two layers, and the boundary between them is the boundary between domain judgment and fixed mathematics:

- The domain adapter (written once per system studied) supplies primitive observations together with the domain-specific representation and measurement mappings required to instantiate the theory — commensuration maps, the outcome space, resource normalization, observational comparability, and similar choices that are mathematically constitutive of the model, not raw data alone. It never computes a quantity the engine's fixed formulas are responsible for deriving.
- The engine (written once, reused across every domain) consumes what the adapter supplies, instantiated through those mappings, and computes every quantity derived from those instantiated primitives according to fixed RICD formulas. The engine has no domain-specific code in it anywhere; if it needs to "know" it's looking at a justice system versus a cell versus a corporation, the adapter boundary has been drawn in the wrong place. This is a real, load-bearing division of labor, not a claim that the adapter's choices are inert: a badly chosen commensuration map or projection operator can distort the model's own state representation in ways no engine-side theorem can catch (§6.4's spurious zero-divergence attractor is one documented instance), which is precisely why those choices are treated as adapter judgment requiring domain expertise, not as raw, judgment-free data.

15.2 Adapter Contract — What Every Domain Implementation Must Supply

Constructing O_o and O_p: a practical caution, not a theoretical requirement. Definition 6.1 places no constraint on how an adapter builds its official and operational projection operators beyond their both being deterministic functions of P — but a poorly chosen pair can create a spurious problem no theorem here will catch. If O_o and O_p happen to agree exactly at some point in Ω, that point becomes a zero-divergence attractor for $\mathscr{F}$'s dynamics (§6.4) regardless of whether it represents genuine, informative convergence for the domain being modeled — the gradient flow will be pulled toward it simply because D_op vanishes there, not because it is where the container's true signal actually lies. This is not a flaw in the theory; Definition 6.1 never promised any two projection operators would behave well together, and constructing them sensibly is exactly the kind of domain-specific judgment the adapter contract exists to hand off. Before trusting an adapter's O_o and O_p, check whether they share an exact-agreement point across Ω, and if so, confirm that point is a genuine, meaningful convergence state for the domain rather than an accidental artifact of how the two scoring functions happen to have been constructed.

Input confidence typing. Every quantity an adapter supplies below falls into one of three types, and downstream calculations should propagate this distinction rather than treat all inputs as equally certain. **Observed**: directly measured from the system (e.g., raw resource levels, transaction counts). **Estimated**: inferred statistically from observed data via a stated model (e.g., a fitted composition proxy). **Reconstructed**: retrospectively inferred from incomplete evidence, most consequentially A_true(t′,t) (§10.5b.18.26), which is rarely observable directly and should be supplied via the best available reconstruction (§16.7's forensic procedure, where applicable) rather than treated as known. A reconstructed input should not enter downstream calculations with the same epistemic status as a directly observed one; where an implementation supports uncertainty propagation, reconstructed inputs are where it matters most.

Propagation requirement. This is a requirement on engine output, not only a caution about input handling: every engine output must carry a provenance/confidence profile inherited from its upstream inputs' types above, rather than being emitted with uniform categorical confidence regardless of what fed into it. A regime classification or collapse-risk assessment computed with a reconstructed A_true(t′,t) among its inputs must be labeled as resting on reconstruction, not presented identically to one computed entirely from observed quantities. This does not require a full Bayesian uncertainty system; the minimum requirement is that the observed/estimated/reconstructed distinction, already tracked at the input side, is carried forward and exposed at the output side, so a forensic inference is never presented as equivalent in certainty to a direct measurement.

For each container C in the system being modeled:

- Ω — the outcome space, stated explicitly (discrete category set is the default assumption throughout this part; continuous Ω requires substituting a continuous-compatible D per §4.2).
- O_o, O_p — two data-extraction functions returning empirical distributions over Ω at time t: the official representation and the operational representation. O_m is not a third data-extraction function of the same kind: per §6.1, P_m is an emergent aggregate over the meta-representation layer's own population of sub-processes, so an adapter supplying meta-representation data is supplying (or the engine is aggregating) per-sub-process readings P_m^(k)(t) and their weights φ_k(t), not a single extraction function's output. §6.1's independence requirement is binding on this aggregation process — it must be sourced through its own dedicated observation, not algebraically derived from O_o and O_p's outputs; this is a measurement requirement, not a claim that the channel reports honestly.

Choosing a P_m proxy: a principled connection, not a convenient stand-in. Where a domain lacks a direct meta-representation measurement and a proxy must substitute, the proxy's connection to genuine self-monitoring capacity should be stated and defensible, not merely convenient or correlated. A concrete instance: an early higher-education adapter proxied P_m with raw admissions-rate trend, explicitly flagged at the time as not the real construction, since nothing about admissions selectivity bears on whether a population can genuinely self-monitor. It was later replaced with contingent-versus-secure staffing composition specifically because job security has a real, direct bearing on whether a population's internal population can dissent honestly (§8.5b.5's form/function material and §11.10.2's contingent-staffing signal both connect to this same underlying fact) — a population dependent on annual reappointment is structurally less able to serve as an honest checking function than one with genuine tenure protection, independent of whether it happens to correlate with any outcome of interest. This is the standard a P_m proxy should be held to generally: does the proxy bear on the population's actual capacity for honest self-monitoring, not merely on whether it happens to predict what the engine is trying to detect. Real, independent, external attestations of governance or self-monitoring failure (Definition 6.1.1) should be layered on top of whatever continuous proxy is used, as a direct override per that definition's own terms, never blended into the continuous proxy as one more weighted input regardless of how well-justified the continuous proxy itself is.

Classifying an observable before modeling it: drift-type versus shock-type. Before an adapter decides how a raw observable should be modeled as a latent process, it needs to ask what kind of process actually generated it, not default to the same continuous-smoothing treatment for everything. A concrete instance: an implementation modeled both a slowly-evolving staffing-composition proxy and a raw balance-sheet debt figure as ordinary Gaussian random walks, treating both as continuous-drift quantities where small, roughly consistent steps accumulate over time. This worked correctly for the staffing proxy and failed badly for debt — a real, sudden debt escalation (a container's total liabilities more than tripling within three years) came back from the fitted model an order of magnitude smaller than the same quantity computed directly from raw data, because a Gaussian random walk has no way to represent "mostly small steps, but rarely a large one": it treats a genuine large step as an improbable outlier and pulls it toward consistency with the surrounding small ones, exactly the behavior it is built to have. This was not a badly-tuned prior; it was the wrong class of model for what the observable actually is.

The distinguishing question: does this quantity evolve through many small, roughly homogeneous increments (drift-type — staffing composition, gradual enrollment change, most slowly-evolving institutional characteristics), or does it evolve through long stretches of near-stability punctuated by rare, large, discrete jumps (shock-type — sudden debt escalation, an abrupt regulatory action, any quantity whose real history looks like Definition 8.8a.3's Drive_B(j,t) landing on it)? A shock-type observable is not merely a drift-type observable with a larger variance; it is structurally the same kind of event this document already formalizes as an external knock, and should be modeled that way rather than smoothed through the same continuous machinery drift-type quantities correctly use. Where an adapter identifies a shock-type observable, its latent trajectory should combine an ordinary small-step drift component with a separate, rare-jump component — a jump-diffusion structure, not a single homogeneous random walk — with its own, independently-fitted jump probability and jump-magnitude distribution (wider than the drift component's own step size, consistent with representing a genuine discrete event rather than an extreme instance of ordinary drift). This is the same construction already licensed by Drive_B(j,t) and the resolution-kick machinery (§8.8b) for coupled, multi-container knocks; a single-container shock-type observable is the same phenomenon without requiring a coupled neighbor as its source, and should draw on the same underlying mathematics rather than being treated as a new problem. An adapter that models every raw observable through the same continuous-drift default, without first asking which class a given quantity belongs to, should expect exactly this failure mode on any genuinely shock-type input — not as a rare edge case, but as the default outcome of applying the wrong model class.

- R(t) — a resource observation, in whatever unit is natural to the container, plus R^max (used in §5.6's normalization and §10.4's rigidity).
- P_prod(t), E_exch(t) — observed production and net boundary exchange, feeding §7.4 directly.

- Raw permeability signals — counts of corrective actions and error signals, and a response-lag measurement, feeding §6.5's Π_f proxy.
- d_A^{max} — a domain-specified ceiling on adaptive capacity (§10.3), and, optionally, a direct feasible-transformation count if the domain supports one.
- For containers that are children of a larger aggregate: σ_i, the commensuration map from that container's Ω onto the parent's shared severity scale (§5.2).
- Size(C) — a structural-scale measure (adapter-defined per domain: sub-container count, physical mass, headcount, organizational tier count), feeding §7.5's maintenance scale-dependence. Optional; defaults to a constant (no scale penalty) if not supplied.
- $\chi_{supp}(C,t)$ — the geometric support ratio (§10.5b.0), a domain-specified measure of structural/organizational base breadth relative to height. Optional; defaults to 1 (maximal support assumed) if not supplied, which understates hazard for narrow-based containers.
- μ_o, μ_p, μ_m — representation-layer mass weights (§10.5b.7), if the domain has a natural measure of relative layer salience. Optional; fitted per container if not supplied.
- τ_{diss} — the characteristic dissipation window for a B2 resolution kick (§8.8b). Optional; fitted if not supplied.
- **$\omega_{meta}^{up}(t)$, $\omega_{meta}^{down}(t)$** — the metamessenger-population composition fractions of §10.5b.18.2 and §10.5b.18.5: the resource-weighted share of the container's meta-representation apparatus currently truth-aligned (transmitting operational reality upward) and demand-aligned (transmitting official expectation downward), respectively. Adapter-supplied where the domain has a direct measurement (internal survey, whistleblower/dissent-channel activity, audit-staffing composition, or an equivalent domain-specific proxy). Optional; each defaults to 1 if not supplied, which assumes the low-entropy baseline configuration and therefore understates hazard in a container that is actually in either non-baseline regime — the same conservative-default failure mode already flagged for χ_{supp}, and worth the same caution here.
- **Purpose-built anchor designed capacity** — for any container or sub-structure serving as a purpose-built anchor (§10.5b.13), the adapter-supplied load the structure was actually engineered to absorb, against which its eligibility for continued anchoring should be assessed. Required only for containers used in this anchor role; exceeding the supplied capacity converts the structure into an ordinary coupled neighbor for all subsequent computation.
- **$R_{ext}(t)$** — the extraction-reward signal driving the extraction-hunger accumulator (§10.5b.19.1). Adapter-supplied and overridable like every other quantity in this contract; defaults to the metabolized extraction fraction $\eta_i \cdot X_{\{i,j\}}(t)$ already computed by the engine from §8.6a if the domain has no more specific reward measure of its own.
- **$\varphi_{fb}(i,j,t)$** — the external feedback channel fraction (§10.5b.18.9): how much of coupled neighbor j's own honestly-measured divergence state reaches container i as transparent feedback. Adapter-supplied where a direct cross-container communication-quality measure exists; defaults to the engine-computed ceiling $\omega_{meta}^{up}(j,t)$ itself (assumes the coupling transmits at the neighbor's own maximum internal honesty) if not otherwise supplied — an optimistic default, unlike the conservative defaults elsewhere in this contract, and should be treated with corresponding caution where coupling quality is not independently known.
- **$r_{supp,min}$** — the fitted resource-surplus threshold below which a neighbor is assumed to have nothing to share via the external support flux (§10.5b.18.10). Optional; fitted per container if not supplied.

- **Ψ_loyal(t)** — the metamessenger loyalty composition (§10.5b.18.12): the fraction of a container's own closed, comfort-aligned population whose closure is protectively (negentropic) rather than self-servingly (entropic) oriented. Adapter-supplied where a direct instrument exists (internal survey, dissent-channel content analysis); defaults to the trailing-slope proxy of Definition 10.5b.18.12 if not supplied, which itself requires κ_loyal and w_loyal (below).
- **A_cap(i,j,t)** — the cross-container capture share (§10.5b.18.28): the fraction of container i's own functional control attributable to steering by coupled neighbor j, beyond ordinary resource extraction. Adapter-supplied where a domain has independent evidence of cross-container steering; optional, defaults to 0 (no capture, recovering the original two-way A(t) partition exactly) if not supplied.
- **f_NC^target** — the domain-specified expected baseline fraction of nonconformist (high-loyalty) containers in a network (§8.6a.2–8.6a.3). Required only where the persecution deficit signal P_persecute(t) is to be computed; no universal value is asserted by this document.
- **φ_patch(t)** — the actively-falsified share of official's content (§10.5b.18.21), as distinct from historical residue φ_fossil(t). Adapter-supplied where event-level observation of leak-triggered rewrites exists; where it does not, only the sum 1 − ω_meta^up(t) is available and the fossil/patch split cannot be individually resolved (§14.2's corresponding limitation).
- **A_true(t′,t)** — the true causal-attribution distribution underlying D_attr (§10.5b.18.26). This is not an ordinary adapter input: it is rarely observable directly, and should be supplied as the best available reconstruction (§16.7's forensic procedure, where applicable) rather than as a known ground truth. Applications should treat D_attr's accuracy as bounded by the quality of this reconstruction.

Implementation constraint (Goal Paradox, §10.5b.0.11). No adapter or engine instantiation of $\mathscr{F}$, Φ, or P_C(t) may include Definition 10.5b.0.8's sustained-convergence conditions as an explicit minimization target, weighted term, or optimization objective. Glimpse-recall (§10.5b.0.10) remains permitted as a passive, memory-triggered restorative mechanism; only the state's use as a directed goal within the engine's own forcing or evaluation functions is excluded.

Editorial note: *h(t) and V(t) are not adapter-supplied. §10.5b.0 derives them endogenously from I(t), which is itself computed by the engine from the adapter's O_o, O_p feeds and the aggregated meta-representation population data (§6.1). Supplying h(t) directly would bypass the alignment-precedes-relaxation mechanism the theory now depends on; an adapter that has independent, direct structural-height data available should treat it as a validation check against the engine's derived V(t), not as a substitute input.*

Warning: apparent volatility is not a stability proxy.

§10.5b's entropy regime classification exists because low apparent variance in the cheaply-observed layer (σ_o, the official/declared channel — usually the easiest and cheapest data to pull for a real domain) is the signature of the high-entropy regime, not evidence of health. An adapter that observes only O_o and treats its low volatility as a stability signal will systematically rank the most dangerous containers in a system as its safest ones, and the genuinely recovering containers (medium-entropy, visibly volatile in O_o while actually stable in O_p) as its most dangerous. Every adapter implementation must pull both O_o and O_p, at comparable observation quality, or the engine's regime classification cannot function as intended.

Hard engine state: regime indeterminate. The warning above is elevated from caution to enforced behavior: where an adapter does not supply O_p at comparable observation quality to O_o, the engine must not produce a regime classification (§10.5b.1) at all. It returns

Regime indeterminate: required observational channels unavailable or non-comparable, and every downstream quantity that depends on regime classification — $\sigma\_valve(t)$, $F\_demand(t)$, $\Psi\_loyal(t)$'s regime-conditioned interpretation, and the full collapse-resolution machinery of §10.5b.9 — is withheld rather than silently computed from an assumed or defaulted regime. This is a stronger requirement than the warning alone: a confident classification produced from insufficient data is worse than no classification, and the engine should never be in a position to produce one.

How partial indeterminacy propagates, made explicit. Regime indeterminate withholds regime-dependent quantities specifically; it does not by itself say whether collapse assessment as a whole must be withheld too, and this needs to be stated rather than left for an implementer to guess. The correct behavior: quantities that do not depend on regime classification — the hull-breach inequality (§10.5a) and any other collapse-relevant quantity computable independently of which regime a container is in — remain reportable even while regime is indeterminate. What becomes indeterminate specifically is resolution: which pathway a confirmed collapse onset would follow (self-arrest, terminal lock, reorganization, or the alignment-typology outcomes of §10.5b.2.16) is regime-dependent machinery and is correctly withheld. The engine's output in this state is accordingly a conditional report — collapse pressure or onset indicated; resolution pathway indeterminate — not a blanket withholding of every collapse-relevant output, and not a confident resolution prediction manufactured from an unavailable regime classification. An implementer who reads "full collapse-resolution machinery is withheld" as requiring the entire collapse assessment to be suppressed, including independently-computable quantities like hull-breach, is applying the rule more broadly than intended.

A fourth engine state: regime unstable. Bounded oscillation between entropy regimes on successive observations is real and expected in some containers (§6.9) and is not, by itself, an error to be smoothed away or forbidden. But an engine that reports a confident H or M label on every single observation, even while that label is flipping on essentially every successive reading, is manufacturing false confidence about a genuinely unstable classification rather than reporting the instability itself. Where the regime label changes more than a fitted κ_flip times within a trailing window of κ_window observations, the engine returns Regime unstable: boundary-oscillatory, a fourth state distinct from a confident H or M classification and distinct from regime indeterminate (data unavailable) — here the data is available and comparable, the classification procedure works, and the container's own true state is genuinely alternating faster than a single confident label can honestly represent. This is itself a real, reportable observation about the container, not a failure state: rapid alternation across the entropy boundary is informative in its own right (§6.9's bounded oscillation, occurring unusually fast) and should be surfaced as such rather than resolved into whichever label happened to win the most recent observation. Regime-dependent downstream quantities are withheld during Regime unstable exactly as during regime indeterminate, since neither state licenses treating either label as the container's settled classification.

Precedence over composition defaults, made explicit. $\omega\_meta\hat{}up(t)$ and $\omega\_meta\hat{}down(t)$ (below) have their own, independent default rule for when composition data is not directly measured, unrelated to whether O_p itself is available. This ordering must be enforced: regime indeterminate status is checked first, and if it holds, no regime-dependent quantity may be computed at all — not from measured composition, and not from ω_meta's own default — regardless of whether that default is otherwise ready to use. A composition default existing and being computable is not authorization to proceed; it is not a back door around the observational requirement above. Only once regime classification has actually succeeded does whether ω_meta was measured or defaulted become the relevant question, governed by the caution below.

Causal normalization, required for every live-tracker variable. Several quantities in this document are normalized against an adapter's "historical range" — $d\_A(t)$'s $\tilde{S}(t)$, $\delta\tilde{R}(t)$,

$\tilde{F}(t)$ (§10.3), and the hull-breach inequality's $\tilde{S}(t)$ (§10.5a) among them. "Historical range" is ambiguous between two mathematics that produce different trajectories: a retrospective range computed over the full dataset, which leaks future observations into a normalization applied to an earlier state, and a causal range computed only from information available no later than the current time t. Retrospective normalization is acceptable for forensic, after-the-fact auditing of a completed trajectory, where the full record is legitimately available. It is not acceptable for a live tracker, where it would let the tracker's read of the present silently depend on data that has not yet occurred. This is the same discipline already required of Ψ_loyal(t)'s estimation window (§10.5b.18.12): any historical normalization used in live inference must be computed from a fixed, pre-specified baseline established before deployment, or from information available no later than the current time t. Future observations may not enter the normalization of an earlier state. This rule applies to every normalized quantity entering live collapse machinery, present or future, not only the ones named above — a retrospective auditor and a prospective live tracker must not silently be running future observations into the normalization of an earlier state.

Which causal range, made explicit. "Computed only from information available no later than t" is satisfied by more than one construction, and these produce genuinely different trajectories: an expanding range (every observation up to t, monotonically widening as more data arrives), a rolling window (only the trailing W observations), a fixed calibration range (established once from an initial dataset and frozen thereafter), or an adaptive range with explicit historical restatement (the engine recognizes when its own scale has shifted and revises past normalized values accordingly, rather than leaving them silently stale). All four are causal in the sense already required; they are not interchangeable, since a rolling window can make the same raw trajectory look different from an expanding one, and a new extremum can otherwise cause a normalized value's meaning to shift under previously-reported observations without any change to the underlying data. The default for this document's own computational instantiations is the expanding range: monotonically widening, never contracting, and never revising a previously-reported normalized value once computed, which is the simplest construction consistent with the causality requirement and avoids the retrospective-jump problem a rolling or adaptive range can otherwise introduce. An adapter substituting a rolling window or an adaptive range for its own good reasons (bounded memory, a genuine belief that old extremes should stop constraining the present scale) should state that choice explicitly, since it changes what "normalized" means for every value already reported, not only future ones.

Minimum range width, required alongside causality. Causal normalization alone is not sufficient for a live tracker to behave sensibly: where a quantity's true trailing range is small and genuinely flat, ordinary min-max normalization against that thin range amplifies routine noise into apparent full-scale [0,1] swings, since any small new extremum instantly redefines the whole scale. This is a distinct failure mode from future leakage — it does not violate causality, but it produces a normalized signal dominated by noise rather than by genuine change. Every causal normalization in this document must therefore also enforce a fitted minimum range width: where the trailing max minus min falls below that floor, widen the range symmetrically around its own trailing mean rather than dividing by a near-zero denominator. This keeps a small genuine fluctuation from being read as a dramatic relative change, without reintroducing any dependence on future data.

Warning: a defaulted ω_meta is not a measured one.

The default of 1 given above for ω_meta^up(t) and ω_meta^down(t) exists so the engine has something to compute with in the absence of adapter data, not because 1 is a plausible value for a container already flagged as high- or medium-entropy by other means. ***Engine rule (revised from a prose warning to an enforced behavior):*** where regime classification (σ_o, σ_p, §10.5b.1) indicates a non-baseline regime and either required ω_meta channel is unmeasured, the engine does not silently substitute the default of 1. It instead returns σ_valve(t)

(§10.5b.18.6) and F_demand(t) (§10.5b.18.8) as explicitly low-confidence or indeterminate, rather than computing a definite value under the same false assumption — an honest, symmetric meta-representation layer — that Definition 6.1's revised independence clause exists to rule out for a non-baseline container. The default of 1 remains available, and is the correct default, only where regime classification itself indicates the low-entropy baseline; it is withheld precisely where it would be most misleading. A regime classification inconsistent with a defaulted composition should be treated as a signal to seek direct composition data before trusting the valve-symmetry diagnostics for that container.

15.3 Engine Reference — Every Derived Quantity, One Formula Each

Organized by originating Part. Every entry in the rightmost column is a free parameter analogous to a, b, c, d, e, r in the companion FDFM document — not guessed, not fixed by the theory, and exactly what the Bayesian state-space layer exists to fit once real data and a chosen system are in hand.

Parts 4–6: Geometry and Internal Dynamics

Symbol	Meaning	Formula	Defined in	Fitted params
g_t	history-dependent metric	g(M_t)	§4.10	—
D_ij	pairwise divergence	smoothed Jensen–Shannon	§6.3	ε
Π_f	feedback permeability	logistic(κ_Π·[corrective/error − lag])	§6.5	κ_Π
I	informational integrity	exp[−α·ΣD_ij]	§6.6	α
S(t)	informational stress	recursive exponential-kernel update	§6.7	λ
ℱ	feedback operator	stochastic gradient flow, grad_{g_{C,t}}Φ_t(P|P_m,M), forced by F_conf, F_boom, F_demand; projected to simplex each step	§6.4	γ_ℱ

Part 5: Recursive Containers

Symbol	Meaning	Formula	Defined in	Fitted params
A	aggregation (parent state)	commensurate then resource-weight-pool	§5.2	—
N_{R,i}	resource normalization	R_i / R_i^max	§5.6	—

Part 7: Resource Dynamics

Symbol	Meaning	Formula	Defined in	Fitted params
W(t)	informational work (total)	W_adaptive + M_maint, jointly clipped to R(t)	§7.3	κ_W, p
M_maint(C,t)	maintenance functional	M_base + κ_M·Size(C)^ν	§7.5	κ_M, ν
Γ_R	resource recovery	ρ·(R_target − R)	§7.7	ρ, R_target
δR(t)	resource debt (pre-recovery)	∫(W − P_prod − E_exch)ds	§7.9	—

Part 8: Adaptive Network Dynamics

Symbol	Meaning	Formula	Defined in	Fitted params
Ξ	edge existence	freq × min(Π_f,i, Π_f,j) vs θ	§8.3	θ
Λ_ij	network coupling	γ_net·W_ij·Π_f,i	§8.5	γ_net
W_ij(t)	edge weight update	β_net·success − γ_net·ignored; dynamics per Γ_W(I(P),Π_f,R,M) (§8.7), general form Γ_W(P,R,M,Π_f) (§9.8)	§8.7, §9.8	β_net, γ_net
R_N, N	network resilience/coherence	λ_2(Laplacian) × mean(Π_f), normalized	§8.9	—
ρ(j)	coupling persistence	logistic(ρ_0(j) + κ_desp·F̄_i(t) + κ_persec·Ψ_loyal(j,t)), F̄_i(t) = 1−exp(−λ_F·ℱ_i(t))	§8.6a, §10.5b.11b, §9.6a.1, Part 8 Patch I	κ_desp, κ_persec, λ_F
X_{i,j}	extraction flux	χ·Λ_ij·ρ(j)·R_j·𝒱(j,t)·(1−κ_self·Ψ_total(j)), χ scaled by (1+κ_χ·Z_hun)	§8.6a, §8.6a.4, §8.6a.5, §10.5b.19.1	χ, κ_χ, κ_self
𝒱(j,t)	target vulnerability	1−[w_Ψ·Ψ_total(j)+w_d·(d_A(j)/d_A^max(j))] clipped [0,1]	§8.6a.4	w_Ψ, w_d
Ω_pool(t)	shared external resource pool	γ_pool·(Ω_pool^max−Ω_pool)−Σ_k D_k(t)	§8.5a.1	γ_pool, Ω_pool^max
D_i(t)	competitive draw	min(demand_i, share_i·Ω_pool), share_i=effort_i/Σ_k effort_k	§8.5a.1	—

Symbol	Meaning	Formula	Defined in	Fitted params
E_embed(i,t)	positional embedding	adapter-supplied; defaults to local low/medium-entropy neighbor density	§8.6a.7	—
Σ_passive(i,t)	passive encounter rate	κ_passive·E_embed(i,t)·κ_density(i,t)	§8.6a.7	κ_passive
𝒜(j,t)	target attractiveness	𝒱(j,t)·R_j(t)	§8.6a.8	—
Θ_threat(i,j,t)	threat signal	max(0, 𝒱_expected(j,t_0)−𝒱(j,t))	§8.6a.9	—
Select(i,t)	target selection among search candidates	argmax_k[w_prox·Prox + w_type·𝒱_{i→k}]	§10.5b.1.1	w_prox, w_type
G(t)	grounding	A_acc(k,C,t)·[1−D(k,C_k)]; undefined if no independent external party available	§6.3.2	—
κ_capture(j)	degree of capture at genesis	v_threshold − v_gen(j,t_0)	§10.5b.2.2	—
Perceived_danger(j,i,t)	external threat perception	A_acc(j,i,t)·Danger_true(i,t)	§10.5b.2.3	—
ℛ(t)	receptivity to corrective feedback	1 − (fraction of official/operational/metamessengers in mutual false-belief agreement)	§10.5b.1.2	—
Anchor(k→C,t)	unidirectional stabilizing feedback	adapter-supplied; distinct from bidirectional Λ_kC	§10.5b.2.6	—
Inject(k→{o,p},t)	metamessenger-sourced injection forcing	adapter/engine-fitted; internal analogue of F_conf, F_demand	§10.5b.2.9	—
χ_struct(t), χ_flow(t)	network structural centralization, resource-flow disproportion	graph-structural measure; benefit-vs-contribution measure	§8.5b.1	—
d_A^prey(t), θ_coevolve	grid-level prey adaptive possibility, coevolution floor	aggregate d_A across prey population; fitted threshold	§10.5b.2.8	θ_coevolve

Symbol	Meaning	Formula	Defined in	Fitted params
ρ_resilience(j)	receiver-side resilience modifier	v_gen,effective = v_gen·(1−ρ_resilience)	§10.5b.2.11	ρ_resilience
Functional judgment accuracy	nonfunctioning-member classification validity	checked via G(t); persistence and population-variance as counterevidence	§8.5b.3	—
Emergency triage validity	when discarding is not an entropy signal	requires both R(t)/δR(t) near floor AND G(t) high for the judgment	§8.5b.4	—
Standing proxy divergence	Goodhart-style drift under sustained optimization	proxy-reality gap accumulates under standing (not bounded) optimization pressure	§8.5b.5	—
H_{i,j}, H_i	vented metabolic-failure heat	(1−η_i)·X_{i,j}, summed over j	§8.6a	η_i
Co(t)	cross-sectional common-cause co-movement	fraction of children shifting simultaneously vs. baseline	§8.9.1	θ_shift, Co_baseline
ℱ_i(t)	accumulated unvented heat	∫ H_i(s)·1[N_i(s)=∅] ds	§8.8a	λ_ℱ, ℱ_threshold
Drive_B(j,t)	medium-entropy launch drive	|Σ_k Λ_kj·û_kj|	§8.8a/10.5b.6	Θ_launch
K_j(t)	resolution kick	impulsive (B1) or decaying (B2) release of M_tip(j)	§8.8b	τ_diss
K_{j→k}(t)	propagated kick	K_j · W_jk/Σ_l W_jl	§8.8b	—
ε_align, θ_align	**discharge alignment tolerance**	**coupling trigger: |t_tip,A − t_tip,B| < ε_align and û_A·û_B > cos(θ_align)**	**§10.5b.11b.1**	**ε_align, θ_align**
A_first(C_new,C_k,t)	**first-contact attraction weight**	**ν_pref·Λ_int(C_k,t)**	**§8.3.1**	**ν_pref**

Symbol	Meaning	Formula	Defined in	Fitted params
f_NC(t)	**network-level nonconformist fraction**	**resource-weighted fraction of network with Ψ_loyal(t) > Θ_NC**	**§8.6a.2**	**Θ_NC**
P_persecute(t)	**persecution deficit signal**	**max(0, f_NC^target − f_NC(t))**	**§8.6a.3**	**—**

Part 9: Memory Dynamics

Symbol	Meaning	Formula	Defined in	Fitted params
Λ_int(C,t)	**internal cohesion**	**resource-weighted mean of Λ_ij over co-resident members of μ(C,t); defined as 0 when μ(C,t)=∅**	**§9.10.1**	**—**
g_C(t)	**structural grain**	**normalize[∫K(t,s)·Σ_k Λ_kC(s)·û_kC(s) ds]; undefined (χ_dir defaults to 1) if no coupling history**	**§4.1.1**	**—**
χ_dir(j,t)	**directional force modifier**	**1 + κ_dir·(1 −	g_j(t)·û_net(j,t)	)**

Part 10: Adaptive Stability and Collapse

Symbol	Meaning	Formula	Defined in	Fitted params
d_A(t)	adaptive degrees of freedom	d_A^max·exp[−β_S S̃ − β_R δR̃ − β_F F̃]	§10.3	β_S, β_R, β_F
d_A(C*)	recursive d_A aggregation	[Σw_i·d_A(C_i)]·(1−H(C*))	§10.3	—
R_g	adaptive rigidity	1 − d_A/d_A^max	§10.4	—
Hull breach	instantaneous breach signal	(1−I) + S̃ > θ_hull·Π_f	§10.5a	θ_hull

Symbol	Meaning	Formula	Defined in	Fitted params
Collapse onset	confirmed onset (event-confirmation rule)	breach sustained for τ_confirm (default), or k consecutive obs., or posterior threshold	§10.5a	τ_confirm
h(t) = V(t)	structural height / verticality	relaxation toward h_target(I(t)); V:=h	§10.5b.0	κ_h, ξ, h_min
W(t)	horizontality	1 − V(t)	§10.5b.0	—
χ_supp	geometric support ratio	adapter-supplied	§10.5b.0	—
Ω_hazard(t)	structural hazard rate	κ_Ω·V^ψ / χ_supp	§10.5b.0	κ_Ω, ψ
σ_o, σ_p	entropy regime classification	trailing-window variance of P_o, P_p	§10.5b.1	window length
v_gen(t)	genesis-branching signal	shock-transmitted or ambient grid field; content = parasitic entitlement, Def. 10.5b.2.1	§10.5b.2	v_threshold
F_conf	confrontation trigger	ζ·Π_f·D_target, active near Π_c crossing	§10.5b.3	ζ, Π_c
κ(t)	coherence factor (Flip A only)	1/(1+D_residual)	§10.5b.8	—
M_tip(A)	tipping momentum, Flip A	∫|F_i|·κ dt	§10.5b.8	—
M_tip(B1), M_tip(B2)	tipping momentum, Flip B	∫(μ_o+μ_m)·Drive_B·χ_dir dt, ∫(μ_p+μ_m)·Drive_B·χ_dir dt	§10.5b.7, §4.11.2	κ_dir
M̃_tip	normalized tipping momentum	M_tip / M_tip^ref (per route)	§10.5b.9	M_tip^ref (per route)
M_lock^eff(t)	loyalty-extended self-arrest threshold	M_lock·(1+κ_Mlock·Ψ_total(t))	§10.5b.9, §4.11 area (M_lock ext.)	κ_Mlock
F_emerg(t), Θ_spike	emergency firewall reinforcement	κ_emerg·Ψ_loyal(t)·(Π_f(t)−Θ_spike)$_+$	§10.5b.14.2	κ_emerg, Θ_spike

Symbol	Meaning	Formula	Defined in	Fitted params
Outcome	tipping-event resolution	self-arrest / terminal lock / full reorg by M̃_tip vs M_lock, M_c	§10.5b.9	M_lock, M_c
μ_o, μ_p, μ_m	representation-layer mass weights	fitted or adapter-supplied; μ_p>μ_o in low-div-apparent	§10.5b.7	μ_o, μ_p, μ_m
Z(t)	resilience accumulator	s(t)·η_Z·Z(1−Z)·Drive_E, bounded to [ε_Z, 1−ε_Z], medium-entropy only	§10.5b.9a	η_Z, ε_Z
c_n, τ_n	backslide cost / duration	c_0(1−sZ), τ_0(1−Z) [medium-entropy]; F_conf(t_n) [high-entropy cost]	§10.5b.9a	c_0, τ_0
γ_ℱ(t)	resilience-modulated drift rate	γ_ℱ,0·(1+κ_Z·Z)	§10.5b.9a	κ_Z
Φ_track(t)	**tracking fidelity (horseshoe diagnostic)**	**max_{δ∈[0,δ_max]} corr_{s∈[t−w,t]}(‖v(s−δ)‖_g, D(s))**	**§10.5b.15.1**	**w, δ_max**
F_boom(t)	**boomerang / compulsive-closure forcing**	**κ_boom·Π_f(t)·1[D(t)<D_low]·(D_low−D(t))**	**§10.5b.16.1**	**κ_boom, D_low**
ω_meta^up(t)	**metamessenger composition, upward (truth) channel**	**adapter-supplied; defaults to 1 (baseline assumed) if not measured**	**§10.5b.18.2**	**—**
M_decep(t)	**self-deception maintenance cost**	**M_decep,int + M_decep,ext; int: κ_decep·(1−ω_meta^up)·D_op; ext: κ_decep,ext·Σ_j Λ_ij·𝟙[active target]**	**§10.5b.18.4**	**κ_decep, κ_decep,ext**
M_diverge(t)	general divergence cost	κ_diverge·D(t), D=D_op+D_om+D_pm	§7.9.1	κ_diverge
M_extract(i,t)	extraction performance cost	κ_extract·Σ_j X_{i,j}(t)	§7.9.1	κ_extract

Symbol	Meaning	Formula	Defined in	Fitted params
M_search(i,t)	search cost	κ_search·Σ_search(i,t)	§7.9.1	κ_search
ω_meta^down(t)	**metamessenger composition, downward (demand) channel**	**adapter-supplied; defaults to 1 (baseline assumed) if not measured**	**§10.5b.18.5**	**—**
σ_valve(t)	**valve symmetry index**	**min(ω_meta^up, ω_meta^down) / max(ω_meta^up, ω_meta^down); = 1 when both are 0**	**§10.5b.18.6**	**—**
F_demand(t)	**demand forcing (downward channel)**	**κ_demand·ω_meta^down(t)·D_op(t)**	**§10.5b.18.8**	**κ_demand**
Z_hun(t)	**extraction-hunger accumulator**	**dZ_hun/dt = η_hun(t)·Z_hun(1−Z_hun)·R_ext(t); η_hun(t) = η_hun,0·[1 + κ_recall·∫K(t,s)R_ext(s)ds / R_ext^ref]**	**§10.5b.19.1, §10.5b.19.2**	**η_hun,0, κ_recall**
φ_fb(i,j,t)	**external feedback channel fraction**	**adapter-supplied; defaults to ω_meta^up(j,t)**	**§10.5b.18.9**	**—**
A_acc(source,i,t)	**feedback accuracy ceiling**	**self: 1−D_om(i,t); third-party: κ_access+(1−κ_access)(1−D_op(i,t))**	**§10.5b.18.9.1**	**—**
κ_access(j,i,t)	**observational depth into target**	**adapter-supplied; defaults to 0 (official-only access)**	**§10.5b.18.9.1**	**—**
Γ_supp(i,j,t)	**external support flux**	**κ_Γ·Λ_ij(t)·max(0, R_j(t)/R_j^max − r_supp,min)**	**§10.5b.18.10**	**κ_Γ, r_supp,min**
F_ext,fb(i,t)	**coupling correction forcing, feedback edge**	**κ_fb·[Σ_j Λ_ij·φ_fb(i,j,t)·A_acc(j,i,t)]·(D_om,floor − D_om(i,t))$_+$**	**§10.5b.18.11, §10.5b.18.9.1**	**κ_fb, D_om,floor**
F_ext,supp(i,t)	**coupling correction forcing, support edge**	**κ_supp·[Σ_j Γ_supp(i,j,t)]·(D_om(i,t) − ε_healthy)$_+$**	**§10.5b.18.11**	**κ_supp**

Symbol	Meaning	Formula	Defined in	Fitted params
Ψ_loyal(t)	**metamessenger loyalty composition**	**adapter-supplied; defaults to σ(κ_loyal·Δω_meta^up(t; w_loyal))**	**§10.5b.18.12**	**κ_loyal, w_loyal**
μ_p^eff(t)	**loyalty-adjusted operational mass (Flip B split)**	**μ_p·(1 + κ_ML·Ψ_loyal(t))**	**§10.5b.18.13**	**κ_ML**
Ω_hazard^eff(t)	**loyalty-adjusted structural hazard (post-collapse)**	**Ω_hazard(t)·(1 − κ_ΩL·Ψ_loyal(t))**	**§10.5b.18.13**	**κ_ΩL**
Ψ_total(t)	**total convergence-favorability**	**ω_meta^up(t) + (1−ω_meta^up(t))·Ψ_loyal(t)**	**§10.5b.18.14**	**—**
Voluntary flood trigger	**self-initiated realignment condition**	**fires when Ψ_total(t) ≥ θ_flood and no active tipping window; drives Π_f(t) toward ceiling over τ_flood**	**§10.5b.18.16**	**θ_flood, τ_flood**
Despair signature	**official-convergence collapse, no redirection available**	**diagnostic co-occurrence: D_om → 0 during knock AND Γ_redir(t) ≈ 0**	**§10.5b.18.17**	**—**
A(t)	**control locus (operational share)**	**adapter-supplied where directly observable; defaults to 1 (operational in charge)**	**§10.5b.18.18**	**—**
δ(t)	**delegation type**	**categorical: staged / ceded; adapter-supplied or inferred from context (staged only under active §8.6a/§10.5b.11b coupling attempt; ceded otherwise when A(t)<1)**	**§10.5b.18.18**	**—**

Symbol	Meaning	Formula	Defined in	Fitted params
Φ_o(t)	**official self-model content**	**categorical: aligned / confidently-false / bewildered / aware-but-inert, per D_om band, M_o,t^false activity, unpatched-leak timing, and regime**	**§10.5b.18.19, §10.5b.18.23**	**—**
Override	**control-plane correction, ceded state**	**fires when δ(t)=ceded and [(1−I(t))+S̃(t)] crosses θ_ovr^eff, or dδR/dt crosses θ_ovr^eff while dR/dt<0; drives A(t)→1 over τ_ovr**	**§10.5b.18.20, §10.5b.19.5**	**θ_ovr, τ_ovr**
φ_fossil(t), φ_patch(t)	**historical residue / active-patch split of official content**	**φ_fossil = [1−ω_meta^up(t)] − φ_patch(t); φ_patch adapter-supplied or event-triggered**	**§10.5b.18.21**	**—**
D_attr(t′,t)	**causal attribution divergence**	**D(A_self(t′,t), A_true(t′,t)) via §6.3’s divergence**	**§10.5b.18.26**	**—**
A_cap(i,j,t)	**cross-container capture share**	**adapter-supplied; defaults to 0**	**§10.5b.18.28**	**—**
Θ_capture	**influence/control threshold**	**A_cap ≥ Θ_capture ⟹ total control; below ⟹ influence**	**§10.5b.18.28b**	**Θ_capture**
g(t)	**glimpse marker**	**binary tag on M_t at moments satisfying Ultimate Convergence briefly**	**§10.5b.0.9**	**—**

Symbol	Meaning	Formula	Defined in	Fitted params
F_glimpse(t)	**glimpse-recall forcing**	**κ_glimpse·∫K_glimpse(t,s)g(s)ds·(D_om(t) − D_om,floor)₊**	**§10.5b.0.10**	**κ_glimpse, K_glimpse**
Γ_glimpse(t)	**cumulative glimpse residue**	**dΓ_glimpse/dt = κ_G·g(t)·(1−Γ_glimpse(t)); conditional gate on permanent V(t) reduction**	**§10.5b.0.10**	**κ_G**
Σ_search(i,t)	**active search intensity**	**κ_search·max(0, M_maint+M_decep−P_prod)**	**§10.5b.11c.1**	**κ_search**
Λ_amb(t), Σ_search^eff(i,t)	**ambient cohesion / suppressed search reach**	**Σ_search·exp[−κ_amb·Λ_anch]**	**§10.5b.11c.2**	**κ_amb**
ζ_mirror(i,j,t)	**mirror-threat suppression**	**exp[−κ_mirror·min(D_op,i, D_op,j)]**	**§10.5b.11c.3**	**κ_mirror**
Λ_mutual, Φ_mutual(t), Γ_mutual(t)	**mutual anchor cluster density**	**cluster-averaged bidirectional φ_fb, Γ_supp over all pairs**	**§10.5b.13.1**	**Λ_mutual, κ_cluster**
Z_hab(t)	**habituation accumulator**	**dZ_hab/dt = η_hab·Z_hab(1−Z_hab)·R_conv(t)**	**§10.5b.19.3**	**η_hab,0, κ_recall,hab**
F_hab(t), θ_ovr^eff(t)	**habituation forcing / raised override threshold**	**κ_hab·Z_hab·D_om; θ_ovr·(1+κ_hab2·Z_hab)**	**§10.5b.19.5**	**κ_hab, κ_hab2**
D_chaos(j,t)	**multi-source chaos pressure**	**∫ Drive_B(j,s) ds over trailing window w**	**§10.5b.9d.1**	**w, Θ_disintegrate**

Part 11: Derived Observables

Symbol	Meaning	Formula	Defined in	Fitted params
C_A	adaptive capacity	d_A (φ = identity)	§11.5	—
F	fragmentation	Σ_k w_k·[D_0^(k) + κ_N·(1−N^(k))]	§11.7	w_k, κ_N
P_C(t)	collapse potential	sigmoid(κ_1(1−I)+κ_2S̃+κ_3(1−Π_f)+κ_4(1−d_A/d_A^max)+κ_5(1−N))	§11.10	κ_1..κ_5
Divergent(t)	severity-trajectory disagreement flag	𝟙[δR̃>θ_sev]·𝟙[trend(d_A)>θ_mild]	§11.10	θ_sev, θ_mild

Part 16: Further Considerations and Uses

Symbol	Meaning	Formula	Defined in	Fitted params
R_class(t)	**classification reliability ratio**	**σ_o(t) / [σ_o(t) + σ_p(t)]**	**§16.9.2**	**—**

Editorial note: The verticality operator V(t) and the genesis-branching signal v_gen(t) (§10.5b.2) are distinct quantities and should not be conflated despite the surface similarity of their names. Each tipping route (A, B1, B2) has its own M_tip formula, using that route's own weighting (coherence-weighted for A, mass-weighted for B), and §10.5b.9's normalization step is required before comparing M_tip across routes. ω_meta^up(t) and ω_meta^down(t) are listed as adapter-supplied per §15.2 rather than engine-derived; their table entries exist so that an implementer working from this reference alone, without separately consulting §15.2, still sees where they enter and what depends on them (M_decep, σ_valve, F_demand).

15.4 What Happens Next (Deliberately Not Done in This Document)

This document stops short of choosing Ω, writing the adapter for any specific system, or fitting any of the coefficients in §15.3 to real data — that work is domain-specific by construction (§15.1) and, per the scope of this revision, comes after the mathematics is settled, not before. The next step is choosing a system, defining its adapter (§15.2) concretely, and building the Bayesian state-space model that infers §15.3's fitted parameters from that system's data — mirroring the companion FDFM document's PyMC layer, generalized to RICD's larger parameter set and corrected so that every latent variable (I, S, d_A, P_C, and, in this revision, V, Z, Ω_hazard, and the valve-symmetry quantities) is computed by exactly one formula throughout, used identically in the generative model, the inference likelihood, and any downstream simulation.

Final Statement

Recursive Information–Container Dynamics advances the proposition that adaptive systems, despite their immense diversity, can be described within a common mathematical framework in which recursively organized information-processing containers evolve through the coupled dynamics of information, resources, memory, boundaries, and networks. Within this framework, persistence is the preservation of adaptive possibility, collapse is its contraction, and recovery is its expansion. Every quantity in that framework, as of this edition, is either computed from real data through a stated formula, supplied through an explicit adapter-supplied input, or explicitly returned as unavailable/undefined when its required evidentiary conditions do not exist (as with G(t) absent a genuinely independent external party, §6.3.2) — every symbol denotes exactly one quantity, the structural form a container takes is itself derived from the same informational dynamics that govern everything else in the theory, and the meta-representation layer's two directions of transmission are now tracked as two independent, comparably-measured quantities rather than one — the theory and the tool are now the same object.

Part 16. Further Considerations and Uses

This part collects several uses of the framework beyond forward-looking collapse tracking, all built from machinery already established in Parts 1–15 rather than new dynamical layers. §16.1–§16.6 develop the first of these — precise, recursive collapse-timing prediction — in full. §16.7 onward extend the same recursive procedure to uses the earlier parts of this document do not otherwise cover: explaining collapses and unusual decisions that have already happened, assessing whether a system's surrounding network helped or actively harmed it, and feeding what is learned back into the theory's own fitted parameters.

16.1 From Trend to Timing

A tracker built on Part 5's pooled aggregation (§5.2) can show that a macro-system is trending toward collapse, because pooling reflects the resource-weighted average of its children's states. It cannot, from pooling alone, show precisely when, because averaging is exactly the operation that hides a single child crossing hull-breach while its siblings remain in ordinary bounded oscillation — the "straw that breaks the camel's back" is invisible in an average until it has already moved the average. Precision requires examining children, and where warranted their own children, individually rather than only through their parent's pooled summary.

This section formalizes that examination as a disciplined recursive procedure: how deep to audit, how a child's individual state should correct rather than simply supplement its parent's computed quantities, and how to distinguish a correction that merely sharpens an existing estimate from one that changes which branch of the theory governs a container entirely.

16.2 Audit Depth and Its Stopping Criterion

Recursive audit cannot proceed indefinitely; a federal justice system audited down to the level of individual personal relationships has been audited past the point of relevance to the macro-system's own trajectory. The stopping criterion is therefore not a fixed depth, but a relevance test applied at each level.

Definition 16.1 (Direct Relevance).

For a candidate child container C_child of a macro-system C^(top), let DirectRel(C_child, C^(top)) be an adapter-supplied boolean: true if C_child's own internal dynamics measurably bear on C^(top)'s trajectory through the recursive coupling operators of Part 5 and Part 8; false if C_child's further subdivision would examine components whose behavior affects only matters internal to C_child, not C^(top). (Example: for a federal justice system as C^(top), regional offices, courts, and prisons typically satisfy DirectRel; individual personal relationships or bodies within them typically do not — subdividing further gains no timing precision on the macro-system and exceeds the audit's proper scope.)

Definition 16.2 (Audit Depth).

The audit recursively examines C^(top)'s children, and their children, down to the smallest container along each branch for which DirectRel still holds — the stopping depth L*(C) need not be uniform across branches of the recursive tree; one sub-lineage may warrant deeper audit than a sibling.

16.3 The Weakest-Link Correction

Pooled aggregation's central limitation for precision timing is that it averages hull-breach margin across children rather than tracking the least stable one. The correction below restores that visibility without discarding the pooled computation, which remains useful as a general-trend indicator in its own right.

Definition 16.3 (Local Margin).

For any audited container C_i, define its hull-breach margin directly from §10.5a:

*margin*_i(t) = θ_hull·Π_f,i(t) − [(1−I_i(t)) + S̃_i(t)].

Positive values indicate distance from breach; margin_i(t) ≤ 0 indicates C_i has crossed or is at hull-breach.

Definition 16.4 (Weakest-Link Margin).

For a parent C* with audited children {C_1,...,C_n} at whatever depth the audit reaches on each branch:

*margin*_WL(C*,t) = min_{i : w_i ≥ w_min} [margin_i(t)],

restricted to children whose resource weight w_i (§5.2) exceeds a materiality threshold w_min, so a negligible child cannot trigger a false alarm while any sufficiently material child's approach to breach is not averaged away. Recursively, where C_i is itself audited to a further depth, margin_i(t) is replaced by its own weakest-link margin from below — the correction propagates from the stopping depth up to the top in one pass.

Definition 16.5 (Corrected Effective Margin).

The precision gain over pooling alone:

*margin*_eff(C*,t) = min[margin_pooled(C*,t), margin_WL(C*,t)],

where margin_pooled is §10.5a applied directly to C*'s own aggregated state. C* is flagged as approaching hull-breach if either its aggregate state or any sufficiently material child's state is close to breach — the condition pooling alone cannot detect.

16.4 Upward Propagation: Refinement versus Reclassification

Not every audit finding changes the parent's computation the same way. Distinguishing the two cases below matters because they carry different implications for how much the parent's prior trajectory estimate should be trusted going forward.

Definition 16.6 (Audit Correction).

Let Δ_audit(C*,t) denote the change to C*'s own computed quantities (P*, R(C*), d_A(C*), and downstream observables) that results from replacing assumed or estimated child-level inputs with directly audited ones, propagated back up through §5.2's aggregation and §10.3's recursive d_A aggregation. Two distinct cases:

- Refinement. The audited values fall within the range the > > parent-level computation had already assumed (e.g., an estimated > > Size(C_child), §7.5, is replaced by a directly measured one > within > prior uncertainty). Δ_audit narrows parameter > uncertainty without > changing functional form.
- Reclassification. The audited values fall outside prior assumptions > > in a way that changes which regime-dependent formulas apply — > > most consequentially, a child previously assumed low-entropy > > baseline is found, on audit, to already be > high-entropy > (§10.5b.1). This is not a parameter > adjustment; it changes > which branch of the theory governs that > child, and therefore > what it contributes to the parent's > aggregate.

A reclassification finding should be treated as informative beyond the single child it was found in: if one child was mis-classified under the parent's prior assumptions, siblings audited only at lower resolution warrant re-examination before the parent's corrected margin is trusted.

16.5 Procedure

The audit is computed bottom-up: at each branch, audit to stopping depth L*(C) first; compute local margins and regime classifications at that depth; propagate upward one level at a time via §5.2/§10.3's aggregation with the weakest-link correction of §16.3 and the Δ_audit correction of §16.4; repeat until the top is reached. This is a single post-order traversal of the recursive container tree.

16.6 The Precision–Depth Tradeoff

Auditing to depth k+1 rather than k should narrow the uncertainty in predicted hull-breach timing at the top level — but the size of that gain is domain-specific and empirical, not derivable in closed form from the theory alone. This document does not commit to a specific precision-per-depth-level functional relationship; an implementation should treat it as a monitored quantity, informing where additional audit depth is worth its cost, consistent with the empirically-open items already flagged in §14.2.

This tradeoff has a practical corollary worth stating directly: audit depth is a resource-allocation decision, not a purely technical one, and §16.2's DirectRel criterion is what keeps that allocation bounded — a system audited past the point of direct relevance to its own macro-trajectory is spending audit resources without a corresponding gain in the precision §16.1 exists to deliver.

The Auditing Principle.

margin_WL is a conservative early-warning statistic, not an assertion that every material child-level breach necessarily propagates to the parent's own collapse timing: a child can be materially weighted, individually approaching breach, and still weakly coupled upward, or can collapse and remain contained without materially affecting C*'s own trajectory. The current formula takes a minimum over materially-weighted children's margins without weighting by each child's actual upward coupling or propagation capacity (Λ_{i→C})*, so it should be read as: recursive auditing recovers a possible-instability signal the aggregate average alone would miss, not a proof that the least stable subsystem determines the parent's timing. This is deliberately conservative rather than causally precise — flagging early on a signal that may overstate risk is the intended failure mode, not silently missing a real one — and does so only as deep as each branch's continued relevance to the macro-system justifies. A future revision could make propagation explicit by weighting each child's margin against its own $\Lambda_{i\to C^*}$; this document does not build that now, absent a substantive reason the conservative statistic is insufficient.

16.7 Forensic (Backward-Facing) Auditing

The recursive procedure of §16.1–§16.6 is stated for live monitoring, but nothing about it requires the data to be current. Applied to historical records instead of a live feed, the same bottom-up, weakest-link procedure reconstructs which specific subsystem actually crossed the modeled hull-breach criterion first in a collapse that has already happened, and when — reconstructing the observed sequence of modeled hull-breach events and generating a testable candidate causal ordering of an event that already occurred, rather than a live prediction of one that has not. Consistent with §16.6's own limitation (the weakest-link margin does not model upward propagation capacity $\Lambda_{i\to C}$, and a materially-weighted child can breach and remain contained), this reconstructed ordering is a falsifiable candidate sequence, not an established causal chain: which subsystem crossed the modeled breach criterion first is not the same claim as which subsystem was causally responsible for the parent's collapse, and the forensic procedure here supports only the former directly. A future revision incorporating $\Lambda_{i\to C}$-weighted propagation, per §16.6's own noted extension, could promote this to a genuine causal claim; absent that, it should be reported as a candidate ordering awaiting corroboration by independent evidence. §16.2's audit-depth criterion and §16.4's refinement/reclassification distinction both apply unchanged: a forensic finding may simply sharpen a previously estimated parameter, or it may reveal that a subsystem's assumed regime classification in the period before the collapse was wrong — exactly as a live audit would, just discovered after the fact rather than before it. The three uses that follow (§16.8–§16.9) are specific applications of this backward-facing procedure; §16.10 addresses what to do with what they find.

16.8 Attributing Self-Sacrifice

§10.5b.11a's mutual-coupling route allows two comparably-struggling containers to redirect a shared discharge asymmetrically, so that one absorbs the impact and the other reaches full reorganization. Some of that asymmetry is explained directly by comparing the two containers as they stood at the time — their relative resource levels, mass, or adaptive capacity. Often, it is not fully explained that way.

Definition 16.7 (Sacrifice Asymmetry Decomposition).

For a mutual-coupling event in which the discharge asymmetry favors one container's survival, decompose the asymmetry into a strength-explained component (recoverable from the two containers' comparative state at the time, by the ordinary mass/resource comparisons already used elsewhere in this document) and a remainder — whatever the strength comparison does not account for.

Hypothesis.

The remainder, where one exists, is hypothesized to be attributable to the sacrificing container's own history of internal messaging — the relationship among its official story, its actual behavior, and its own self-monitoring, in relation to its neighbor — in the period before the event. The backward-facing procedure below provides a means of determining whether an identifiable pre-event messaging pattern accounts for that remainder. This hypothesis holds regardless of which regime either container occupied at the time, and regardless of what caused the sacrificing container’s own difficulty in the first place; nothing in this framework restricts which pairing of regimes can produce this outcome, or requires the sacrificing container’s trouble to have originated with the neighbor it sacrifices for.

This document does not attempt to enumerate every specific historical pattern that could be responsible in a given case — doing so would mean working out every real-world instance in advance, which a general theory cannot do. §10.5b.14’s account of a small, independently-acting part of a container’s self-monitoring apparatus is one documented pattern capable of producing this kind of outcome, but it is offered here as one known example, not as the mechanism; other patterns, not yet catalogued, may be responsible in other cases. What this document commits to is only that the explanation, where one exists, is findable in that container’s own messaging record, using the backward-facing procedure of §16.7.

Confirmation standard.

An attribution should be reported as confirmed only when the explanatory pattern is independently identifiable in the container’s messaging record from before the sacrifice occurred — not fitted to that record after the outcome is already known. Where a candidate explanation cannot be shown to have been present and detectable prior to the event, it should be reported as a hypothesis consistent with the outcome, not as a confirmed mechanism. This standard exists specifically to keep an explanation that merely fits a known outcome from being reported with more confidence than it has actually earned.

16.9 Assessing the Surrounding Network's Role

A container’s own internal dynamics can indicate it was on track to recover — its official story, its actual behavior, and its self-monitoring converging in the manner §10.5b.9 associates with successful reorganization — and yet the container fails to recover regardless. This section addresses how to determine whether that gap is attributable to the container itself, or to active harm from its surrounding network during the attempt.

Definition 16.8 (Recovery Baselines).

For a historical container whose pre-collapse internal dynamics indicated it was on track for full reorganization, define two comparison baselines for the outcome that should have resulted:

- the isolated-prediction baseline: what the container’s own dynamics > > predicted in isolation, ignoring all neighbors entirely;
- the neutral-network baseline: what should have resulted had the > > surrounding containers behaved neutrally during the recovery > > window — neither extracting from it nor striking it, but also > > not anchoring or redirecting on its behalf.

Definition 16.9 (Active-Harm Signature).

Where a container’s actual outcome falls short of both baselines, examine the extraction flux and striking behavior of its neighbors (§8.6a, §8.8a — already defined for live use, applied here retrospectively) specifically within the window during which the container was attempting its recovery. Sustained or intensified hostile activity within that window — continued extraction, continued striking, or active refusal of an available redirection channel (§10.5b.11a) — rather than the simple absence of assistance, is the signature that distinguishes active predation by the surrounding network from ordinary neglect as the explanation for a recovery that should have happened but did not.

The distinction matters beyond attribution in any single case: a container failing despite favorable internal dynamics because of neglect and one failing because of active, sustained hostility during its recovery window point to different lessons for the systems around it, and only the retrospective comparison against both baselines, within the correct window, can tell the two apart.

Proposition 16.9.1 (The Decoy Check).

Before attributing a recovery failure solely to the audited container's own dynamics, or to a single identified hostile neighbor, examine whether a quieter neighbor coupled to the container during the recovery window shows elevated outbound extraction flux (§8.6a's $X_{j,k}$) despite presenting calm official-layer statistics. A high-entropy neighbor is, by regime definition (§10.5b.1), the one most likely to escape scrutiny in exactly this kind of review, since its calm presentation is the regime's defining signature, not evidence against involvement. Where such a neighbor is found, the visible, distressed neighbor originally suspected of causing or accelerating the collapse may instead be a co-victim rather than a cause, and the calm neighbor's presentation should be read as consistent with, not contrary to, active involvement. This check should be applied as a standing step in every use of Definition 16.9, not only where a hostile neighbor is not otherwise found, since a hidden predator's chief advantage is exactly that a visible neighbor gets blamed first — the same mechanism, viewed from outside, as §10.5b.17's internal binary-trust inversion.

16.9.2 The Classification Reliability Ratio

Definition 16.9.2 (Classification Reliability Ratio).

$R_{class}(t) = \sigma_o(t) / [\sigma_o(t) + \sigma_p(t)] \in [0,1]$,

the fraction of a container's total observable variance, across both channels, contributed by the official layer alone. This requires both σ_o and σ_p already be measured to compute — it is an auditor's diagnostic, built for someone with access to both channels per §15.2's mandate, not a substitute for obtaining O_p in the first place.

Proposition 16.9.2 ($R_{class}(t)$ Locates the Two Failure Directions).

$R_{class}(t)$'s three characteristic values correspond directly to §10.5b.1's regime classification, and its two extremes mark opposite failure modes for an observer relying on official signal alone:

- Low-entropy baseline: $R_{class}(t) \approx 0.5$, no channel dominant.
- High-entropy (official-shielded): $R_{class}(t) \to 0$. Official variance > is a small fraction of the total; an official-only observer sees > almost none of what is actually happening, systematically > undersampling the container's real instability.
- Medium-entropy (official-exposed): $R_{class}(t) \to 1$. Official variance > dominates the total; an official-only observer sees most of the > container's true variance but misattributes its source, reading > operational turbulence that isn't there and missing operational > stability that is.

$R_{class}(t)$ near either extreme is precisely where official-only observation is most dangerous, for opposite reasons in each direction — near 0, the danger is missing the instability entirely; near 1, the danger is correctly seeing instability but locating it in the wrong layer. This gives §15.2's existing warning and §16.9.1's decoy check a shared, continuous measure rather than two separate qualitative cautions: a low $R_{class}(t)$ should heighten suspicion of exactly the concealment §16.9.1 checks for, while a high $R_{class}(t)$ should heighten suspicion of exactly the misattribution §10.5b.1's regime account already warns against, and both readings are available from the same single ratio.

16.10 Feeding Findings Back Into the Model

Findings from §16.7–§16.9 will accumulate across many independently-studied historical cases, potentially examined by different researchers working on different domains. A simple procedure keeps those findings usable as evidence for the theory itself, rather than isolated case reports:

- Record each finding — an attribution (§16.8), a network assessment > > (§16.9), or any parameter estimate arising from a forensic > audit > (§16.7) — against the specific quantity or functional > form in > this document it bears on.
- Where independent cases agree on a finding, treat it as evidence for > > updating the relevant fitted coefficient through the Bayesian > > state-space fitting layer already described in §15.4, rather > than > adjusting a coefficient on the strength of a single case.
- Where a finding cannot be reconciled with an existing functional > > form under any choice of its coefficients, treat that as evidence > > that the functional form itself — not merely its constants > — > needs reconsideration. This is a structural revision, in > the same > sense §16.4 already distinguishes reclassification > from ordinary > refinement for live audits, and should be flagged > and reported as > such rather than forced into a parameter fit > that does not really > hold.
- Maintain a running case log per domain. A single case rarely > > justifies revising the theory; a consistent pattern across several > > independently-studied cases is exactly the evidence this > framework > is built to accumulate toward that end.

16.11 Structural Baseline Tracking

Distinguishing a genuinely healthy, stable container from a frozen or compulsively-closed one (§10.5b.15's Axiom C5) requires a baseline to compare against — and, as with most comparisons in this document, "healthy" is not an absolute, domain-independent value but something established relative to comparable containers of the same general type and to the specific container's own history, including its own pre- and post-reorganization periods.

Definition 16.10 (Structural Baseline).

For a container C, define its structural baseline as a resource-weighted average of $V(t)$ (§10.5b.0) and related quantities (Ω_{hazard}, χ_{supp}) computed over a reference set: either a population of comparable containers of the same adapter-defined type, or C's own historical trajectory, itself segmented at its own recorded hull-breach crossings and reorganization events — §16.7's forensic procedure supplies these segmentation points directly, where available.

This document does not commit to a single closed-form procedure for combining population and self-history baselines, or for weighting recency against long-run history within the latter — this is flagged explicitly as an area needing further development, structurally similar to the way §16.6 leaves the precision-depth relationship as a monitored empirical quantity rather than a fixed formula. What this section commits to is only that such a baseline is necessary before Axiom C5's tracking-fidelity diagnostic (§10.5b.15.1) can be meaningfully applied to a real case, since "pinned near its floor" and "pinned near its ceiling" are only meaningful relative to a container's own typical range, not a universal scale.

A related observation for interpreting a baseline once computed: a container moving from high toward medium entropy characteristically looks worse to an outside observer for a sustained period before it looks better, since §10.5b.9a's backslide dynamics and the surface-level turbulence of ongoing integration are not the same thing as continued decline. Whether an observer trusts an eventual low-entropy outcome, once reached, may depend heavily on how well they knew the container during that harder-looking middle stretch — a real source of disagreement between observers working from the same eventual outcome but different historical vantage points, worth anticipating rather than treating as a data error.

The Further-Uses Principle.

This framework's recursive and messaging-based machinery does not stop applying once a collapse has already happened, or once a decision looks inexplicable in isolation. The same tools that predict instability forward can explain it backward — including decisions, such as self-sacrifice, that appear to have no accounting until the deciding container's own history is actually examined — and the accumulated findings of that backward-looking work are what the theory's own fitted parameters, and where necessary its functional forms, should ultimately be built from.